

CANCER MONOGRAPH

(Compendium of ICMR's Cancer Research Activities)



Division of Non Communicable Diseases

Indian Council of Medical Research,
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2019

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स्वास्थ्य एवं परिवार कल्याण मंत्रालय एवं

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Foreword

I am delighted to write this Foreword for Cancer Monograph. This monograph outlines the data and evidence-based scientific knowledge on Cancer in India. The ICMR as the apex body realized that there was a need for a compendium that could give insight into the cancer research in India and help in providing government authorities with evidence-based scientific opinion on cancer. This monograph is an attempt to solve the overall quandary of cancer in Indian context. Different authors, with expertise in their fields have contributed towards fruition of the monograph. Much of its content is derived from work undertaken on cancer at different ICMR laboratories. Each writer deserves appreciation and acknowledgement for their dedicated review, revision and efforts towards its successful completion.

The aim of this publication is to succinctly provide information on incidence, leading sites, mortality, treatment and other measures of cancer burden in India. As we release this monograph, I envision that it would serve a variety of audiences. Not only will it be a guide for policymakers in health and related fields, but will also be of interest to health ministries and academic institutions and, more generally to oncologists/researchers and other health professionals who need to be aware of statistics and developments in cancer control. The media, educators and members of the public with an interest in cancer may also find this monograph valuable. I am sure this will stimulate further research on cancer in our country.



Balram Bhargava

(Dr. Balram Bhargava)

Preface

Cancer. This is not a word but an emotion, as it conjures up deep fear in people and has become synonymous to trauma, stress and pain because the constraints and difficulties that one faces after falling prey to the disease outweigh the opportunities to cure and relieve this suffering. Majority of the cancer patients live in the developing world. According to the World Health Organization (WHO), death from cancer cases in India is projected to rise to 13.1 million by the year 2030 and the burden is expected to increase further. There is thus a very urgent need to not get overwhelmed by the figures but to plan concrete programs/policies for prevention, treatment and management of cancer.



This monograph was conceptualized when the Indian Council of Medical Research, as the apex body, recognized the need for a compendium that could give insight into the cancer research in India and help in providing government authorities with evidence-based scientific opinion on cancer. This monograph is an attempt to solve the cancer conundrums in Indian context and tries to fill the gaps in literature by offering views of national experts on the subject.

Divided into nine chapters, this monograph, is a compilation of work that gives overview of cancer burden in India, the leading sites, major causes and research being undertaken. The major impetus on cancer research in India was in the early 1980s. Chapter 1 & 2 highlight how as first step; data collected through the Population Based Cancer Registries (PBCRs) across different regions in India was used to identify leading sites of cancer among males and females. Subsequent NCRP data has demonstrated a wide demographic variation in incidence of cancer in India. Cancer in North-East region has garnered special mention in Chapter 5 as this region has recorded highest incidence rates of cancer in the country because of peculiar dietary habits and lifestyle. The role of Tobacco related products as the single largest but preventable cause of various cancers is discussed in Chapter 6. The role of diet in causation of cancer and also the preventive aspects have been described in Chapter 8. The monograph concludes with Chapter 9 which highlights extramural research by ICMR across the country by setting up centres for advanced research, promoting research in different medical colleges and initiating Task Force on cancer.

I believe that this document will provide useful insight to students, researchers and oncologists; and will help direct future policies for cancer prevention and control programs in India as it contains many interesting facts and is also a source of new and intriguing questions for translating knowledge into action.



(Dr. R.S. Dhaliwal)
Head, NCD

Acknowledgement

Cancer has been a priority area of research for the Indian Council of Medical Research (ICMR). Cancer research received a boost in early 1980s, when besides etiological aspects, the work was initiated on assessment of magnitude of cancer as well as on cancer control related activities. Application of leads from research projects has been a major plank for cancer research and control activities. The Indian Council of Medical Research realized that there was a need for a concise monograph giving conspectus of cancer in Indian context. This monograph of cancer is a concerted outcome of efforts made by experts of varied disciplines of oncology across the nation. The cancer research activities supported by ICMR and undertaken in the form of centrally commissioned projects, adhoc research schemes and major research programmes undertaken by ICMR institutes have now been compiled in this document.



I would like to take this opportunity to thank Dr Balram Bhargava, Secretary, Department of Health Research and Director General, Indian Council of Medical Research for taking special interest and understanding the need for this compendium. I am indebted to Dr VM Katoch and Dr Soumya Swaminathan, Ex-Secretary, Department of Health Research and Director General, Indian Council of Medical Research for their encouragement in taking up this initiative. I am grateful to Dr RS Dhaliwal, Head, Division of Non Communicable Diseases, for his motivation and support in finalizing the document. The suggestions and inputs provided by all the expert members from time to time are deeply appreciated. It is my pleasure to acknowledge the dedication and determination of each author who worked tirelessly in completion of this document. The background information provided by Dr Kishore Chaudhry is gratefully acknowledged. My sincere thanks to research staff –Ms Ojasvi Mehta, Ms V Sucharita, Mr Rahul Sharma and Ms Jyoti Sharma for their efforts in compiling the information and editing the document.

The purpose of this document is to provide a consolidated ready reference of the work supported by ICMR in cancer research till 2016. All efforts have been made to compile available information. However, some individual projects might have been missed. It is hoped that Cancer Monograph would help researchers for planning future areas in cancer research and would also provide a database for the policy makers for amendments and resource allocation.



(Dr. Tanvir Kaur)
Scientist 'F'

Programme Officer and Coordinator

Contents

Foreword	3
Preface	4
Acknowledgement	5
Cancer Research in ICMR R.S. Dhaliwal and Tanvir Kaur	9
1. Population and Cancer Incidence A. Nandakumar and Tanvir Kaur	28
2. Leading Sites of Cancer A. Nandakumar and M. Chaturvedi	36
3. Breast Cancer Sunita Saxena, Hemant Malhotra and Tanvir Kaur	54
4. Cervical Cancer Research Ravi Mehrotra, A. Nandakumar and Tanvir Kaur	79
5. Cancer Research in North-East J. Mahanta, Sunita Saxena, R.K. Phukan and Tanvir Kaur	112
6. Tobacco Related Cancers Ravi Mehrotra, Sunita Saxena, J. Mahanta and Tanvir Kaur	137
7. Prostrate Cancer V. M. Katoch, Ravi Mehrotra and Sunita Saxena	186
8. Diet and Cancer T. Longvah and V. M. Katoch	195
9. Extramural Research Activities R.S. Dhaliwal and Tanvir Kaur	201

Cancer Research in ICMR

R.S. Dhaliwal and Tanvir Kaur

Cancer has been a priority area of research for the Indian Council of Medical Research (ICMR). The major impetus on cancer research was in early 1980s, when besides aetiological aspects, the work was initiated on assessment of magnitude of cancer as well as on cancer control related activities. Application of leads from research projects has been a major plank for cancer research. Assessment of the work of existing cancer registries and initiation of a network of cancer registries helped tremendously in identifying the dimensions of the problem and suggest for the need and strategy for a central cancer control programme. The data from the National Cancer Registry Programme (NCRP) indicated cervical and tobacco related cancers to be most common against which urgent action was needed. Therefore, several of the Task Force projects in the years to come focused on these cancer entities. ICMR served as the secretariat for development of the National Cancer Control Programme of India. In view of the national priorities, the focus of research in the field of cancer has been on the aetiology with identification of preventable risk factors; understand the mechanism of carcinogenesis; assessment of the magnitude and trends in cancer occurrence; and on operational research for control of tobacco use and common cancers through existing infrastructures. The multi-disciplinary research involved clinical, epidemiological as well as basic sciences including modern molecular techniques. These aspects were addressed through a task force approach with close planning and monitoring by ICMR. However, high importance was also accorded to supporting projects submitted by individual scientists, which spanned practically all specialties concerned with cancer and different sites in the body.

National Cancer Control Programme

Cancer research at ICMR provided important information on the magnitude of cancer problem in India. The data suggested cancer to be a major public health problem which cannot be effectively managed through the existing health services at state level. Therefore, ICMR suggested initiation of a centrally supported National Cancer Control Programme (NCCP). The Ministry of Health & Family Welfare constituted a committee for this purpose which finalized the document in 1984. ICMR was the secretariate for this committee. The NCCP document provided epidemiological information including the overall magnitude, common cancers in different geographic areas; objectives of NCCP; common cancer sites for intervention; general principles & strategy for undertaking control measures; training requirements; monitoring mechanisms; funding structure; structure & functions of national and state level advisory boards; functions of different level personnel of health services and role of Regional Cancer Centres.

The ICMR provides technical support and advice to the Ministry of Health & Family Welfare at central and state levels for implementation of various facets of cancer control activities. Regional Cancer Centres were encouraged to initiate facilitation and training for control activities. Task force projects were carried out to assess the feasibility of involving existing health infrastructure in undertaking screening for specific

cancers. The ICMR project on pain relief with morphine provided important initial data on possible tablet strength and also allayed the fear of addiction to morphine. ICMR's technical support and advice helped the Government in review and expansion of NCCP activities at various time periods. ICMR was also involved in developing the strategy and magnitude for NCCP during various five-year plan periods.

1981-2015

National Cancer Registry Programme:

The main broad and overall objective is to sustain and develop a national research data-base on cancer, diabetes, CVD and stroke through recent advances in electronic information technology with a national collaborative network, so as to undertake aetiological, epidemiological and clinical research in these areas.

National Cancer Registry Programme (NCRP) as an activity of National Center for Disease Informatics and Research

The Coordinating Unit of the National Cancer Registry Programme (NCRP) that has now emerged as the new permanent institute of ICMR – National Centre for Disease Informatics and Research was commenced with the following objectives:

1. To generate reliable data on the magnitude and patterns of cancer;
2. To undertake epidemiologic research studies in the form of case control and cohort studies based on observations of registry data;
3. Provide research base for developing appropriate strategies to aid in National Cancer Control Programme of the Government of India; this would be in the form of planning, monitoring and evaluation of activities under this programme;
4. Develop human resource in cancer registration and epidemiology.

With these objectives; three Population Based Cancer Registries (PBCRs) at Bangalore, Chennai and Mumbai and three Hospital Based Cancer Registries (HBCRs) at Chandigarh, Dibrugarh and Thiruvananthapuram were commenced from 1st January 1982. NCRP also developed a unique concept of an Atlas of Cancer in India (NCRP, 2004b) covering several regions in the country by use of internet for data collection for the year 2001-2002 thus mapping the patterns of cancer region-wise. The outcome of this activity was extensive use of information technology in subsequent activities of NCRP and the birth of newer registries and commencement of Patterns of Care and Survival Studies (POCSS).

The PBCRs have gradually expanded over the years and as of now there are 29 PBCRs and 29 HBCRs under the NCRP network.

Historical Background

The NCRP is a long term activity of the Division of Non-communicable Diseases. Dr Usha K. Luthra who was the then (during 1970-92) Chief of the Division of NCD and Dr L.D. Sanghvi who was the first Project Officer were the founder members of the NCRP. Dr G.K. Rath, Chief of the Institute Rotary Cancer Hospital and Head Radiation Oncology, AIIMS, New Delhi is the current chairman of the Steering Committee. Dr P.C. Gupta, currently the Director, Healis-Sarkaria Institute of Public Health, Mumbai is the present chairman of the Monitoring Committee. However, Dr P.C. Gupta has been associated with the NCRP and its functioning in one capacity or the other since its inception. Likewise Mr P. Gangadharan, Consultant, Oncology Centre, Amrita Institute of Medical Sciences, Kochi and Chairman of

the Translational Research Committee of NCRP has been involved with this programme actively both as a senior faculty of the Mumbai and Trivandrum hospital based registries and the Karunagapally population based cancer registry. Other members of the Steering Committee who have actively contributed towards the development of the NCRP include Dr P.S.S. Sundar Rao, Dr S. Radhakrishna, Dr N.C. Misra and Dr Padam Singh. The NCRP in its earlier years received guidance, encouragement and support from several WHO consultants as external evaluators. These include Dr C.S. Muir, Dr T. Hirayama, Dr J. Young and Dr M. Hakama.

Historical Background

The first cancer registry was commenced in Bombay (Mumbai) in the year 1964 under the auspices of the Indian Cancer Society by the late Dr D.J. Jussawalla, renowned cancer surgeon and former Director of the Tata Memorial Hospital. With the control of infectious diseases and increased longevity of the growing population in a country like India the spectrum of disease pattern has changed and the burden of cancer is on the increase. Of the ten million new cases of cancer diagnosed every year; over half are from the developing world. It is estimated that by the year 2020, over 10 million people worldwide would die of cancer every year and that 70 percent of these would be from the developing world. In the late 1970s and early 1980s, there was widespread recognition of the need for organizing new cancer registries in different regions of the country. The Indian Council of Medical Research constituted a task force for strengthening the existing registries and initiating a network of cancer registries in the country. Realising the need and importance of the problem of cancer, the National Cancer Registry Programme (NCRP) was set-up in December 1981.

Overall Functioning

NCRP plans, directs, develops, supports, coordinates and evaluates a national programme of cancer surveillance (electronic or otherwise) involving the collection and analysis of reliable data on magnitude and patterns of cancer so as to answer key questions about cancer incidence and mortality in different demographic and population settings. It designs, undertakes and implements multi-registry collaborative aetiological research studies in keeping with recent advances in epidemiological research.

The Scientific Advisory Committee of the institute and the Research Area Panel on Cancer meet periodically to oversee and guide the functioning of the NCRP. In more recent times with ICMR's thrust on translational research, this committee also meets every three months to closely follow the outcome.

Cancer registration in India is active and staffs of all registries visit hospitals, pathology laboratories and all other sources of registration of cancer cases on a routine basis. Death certificates are also scrutinized from the municipal corporation units and information collected on all cases where cancer is mentioned on the death certificates. This data is collected by means of a 'Standard Core Proforma' designed by NCRP conforming to the data fields as suggested by International norms. The proforma is filled by trained registry workers based on interview/ hospital medical records/ supplementing data by inputs from treating surgeons/radiation oncologists/involved physicians/pathologists. The contents of the proforma are entered into specifically created software and transmitted electronically to the coordinating center at Bangalore. At the center, the data is subjected to range and consistency checks and is published periodically as Consolidated Reports by NCRP (NCDIR). The probable duplicates are removed after due consultation with registries.

A review meeting and workshop are held annually (Annual Review Meeting) where the Principal Investigators and staff of the registries present data and participate in the discussions.

The Pre ARM workshop is 'hands on' activity with discussion on the various aspects of working of the registry, problematic cases, use of coding and discussion on medical terminology, statistical and epidemiologic methods. About 2-3 senior and junior staff of all the registries participate in the workshop.

Milestones

1. An Atlas of Cancer in India (2001-02)

The overall aim of the study was to get to know the similarities and differences in patterns of cancer across the country in a relatively cost-effective way using recent advances in computer and information technology transmission. Knowing patterns of cancer across the country would provide important leads in undertaking aetiological research, in targeting cancer control measures and in examining clinical outcomes.

The rationale of such an atlas was that there were large areas of the population, particularly the rural areas remain largely uncovered and therefore the patterns of cancer in several urban centres and rural areas remain largely unknown. India is a vast country with populations having varied cultures, customs and habits. The environment differs and so does dietary practices, and socio-economic status. Important differences exist in the ways of living of the urban and rural populations. Geographic differences in patterns of cancer have already been observed among the different registries.

The findings of cancer atlas were consolidated in form of a report. The report covers data accrued over a two-year period (1 January 2001 to 31 December 2002) and over 100 centres (including the functioning cancer registries) have contributed and collaborated in the project. The report is therefore a singular example of utilising advances in Information Technology for creating a ready database for research.

In all 217174 cancers for the two year period (1st January 2001 to 31st December 2002) were received. Reliable patterns of cancer including microscopic incidence rate could be deciphered in 82 of 593 districts in the country. Some of the highlights of the findings include:

1. Overall high incidence rates of cancer were observed in both males and females in the districts of Mizoram. Comparable high incidence rates in the districts of other states especially in the North East (Manipur, Sikkim, Assam) were also seen.
2. Cancers of the stomach were particularly high in all the districts of the Mizoram, Nagaland, Sikkim and Manipur in both males and females.
3. Cancers of the oesophagus in both males and females were particularly high in Assam.
4. Cancer of the nasopharynx was high in Nagaland and Meghalaya.
5. Lung cancer in females and males were several fold higher in the districts of Mizoram and Manipur.
6. The coincidence of high incidence of cancer of the cervix and penis were noted in the northern districts of Tamil Nadu state.
7. The south western districts of the country from Goa to Kanyakumari showed a high incidence of thyroid cancer.

Some leads provided on incidence and patterns of cancer in the North east by the above project. Another project "Development of an Atlas of Cancer in the North East" was initiated covering North Eastern States. As an outcome of the leads provided at 2 PBCRs in Arunachal Pradesh, 3 PBCRs in Assam and one each in Sikkim, Meghalaya, Nagaland, Manipur, Tripura and Mizoram were initiated. Some of these registries are covering entire state.

2. Cancer Atlas in Punjab (2012-13)

All medical colleges, pathology laboratories, civil hospitals and individual oncologists throughout the state of Punjab were contacted for their interest to collaborate in the project. Those who responded were supplied with core forms for collecting basic information (mainly patient identification details including area of living, and site and morphology of tumour) and provided guidelines for collecting this information on all malignant cases reported in the department of pathology from 2011. Visits were made to these potential collaborating centres and on the spot instructions given. During the visits their need for support was assessed depending on the infrastructure and average number of malignancies reported per annum and data collation provided. Intense training workshops in different regions of the state were held. Principles of cancer registration, data collation, transmission and fundamentals of epidemiology constituted the thrust areas of training at the workshops. The workshops and visits contributed a great deal to the success of the project.

The incidence rates of multiple myeloma along with other lymphoid and haemopoietic malignancies appeared to be higher in some districts of Punjab compared to that seen in other PBCRs. Here also one needs to undertake case control studies. Some of the other sites of cancer that show higher or comparably high incidence rates with the highest seen in other PBCRs in India are: in males: penile cancer (in Faridkot and Bathinda), prostate (Jalandhar), urinary bladder (Kapurthala, Jalandhar) and brain (Mohali and Mansa); in females: breast (Bathinda, Mohali), vagina (Bathinda, Ludhiana), ovary (Mohali) and brain (Moga and Mohali).

3. Cancer in Bhopal - Comparison of Cancer Patterns in MIC Affected and Un-Affected Areas (1988-2007).

Indian Council of Medical Research initiated many studies among the gas exposed population to assess the ill effects of toxic gas. One of these were Population based Cancer Registry (PBCR) that was set up in the Department of Pathology, Gandhi Medical College, Bhopal. This was a special purpose registry set up to observe occurrence of cancer as an effect of gas in the affected and unaffected areas. The registry started functioning on 01 January 1986 registering all cancer cases among residents of Bhopal. The efforts of data collection by the registry culminated into a report. The report underwent considerable discussions by the expert committee meetings (both of NCRP and that constituted specifically for various studies on Bhopal MIC exposure). The salient finding of the report were- cancers of tongue, mouth, hypopharynx, oesophagus and lung showed higher incidence in the affected area when compared to MIC unaffected area. However, when tobacco habits of the populations of two areas were taken into account, the higher rate observed in MIC affected area was essentially neutralised. Among females higher incidence of cancer cervix was observed in affected area while cancer breast was found high in the unaffected area. Cancer cervix is known to have higher incidence rates in lower socioeconomic strata of the society and *vice versa* for cancer breast. Majority of the population in affected area is from lower socioeconomic strata when compared to unaffected area though the specific data defining the socioeconomic status is not available. The report also gave trends in AARs over 20 years in the affected and unaffected areas. Cancers of all sites for both males and females have overall shown a statistically significant increasing trend in AARs from 1987 to 2007.

NATIONAL CENTRE FOR DISEASE INFORMATICS AND RESEARCH
NATIONAL CANCER REGISTRY PROGRAMME
Indian Council of Medical Research

Map showing network of centres that are transmitting data

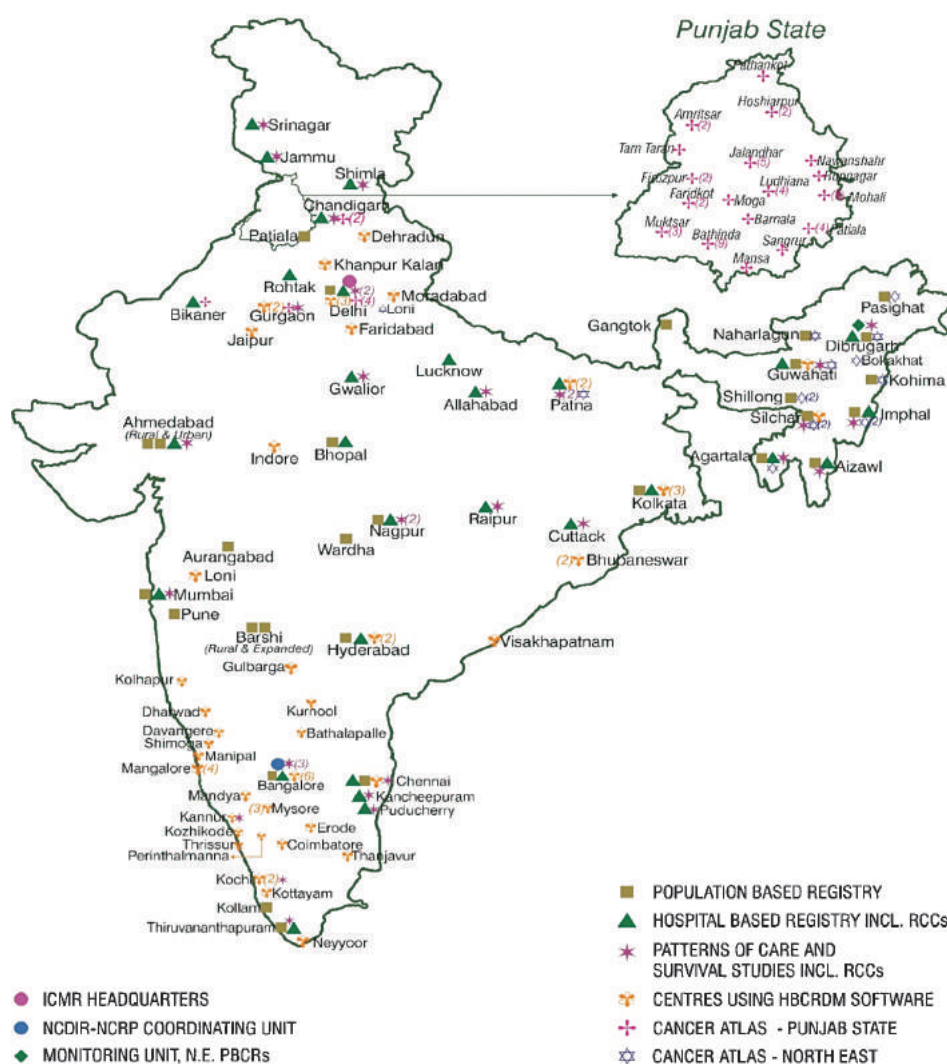


Fig. 1: Map of India depicting PBCRs, HBCRs and other centers under the NCRP Network.

Cancer Registration

Cancer Registration is the process of systematically and continuously collecting information on malignant neoplasms reported. Broadly there are two types of cancer registries - population based (PBCR) and hospital based (HBCR) cancer registry.

Population based cancer registries collect information of all new cancer cases reported from the population residing in a defined geographic area. Population cancer registries are concerned with providing information on cancer incidence and mortality, trends over time and constitute a base for carrying out research studies on cancer etiology through epidemiologic studies. Population based cancer registries provide reliable data on the magnitude of the cancer problem and also about the patterns of cancer – the types and changes in the rates of these cancers over time. To initiate, establish and sustain population based cancer registries as per international standards requires meticulous planning, cooperation of medical institutions in the area, dedicated personnel and adequate funding. There are several sources from where

staff of registries collect information on cancer cases. These include pathology reports, medical records, radiology and radiotherapy departments and death certificates to name a few. The availability of updated investigation/diagnostic facilities, well maintained medical records using International Classification of Diseases together with an efficient death registration system are essential for good quality cancer registration.

Hospital based cancer registries collect information of cancers reported in a given hospital, regardless of where that person with cancer resides. Hospital based registries help in assessing cancer patient care in a given hospital and aid in hospital administration including the individual hospital's cancer control programmes. Hospital registries also collect information pertaining to clinical stage, treatment and follow-up. Hospital registries help in planning and monitoring survival studies and clinical trials. In many instances, hospital registries contribute cases to the population based cancer registry in the area.

Cancer patterns vary not only throughout the world but also between different population groups within the same country. The preliminary and subsequent reports of the National Cancer Registry Programme (NCRP) have shown cancer of the stomach as a consistent leading site of cancer among males in Bangalore and Chennai, whereas it was lower down among the leading sites of cancer in Bhopal, Delhi or Mumbai. Similarly, cancer of the gall bladder is a leading site of cancer especially among women in Bhopal and Delhi, but was hardly reported seen in Bangalore and Chennai.

Studying the magnitude and patterns of cancer would be the first step in determining clues to the cause of cancer and having a baseline to plan and assess control measures. Epidemiologic studies based on these help in knowing what are happening and what can be done about it. Cancer registries provided the needed information to undertake such investigations.

Summary of the Three-Year Report of PBCRs: 2009-2011

This report of 25 Population Based Cancer Registries (PBCRs) includes the data of four new PBCRs (Meghalaya, Nagaland, Tripura and Wardha) that have commenced working and included under the NCRP network since the last report for the years 2006-2008 was published. Three other PBCRs at Patiala, Naharlagun and Pasighat have commenced collating data in 2010 and therefore, only a description of their working was included in this report.

The 25 PBCRs that comprise the basis of this report cover 7.45% of the population of India, with partial or complete representation of 16 States and one Union Territory. Eleven of the PBCRs are in urban localities and 2 in wholly rural regions. The remaining 11 correspond to both urban and rural parts by virtue of being state or district registries. Though the geographic area and population covered were small compared to the vastness of India and its population, they gave a fair idea of the cancer problem in the country.

Highlights

Cancer incidence rate is generally expressed as age adjusted or age standardized rate (AAR) (according to world standard population) per 100,000 persons. As in the previous report (NCRP: 2006-08), Aizawl district in Mizoram state showed the highest AAR in both males and females. In males, Aizawl district was followed by East Khasi Hills district of Meghalaya state and in females by Kamrup Urban district. The incidence rates by each PBCR are given in Chapter 1 and Chapter 2 of the report describes the leading sites of cancer in the different PBCRs.

Among males; cancers of lung, mouth, oesophagus and stomach were the leading sites across all the registries. Lung cancer was the leading site in Bangalore, Chennai, Delhi, Kolkata, Tripura, Kollam and

Thiruvananthapuram. All the PBCRs in Gujarat and Maharashtra states and Bhopal PBCR have mouth cancer as the leading site of cancer. Cancer of the oesophagus was the leading site in the registries in the states of Assam and Meghalaya. Stomach cancer was the leading site in Sikkim and Mizoram while cancer of the nasopharynx is the leading site in Nagaland. Among females, cancer of the breast and cervix were the leading sites of cancer in 18 of 25 PBCRs. Cancer of the gallbladder and cancer of the oesophagus followed by cancer of breast as the leading site in Dibrugarh and Kamrup respectively. Lung was the leading site in Manipur and Mizoram. Cancer of the oesophagus led the list of cancers in Meghalaya. Cancer of the thyroid followed by cancer of breast in the two PBCRs at Kollam and Thiruvananthapuram in Kerala state.

Chapter 3 of report gives the proportion of cancers associated with the use of tobacco. East Khasi Hills district of Meghalaya had the highest relative proportion with 69.3% and 43.0% for males and females respectively. Chapter 4 deals with different types of cancers in childhood. The proportion of cancers in childhood relative to cancers of all ages in both sexes varied from 1.1% in Meghalaya to 4.4% in Delhi.

Chapter 7 provides a comparison of cancer incidence and patterns in 25 PBCRs along with 5 sub areas (districts or combination of districts) within the states of Manipur, Mizoram and Meghalaya.

The data of the newer PBCRs that were being reported for the first time in this report are Nagaland, Meghalaya including Khasi Hills district, Tripura and Wardha. Among these new PBCRs, males in Nagaland showed the highest AAR of 21/100,000 for nasopharyngeal cancer. East Khasi Hills district and Meghalaya state as a whole showed the highest AAR in cancers of the oesophagus (in both males and females) and cancers of the hypopharynx and larynx in males. Nonetheless, the states of the North East had already reported high incidence rates of cancers of the upper aero-digestive tract that would include anatomical sites such as, oral cavity, pharynx, hypopharynx, nasopharynx, larynx, oesophagus and even stomach. The international comparisons of the incidence rates of these sites of cancer (Chapter 8) reveal that the PBCRs at Meghalaya (especially East Khasi Hills district) and Mizoram (Aizawl district) and Kamrup in Assam reported the highest AARs of cancers of these sites, particularly oesophagus, hypopharynx and larynx.

The details of individual registry write-up including that of Patiala, Naharlagun and Pasighat PBCRs were provided at the end of the report.

Quality of Data

The authenticity of the data depends on its quality and the completeness of coverage of cancer cases in the geographical area (quantity) and timeliness in data abstraction and is comprehensively covered in Chapter 9. This chapter along with Chapter 5 that addressed the most valid basis of diagnosis of cancer and Chapter 6 dealing with mortality data show the challenges and limitations of cancer registration in the Indian context vis-à-vis International comparisons. Every effort was made by both the individual PBCRs and the coordinators at NCDIR to ensure that the data reported was as correct and as complete as possible. In more recent years the PBCR software applications programme has greatly helped in enhancing the speed of data submission and its quality.

Use of Electronic Information Technology

Software programmes developed in-house have to a great extent contributed to improving the quality and coverage. Data that is keyed is checked as per international norms. In terms of completeness of coverage a new feature in the software monitors the data received from different sources of a particular registry and identifies paucity of data received from any given source for a desired period of time. The full potential

and thereby the benefit of the software will be felt if the registries could quickly get into doing real-time data entry. This real-time data entry in other words mean, data abstraction, collation and entry onto the computer of currently diagnosed cancer cases (say within one week). Six PBCRs were able to report the data of 2011. This brought down the gap between the reporting year and the calendar year to only one year for these PBCRs.

Priorities for Cancer Research and Control

The results provided a lead to set priorities for cancer research and identified target sites – both anatomic and geographic, for cancer control measures.

The upper aero-digestive tract cancers as a group (oral cavity, hypopharynx etc., larynx, oesophagus and stomach) have emerged as important sites for undertaking risk factor research and implementing early detection programme. The PBCRs in the North East had shown the highest or some of the highest incidence rates in the world of these sites of cancer and also comprise a substantial proportion of all cancers. They comprised 37.6% of all cancers in both genders (50.5% in males and 24.7% in females) in the registries in the North East. Investigation into the type of tobacco consumption and dietary factors (for cancers of the oesophagus and stomach) is an obvious necessity. Aside from the unmistakable need for tobacco control and possible elimination of the habit (for primary prevention), early detection and screening exercises specifically for the cancers of these anatomical sites as an entity is an immediate requirement. Such plans are vital across the North East and in other registry areas like Bangalore that has shown consistent high AARs over the years for cancer of the oesophagus in both sexes. Programmes using endoscopy and laryngoscopy should be tried out initially as pilot studies in the Indian context and then depending on the feasibility, validity and acceptability extend state/region-wise. Simultaneously, education of the public about these sites of cancer and the need for early detection including the benefits of undergoing endoscopy should be widely publicised. In patients who do come and undergo endoscopy, facilities should be created for prompt diagnosis and wherever necessary, adequate treatment.

Thus, this important feature of this report described above should be used to undertake research studies and implement cancer control measures. If done right, one would have evidence of the effectiveness of such measures over the years through the registry data.

There are several other anatomical sites of cancer that continue to be a challenge in terms of aetiology and control. Cancers due to use of tobacco including cancers of the lung, were at the top of the list followed by cancers of the breast and cervix in women. Cancers of the gallbladder and thyroid also in women, too required special attention in terms of identifying risk factors and initiating early detection schemes. The former was seen in the north and northeast and the latter in the south and northeast of the country.

Consolidated Report of Hospital Based Cancer Registries: 2007-2011

The objectives of Hospital Based Cancer Registries (HBCRs) have been detailed out in an earlier report of the NCRP including principles and ICD coding.

This report of seven Hospital Based Cancer Registries (HBCRs) included the data of two new HBCRs (Dr. B. Borooah Cancer Institute, Guwahati and Postgraduate Institute of Medical Education and Research, Chandigarh) that commenced working and contributed data to this report. Thus, the five year (2007-2011) report of the seven HBCRs was the contribution of data from the following hospitals, namely,

1. Tata Memorial Hospital, Mumbai
2. Kidwai Memorial Institute of Oncology, Bangalore

3. Cancer Institute, Adyar, Chennai
4. Regional Cancer Centre, Thiruvananthapuram
5. Assam Medical College, Dibrugarh
6. Dr B. Borooah Cancer Institute, Guwahati
7. Postgraduate Institute of Medical Education and Research, Chandigarh

Quality of Data

The accuracy of the data depends on its quality, timeliness in data abstraction in a given hospital and is comprehensively covered in Chapter 14. The quality of data from each registry will undergo vigorous checks during data entry and subsequently. Efforts are made by both the individual HBCRs and the coordinators at NCDIR to ensure that the data reported is as correct and as complete as possible. In more recent years the HBCRDM software application programme has greatly helped in enhancing the speed of data submission and its quality.

Use of Electronic Information Technology

Software programmes developed in-house have contributed to a great extent in improving the quality and coverage. Data that is keyed is checked as per international norms. In hospital based cancer registry setup, the core form is mainly divided into three portions namely, Patient Identifying Information, Diagnostic Details and Details of Stage & Treatment which requires recalling the patient record more than one time to complete the form. In order to facilitate the above, partial saving of a record has been provided.

Time Trends in Cancer Incidence Rates: 1982-2010

The National Cancer Registry Programme (NCRP) published the first report on Time Trends in Cancer Incidence Rates 1982-2005 in 2009. The second report published in July 2013 covered additional five years (2006-2010) data for the older PBCRs at Bangalore, Bhopal, Chennai, Delhi, Mumbai and Barshi and provides an insight on the trends in seven additional PBCRs at Thiruvananthapuram, Dibrugarh, Kamrup Urban District, Imphal West District, Ahmedabad Rural District and the states of Mizoram and Sikkim. These seven PBCRs have contributed at least 7 years data. Some information on trends in incidence rates is also provided for the PBCRs at Kolkata, Kollam, Ahmedabad Urban, Pune, Aurangabad and Nagpur that have data for 5-6 years. The older PBCRs have been functioning for a much longer period of time. Further one is able to examine the time trend based on five year annual average age adjusted incidence rates (AAR) for at least five such periods, thus providing potency to the findings. Therefore, this summary deals predominantly with the results from these centres. Notable, outcomes from other PBCRs do find mention. This second report has also examined differences if any in the rise in certain sites of cancer by broad age groups, particularly the younger age groups and some of the findings are also highlighted.

The population of India in general and that of the areas covered by the registries in particular, have displayed rapid changes in life styles, dietary practices and socio-economic milieu. Diagnostic and detection technologies have improved and more of the population has not only access, but can also afford the same. Pathology laboratories with amenities for microscopic diagnosis of cancer and radiation oncology treatment facilities have multiplied, increasing the sources of data collection for registry workers underscoring the greater need of cooperation from more persons and institutions. Software applications programmes have considerably enhanced the quality of the data. On the downside, the stringent and varied methods used by the de-duplication module in the software applications programme for recent years' data could exclude those cases that would have otherwise been included in earlier years. Thus,

the above factors involving population dynamics, registry practices and use of electronic information technology could all affect the incidence rates in different ways. These need to be kept in mind when interpreting the results.

Among males, cancers of the tongue, mouth, colon, rectum, liver, lung, prostate, brain, non-hodgkin's lymphoma and lymphoid leukaemia have shown statistically significant increase in incidence rates. Some additional sites of cancer for males that have shown a rising trend and included in this report are gallbladder, pancreas, kidney and urinary bladder.

In females, cancers of the gall bladder, lung, breast, corpus uteri, ovary, thyroid, brain, non-Hodgkin's lymphoma and myeloid leukaemia have shown a statistically significant increase in AARs.

In terms of highest Annual Percentage Change (APC %), cancer of the corpus uteri (Bangalore: 6.3%; Delhi: 3.6%) and gall bladder in females (Chennai: 6.2%) stand out. Cancer of the corpus uteri is significantly increasing in all four metro PBCRs both overall and in the 45+ age groups. In addition in Bangalore, Chennai and Delhi the younger 35-44 year age group also showed a significant increase. Cancer of the gall bladder in females was increasing in both the southern PBCRs at Chennai and Bangalore and also in Bhopal. In Delhi the increase was seen only in males and in Mumbai in both males and females. In Chennai and Bhopal, the increase was significantly seen in younger women (25-34 and 35-44 age groups). Increased risk of cancer of the corpus uteri was associated with obesity, low parity and use of unopposed or non-cyclical oestrogen replacement therapy. Cancer of the lung in females had shown an increase in all the four metros and also in Kamrup urban district. In females in Chennai there was significant increase in lung cancer in the younger age groups of 25-34 and 35-44. Tumours of the brain that had shown an increase in Bangalore and Chennai were also significantly increasing in the younger 25-44 age group in both males and females in Bangalore and in females in Chennai. Likewise the increasing incidence of cancer of the thyroid in females in Chennai and Delhi was also seen in the younger age groups.

The decline in the incidence of cancer cervix is seen across all registries including the rural registry at Barshi. This decline was observed in the absence of any organized screening or early detection programmes in the registry areas. The same factors like later age at marriage, fewer children that could be contributing to increased breast cancer could possibly be responsible for the decline in the incidence of cancer cervix. Another possible reason for the decline could be due to the increased number of child births at institutions (as opposed to home deliveries) culminating in improved maternal and maternity care including genital hygiene.

The last chapter of this report estimates and projects the number of cancer cases overall and by selected leading sites. This was based on the trends in the incidence rates published in this report and the latest three-year (2009-2011) report of the population based cancer registries. Only simple methods of projections based on the Annual Percentage Change in the incidence rates were used and no sophisticated statistical models were applied for arriving at the numbers. Nonetheless, the chapter provided a reasonably accurate estimate of the future burden of cancer.

Summary of Ongoing Projects

1. Patterns of Care and Survival Studies

The study was launched with the following objectives:

- a. To obtain the details of clinical stage and types of treatment in the Hospital Based Cancer Registries under the NCRP and in other selected centres and medical colleges across the country;

- b. To obtain clinical stage and treatment based survival in all registered and treated patients at these centres;

Following a series of meetings/workshops the patient information form for each of the above three sites were finalized. Procedure manuals for completing the forms were also developed. A website – hbccrindia.org has been functioning since September 2005. This site allows centres to open the Patient Information Form through a unique Login ID and Password and enter information from the completed hard copy of the form on to that in the web-site and transmit the same to the Coordinating Centre of the NCRP in Bangalore. Several features have been incorporated in the software. These include checks on data, listing of cases in varying states of follow-up, summary tables on types of treatment given and so on. Sixteen institutions are participating in this study and so far information on about 50,000 cases on these three sites of cancer has accrued. The centers would pursue active follow-up of all registered treated patients in their respective institutions.

Software aspects-Integration with HBCR

The main objective of this project was to centralize the registration by the combination of both HBCR and POCSS Application. Alter the HBCR Web Application by adding the additional information from POCSS. The benefit would be that by adding the additional fields of POCSS to HBCR Website will help to optimize the Data Entry time.

2. NCRP and Development of Software Applications Programmes

The project on “Development of an Atlas of Cancer in India” not only gave the stimulus for further PBCRs across the country but also revealed and provided the impetus for the development of various software applications programmes culminating in the setting up of a new permanent ICMR “National Centre for Disease Informatics and Research” in Bangalore. Therefore, the thrust of work of the Coordinating Unit of NCRP in recent times has been to harness the power of electronic information technology. Over the years, the NCRP has developed software to aid cancer research and development. Web-based data capture allows individual centres to track and get listing of relevant summary tables. NCRP presently monitors and manages following datasets.

Independent Software Modules

The efforts to develop an interface between the cancer research and software have culminated into development of independent software modules which are presently available free of cost for widespread usage. The features of these modules are -

- Paperless or Option to obtain blank format printouts or Obtain printout of entered Information
- Flexi Data Entry Module
- Separate Login based on type of user (Eg: Data Entry Operator, Social Investigator, Physician/Oncologist)
- Site Specific
- Case wise Quality Check Reports
- Research Outputs.
- Administration Outputs

PBCRDM Software

The PBCR Data Management software has been developed in-house and distributed to most of the 29 PBCRs. This has greatly helped in further systematizing data checking, processing and analysis.

HBCR Data Management Software

In-house software for widespread usage for all hospital based cancer center across the country was developed. This software is presently being used by 86 centers across the country.

Pathology Software

Since majority of cancer cases are microscopically diagnosed, this software was developed keeping in view the requirements of pathologists. Presently 126 centers use the software.

Radiotherapy Software

A similar module was developed for radiotherapy for widespread usage. The module is machine specific, with inclusion of radiotherapy chart and note sheet for radiation oncology. For RCC JIPMER, Puducherry, customized RT module for data entry has been developed and deployed in JIPMER server. It also includes discharge summary. Once the patient information is entered in this software automatic data transfer is done in HBCR-POCSS database.

Karnataka State Registry Programme

Government of Karnataka made cancer as a notifiable disease

A registration module has been developed and uploaded at HBCR website. The centres already registered under HBCRDM and SAST has been linked for this project. An e-Monitoring software is also being developed for viewing the status of registration, data collection and centres working online for administrators. A two page simple basic proforma with less number of checks during data entry is being developed.

3. Cancers of Sites associated with use of Tobacco

Sites of cancer that have been associated with use of tobacco (Tobacco Related Cancers – TRCs) include oral cavity, pharynx (including oropharynx & hypopharynx) oesophagus, larynx, lung and urinary bladder (IARC, 1987).

The most recent reports of the PBCRs show the number and relative proportion of Tobacco Related Cancers (TRCs), in different PBCRs (Fig. 2). East Khasi Hills district of Meghalaya had the highest relative proportion of cancers associated with the use of tobacco for males and females at 69.3% and 43.0% respectively when compared to all sites of cancers. The lowest proportion of tobacco related cancers was among females in Thiruvananthapuram with 10.3% of cancers being associated with the use of tobacco.

Among HBCRs, the highest percentage of TRC was observed in Guwahati: both in males (59.2%) and in females (26.7%). In other registries, it varied from 38.8% to 56.0% of all cancers in males and from 12.1% to 25.6% in females.

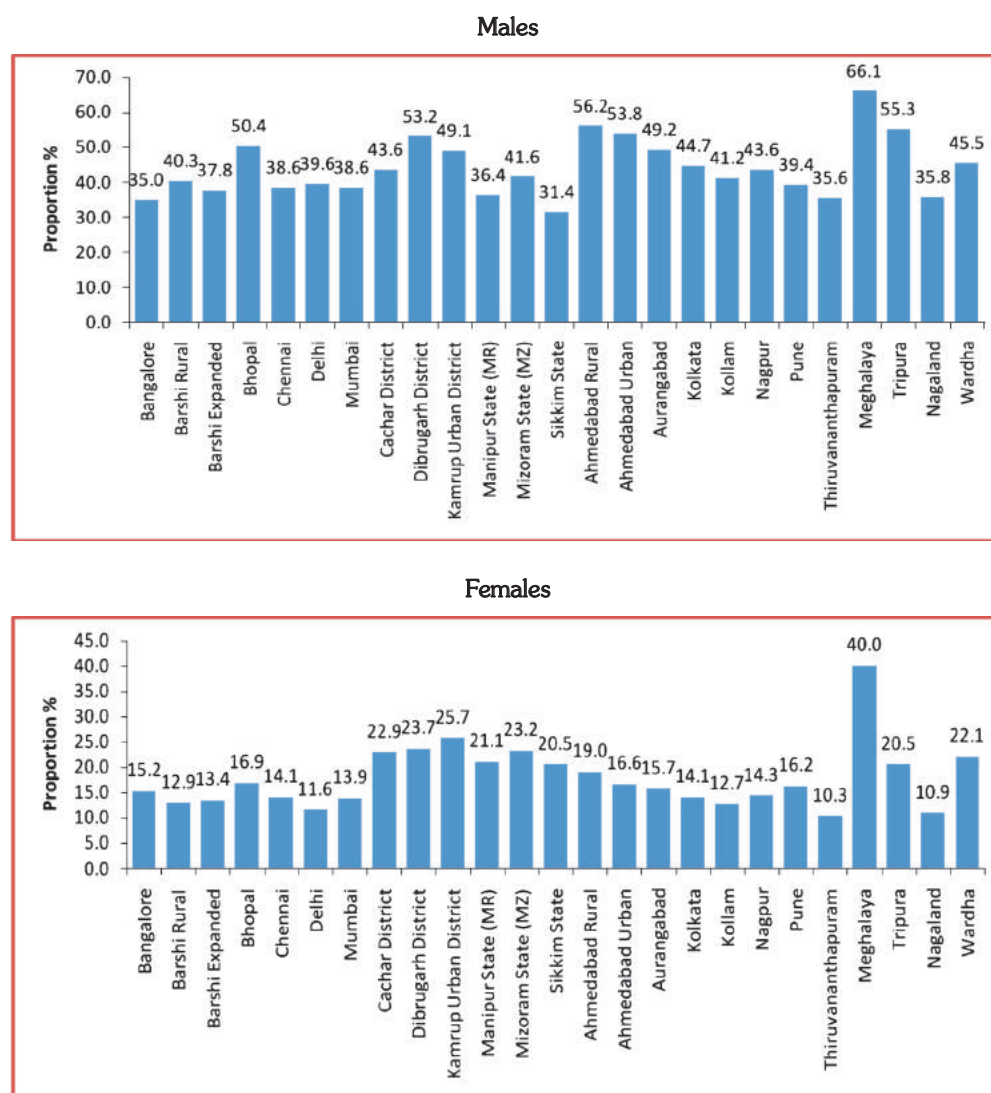


Fig. 2: Proportion (%) of Tobacco Related Cancers (TRCs) Relative to All Sites of Cancers across PBCRs (2009-11).

Cancer Registration and Cancer Research

Cancer registration is a means to a purpose and not a purpose in itself. It is the forerunner of studies in descriptive epidemiology of cancer, which in turn generate specific scientific hypotheses. Studying the magnitude and patterns of cancer would be the first step in determining clues to the cause of cancer and having a baseline to plan and assess control measures. Epidemiologic studies help in knowing what is happening and what can be done about it. Cancer registries provide the needed information to undertake such investigations. The registries under the NCRP have highlighted the need to undertake aetiological studies in several sites of cancer. The data of the population based cancer registries has been published² in successive volumes of the WHO publication – Cancer Incidence in Five Continents, published every five years by the International Agency for Research on Cancer – the cancer research arm of the WHO. Data from registries on Childhood cancer and cancer survival have also appeared in the agency's publications¹. A number of research investigations have been undertaken based on the cancer registry data and many of the results have been published in national and international journals. These include, for the first time in the context of a developing country population based survival studies on selected sites of cancer have been published^{3,4}. Others include case control studies on cancers of the stomach, oesophagus, gall bladder etc. The registries provide an opportunity to embark on studies that would combine laboratory

components with epidemiologic design such as looking at the influence of *Helicobacter pylori* in the occurrence of stomach cancer, typing of HPV in relation to cervical cancer, gene environment interaction in breast cancer and so on.

Priorities for Anatomical Site Specific Cancer Research: In the present background, cancer research could comprise epidemiological research, basic laboratory research or cancer control research or a combination of these. Risk factors, such as tobacco and its association with specific anatomical sites of cancer are well known. However, in the Indian context some of the tobacco habits (like types of tobacco chewed, beedi smoking) vary in different parts of the country and India specific tobacco related cancer research is required. Besides, the report on time trends has shown several specific sites of cancer that are rising in incidence rates. The recently included PBCRs for trend reports show a significant increase in cancers of mouth, stomach, liver, colon, prostate, gall bladder for males while lung and corpus uteri. In both sexes, tumours of the brain and nervous system and kidney have shown a rise. This calls for systematic epidemiological research preferably taking into account laboratory parameters for each of these sites of cancer. One of the outstanding features noted relate to the contrasting incidence of certain cancers in relation to geographic locations. Cancers of the gall bladder and thyroid are prominent examples. These provide unique opportunities for multi-centric case control studies which have to be systematized now. Similarly, there are no definitive directly replicable models for early detection / screening of cancers of the thyroid, gall bladder, rectum, liver, ovary and corpus uteri. Thus research into various modules for cancer control of these latter sites that would be applicable to Indian conditions needs to be done.

Priorities for Anatomical Site Specific Clinical Research: There are international and standard guidelines for clinical staging and protocols for treatment of most cancers, including the ones that are showing an increase in the report on time trends in incidence rates summarized earlier. Their applicability in our settings needs detailed studies and investigations. This seems particularly important for cancers of the prostate, colon and rectum in men and for cancers of the breast, ovary, corpus uteri and thyroid in women.

Cancer Registries in Cancer Control

The cancer registry is central to any rational programme on cancer control. To this end the registries under the NCRP have provided baseline parameters with data of high quality and scientific validity. Systematic collation of cancer information albeit in a few centers, has given pointers to the burden and patterns of cancers amenable to cancer control measures. The population based survival studies conducted by the registries, for the first time in the context of a developing country have made known additional factors in the form of age and stage specific survival and mortality.

Based on this important information, the Government of India formulated the National Cancer Control Programme in 1985 with the following main objectives:

1. Primary Prevention of Tobacco Related Cancers;
2. Early detection and treatment of cancer of the uterine cervix;
3. Distribution and extension of cancer management and control services through Regional Cancer Centres, medical and dental colleges.

Priorities for Cancer Control: With information on the North East PBCR reports and that on “Trends over Time in Incidence Rates”, there appears to be a need for reassessing priorities for cancer control. The overall thrust of the cancer control programme towards anti tobacco education / legislation and control of cancer cervix is largely unchanged. However, certain specific pointers emerge from these reports.

In males, mouth cancer was the leading site in ten registries, its contribution ranging from 22.6% in Barshi Expanded to 39.6% in Wardha, oesophagus cancer was the leading site in seven registries, its contribution ranging from 19.9% in Cachar District to 45.1% in Meghalaya, lung cancer was the leading site in eight registries, its contribution ranging from 24.7% in Delhi to 48.9% in Manipur state.

In females, mouth cancer, oesophageal cancer and lung cancer were the leading sites among the tobacco related cancers. Mouth cancer and lung cancer contributed to 48.2% of TRCs in Chennai. Similarly, mouth cancer and oesophageal cancer contributed to 66.6% of TRCs in Nagaland. Lung cancer contributed to 67.9% and 72.8% of TRCs in Manipur State and Mizoram State respectively. In Meghalaya, 54.7% of TRCs belong to oesophageal cancer. The incidence of cancer of the stomach is extremely high in Mizoram State and on the higher side in Sikkim and Manipur. The education of the public about stomach cancer and the need for early detection including the benefits of undergoing endoscopy should be widely publicized. In patients who do come and undergo endoscopy, facilities should be created for prompt diagnosis and wherever necessary, adequate treatment. A similar programme as above for oesophageal cancer in Assam is also essential. Since lung cancer is also very common and of high incidence in both men and women in these states, campaigns on harmful effects of tobacco as also early detection programmes for lung cancer should be instituted. The value of a simple chest X-ray in early detection or at least picking early stage, lung cancer should be tried out as a pilot project and subsequently extended. Besides, cancer of the gall bladder especially in women also showed a higher incidence in the North East. Little is known about the specific risk factors for gall bladder cancer. Cancer of the nasopharynx is also high in these North East States. Nasal and Oropharyngeal examination by trained medical personnel could help in the early detection of these cancers also.

The report on time trends in incidence rates had given several pointers. The rate of increase in the incidence of cancer of the lung among women is striking calling for systematic evidence based and focused anti-tobacco campaigns targeting the urban woman. Two other cancers in women that are increasing and for which, promotion of awareness for early detection is required are cancers of the breast and corpus uteri. Among males, screening programmes for cancer of the prostate, colon and rectum should be looked into for its applicability and efficacy in the Indian context. While control of tobacco use has to be constantly emphasised, the importance of obesity control and genital hygiene could be propagated. The necessity for early detection should be increasingly highlighted along with public education and self examination methods. The role of registries would ultimately be made applicable with overall trend studies as well as trends of stage at disease presentation.

NCRP and Human Resource Development

The commencement of the NCRP by the ICMR in 1981 has attracted a variety of talent in both medical and other disciplines. This has paved the way in providing several opportunities for training in cancer registration and epidemiology to many personnel working under this programme. Staff of registries have acquired higher training and qualifications in universities in Finland, United Kingdom, Canada, Australia and the United States. The trained staff have since returned and contributed to the development of the programme in terms of numerous research publications in indexed international journals, undertaking newer research projects and in contributing to training of other staff of registries. Such training programmes are continuing at present.

Burden of Cancer in India

Projection of cancer burden means a systematic way of prediction of number of cases for all anatomical sites of cancer or for a specific site and for specified period of time. The NCRP data that has accrued over the years is essentially that of urban centers and some rural registries that cover part of a district is

represented. Therefore it would be difficult to provide valid estimates of the burden of cancer of the entire country with over 70% of the population of India residing in the rural areas. Table 1 and Table 2 below summarize the projected population estimates.

Table 1: Projected Cancer cases in India (All sites:C00-C97)

Year	Males	Females	Total
2013	522164	564619	1086783
2014	535324	581945	1117269
2015	548844	599847	1148692
2020	622203	698725	1321928

Cancer Burden by Broad Anatomical Sites-2013 and 2015

Table 2: Projected Cancer cases in India (All sites:C00-C97)

Year	2013		2015	
	No. of cases	Approx. Relative Proportion	No. of cases	Approx. Relative Proportion
All sites	1086783		1148692	
Tobacco Related Cancers	298457	27	318186	28
Gastrointestinal tract	207861	19	221526	19
Cervix	92731	09	94857	08
Breast	94208	09	100611	09
Corpus Uteri & Ovary	57298	05	63005	05
Lymphoid & Haemopoietic Malignancies	109484	10	115525	10
Prostate	35029	03	39200	03
Central Nervous System	25660	02	25811	02

Environmental Carcinogen Testing Units

The concept of Environmental Carcinogen Testing Units (ECTU) in early 1990s was to understand the role of environmental carcinogens in aetiology of major cancers in India. However, stringent budgetary position at that time did not permit initiation of an elaborate programme, which the subject deserved. Therefore, some ICMR institutes were requested to work in this area. The National Institute of Occupational Health, Ahmadabad and National Institute of Nutrition, Hyderabad started work in this direction. The work was carried out till late 1990s.

The ECTUs worked towards understanding the carcinogenic potential of various suspected carcinogens under Indian conditions, and on monitoring of known carcinogens.

The National Institute of Occupational Health, Ahmedabad worked on chemicals in work environment, while Drug Toxicology Research Centre, National Institute of Nutrition (NIN), Hyderabad, studied food items. The studies carried out by the National Institute of Occupational Health, Ahmedabad, included carcinogenicity of DDT and HCH studies on workers exposed to benzidine dyes, carcinogenic potential of HCH in animals exposed to aflatoxin, presence of green symptoms in agriculture tobacco workers, role of black tea extract on carcinogenesis in animals, development of microbial systems for assessing the genotoxicity, genotoxic potential of air samples from high air pollution areas, and chemical analysis

of pan masala.

The work carried out at ECTU at NIN, Hyderabad included studies on pan masala, nitrosamines in foods, experimental iron deficiency and gastrointestinal tract tumours, screening for protective factors in foods and biomarkers of genotoxicity, determination of levels of nitrosamines in certain food groups and measure the quantity of volatile nitrosamines formed from foods under stimulated gastric conditions.

While the activities were appreciated by the scientific community, the limited availability of funds was a constraint in quick initiation of studies. Suggestions from the ICMR expert group were on gaps in knowledge regarding mutagenicity / carcinogenicity of various substances was provided to scientists. The group had recommended preparation of standard guidelines on methodology for certain laboratory techniques, review of literature for role of biomass fuel, plastic material used for food packaging, plain pan masala and edible oils and condiments, in mutagenicity / carcinogenicity. These suggestions were to help individual scientist to plan a direction for research in the field.

Cancer Management Guidelines

Cancers are an important disease entity in India, accounting for about 950,000 new cases in the country. Although a lot of progress has been made in treatment of cancers, a lot more research and systematic application of research leads is needed. While survival after treatment in certain sites of cancer is good, the response in many other common sites (like lung, oesophagus, stomach, etc.) is extremely poor. This is especially true about India, where despite availability of leads from western world, survival is lower. Several Indian oncologists have pointed out the possibility of variable results of treatment of cancers in India, in terms of differences in cancer spectrum & response to treatment and many kind of Indian case presentations not being covered in western guidelines. In view of the above, a task force on management of cancers was constituted.

The aim of the task force was to develop and recommend most appropriate management strategy for common cancers of India, based on the review of literature and the results from Indian studies and experience.

Initially two sub-committees were constituted to review the literature on cancers of buccal mucosa and chronic myeloid leukemia. Subsequently, 18 more sub-committees were constituted to review the literature related to guidelines for management of various sites of cancers. The selected cancer sites were that of lung, breast, esophagus, cervix, uterus, stomach, gall bladder, soft tissue sarcoma and osteo-sarcoma, tongue, acute myeloid leukaemia, ALL, CLL, NHL-high grade, NHL-low grade, HD, MM, MDS, and paediatric lymphoma. Chairperson of each of the sub-committee moderated the review and discussion process, which in the initial stage was through email. Each member reviewed the assigned literature and prepared an abstract as per the format prepared for this purpose. All aspects related to treatment were considered including specific anti-cancer treatment, supportive care, palliative care, molecular markers, epidemiological and clinical aspects. Based on the review of literature, the subcommittees indicated management plan along with situation-based variations (depending on the limitations of equipments and expert manpower). The draft report of these sub-committees were discussed in a meeting of the sub-committee for modification/finalization. A total of 132 scientists were connected with the development of guidelines for management of cancer of 20 sites.

The guidelines for thirteen cancer sites (Breast Cancer, Buccal Mucosa Cancer Research, Cervix Cancer, Colorectal Cancer, Esophageal Cancer, Gastric Cancer, Gallbladder Cancer, Larynx and Hypopharynx Cancers, Multiple Myeloma, Non-Hodgkin's Lymphoma (High Grade), Soft Tissue Sarcoma and Osteosarcoma, Tongue Cancer) have been formulated as per the above procedure and are available on

ICMR website. Summary is also published in reputed journals. The drafts reports of guidelines for other sites are at different stages of preparation.

In summary, the study demonstrated how the internet (and email) approach can gainfully be achieved in conduct of a task force activity. The study was also carried out without any expenses on part of ICMR, as the scientists did not need any funds to review the literature. The study provided a suggested management plan which may be useful for most of the patients. However, it is not the intention of ICMR to make it compulsory as a first step. It is realized that each patient may have its own intricacies, side-effects and biological characteristics. Therefore, these guidelines should be considered as a tentative plan for understanding of the physicians, but the final decision on the treatment plan should be taken by the physicians only.

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The major concern of Population Based Cancer Registries (PBCRs) is to calculate cancer incidence rates, study the rates of individual cancers by comparing cancer incidence and patterns in other registries and in different subgroups of population in the respective areas.

PBCRs systematically collect information on all reportable neoplasms occurring in a geographically defined population from multiple sources. The comparison and interpretation of population based cancer incidence data support population-based actions aimed at reducing the cancer burden in the community.

The systematic ascertainment of cancer incidence from multiple sources can provide an unbiased profile of the cancer burden in the population and how it is changing over time.

These registries have a unique role in planning and evaluating cancer control programmes.

This report on the occurrence of cancer in a defined population provides a framework for assessing and controlling the impact of cancer in the community.

Geographical Area and Population at Risk

The data from Meghalaya PBCR belong to the districts of East Khasi Hills, West Khasi Hills, Jaintia Hills and Ri Bhoi; that of Nagaland PBCR belong to the districts of Kohima and Dimapur.

Two PBCRs namely, Pasighat and Naharlagun PBCRs from Arunachal Pradesh cover different districts from the state. Pasighat PBCRs data belongs to the districts of East Siang and Upper Siang. Naharlagun PBCR's data belongs to 8 districts namely, Tawang, West Kameng, East Kameng, Upper Subansiri, Lower Subansiri, Kurng Kumey, Papumpare and West Siang. The data has been analysed for Naharlagun, Papumpare District and Naharlagun excluding Papumpare District.

Patiala PBCR's data covers Patiala District.

The registry area covered by Thiruvananthapuram PBCR has increased from Thiruvananthapuram Taluk (2005-2011) to Thiruvananthapuram District, 2012 onwards.

The growth rate of the population between the census years 2001 and 2011 has been used (through the Difference Distribution Method of Takiar and Shobana, 2009) to estimate the mid-year populations (five year age group and total) for the years of the report, viz., 2012 to 2014. The same has been followed for 25 PBCRs, except for Kamrup Urban and Nagaland PBCRs. Factors such as net migration, birth rate and death rate have not been considered.

The growth rate of the population between the census years 1991 and 2001 has been used to estimate the mid-year populations (five year age group and total) for the years of the report, viz., 2012 to 2014

for Kamrup Urban PBCR, because of certain concerns in population enumeration in north-eastern states by the Census of India.

Table 1.1 shows the number of male and female population covered by 27 PBCRs and provides information from 34 geographical areas. The average population covered per year ranged from 1.38 lakhs in Pasighat PBCR (East Siang and Upper Siang districts) from Arunachal Pradesh to 168.9 lakhs in Delhi Registry.

Table 1.1: Estimated Population of all PBCRs

Registry	Area (sq.km.)	Male	Annual Average	Female	Annual Average	Total Pop.	Annual Average
Bangalore (2012)	741	4616478	4616478	4269020	4269020	8885498	8885498
Barshi Rural (2012-2014)	3713	798979	266326	719153	239718	1518132	506044
Barshi Expanded (2012)	18262	2257996	2257996	2070963	2070963	4328959	4328959
Bhopal (2012-2013)	350	2088045	1044022	1931368	965684	4019413	2009706
Chennai (2012-2013)	170	4715650	2357825	4693785	2346893	9409435	4704718
Delhi (2012)	1157	9012555	9012555	7884429	7884429	16896984	16896984
Mumbai (2012)	603	6728832	6728832	5777374	5777374	12506206	12506206
Cachar District (2012-2014)	3786	2770786	923595	2668126	889375	5438912	1812970
Dibrugarh District (2012-2014)	3381	2076093	692031	2009170	669723	4085263	1361754
Kamrup Urban District (2012-2014)	336	2141950	713983	1940459	646820	4082409	1360803
Manipur State (MR) (2012-2014)	22327	4600029	1533343	4539808	1513269	9139837	3046612
Imphal West District (2012-2014)	519	790580	263527	820147	273382	1610727	536909
MR - Excl. Imphal West (2012-2014)	21808	3809449	1269816	3719661	1239887	7529110	2509703
Mizoram State (MZ) (2012-2014)	21087	1741869	580623	1716527	572176	3458396	1152799
Aizawl District (2012-2014)	3576	623139	207713	637360	212453	1260499	420166
MZ - Excl. Aizawl (2012-2014)	17511	1118750	372917	1079174	359725	2197924	732642
Sikkim State (2012-2014)	7096	995202	331734	889210	296403	1884412	628137
Ahmedabad Urban (2012-2013)	364	6263952	3131976	5639544	2819772	11903496	5951748
Aurangabad (2012-2014)	148	1980880	660293	1850731	616910	3831611	1277203
Kolkata (2012)	185	2338295	2338295	2149037	2149037	4487332	4487332
Kollam District (2012-2014)	2491	3739048	1246349	4203108	1401036	7942156	2647385

Nagpur (2012-2013)	237	2620014	1310007	2532597	1266299	5152611	2576306
Pune (2012-2013)	613	5477270	2738635	4956064	2478032	10433334	5216667
Thi'puram District (2012-2014)	2192	4753308	1584436	5198769	1732923	9952077	3317359
Meghalaya (2012-2014)	14262	2964519	988173	2968522	989507	5933041	1977680
East Khasi Hills District (2012-2014)	2748	1293767	431256	1316769	438923	2610536	870179
Tripura State (2012-2014)	10492	5799657	1933219	5584512	1861504	11384169	3794723
Nagaland (2012-2014)	2390	1091662	363887	1016933	338978	2108595	702865
Wardha District (2012-2014)	6309	2026323	675441	1922291	640764	3948614	1316205
Naharlagun (NH) (2012-2014)	42095	1256000	418667	1203625	401208	2459625	819875
Papumpare District (2012-2014)	3462	288990	96330	288861	96287	577851	192617
NH - Excl. Papumpare (2012-2014)	38633	967010	322337	914764	304921	1881774	627258
Pasighat (2012-2014)	10193	210521	70174	203495	67832	414016	138006
Patiala District (2012-2014)	3325	3130053	1043351	2800456	933485	5930509	1976836

Table 1.2: Total Number of Cases Registered for all PBCRs*(Calendar years of data shown in parentheses for each registry)*

Registry	Male	Female	Total Cases
Bangalore (2012)	3824	4547	8371
Barshi Rural (2012-2014)	454	475	929
Barshi Expanded (2012)	901	1131	2032
Bhopal (2012-2013)	1718	1746	3464
Chennai (2012-2013)	5447	6212	11659
Delhi (2012)	10148	9598	19746
Mumbai (2012)	6598	6759	13357
Cachar District (2012-2014)	2666	2100	4766
Dibrugarh District (2012-2014)	1498	1345	2843
Kamrup Urban District (2012-2014)	3071	2392	5463
Manipur State (MR) (2012-2014)	2081	2542	4623
Imphal West District (2012-2014)	640	823	1463
MR - Excl. Imphal West (2012-2014)	1441	1719	3160
Mizoram State (MZ) (2012-2014)	2567	2089	4656
Aizawl District (2012-2014)	1275	1066	2341
MZ - Excl. Aizawl (2012-2014)	1292	1023	2315
Sikkim State (2012-2014)	707	678	1385
Ahmedabad Urban (2012-2013)	5477	4117	9594
Aurangabad (2012-2014)	1123	1118	2241
Kolkata (2012)	2777	2596	5373
Kollam District (2012-2014)	5534	5478	11012
Nagpur (2012-2013)	2236	2417	4653
Pune (2012-2013)	3417	3686	7103
Thi'puram District (2012-2014)	7638	8002	15640
Meghalaya (2012-2014)	2632	1616	4248
East Khasi Hills District (2012-2014)	1624	988	2612
Tripura State (2012-2014)	3628	2702	6330
Nagaland (2012-2014)	815	546	1361
Wardha District (2012-2014)	1306	1424	2730
Naharlagun (NH) (2012-2014)	735	704	1439
Papumpare District (2012-2014)	299	333	632
NH - Excl. Papumpare (2012-2014)	436	371	807
Pasighat (2012-2014)	175	159	334
Patiala District (2012-2014)	2853	3158	6011

**Table 1.3: Crude Rate (CR), Age Adjusted (AAR) and Truncated (TR) Incidence
Rates per 100,000 population in different PBCRs**

Registry	Male			Female		
	CR	AAR	TR	CR	AAR	TR
Bangalore (2012)	82.8	105.4	160.3	106.5	125.9	243.2
Barshi Rural (2012-2014)	56.8	53.9	84.6	66.0	60.4	125.3
Barshi Expanded (2012)	39.9	40.9	76.7	54.6	52.0	114.4
Bhopal (2012-2013)	82.3	101.5	181.0	90.4	108.3	223.5
Chennai (2012-2013)	115.5	116.1	181.0	132.3	126.2	252.7
Delhi (2012)	112.6	149.4	235.8	121.7	144.8	286.9
Mumbai (2012)	98.1	113.1	160.4	117.0	118.5	209.8
Cachar District (2012-2014)	96.2	125.4	227.7	78.7	95.2	210.1
Dibrugarh District (2012-2014)	72.2	92.8	155.7	66.9	78.6	178.8
Kamrup Urban District (2012-2014)	143.4	206.0	321.8	123.3	174.0	325.3
Manipur State (MR) (2012-2014)	45.2	60.5	88.7	56.0	68.6	126.1
Imphal West District (2012-2014)	81.0	92.4	119.6	100.3	103.6	184.6
MR - Excl. Imphal West (2012-2014)	37.8	52.2	81.0	46.2	59.2	110.4
Mizoram State (MZ) (2012-2014)	147.4	211.5	372.1	121.7	165.8	296.9
Aizawl District (2012-2014)	204.6	270.7	499.9	167.3	207.7	356.7
MZ - Excl. Aizawl (2012-2014)	115.5	175.0	293.4	94.8	136.6	255.6
Sikkim State (2012-2014)	71.0	90.7	133.0	76.2	100.3	175.7
Ahmedabad Urban (2012-2013)	87.4	98.5	181.0	73.0	76.5	155.7
Aurangabad (2012-2014)	56.7	72.0	122.8	60.4	73.0	153.2
Kolkata (2012)	118.8	100.9	158.5	120.8	103.4	205.6
Kollam District (2012-2014)	148.0	120.5	189.0	130.3	101.7	195.8
Nagpur (2012-2013)	85.3	89.4	151.6	95.4	94.5	194.1
Pune (2012-2013)	62.4	77.6	116.2	74.4	84.9	166.0
Thi'puram District (2012-2014)	160.7	132.0	203.7	153.9	120.4	231.5
Meghalaya (2012-2014)	88.8	169.6	376.1	54.4	94.4	199.3
East Khasi Hills District (2012-2014)	125.5	218.3	482.4	75.0	117.0	241.3
Tripura State (2012-2014)	62.6	76.4	136.5	48.4	54.9	120.0
Nagaland (2012-2014)	74.7	125.8	227.9	53.7	84.9	179.9
Wardha District (2012-2014)	64.5	60.2	105.9	74.1	66.7	145.1
Naharlagun (NH) (2012-2014)	58.5	103.5	207.4	58.5	100.5	227.8
Papumpare District (2012-2014)	103.5	230.4	412.4	115.3	249.0	560.6
NH - Excl. Papumpare (2012-2014)	45.1	76.9	159.7	40.6	66.6	150.3
Pasighat (2012-2014)	83.1	107.4	194.3	78.1	101.4	239.3
Patiala District (2012-2014)	91.1	97.9	175.3	112.8	111.2	244.2

Table 1.2 gives the number of cases registered for the calendar year(s) (1 January to 31 December) by each PBCR from 2012 to 2014. The top five PBCRs to register maximum number of cases were Delhi (19746), Thiruvananthapuram District (15640), Mumbai (13357), Chennai (11659) and Kollam (11012) PBCRs. Cancer cases in males were more in Delhi, Cachar District, Dibrugarh District, Kamrup Urban District, Mizoram State, Sikkim State, Ahmedabad Urban District, Kolkata, Kollam District, Meghalaya State, Tripura State, Nagaland and Naharlagun PBCRs. Female cancers were more in Bangalore, Barshi Expanded, Bhopal, Chennai, Mumbai, Manipur State, Nagpur, Pune, Thiruvananthapuram District, Wardha and Patiala PBCRs. Slight variation was observed in numbers between the sexes in the PBCRs at Barshi Rural, Aurangabad and Pasighat.

Crude Rate (CR)

Table 1.3 gives the Crude Rate (CR), Age Adjusted (AAR) and Truncated (TR) Incidence Rates per 100,000 population in different PBCRs.

The first five highest CR per 100,000 population among males was observed in Aizawl District (204.6), followed by Thiruvananthapuram (160.7), Kollam (148.0), Mizoram State (147.4) and Kamrup Urban District (143.4) PBCRs.

Similarly, among females, the first five highest CR was observed in Aizawl District (167.3) followed by Thiruvananthapuram (153.9), Chennai (132.3), Kollam (130.3) and Kamrup Urban District (123.3) PBCRs.

Aizawl District showed the highest CR in both males and females. Two PBCRs from the state of Kerala, namely Thiruvananthapuram and Kollam had high CRs in both males and females, Kamrup Urban District from the north east too showed high CRs in both males and females and thus they were listed within the top 5 PBCRs.

Age Adjusted Rates (AAR)

The AAR per 100,000 population in males ranged from 40.9 in Barshi Expanded to 270.7 in Aizawl District of Mizoram State followed by Papumpare District (230.4) under Naharlagun PBCR in Arunachal Pradesh. In females, it ranged from 52.0 in Barshi Expanded to 249.0 in Papumpare District under Naharlagun PBCR followed by Aizawl District (207.7).

Truncated Rates (TR)

In males, the TR per 100,000 population ranged from 76.7 in Barshi Expanded to 499.9 in Aizawl District followed by East Khasi Hills District (482.4). Similarly, in females, it ranged from 110.4 in MR - excluding Imphal West to 560.6 in Papumpare District of Arunachal Pradesh.

Cumulative Rate and Risk

Day (1987) proposed the Cumulative Rate (CR) as another age standardised incidence rate. The Cumulative risk is the probability that an individual will be diagnosed with cancer during a certain age period in the absence of any competing cause of death and assuming that the current trends prevail over the time period. For practical purposes, Cumulative rate is a good approximation of Cumulative risk over the defined period of time.

Cumulative Risk (%) in 0-64 years

This cumulative risk (%) gives an idea about a person developing cancer during the life period of 0-64 years of age. Among males, the cumulative risk ranged from 2.8% in Barshi Expanded to 15.9% in Aizawl

District of Mizoram State followed by East Khasi Hills (14.9%) of Meghalaya (Table 1.4). Hence 15.9% of males in the age group of 0-64 years in Aizawl District are likely to develop cancer in their life time and the males in the age group of 0-64 years in Barshi Expanded at 2.8% are the least at risk.

Among females, the cumulative risk ranged from 3.8% in Barshi Expanded to 17.9% in Papumpare District of Arunachal Pradesh. This means in Papumpare District, on an average about 17.9% of females in the 0-64 age group are likely to develop cancer in their life time whereas in Barshi Expanded the risk is 3.8%.

Cumulative Risk (%) in 0-74 years

Among males, the cumulative risk (%) in the 0-74 year age group ranged from 4.5% in Barshi Expanded to 27.0% in Aizawl District of Mizoram, followed by Papumpare District of Arunachal Pradesh at 25.2% and East Khasi Hills in Meghalaya at 22.9%.

Among females, the cumulative risk percentage in the 0-74 age group ranged from 5.7% in Barshi Expanded to 25.3% in Papumpare District of Arunachal Pradesh, followed by Aizawl District of Mizoram at 21.1%.

Table 1.4: Cumulative Incidence Rate, Cumulative Risk and Possibility of one in number of Persons Developing Cancer of Any Site (ICD-10): C00-C97 for all PBCRs (2012-2014)
Calculation based on age specific rates from 0-64 and 0-74 years of age

Registry	Cumulative Rates (%)				Cumulative Risk (%)				Possibility of one in number of persons developing cancer			
	0-64 Years		0-74 Years		0-64 Years		0-74 Years		0-64 Years		0-74 Years	
	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
Bangalore (2012)	5.9	8.5	12.7	14.7	5.8	8.2	11.9	13.7	17	12	8	7
Barshi Rural (2012-2014)	3.3	4.3	6.4	7.0	3.2	4.2	6.2	6.8	31	24	16	15
Barshi Expanded (2012)	2.8	3.9	4.6	5.8	2.8	3.8	4.5	5.7	36	26	22	18
Bhopal (2012-2013)	6.5	7.7	11.8	12.6	6.3	7.4	11.2	11.8	16	14	9	8
Chennai (2012-2013)	6.7	8.8	13.3	14.4	6.5	8.4	12.5	13.5	16	12	8	7
Delhi (2012)	8.9	10.2	17.3	16.6	8.5	9.7	15.9	15.3	12	10	6	7
Mumbai (2012)	6.0	7.3	13.1	13.6	5.8	7.1	12.2	12.8	17	14	8	8
Cachar District (2012-2014)	8.1	7.3	15.2	10.2	7.8	7.0	14.1	9.7	13	14	7	10
Dibrugarh District (2012-2014)	5.5	6.2	11.2	9.0	5.3	6.0	10.6	8.6	19	17	9	12
Kamrup Urban District (2012-2014)	11.6	11.5	24.2	20.5	11.0	10.8	21.5	18.6	9	9	5	5
Manipur State (MR) (2012-2014)	3.3	4.6	7.5	8.0	3.2	4.5	7.2	7.7	31	22	14	13
Imphal West District (2012-2014)	4.4	6.9	11.6	11.7	4.3	6.6	10.9	11.1	23	15	9	9
MR - Excl. Imphal West (2012-2014)	3.0	4.0	6.4	7.0	3.0	3.9	6.2	6.7	34	26	16	15

Mizoram State (MZ) (2012-2014)	12.9	10.3	25.0	19.4	12.1	9.8	22.1	17.6	8	10	5	6
<i>Aizawl District</i> (2012-2014)	17.3	12.3	31.4	23.7	15.9	11.6	27.0	21.1	6	9	4	5
<i>MZ - Excl. Aizawl</i> (2012-2014)	10.2	8.9	21.1	16.4	9.7	8.5	19.0	15.1	10	12	5	7
Sikkim State (2012-2014)	4.7	6.2	10.8	11.7	4.6	6.0	10.2	11.1	22	17	10	9
Ahmedabad Urban (2012-2013)	6.5	5.4	11.5	8.8	6.3	5.3	10.8	8.5	16	19	9	12
Aurangabad (2012-2014)	4.6	5.4	8.5	8.6	4.5	5.3	8.1	8.3	22	19	12	12
Kolkata (2012)	5.9	7.2	11.9	11.7	5.7	6.9	11.2	11.1	18	14	9	9
Kollam District (2012-2014)	7.0	7.1	14.5	11.1	6.8	6.9	13.5	10.5	15	15	7	9
Nagpur (2012-2013)	5.7	6.8	10.3	10.4	5.5	6.6	9.8	9.8	18	15	10	10
Pune (2012-2013)	4.2	5.7	9.1	9.9	4.1	5.5	8.7	9.4	24	18	11	11
Thi'pura District (2012-2014)	7.6	8.4	15.7	13.4	7.3	8.0	14.5	12.6	14	12	7	8
Meghalaya (2012-2014)	12.5	6.8	20.0	11.1	11.8	6.6	18.1	10.5	8	15	6	10
<i>East Khasi Hills District</i> (2012-2014)	16.1	8.3	26.0	14.0	14.9	8.0	22.9	13.0	7	13	4	8
Tripura State (2012-2014)	4.9	4.2	9.4	6.3	4.8	4.1	9.0	6.1	21	24	11	16
Nagaland (2012-2014)	8.0	6.1	15.1	9.7	7.7	5.9	14.0	9.2	13	17	7	11
Wardha District (2012-2014)	4.0	5.0	6.8	7.2	4.0	4.9	6.6	7.0	25	20	15	14
Naharlagun (NH) (2012-2014)	7.4	8.0	12.7	11.1	7.2	7.7	11.9	10.5	14	13	8	10
<i>Papumpare District</i> (2012-2014)	15.1	19.7	29.0	29.2	14.0	17.9	25.2	25.3	7	6	4	4
<i>NH - Excl. Papumpare</i> (2012-2014)	5.6	5.2	9.5	7.2	5.5	5.1	9.1	7.0	18	20	11	14
Pasighat (2012-2014)	7.0	8.0	12.4	11.3	6.8	7.7	11.6	10.7	15	13	9	9
Patiala District (2012-2014)	6.3	8.4	11.1	12.4	6.1	8.1	10.5	11.7	16	12	9	9

Publication

1. National Cancer Registry Programme (NCRP) Three-year Report of Population Based Cancer Registries: 2012-2014. Incidence, Disutribution, Trends in Incidence Rates and Projections of Burden of Cancer. Report of 27 PBCRs in India, (Indian Council of Medical Research), Bangalore, 2016.

2

Leading Sites of Cancer

A. Nandakumar and M. Chaturvedi

This chapter depicts the leading sites of cancer in the different PBCRs using Figure 2.1 to Figure 2.27. The leading sites of cancer for each gender were decided on the basis of proportion relative to all sites of cancer or in other words based on crude incidence rates.

In the description given below, the relative proportions (%) of leading sites are given in parentheses after the name of each leading site. Likewise, the respective Crude Rate (CR) and Age Adjusted Rate (AAR) per 100,000 population are also shown in parentheses after the name of each site.

BANGALORE

Males: The leading sites of cancer were: lung (10.6%), stomach (7.6%), prostate (7.0%), oesophagus (6.0%) and brain, NS (5.0%). The respective CR and AAR per 100,000 population for these sites were: lung (8.8 and 11.7), stomach (6.3 and 8.2), prostate (5.8 and 8.2), oesophagus (5.0 and 6.7) and brain, NS (4.1 and 4.6).

Females: Breast (27.5%) was the leading site of cancer in Bangalore registry area. The other leading sites were cervix uteri (12.3%), ovary (5.3%), thyroid (4.0%) and mouth (3.9%). The respective CR and AAR per 100,000 population for the sites were: breast (29.3 and 34.4), cervix uteri (13.1 and 15.3), ovary (5.6 and 6.5), thyroid (4.3 and 4.1) and mouth (4.2 and 5.4).

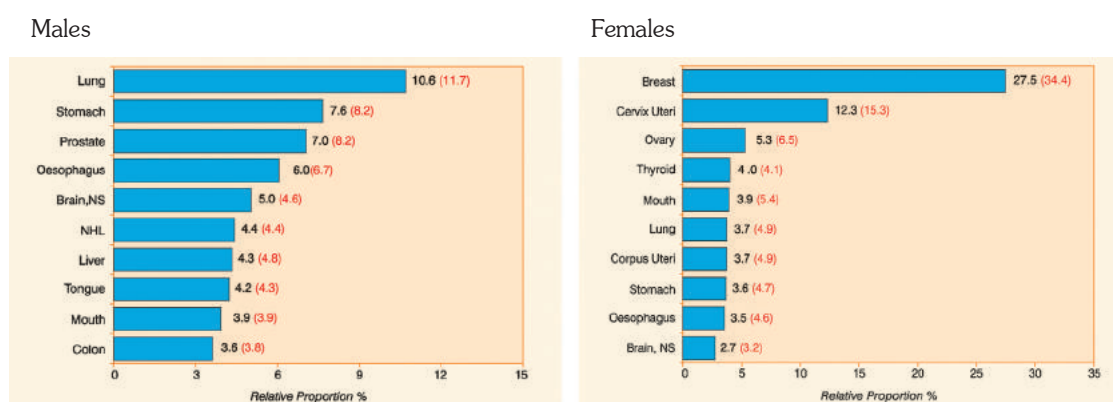


Fig. 2.1: Ten Leading Sites of Cancer (2012). Age Adjusted Incidence Rates given in parentheses

BARSHI RURAL

Males: The leading sites of cancer were: mouth (8.6%), oesophagus (7.7%), liver (6.2%), rectum (4.4%), tongue (4.4%), NHL (4.4%) and prostate (4.4%). The respective CR and AAR per 100,000 population for these sites were: mouth (4.9 and 5.1), oesophagus (4.4 and 4.0), liver (3.5 and 3.7), rectum (2.5 and 2.4), tongue (2.5 and 2.3), NHL (2.5 and 2.2) and prostate (2.5 and 2.1).

Females: Cervix uteri (26.7%) was the leading site of cancer in Barshi Rural registry area. The other leading sites were breast (20.0%), ovary (4.6%), oesophagus (3.8%) and lung (3.6%). The respective CR and AAR per 100,000 population for the sites were: cervix uteri (17.7 and 16.1), breast (13.2 and 12.4), ovary (3.1 and 2.9), oesophagus (2.5 and 2.2) and lung (2.4 and 2.1).

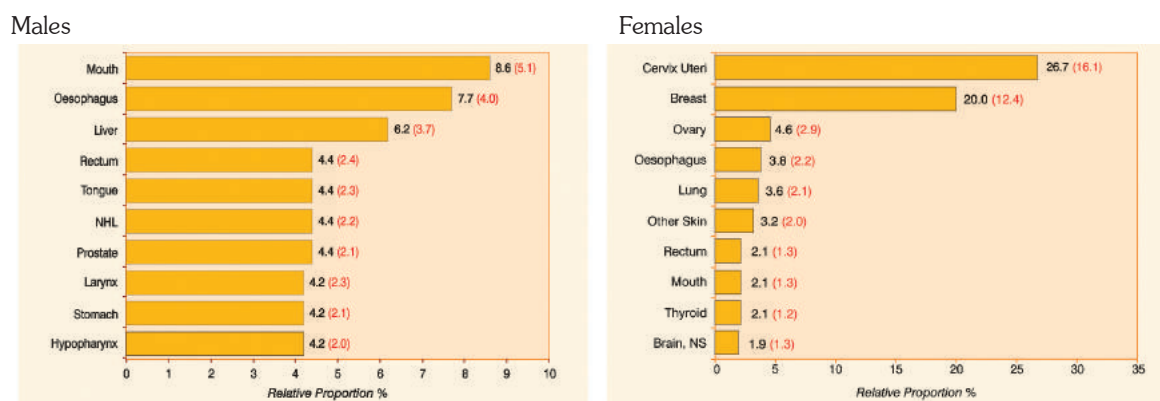


Fig. 2.2: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

BARSHI EXPANDED

Males: The leading sites of cancer were: mouth (9.0%), tongue (8.1%), oesophagus (6.7%), lung (5.0%) and liver (4.3%). The respective CR and AAR per 100,000 population for these sites were: mouth (3.6 and 3.8), tongue (3.2 and 3.5), oesophagus (2.7 and 2.7), lung (2.0 and 2.1) and liver (1.7 and 1.8).

Females: Cervix uteri (28.6%) was the leading site of cancer in Barshi Expanded registry area. The other leading sites were breast (21.8%), ovary (4.6%), tongue (2.8%) and stomach (2.7%). The respective CR and AAR per 100,000 population for the sites were: cervix uteri (15.6 and 14.7), breast (11.9 and 11.7), ovary (2.5 and 2.3), tongue (1.5 and 1.5) and stomach (1.4 and 1.3).

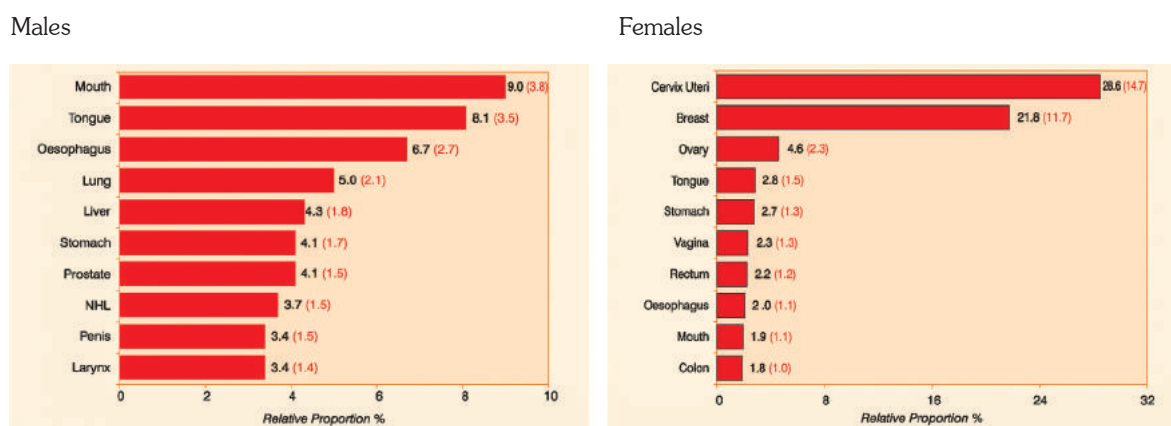


Fig. 2.3: Ten Leading Sites of Cancer (2012). (AARs given in parentheses).

BHOPAL

Males: The leading sites of cancer were: mouth (15.3%), lung (11.3%), tongue (9.1%), larynx (5.8%) and prostate (4.8%). The respective CR and AAR per 100,000 population for these sites were: mouth (12.6 and 14.3), lung (9.3 and 12.4), tongue (7.5 and 8.4), larynx (4.8 and 6.3) and prostate (4.0 and 5.6).

Females: Breast (31.2%) was the leading site of cancer in Bhopal registry area. The other leading sites were cervix uteri (12.5%), ovary (7.8%), gall bladder (5.8%) and mouth (4.9%). The respective CR and

AAR per 100,000 population for the sites were: breast (28.2 and 33.1), cervix uteri (11.3 and 13.8), ovary (7.0 and 8.4), gall bladder (5.2 and 6.4) and mouth (4.4 and 5.5).

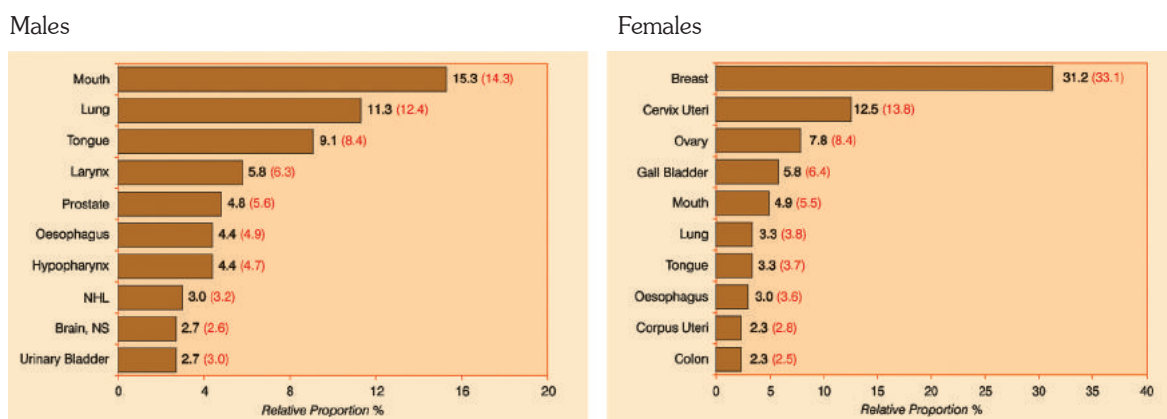


Fig. 2.4: Ten Leading Sites of Cancer (2012-2013). (AARs given in parentheses).

CHENNAI

Males: The leading sites of cancer were: lung (9.8%), stomach (9.3%), mouth (8.0%), tongue (7.0%) and prostate (5.4%). The respective CR and AAR per 100,000 population for the sites were: lung (11.3 and 11.6), stomach (10.8 and 10.8), mouth (9.2 and 8.5), tongue (8.1 and 7.4) and prostate (6.2 and 6.7).

Females: The leading cancer sites among females were: breast (30.7%) followed by cervix uteri (12.6%), ovary (6.6%), corpus uteri (4.6%) and stomach (4.1%). The respective CR and AAR per 100,000 population for the sites were: breast (40.6 and 37.9), cervix uteri (16.7 and 15.9), ovary (8.7 and 8.2), corpus uteri (6.0 and 6.0) and stomach (5.4 and 5.1).

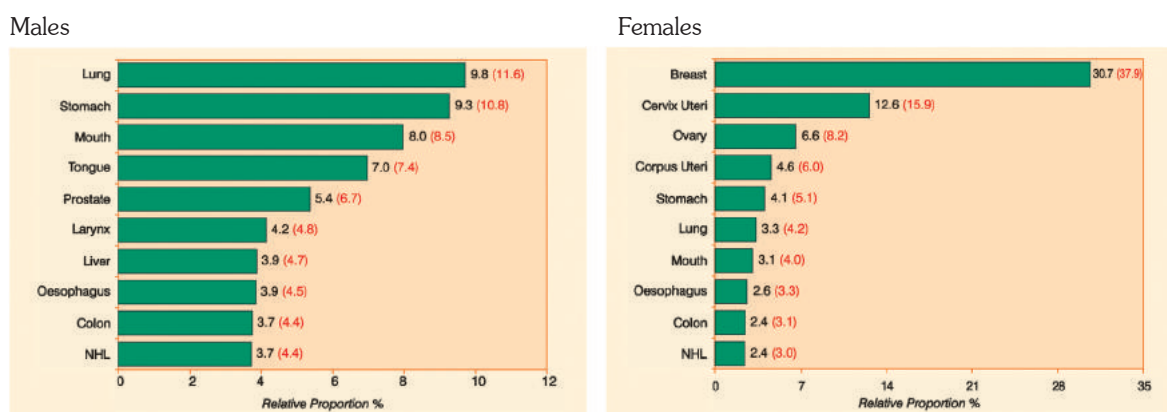


Fig. 2.5: Ten Leading Sites of Cancer (2012-2013). (AARs given in parentheses).

DELHI

Males: The leading sites of cancer were: lung (10.5%), mouth (6.9%), prostate (6.7%), tongue (6.5%) and larynx (5.7%). The respective CR and AAR per 100,000 for the sites were: lung (11.8 and 17.2), mouth (7.8 and 9.5), prostate (7.6 and 12.4), tongue (7.3 and 9.3) and larynx (6.4 and 8.9).

Females: The leading cancer sites among females were: breast (28.6%) followed by cervix uteri (10.8%), gall bladder (7.9%), ovary (7.2%) and corpus uteri (3.5%). The respective CR and AAR per 100,000 population for the sites were: breast (34.8 and 41.0), cervix uteri (13.2 and 15.5), gall bladder (9.6 and 11.8), ovary (8.7 and 10.0) and corpus uteri (4.3 and 5.5).

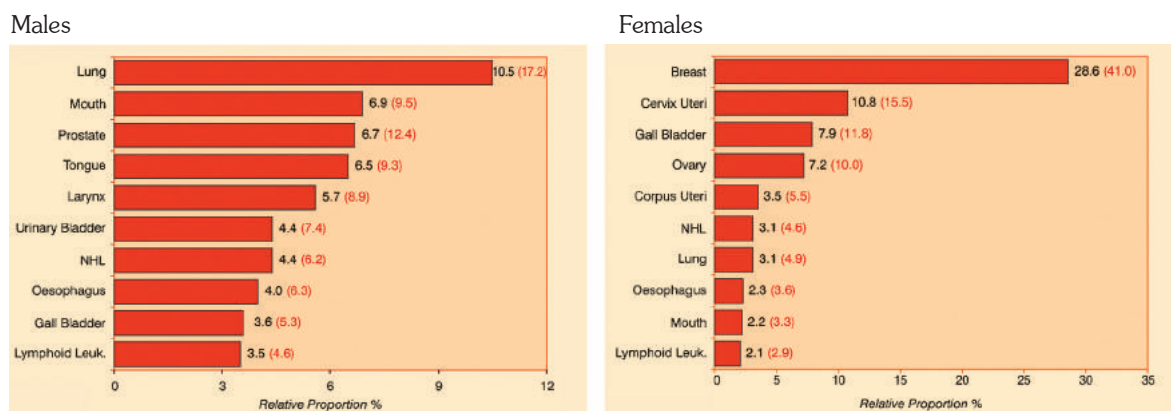


Fig. 2.6: Ten Leading Sites of Cancer (2012). (AARs given in parentheses).

MUMBAI

Males: The leading sites of cancer were: lung (10.4%), mouth (8.8%), prostate (7.9%), liver (5.7%) and tongue (4.9%). The respective CR and AAR per 100,000 population for the sites were: lung (10.2 and 12.4), mouth (8.6 and 9.1), prostate (7.7 and 9.8), liver (5.6 and 6.6) and tongue (4.8 and 5.1).

Females: The leading cancer sites among females were: breast (28.8%) followed by cervix uteri (7.7%), ovary (6.9%), lung (5.0%) and gall bladder (3.5%). The respective CR and AAR per 100,000 population for the sites were: breast (33.6 and 33.6), cervix uteri (9.0 and 9.0), ovary (8.1 and 8.1), lung (5.8 and 6.0) and gall bladder (4.1 and 4.1).

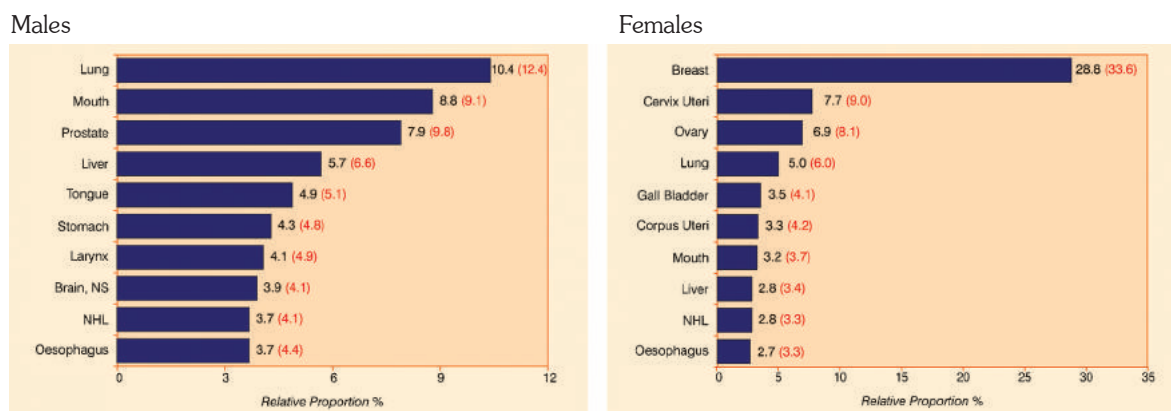


Fig. 2.7: Ten Leading Sites of Cancer (2012). (AARs given in parentheses).

CACHAR DISTRICT

Males: The leading sites of cancer were: oesophagus (10.2%), hypopharynx (8.6%), lung (8.4%), mouth (5.7%) and larynx (5.4%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (9.9 and 12.9), hypopharynx (8.3 and 11.1), lung (8.1 and 11.5), mouth (5.5 and 7.1) and larynx (5.2 and 6.9).

Females: The leading cancer sites among females were: breast (14.3%) followed by cervix uteri (13.9%), gall bladder (10.3%), oesophagus (6.8%) and ovary (5.1%). The respective CR and AAR per 100,000 population for the sites were: breast (11.3 and 12.8), cervix uteri (10.9 and 12.7), gall bladder (8.1 and 10.2), oesophagus (5.3 and 7.1) and ovary (4.0 and 4.1).

Fig. 2.8: Ten Leading Sites of Cancer (2012-2014)
(AARs given in parentheses)

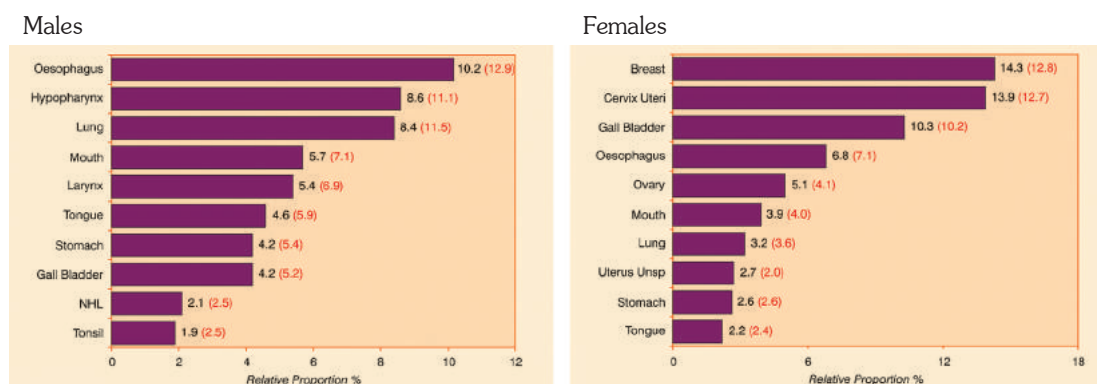


Fig. 2.8: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

DIBRUGARH DISTRICT

Males: The leading sites of cancer were: oesophagus (15.3%), hypopharynx (11.6%), stomach (7.9%), mouth (6.8%) and lung (5.1%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (11.0 and 14.6), hypopharynx (8.4 and 10.9), stomach (5.7 and 7.6), mouth (4.9 and 6.3) and lung (3.7 and 5.0).

Females: The leading cancer sites among females were: breast (19.0%), gall bladder (10.7%), oesophagus (9.4%), ovary (8.9%) and cervix uteri (6.4%). The respective CR and AAR per 100,000 population for the sites were: breast (12.7 and 13.9), gall bladder (7.2 and 8.6), oesophagus (6.3 and 8.1), ovary (6.0 and 6.5) and cervix uteri (4.3 and 4.9).

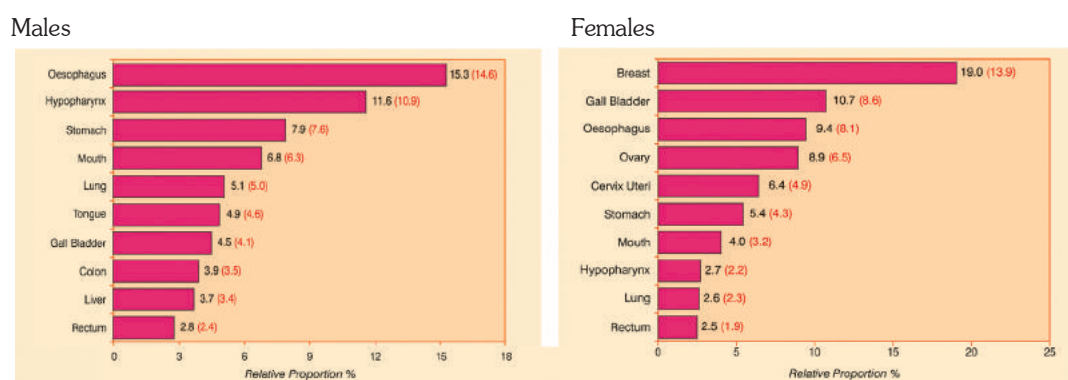


Fig. 2.9: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

KAMRUP URBAN DISTRICT

Males: The leading sites of cancer were: oesophagus (14.3%), hypopharynx (8.9%), lung (8.4%), stomach (6.7%) and mouth (5.2%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (20.5 and 29.1), hypopharynx (12.7 and 17.5), lung (12.0 and 18.7), stomach (9.6 and 14.1) and mouth (7.5 and 10.5).

Females: The leading cancer sites among females were: breast (17.5%) followed by oesophagus (10.2%), gall bladder (9.3%), cervix uteri (8.6%) and ovary (5.5%). The respective CR and AAR per 100,000 population for the sites were: breast (21.5 and 27.1), oesophagus (12.6 and 20.7), gall bladder (11.4 and 17.1), cervix uteri (10.6 and 14.5) and ovary (6.8 and 8.7).

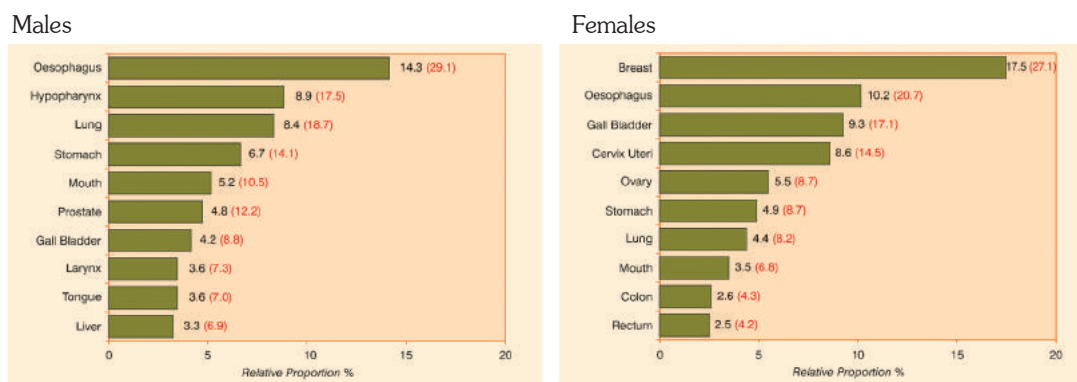


Fig. 2.10: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

MANIPUR STATE

Males: The leading sites of cancer were: lung (17.2%), stomach (7.1%), nasopharynx (5.9%), oesophagus (5.2%) and NHL (4.7%). The respective CR and AAR per 100,000 population for the sites were: lung (7.8 and 11.4), stomach (3.2 and 4.4), nasopharynx (2.7 and 3.4), oesophagus (2.4 and 3.3) and NHL (2.1 and 2.6).

Females: The leading cancer sites among females were: breast (15.3%) followed by lung (13.9%), cervix uteri (9.2%), thyroid (8.0%) and ovary (5.7%). The respective CR and AAR per 100,000 population for the sites were: breast (8.6 and 9.7), lung (7.8 and 10.9), cervix uteri (5.1 and 6.1), thyroid (4.5 and 4.9) and ovary (3.2 and 3.5).

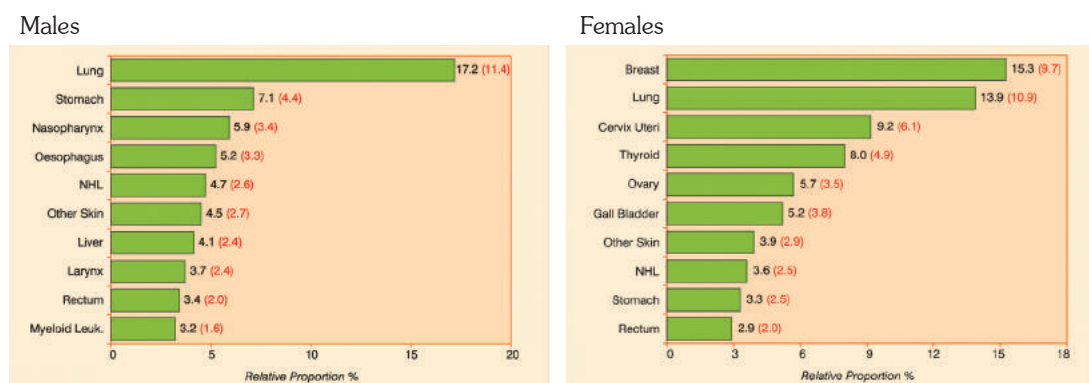


Fig. 2.11(a): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

IMPHAL WEST DISTRICT

Males: The leading sites of cancer were: lung (16.6%), oesophagus (6.4%), other skin (5.9%), liver (5.6%) and colon (4.4%). The respective CR and AAR per 100,000 population for the sites were: lung (13.4 and 15.8), oesophagus (5.2 and 6.2), other skin (4.8 and 5.4), liver (4.6 and 5.0) and colon (3.5 and 4.2).

Females: The leading cancer sites among females were: breast (16.5%) followed by lung (14.1%), thyroid (8.9%), cervix uteri (8.4%) and ovary (6.2%). The respective CR and AAR per 100,000 population for the sites were: breast (16.6 and 16.2), lung (14.1 and 15.4), thyroid (8.9 and 8.6), cervix uteri (8.4 and 8.6) and ovary (6.2 and 6.1).

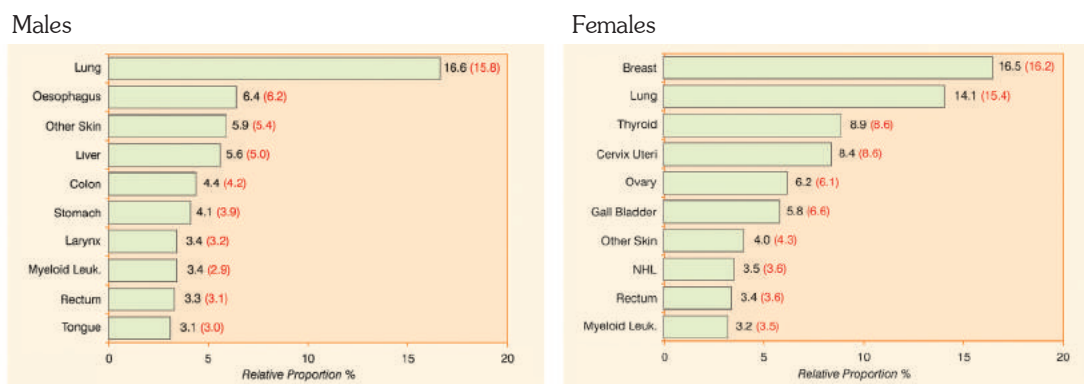


Fig. 2.11(b): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

MANIPUR STATE EXCLUDING IMPHAL WEST DISTRICT

Males: The leading sites of cancer were: lung (17.5%), stomach (8.4%), nasopharynx (7.3%), NHL (5.5%) and oesophagus (4.7%). The respective CR and AAR per 100,000 population for the sites were: lung (6.6 and 10.2), stomach (3.2 and 4.5), nasopharynx (2.8 and 3.6), NHL (2.1 and 2.7) and oesophagus (1.8 and 2.6).

Females: The leading cancer sites among females were: breast (14.8%) followed by lung (13.9%), cervix uteri (9.5%), thyroid (7.6%) and ovary (5.5%). The respective CR and AAR per 100,000 population for the sites were: breast (6.8 and 8.0) followed by lung (6.4 and 9.6), cervix uteri (4.4 and 5.5), thyroid (3.5 and 3.9) and ovary (2.5 and 2.8).

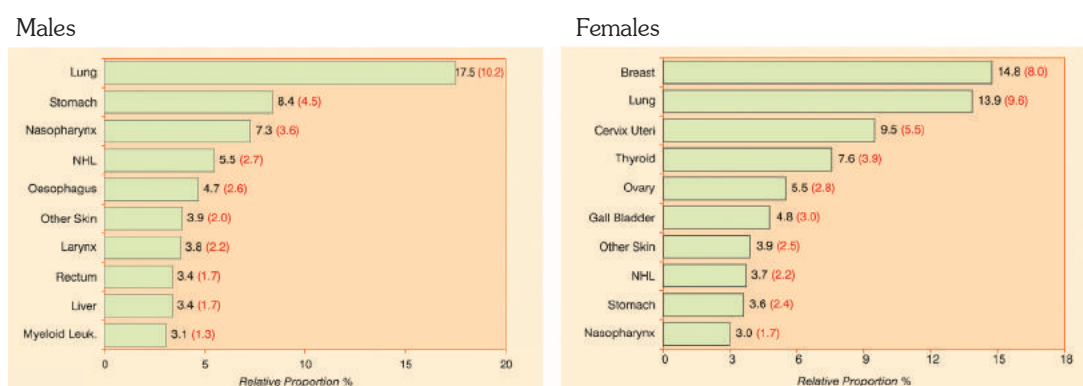


Fig. 2.11(c): Ten Leading Sites of Cancer – (2012-2014). (AARs given in parentheses).

MIZORAM STATE

Males: The leading sites of cancer were: stomach (18.5%), oesophagus (16.2%), lung (14.1%), liver (5.5%) and hypopharynx (5.0%). The respective CR and AAR per 100,000 population for the sites were: stomach (27.3 and 41.1), oesophagus (23.9 and 32.6), lung (20.8 and 32.6), liver (8.0 and 11.3) and hypopharynx (7.4 and 10.2).

Females: The leading cancer sites among females were: cervix uteri (15.9%), lung (15.6%), breast (13.0%), stomach (11.3%) and oesophagus (4.2%). The respective CR and AAR per 100,000 population for the sites were: cervix uteri (19.4 and 23.1), lung (19.0 and 29.3), breast (15.8 and 19.9), stomach (13.7 and 20.2) and oesophagus (5.1 and 7.6).

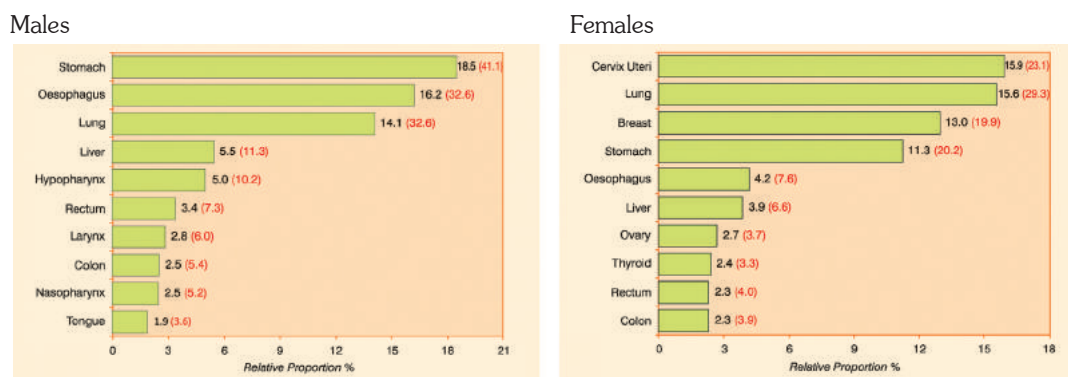


Fig. 2.12(a): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

AIZAWL DISTRICT

Males: The leading sites of cancer were: oesophagus (19.2%), stomach (15.5%), lung (12.6%), hypopharynx (6.5%) and liver (5.2%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (39.3 and 49.9), stomach (31.8 and 43.9), lung (25.8 and 37.9), hypopharynx (13.3 and 17.2) and liver (10.6 and 13.6).

Females: The leading cancer sites among females were: lung (17.8%), cervix uteri (15.6%), breast (14.5%), stomach (10.4%) and oesophagus (4.5%). The respective CR and AAR per 100,000 population for the sites were: lung (29.8 and 40.8), cervix uteri (26.0 and 28.0), breast (24.3 and 28.0), stomach (17.4 and 23.7) and oesophagus (7.5 and 10.3).

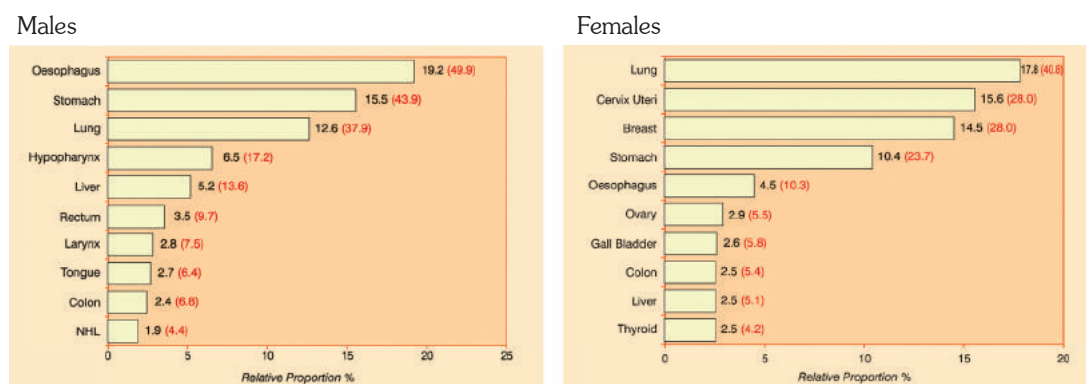


Fig. 2.12(b): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

MIZORAM STATE EXCLUDING AIZAWL DISTRICT

Males: The leading sites of cancer were: stomach (21.4%), lung (15.6%), oesophagus (13.2%), liver (5.7%) and hypopharynx (3.6%). The respective CR and AAR per 100,000 population for the sites were: stomach (24.8 and 39.3), lung (18.0 and 29.3), oesophagus (15.3 and 22.0), liver (6.6 and 9.9) and hypopharynx (4.1 and 5.8).

Females: The leading cancer sites among females were: cervix uteri (16.3%), lung (13.3%), stomach (12.1%), breast (11.3%) and liver (5.3%). The respective CR and AAR per 100,000 population for the sites were: cervix uteri (15.5 and 19.6), lung (12.6 and 21.1), stomach (11.5 and 17.7), breast (10.7 and 14.2) and liver (5.0 and 7.6).

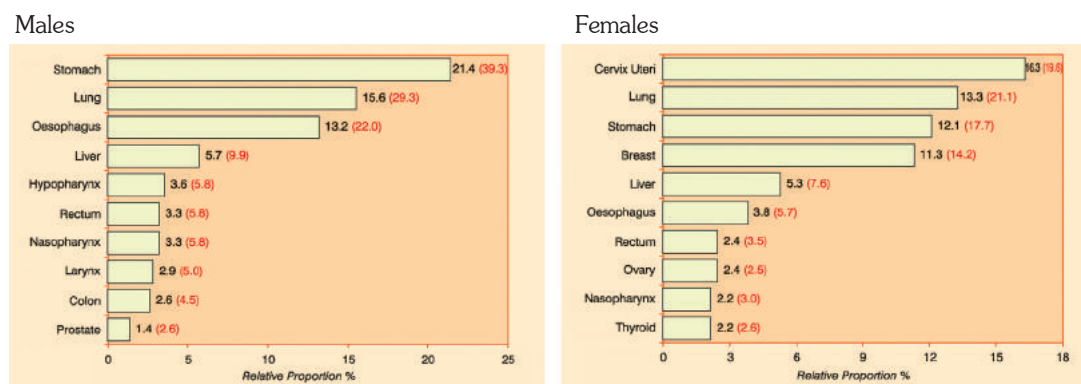


Fig. 2.12(c): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

SIKKIM STATE

Males: The leading sites of cancer were: stomach (15.8%), liver (7.6%), lung (7.2%), oesophagus (6.8%) and nasopharynx (5.4%). The respective CR and AAR per 100,000 population for the sites were: stomach (11.3 and 14.8), liver (5.4 and 7.4), lung (5.1 and 6.7), oesophagus (4.8 and 6.4) and nasopharynx (3.8 and 4.4).

Females: The leading cancer sites among females were: breast (12.7%), cervix uteri (10.0%), lung (6.8%), gall bladder (6.3%), stomach (5.9%) and ovary (5.9%). The respective CR and AAR per 100,000 population for the sites were: breast (9.7 and 11.1), cervix uteri (7.6 and 10.1), lung (5.2 and 7.8), gall bladder (4.8 and 6.7), stomach (4.5 and 6.8) and ovary (4.5 and 4.9).

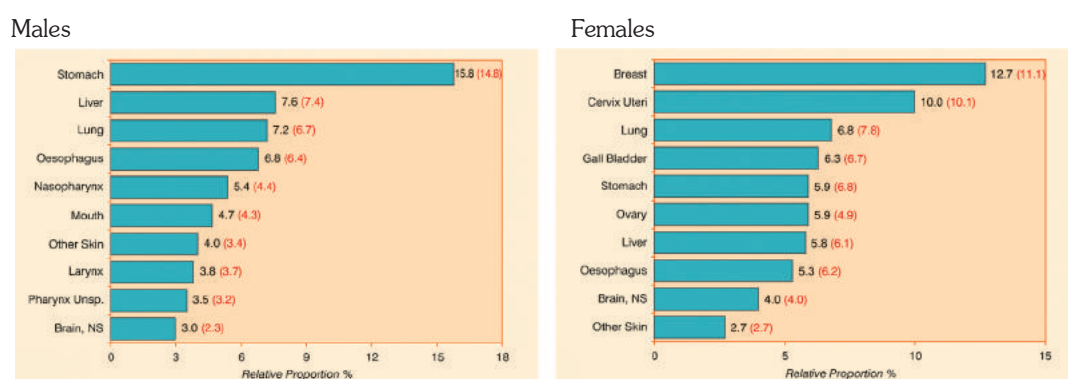


Fig. 2.13: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

AHMEDABAD URBAN

Males: The leading sites of cancer among males were: mouth (20.3%), tongue (11.5%), lung (8.4%), oesophagus (5.6%) and prostate (3.7%). The respective CR and AAR of the sites per 100,000 population for the sites were: mouth (17.8 and 18.1), tongue (10.0 and 10.4), lung (7.3 and 9.1), oesophagus (4.9 and 5.9) and prostate (3.3 and 4.4).

Females: Breast (31.5%) was the leading site followed by cervix uteri (9.3%) in Ahmedabad Urban area. The other leading sites were: ovary (5.3%), mouth (4.8%) and tongue (4.6%). The respective CR and AAR per 100,000 population were: breast (23.0 and 23.8), cervix uteri (6.8 and 6.9), ovary (3.9 and 4.0), mouth (3.5 and 3.6) and tongue (3.3 and 3.4).

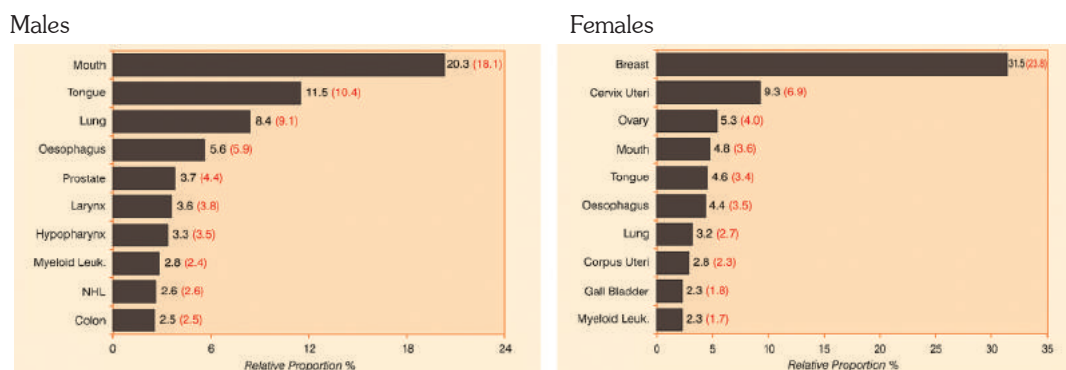


Fig. 2.14: Ten Leading Sites of Cancer (2012-2013). (AARs given in parentheses).

AURANGABAD

Males: The leading sites of cancer among males were: lung (12.3%), mouth (11.8%), tongue (9.6%), oesophagus (6.8%) and larynx (6.6%). The respective CR and AAR of the sites per 100,000 population for the sites were: lung (7.0 and 9.7), mouth (6.7 and 7.4), tongue (5.5 and 6.3), oesophagus (3.8 and 5.4) and larynx (3.7 and 5.2).

Females: The leading sites of cancer among females were: breast (30.6%) followed by cervix uteri (19.9%), ovary (6.2%), lung (4.4%), oesophagus (3.3%) and tongue (3.3%). The respective CR and AAR per 100,000 population were: breast (18.5 and 22.6), cervix uteri (12.0 and 14.3), ovary (3.7 and 4.7), lung (2.6 and 3.4), oesophagus (2.0 and 2.6) and tongue (2.0 and 2.4).

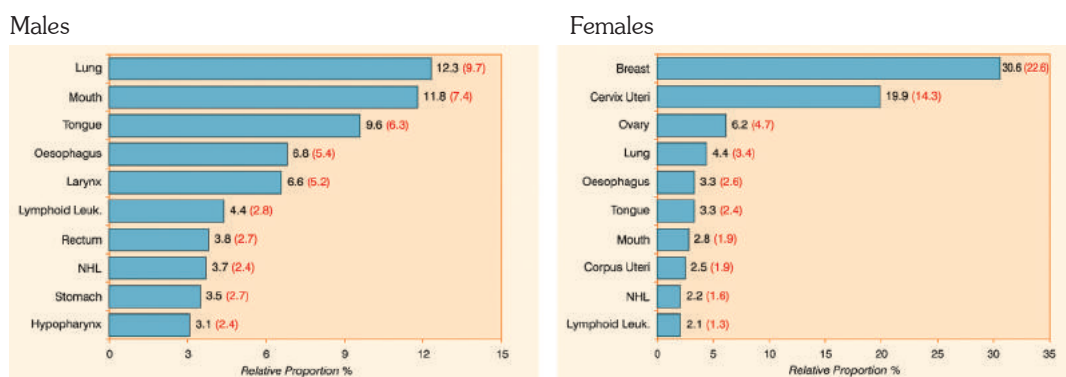


Fig. 2.15: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

KOLKATA

Males: The leading sites of cancer were: lung (18.9%), prostate (8.2%), mouth (6.9%), tongue (5.5%) and larynx (5.4%). The respective CR and AAR per 100,000 population for the sites were: lung (22.4 and 19.1), prostate (9.8 and 8.2), mouth (8.2 and 6.8), tongue (6.5 and 5.4) and larynx (6.4 and 5.6).

Females: The leading cancer sites among females were: breast (25.4%) followed by cervix uteri (10.1%), ovary (7.8%), gall bladder (7.4%) and lung (7.1%). The respective CR and AAR per 100,000 population for the sites were: breast (30.7 and 25.5), cervix uteri (12.2 and 10.4), ovary (9.4 and 8.0), gall bladder (8.9 and 7.7) and lung (8.6 and 7.4).

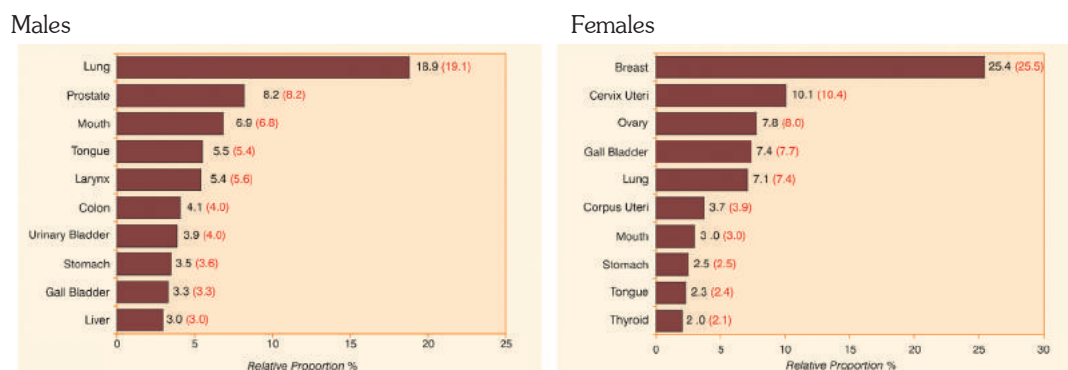


Fig. 2.16: Ten Leading Sites of Cancer (2012). (AARs given in parentheses).

KOLLAM DISTRICT

Males: The leading sites of cancer among males were: lung (18.2%), mouth (5.6%), prostate (5.2%), liver (4.7%), stomach (4.7%) and larynx (4.5%). The respective CR and AAR of the sites per 100,000 population for the sites were: lung (26.9 and 21.5), mouth (8.3 and 6.6), prostate (7.6 and 6.1), liver (7.0 and 5.5), stomach (6.9 and 5.5) and larynx (6.7 and 5.3).

Females: The leading sites of cancer among females were: breast (27.9%) followed by thyroid (11.0%), cervix uteri (6.8%), ovary (5.3%), mouth (3.9%) and lung (3.8%). The respective CR and AAR per 100,000 population were: breast (36.3 and 27.7), thyroid (14.3 and 12.0), cervix uteri (8.9 and 6.7), ovary (6.9 and 5.4), mouth (5.1 and 3.7) and lung (5.0 and 3.8).

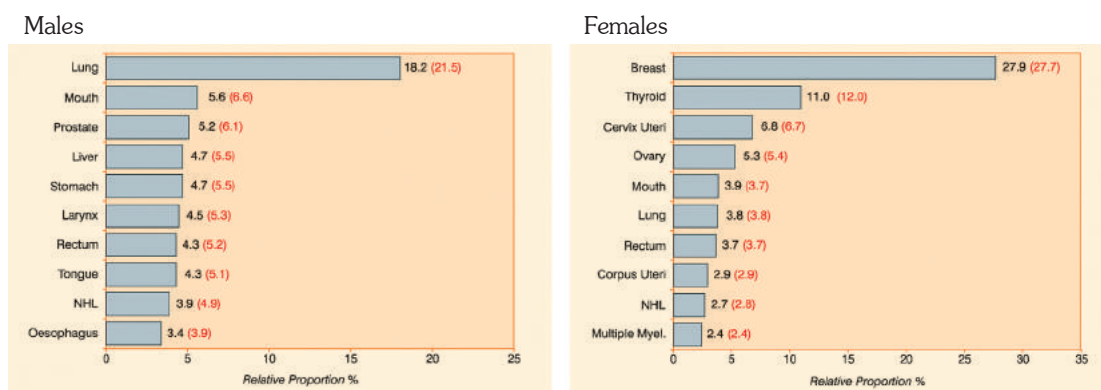


Fig. 2.17: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

NAGPUR

Males: The leading sites of cancer were: mouth (15.7%), tongue (8.0%), oesophagus (7.0%), lung (6.8%) and larynx (6.0%). The respective CR and AAR per 100,000 population for the sites were: mouth (13.4 and 12.8), tongue (6.8 and 6.4), oesophagus (6.0 and 6.7), lung (5.8 and 6.5) and larynx (5.1 and 5.8).

Females: The leading cancer sites among females were: breast (31.9%) followed by cervix uteri (13.5%), ovary (6.3%), mouth (5.1%) and oesophagus (4.1%). The respective CR and AAR per 100,000 population for the sites were: breast (30.4 and 29.3), cervix uteri (12.9 and 12.9), ovary (6.0 and 6.0), mouth (4.9 and 4.9) and oesophagus (3.9 and 4.1).

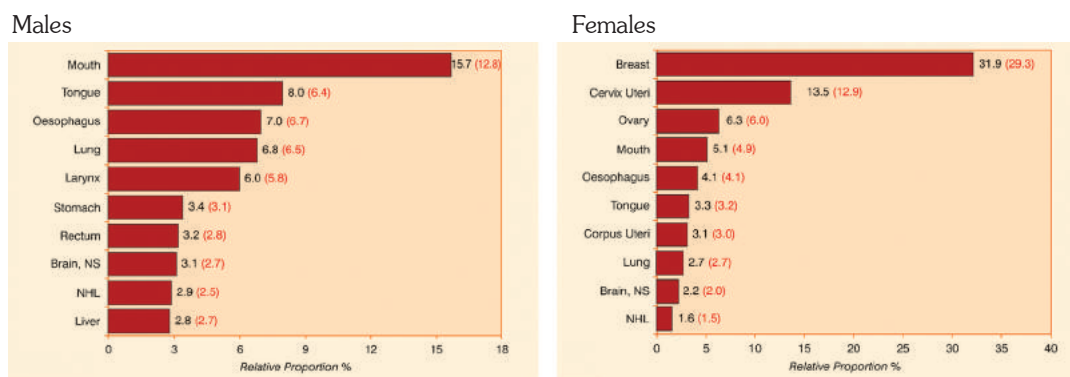


Fig. 2.18: Ten Leading Sites of Cancer (2012-2013). (AARs given in parentheses).

PUNE

Males: The leading sites of cancer were: mouth (10.6%), lung (8.3%), prostate (7.3%), tongue (5.7%) and NHL (4.7%). The respective CR and AAR per 100,000 population for the sites were: mouth (6.6 and 7.7), lung (5.2 and 6.9), prostate (4.5 and 6.3), tongue (3.6 and 4.0) and NHL (2.9 and 3.7).

Females: The leading cancer sites among females were: breast (31.4%) followed by cervix uteri (10.6%), ovary (7.4%), mouth (4.4%) and lung (4.3%). The respective CR and AAR per 100,000 population for the sites were: breast (23.4 and 26.4), cervix uteri (7.8 and 9.0), ovary (5.5 and 6.3), mouth (3.3 and 3.7) and lung (3.2 and 3.9).

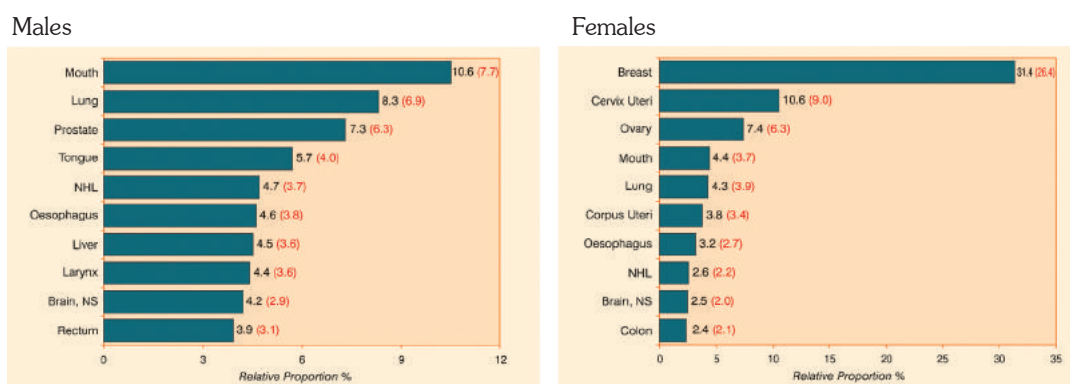


Fig. 2.19: Ten Leading Sites of Cancer (2012-2013). (AARs given in parentheses).

THIRUVANANTHAPURAM DISTRICT

Males: The leading sites of cancer were: lung (13.0%), prostate (7.2%), mouth (5.1%), tongue (5.1%), rectum (4.2%) and larynx (4.2%). The respective CR and AAR per 100,000 population for the sites were: lung (20.8 and 16.8), prostate (11.5 and 9.4), mouth (8.2 and 6.6), tongue (8.1 and 6.5), rectum (6.8 and 5.5) and larynx (6.7 and 5.4).

Females: The leading cancer sites among females were: breast (28.5%) followed by thyroid (10.4%), cervix uteri (6.0%), ovary (5.9%) and corpus uteri (4.3%). The respective CR and AAR per 100,000 population for the sites were: breast (43.9 and 33.7), thyroid (16.0 and 13.3), cervix uteri (9.3 and 7.0), ovary (9.0 and 7.0) and corpus uteri (6.6 and 5.1).

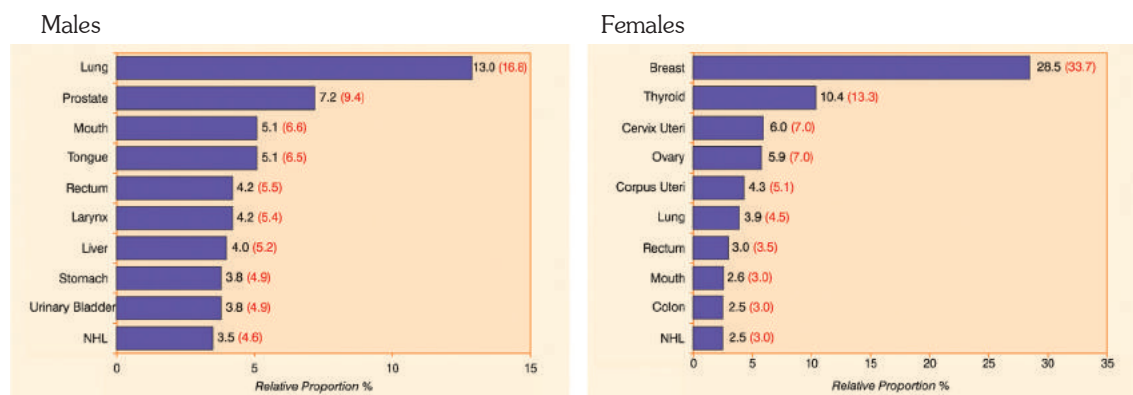


Fig. 2.20: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

MEGHALAYA

Males: The leading sites of cancer were: oesophagus (30.8%), hypopharynx (9.0%), stomach (6.8%), lung (6.4%) and tongue (5.7%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (27.3 and 51.2), hypopharynx (8.0 and 15.1), stomach (6.1 and 12.5), lung (5.7 and 12.8) and tongue (5.1 and 9.2).

Females: The leading cancer sites among females were: oesophagus (22.2%) followed by cervix uteri (11.1%), breast (7.9%), mouth (7.7%) and stomach (7.4%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (12.1 and 22.7), cervix uteri (6.0 and 9.6), breast (4.3 and 6.7), mouth (4.2 and 7.7) and stomach (4.0 and 6.8).

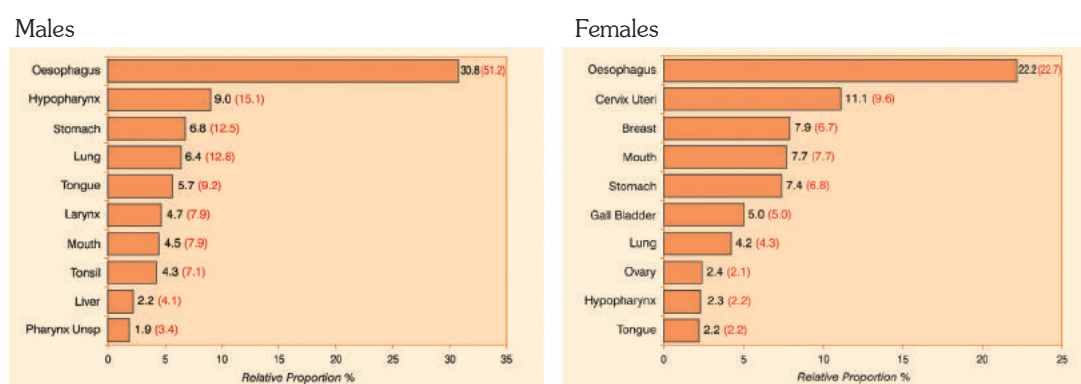


Fig. 2.21(a): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

EAST KHASI HILLS DISTRICT

Males: The leading sites of cancer were: oesophagus (33.7%), hypopharynx (10.4%), stomach (5.9%), lung (5.5%), tongue (5.5%) and larynx (5.1%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (42.4 and 71.2), hypopharynx (13.1 and 22.2), stomach (7.3 and 14.3), lung (7.0 and 14.4), tongue (7.0 and 11.7) and larynx (6.4 and 10.8).

Females: The leading cancer sites among females were: oesophagus (26.5%) followed by cervix uteri (9.6%), breast (8.4%), mouth (7.5%) and stomach (6.8%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (19.9 and 33.0), cervix uteri (7.2 and 10.3), breast (6.3 and 8.9), mouth (5.6 and 9.1) and stomach (5.1 and 7.6).

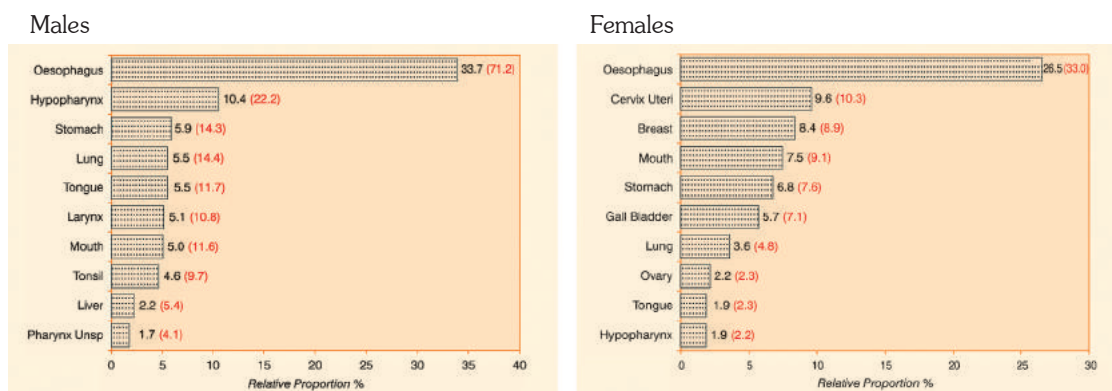


Fig. 2.21(b): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

Males: The leading sites of cancer were: lung (17.6%), oesophagus (7.9%), larynx (6.6%), stomach (6.1%) and mouth (6.0%). The respective CR and AAR per 100,000 population for the sites were: lung (11.0 and 14.2), oesophagus (5.0 and 6.2), larynx (4.1 and 5.1), stomach (3.8 and 4.6) and mouth (3.7 and 4.5).

Females: The leading cancer sites among females were: cervix uteri (16.8%) followed by breast (13.7%), gall bladder (9.3%), ovary (6.1%) and oesophagus (5.8%). The respective CR and AAR per 100,000 population for the sites were: cervix uteri (8.1 and 9.2), breast (6.6 and 6.9), gall bladder (4.5 and 5.3), ovary (2.9 and 3.2) and oesophagus (2.8 and 3.5).

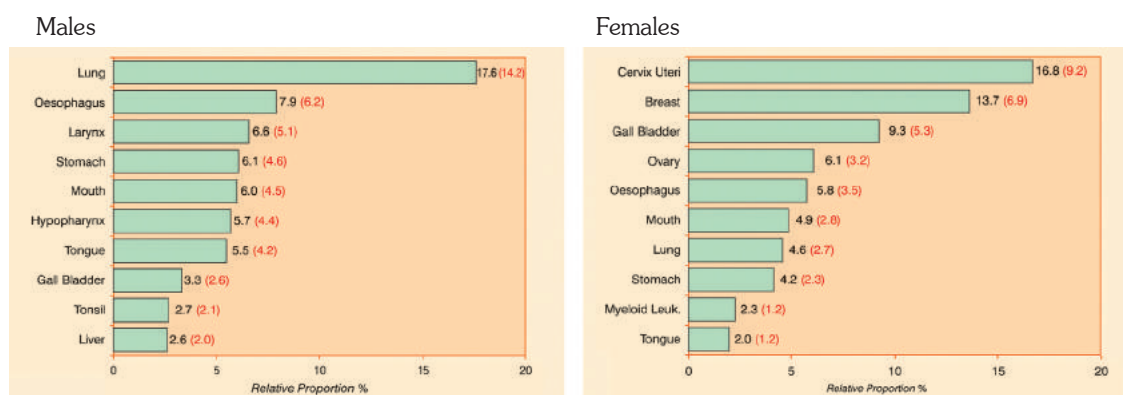


Fig. 2.22: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

NAGALAND

Males: The leading sites of cancer were: nasopharynx (15.3%), stomach (12.3%), oesophagus (10.4%), lung (6.1%), hypopharynx (5.9%) and larynx (5.9%). The respective CR and AAR per 100,000 population for the sites were: nasopharynx (11.5 and 15.2), stomach (9.2 and 17.8), oesophagus (7.8 and 13.8), lung (4.6 and 7.9), hypopharynx (4.4 and 7.8) and larynx (4.4 and 7.8).

Females: The leading cancer sites among females were: cervix uteri (16.7%) followed by breast (12.6%), stomach (11.0%), nasopharynx (10.8%) and thyroid (5.3%). The respective CR and AAR per 100,000 population for the sites were: cervix uteri (8.9 and 13.1), breast (6.8 and 9.1), stomach (5.9 and 11.1), nasopharynx (5.8 and 6.8) and thyroid (2.9 and 4.2).

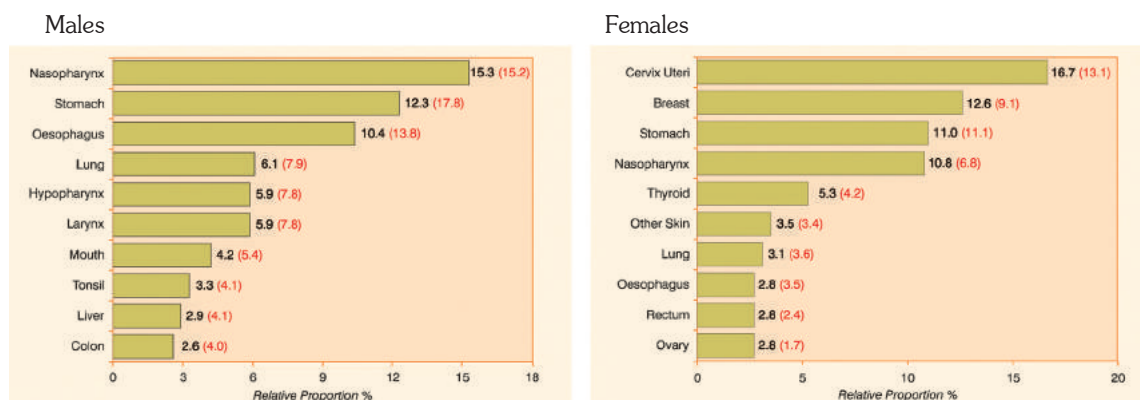


Fig. 2.23: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

WARDHA DISTRICT

Males: The leading sites of cancer were: mouth (14.6%), oesophagus (7.0%), lung (6.4%), liver (6.2%) and tongue (5.2%). The respective CR and AAR per 100,000 population for the sites were: mouth (9.4 and 8.6), oesophagus (4.5 and 4.1), lung (4.1 and 3.8), liver (4.0 and 3.8) and tongue (3.4 and 3.0).

Females: The leading cancer sites among females were: breast (28.2%) followed by cervix uteri (13.0%), ovary (6.6%), mouth (5.7%), oesophagus (4.6%) and thyroid (4.6%). The respective CR and AAR per 100,000 population for the sites were: breast (20.8 and 18.8), cervix uteri (9.6 and 8.6), ovary (4.9 and 4.4), mouth (4.3 and 3.7), oesophagus (3.4 and 3.0) and thyroid (3.4 and 2.9).

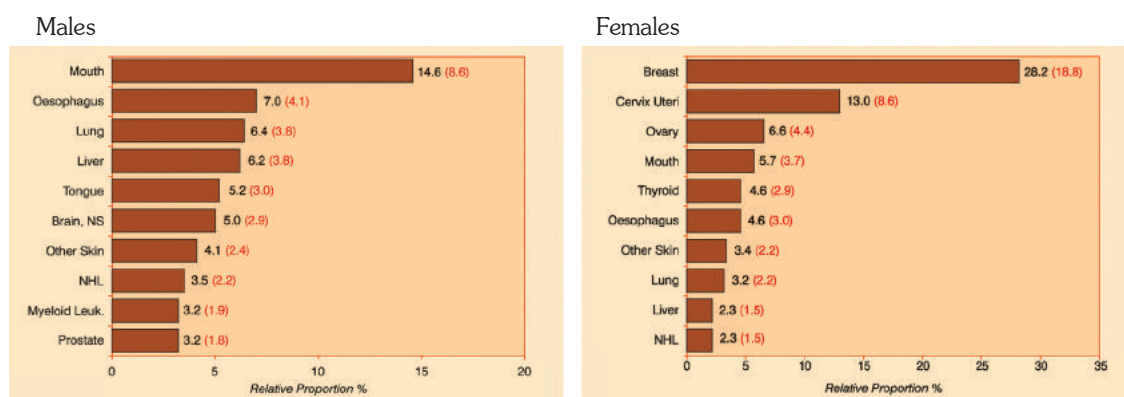


Fig. 2.24: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

NAHARLAGUN

Males: The leading sites of cancer were: stomach (24.5%), liver (19.9%), oesophagus (7.8%), lung (6.8%) and nasopharynx (4.4%). The respective CR and AAR per 100,000 population for the sites were: stomach (14.3 and 26.7), liver (11.6 and 21.4), oesophagus (4.5 and 8.7), lung (4.0 and 7.4) and nasopharynx (2.5 and 4.5).

Females: The leading cancer sites among females were: stomach (13.8%) followed by cervix uteri (12.9%), breast (10.8%), thyroid (9.2%), liver (6.8%) and ovary (6.8%). The respective CR and AAR per 100,000 population for the sites were: stomach (8.1 and 16.3), cervix uteri (7.6 and 11.3), breast (6.3 and 9.4), thyroid (5.4 and 7.4), liver (4.0 and 7.1) and ovary (4.0 and 6.7).

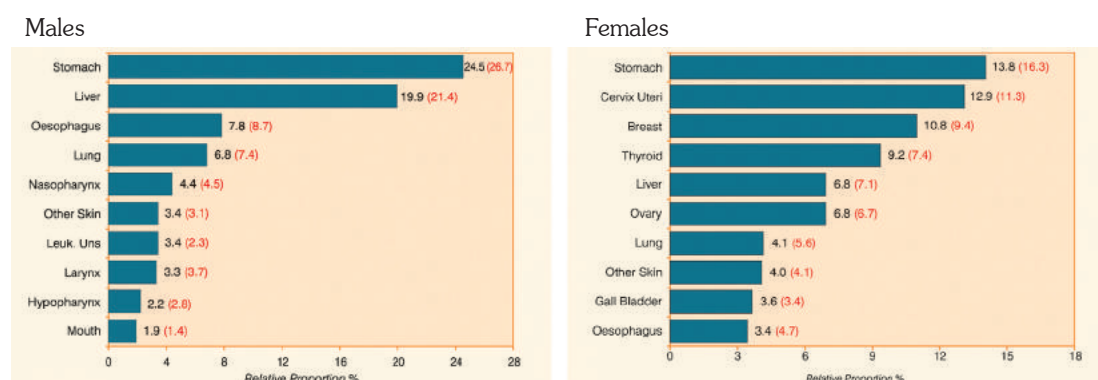


Fig. 2.25(a): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

PAPUMPARE DISTRICT

Males: The leading sites of cancer were: stomach (20.7%), liver (14.4%), oesophagus (10.7%), lung (9.7%) and other skin (5.0%). The respective CR and AAR per 100,000 population for the sites were: stomach (21.5 and 50.2), liver (14.9 and 38.0), oesophagus (11.1 and 26.5), lung (10.0 and 27.8) and other skin (5.2 and 11.0).

Females: The leading cancer sites among females were: breast (15.0%) followed by cervix uteri (13.5%), thyroid (11.1%), stomach (9.9%) and ovary (6.3%). The respective CR and AAR per 100,000 population for the sites were: breast (17.3 and 29.2), cervix uteri (15.6 and 30.2), thyroid (12.8 and 20.7), stomach (11.4 and 29.2) and ovary (7.3 and 15.2).

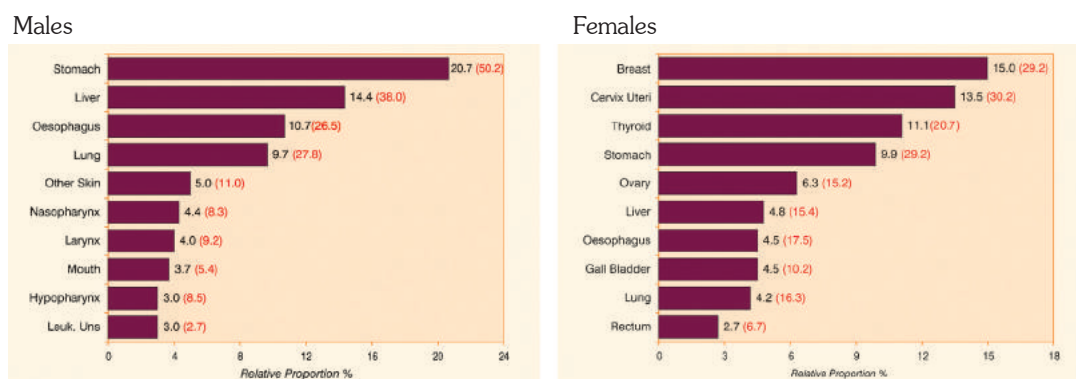


Fig. 2.25(b): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

NAHARLAGUN EXCLUDING PAPUMPARE DISTRICT

Males: The leading sites of cancer were: stomach (27.1%), liver (23.6%), oesophagus (5.7%), lung (4.8%) and nasopharynx (4.4%). The respective CR and AAR per 100,000 population for the sites were: stomach (12.2 and 21.7), liver (10.7 and 18.4), oesophagus (2.6 and 4.8), lung (2.2 and 3.4) and nasopharynx (2.0 and 3.6).

Females: The leading cancer sites among females were: stomach (17.3%) followed by cervix uteri (12.4%), liver (8.6%), thyroid (7.6%) and ovary (7.3%). The respective CR and AAR per 100,000 population for the sites were: stomach (7.0 and 13.5), cervix uteri (5.0 and 6.9), liver (3.5 and 5.4), thyroid (3.1 and 4.1) and ovary (3.0 and 4.7).

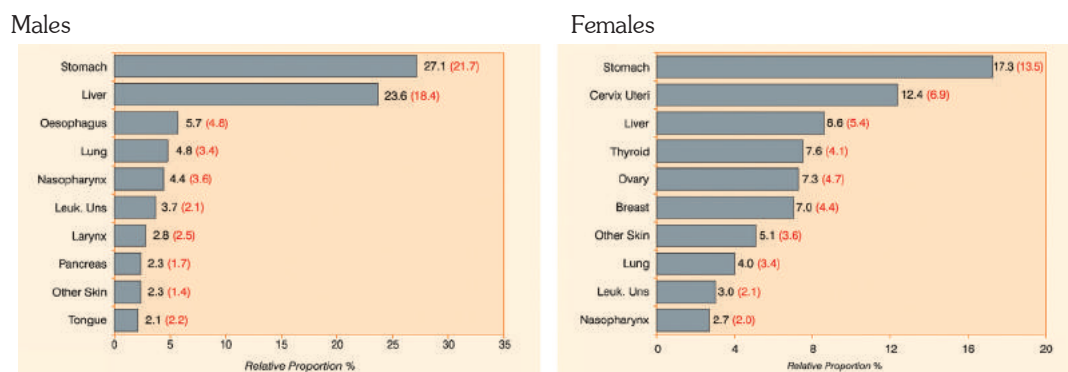


Fig. 2.25(c): Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

PASIGHAT

Males: The leading sites of cancer were: stomach (19.4%), liver (7.4%), hypopharynx (6.3%), oesophagus (5.1%), other skin (5.1%), rectum (5.1%) and lung (4.6%). The respective CR and AAR per 100,000 population for the sites were: stomach (16.2 and 22.8), liver (6.2 and 8.5), hypopharynx (5.2 and 6.1), oesophagus (4.3 and 5.7), other skin (4.3 and 5.6), rectum (4.3 and 5.1) and lung (3.8 and 5.5).

Females: The leading cancer sites among females were: cervix uteri (23.9%) followed by breast (17.6%), stomach (10.7%), ovary (7.6%), liver (3.8%) and other skin (3.8%). The respective CR and AAR per 100,000 population for the sites were: cervix uteri (18.7 and 22.5), breast (13.8 and 17.4), stomach (8.4 and 12.2), ovary (5.9 and 8.1), liver (2.9 and 4.9) and other skin (2.9 and 3.5).

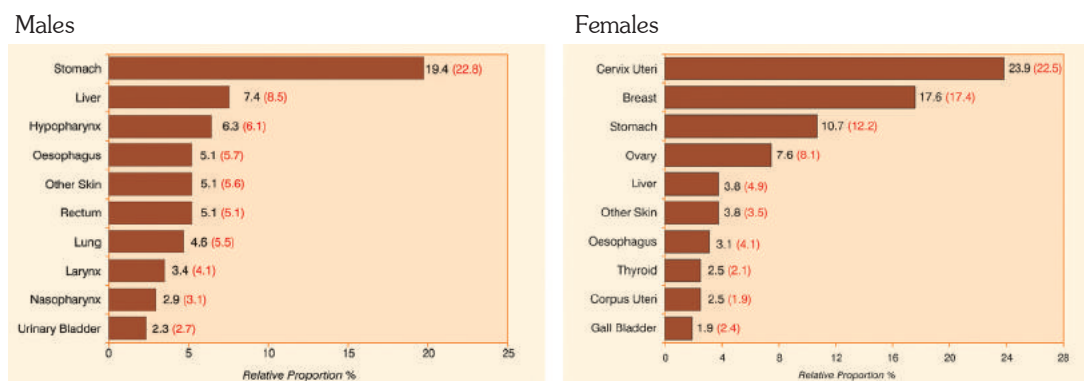


Fig. 2.26: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

PATIALA DISTRICT

Males: The leading sites of cancer were: oesophagus (11.7%), lung (7.3%), prostate (6.3%), tongue (4.6%) and mouth (4.4%). The respective CR and AAR per 100,000 population for the sites were: oesophagus (10.6 and 11.5), lung (6.7 and 7.4), prostate (5.8 and 6.3), tongue (4.2 and 4.4) and mouth (4.0 and 4.3).

Females: The leading cancer sites among females were: breast (30.3%) followed by cervix uteri (10.5%), oesophagus (5.9%), ovary (4.9%), gall bladder (2.8%) and uterus unspecified (2.8%). The respective CR and AAR per 100,000 population for the sites were: breast (34.2 and 33.1), cervix uteri (11.8 and 11.5), oesophagus (6.6 and 6.7), ovary (5.5 and 5.5), gall bladder (3.2 and 3.2) and uterus unspecified (3.1 and 3.0).

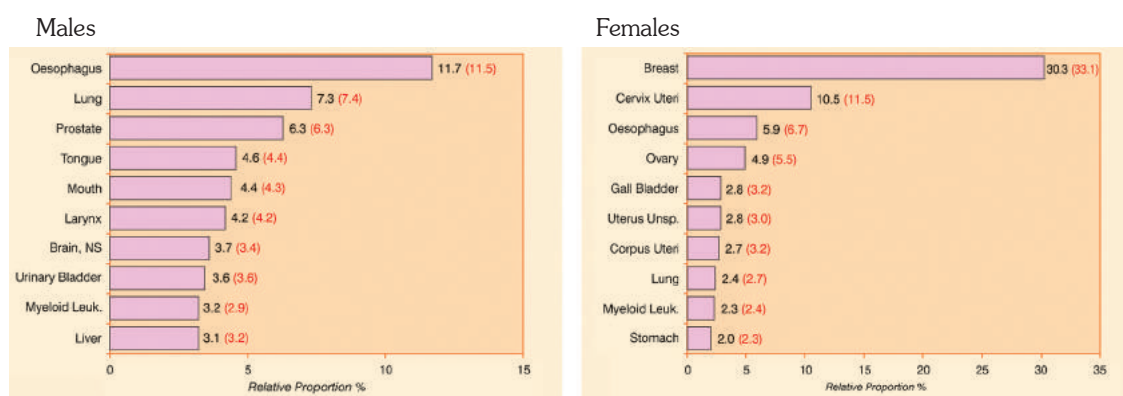


Fig. 2.27: Ten Leading Sites of Cancer (2012-2014). (AARs given in parentheses).

Publication

1. National Cancer Registry Programme (NCRP) Three-year Report of Population Based Cancer Registries: 2012-2014. Incidence, Distribution, Trends in Incidence Rates and Projections of Burden of Cancer. Report of 27 PBCRs in India, (Indian Council of Medical Research), Bangalore, 2016.

Breast cancer is the commonest cancer in women worldwide with a widely variable incidence between countries and regions. The developed countries with a small proportion of the world population account for almost 50% of breast cancers diagnosed worldwide. The lowest breast cancer incidence is reported from far Eastern and South-East Asian countries. In the developing countries of Asia, the health care burden on account of breast cancer has been steadily mounting. It is expected that in the coming decades, these countries would account for majority of new breast cancer patients diagnosed globally. Over 100,000 new breast cancer patients are estimated to be diagnosed annually in India. As per the ICMR-PBCR data, breast cancer is the commonest cancer among women in urban registries of Delhi, Mumbai, Ahmedabad, Kolkatta, and Trivandrum where it constitutes > 30% of all cancers in females. The age standardized incidence rates (AARs) range from 6.2 to 39.5 per 100,000 Indian women. The incidence of this disease has been consistently increasing, and it is estimated it has risen by 50% between 1965 and 1985. The rise in incidence of 0.5–2% per annum has been seen across all regions of India and in all age groups but more so in the younger age groups (< 45 years). In general, breast cancer has been reported to occur a decade earlier in Indian patients compared to their western counterparts. While the majority of breast cancer patients in western countries are postmenopausal and in their 60s and 70s, the picture is quite different in India with premenopausal patients constituting about 50% of all patients. More than 80% of Indian patients are younger than 60 years of age. The average age of patients in 6 hospital-based cancer registries ranged from 44.2 years in Dibrugarh, 46.8 years in Delhi, 47 years in Jaipur, to 49.6 years in Bangalore and Chennai. The average age of breast cancer patients has been reported to be 50–53 years in various population-based studies done in different parts of the country. A significant proportion of Indian breast cancer patients are younger than 35 years of age. This proportion varies between 11% (Tata Memorial Hospital (TMH) Mumbai) to 26% (SGPGIMS Lucknow). Young age has been associated with larger tumor size, higher number of metastatic lymph nodes, poorer tumor grade, low rates of hormone receptor-positive status, earlier and more frequent loco-regional recurrences, and poorer overall survival.

Breast cancer is the most common malignancy in women worldwide with an age standardized incidence rate of 46.3 per 100,000. It is also the most common cause of cancer deaths among women with 626,679 deaths occurring worldwide according to GLOBOCON 2018. In India breast cancer represents a major health concern with an incidence of 27.7%. There is a rising trend in breast cancer incidence with an average of 162468 women being diagnosed every year and about 87090 succumbing to the disease. Carcinoma breast has surpassed cervical cancer as the most common cancer in twelve population based cancer registries (PBCR) across India except at rural registry at Barshi where latter is still the commonest cancer among women. However, it is estimated that by 2020 breast cancer will emerge as the leading cancer in Indian women. Also, there is a high proportion of early onset breast cancer among Indian women. Incidence of breast cancer in India is significantly low (AAR 21-37/100,000) as compared to western countries (AAR 75-90/100,000), however the incidence of breast cancer in younger women (<40 years) remains more or less same suggesting a high proportion of early onset of breast cancer in Indian women.

Marriage at an early age, early and multiple childbirths and breastfeeding of all children for a long period of time is the norm in most Indian societies. However, the urban educated class is moving away from this trend, with late-age childbirth and little or no breastfeeding due to changing social values and the demands of jobs on working women. These changes may be partly responsible for the increasing trend of breast cancer incidence. Nulliparity and late age at first childbirth are consistently observed reproductive risk factors. A case control study in Mumbai indicated that compared to married women, single women had a 4–5-fold higher risk for developing breast cancer in the age group of 40–54 and above. In another study, nulliparous women had a 2.2-fold higher risk than parous women.

Molecular Basis of Genesis of Breast Cancer

The Director General, Indian Council of Medical Research constituted a Sub-Committee on Breast Cancer. The situational analysis paper in different facets of Breast Cancer such as epidemiology, morphology, clinical management and molecular oncology were debated. The situational analysis papers in each subject area were presented and committee recommended the formulation of two projects “Clinico-Epidemiological Study in Breast Cancer among Rural and Urban areas,” and “Establishment of Breast Cancer Cell line.”

The proposal on, “Establishment of Breast Cancer Cell line,” was undertaken with the aim to develop cell line from primary breast tumors. The study was aimed to (i) establish cell lines(s) from human primary breast tumors and (ii) characterize established breast cancer cell lines. The breast tumor specimens were collected and primary cultures were processed. Expression of epithelial markers such as ESA, cytokeratins, mesenchymal marker vimentin, estrogen receptor- was analyzed in the established cultures by either immune-cytochemistry. The tumorigenicity in nude mice was established in collaboration with ICMR’s National Institute of Nutrition, Hyderabad. The breast tumors were collected and used for initiation of primary cultures. Primary cultures were established by subjecting the primary tumor pieces to enzymatic disaggregation (using trypsin and collagenase), single cell suspension obtained was seeded into tissue culture flasks containing DMEM and growth factors. Of the total samples collected, 28 were from early onset (< 45 years) and 24 are from late onset (>50 years) breast cancers. Together, the primary cultures were established. Two primary cell lines were completely purified and passaged several times and hence considered as established continuous cell lines. Karyotyping of both the cell lines has been done at both early and late passages to facilitate cytogenetic examination and comparison of the karyotypes between early and late passages. However, no karyotype changes have been observed among them; indicating that both established cell lines were quite stable. The project on “Comparative Study of Genetic, Clinical and Epidemiological factors of Breast Cancer in Rural and Urban Area of India,” is aimed to understand prevalence of breast cancer among rural and urban populations, reasons of variation, if any; analyze the genetic susceptibility by single nucleotide polymorphisms (SNP) in familial and sporadic breast cancer. The identified centers are rural and urban areas with centers at ICPO, Noida; IOP, Delhi; AIIMS, Delhi; Regional Cancer Centre (RCC), Thiruvananthapuram and Regional Medical Research Centre (RMRC), Dibrugarh. Preliminary data showed increase in cancer cases in young women associated with risk factors such as late age of marriage, nulliparity etc. Some of data is published.

The Indian Council of Medical Research’s the National Institute of Pathology has made significant advances in its activities in tumour biology along with development and strengthening of its infrastructure facilities to include high throughput research technologies mainly for genomics. The major thrust is on translational research such as identifying genes that are associated with risk or serve as diagnostic tools, targets for therapy or predicting the drug response. Molecular strategies such as genomics and bioinformatics have been used to generate information that may be used for targeting the result into daily medical practice with a view to understand the biology, improving diagnosis, enhancing treatment and reducing mortality

associated with cancer. The Institute is on the verge to establish proteomics, nanotechnology and stem cell facility to realize the research goals. The tumor biology/oncopathology division which was established in 1992 with the aims of morphologic, immune-histochemical and electron microscopic characterization of undifferentiated tumors has been recognized as a referral center for diagnosis and immune-histological characterization of undifferentiated and special tumors like lymphomas/leukemia. Thrust areas of research have been on breast cancer, including genetics of breast cancer in Indian population, urologic malignancies including prostate and urinary bladder cancer, haematopoietic-lymphoid malignancies and melanotic tumors with a view to understand the molecular mechanism of carcinogenesis. More recently, projects on cancers in Northeast region in India have given interesting leads as to the reason for the high prevalence of cancers in this region of India.

Studies on breast cancer were initially focused on determinants of cell behavior in breast cancer including steroid hormone receptor viz estrogen and progesterone receptors, study of molecular events leading to immortalization / senescence to human mammary epithelial cancer cells *in vitro*. Studies on time trends and clinico-epidemiologic features of breast cancer in Indian women showed the occurrence of breast cancer in younger women (<40 years) with significant annual percentage increase in incidence at the younger age (4.24%) as compared to older patients (0.8% to 1.6%). For the first time, it was reported that 22.4% familial incidence of breast cancer in Indian women indicating involvement of genetic risk factors. Studies on contribution of genetic risk factors showed low prevalence of BRCA1/2 mutations (2.94%) in Indian patients as compared to other Asian and Western countries. However, other genetic alterations such as CYP17 A2 allele, VDR Poly A L allele and ≥ 20 CAG repeats in AR gene showed putative high risk. Women carrying genotype of three of the above alleles were found to have increased risk for developing breast cancer (OR=4.68). Genome wide scanning of gene expression and hyper methylation profile delineated the molecular functional pathways involved in early and late onset breast cancer. Since majority of breast cancer patients in India present with locally advanced stage requiring neo adjuvant chemotherapy for down staging and de bulking of tumor, the role of type I growth factor receptor genes (EGFR, C-erbB2, C-erbB3), androgen receptor gene, p-glycoprotein gene and mismatched repair genes as biomarker to predict response to preoperative neoadjuvant chemotherapy and showed androgen receptor to be an independent predictor for drug response.

Studies on the 'Role of pesticide exposure for high incidence of Breast Cancer and Non-Hodgkin lymphomas in North-East India' for which mutation analysis for high penetrance genes BRCA 1 & 2 and low penetrance genes associated with estrogen metabolism and detoxification enzymes for detoxification of carcinogens viz. CYP17 and GST genes in breast cancer did not show significant contribution to the risk, however significant association of betel-quid chewing has been demonstrated with risk of breast cancer in this population. A genome wide copy number analysis has demonstrated chromosomal changes specific to betel quid chewing in these patients.

Breast Cancer in Indian Women-Time Trends and Clinico-Epidemiological Features

The cancer registry data demonstrates the magnitude of the current health problem associated with breast cancer in the Indian population. Understanding its pathogenesis, morphological features and epidemiological factors including various risk factors, family history etc holds a great promise for the treatment, early detection and prevention of this cancer. The study was undertaken to describe some of the clinico-pathological features of the breast cancer cases seen at Safdarjang Hospital-a tertiary level hospital in the city of Delhi. A detailed analysis of 578 breast cancer cases diagnosed during the years 1989-2003 was done for various parameters such as age, religious groups, morphological patterns of tumors, lymph node status, TNM staging and familial history.

Average age of the patient at presentation was found to be 47.9 years and 57.9% patients were under fifty years of age. A positive family history had been found in 22.4% cases, more in patients under 45 years of age, breast cancer being the commonest cancer type among the family members [85%]. TNM staging revealed 78.6% patients presented at locally advanced stage (IIIB in 36.1%, III A in 27.1%, IIB in 16.3%) and metastasis (IV) was found in 7.9% cases. Since high percentage of women presented at locally advanced stage, educating the population about the risk factors and self breast examination would be helpful in detection of breast cancer at early stage and proper management.

These studies showed the occurrence of breast cancer at younger age in Indian women with significant annual percentage increase in incidence at younger age. The significant number of cases showing family history indicates genetic component playing a major role in the etiology.

Genetic Risk Factors for Breast Cancer in Indian Women

Several environmental risk factors that may contribute to the development of breast cancer include mainly lifestyle and reproductive factors. These may account for the majority of observed trends and the variation in incidence rates between developed and developing countries. Indian women develop breast cancer at younger age despite having children at younger age and with long duration of breast feeding. This is an enigma which needs elucidation of the etio-pathogenesis and identification of risk factors.

Genetic susceptibility to cancer is triggered in several ways; the best understood causal mechanism being due to inactivating germ line mutations in tumor suppressor and DNA repair genes, which lead to an accumulation of mutations in oncogenes and cell-cycle checkpoints that are required for uncontrolled cell division. About 5–10% of breast and ovarian cancers occur as a result of highly penetrant germ line mutations.

Two major breast cancer susceptibility genes are *BRCA1* (MIM 113705, Gene bank accession no. U14680) and *BRCA2* (MIM 600185, Gene bank accession no. U43746), located on long arms of chromosomes 17 and 13, respectively, and both apparently function as tumour suppressor genes. Mutations in the *BRCA1* and *BRCA2* genes were first reported in conjunction with their identification in 1994 and 1996. Germline mutations in the *BRCA1* or *BRCA2* tumour-suppressor genes are strong predictors of breast and/or ovarian cancer development. Mutation prevalence varies among different ethnic groups and may be influenced by the potential modifying effects of the patient's own genetic and environmental background.

Prevalence of BRCA Genes Mutations in North Indian Population

A large number of distinct mutations in the *BRCA1* and *BRCA2* genes have been reported worldwide, but little is known regarding the role of these inherited susceptibility genes in breast cancer risk among Indian women. Study of distribution and the nature of *BRCA1* and *BRCA2* germ line mutations and polymorphisms in a cohort of 204 Indian breast cancer patients and 140 age-matched controls was undertaken at National Institute of Pathology, New Delhi. All coding regions and exon-intron boundaries of the *BRCA1* and *BRCA2* genes were screened by heteroduplex analysis followed by direct sequencing of detected variants.

In total, 18 genetic alterations were identified (Fig. 3.1). Three deleterious frame-shift mutations (185delAG in exon 2; 4184del4 and 3596del4 in exon 11) were identified in *BRCA1*, along with one missense mutation (K1667R), one 5'UTR alteration (22C>G), three intronic variants (IVS10-12delG, IVS13+2T>C, IVS7+38T>C) and one silent substitution (5154C>T). Similarly three pathogenic protein-truncating mutations (6376insAA in exon 11, 8576insC in exon 19, and 9999delA in exon 27) along with one missense mutation (A2951T), four intronic alterations (IVS2+90T>A, IVS7+75A>T, IVS8+56C>T, IVS25+58insG) and one silent substitution (1593A>G) were identified in *BRCA2*. Four previously reported polymorphisms

(K1183R, S1613G, and M1652I in *BRCA1*, and 7470A>G in *BRCA2*) were detected in both controls and breast cancer patients. Of these, six were pathogenic protein truncating mutations. In addition, several variants of uncertain clinical significance were identified. Among these are two missense variants, one alteration of a consensus splice donor sequence, and a variant that potentially disrupts translational initiation. Of these 18 mutations, six clearly deleterious sequence variants were detected in 2.94% of the tested patients and 4 of the 9 *BRCA1* (44%) and 7 of the 9 *BRCA2* mutations (78%) were found unique to the Indian population and are distributed throughout the exons of *BRCA1* and *BRCA2* gene.

The prevalence of *BRCA1/2* mutations in Indian patient series appears to be low compared to other Asian countries but is comparatively similar to that reported from Shanghai China. However, a significant proportion of women who had breast cancer diagnosed at age 40 years without any family history were carriers (all variants: 14.2%, known deleterious mutations: 3.8%) which is comparable to other Asian countries. In addition, the present study is in agreement with the findings from a pilot study done on a small independent group of 20 breast cancer patients where 3 out of 5 cases with mutations in *BRCA1/2* had early onset disease. Thus, it is reasonable to postulate that women with early-onset disease without family history are likely to have a disease associated alteration of the *BRCA1* or *BRCA2* gene. In this study, the identified mutations account for a comparatively small proportion of the familial risk of breast/ovarian cancer (all variants: 11.7%, known deleterious mutations: (2.9%).

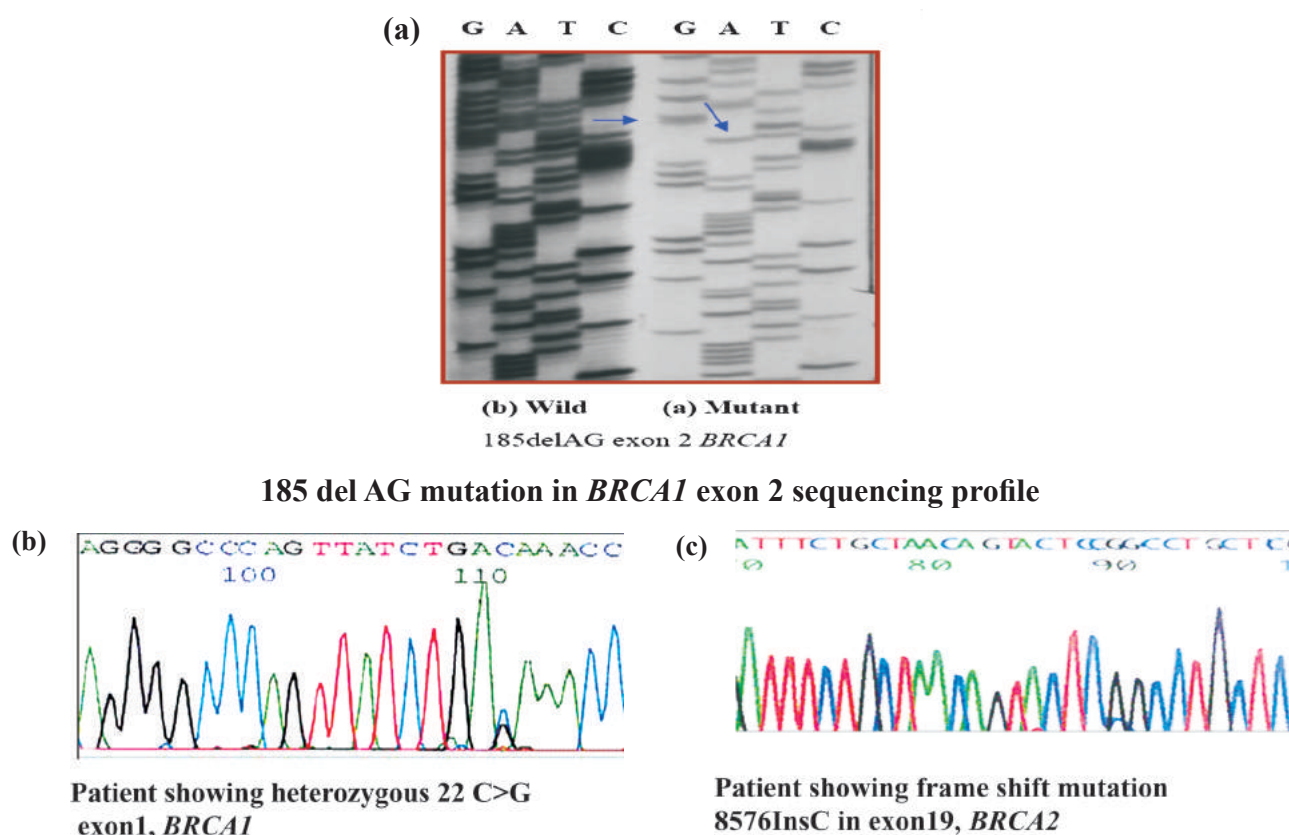


Fig. 3.1: *BRCA1* and *BRCA2* mutational analysis **a.** *BRCA1* 185 del AG by manual Sequencing **b.** Electropherogram showing *BRCA1* 22C>G mutation **c.** Electropherogram showing *BRCA2* 8576insC by automated sequencing.

The study suggested one of the several possibilities with respect to genetic predisposition in the North

Indian population. First, there may be a significant proportion of BRCA1/2 mutations that are large germ line rearrangements, which would not have been detected by the method of mutation screening employed. Second, it is possible that there are some unknown genes, which may contribute more significantly to familial breast carcinoma in this population than do BRCA1 and BRCA2.

Contribution of Low Penetrance Genes to Breast Cancer Susceptibility in Indian Women

Contribution of mutations in highly penetrant genes BRCA1/2 to only 5-10% of total breast cancer cases suggests that there are other unidentified genes involved which may also play role in conferring susceptibility to develop breast cancer at larger scale. Analysis of genetic risk of cancer has shown that most non-hereditary, sporadic cancers develop in genetically predisposed individuals and this predisposition is the result of several low penetrance genes rather than single high penetrant gene mutation. A number of classes of low penetrance genes polymorphisms with potential functional significance have been evaluated, including genes involved in (i) carcinogen metabolism (GST family), (ii) estrogen metabolite biosynthesis (CYP17), (iii) steroid hormone receptor activation (VDR, AR), (iv) DNA damage response and replication checkpoint (CHEK2), (v) DNA repair (XRCC1) and (vi) alcohol metabolism (ADH3). Because endogenous steroid hormone exposure is known to influence breast cancer risk, genes responsive to such hormones are currently being considered as plausible candidates for low-risk breast cancer genes. Steroid hormones viz estrogen, progesterone, androgen and vitamin D are key factors in the development and growth of tumors in hormone dependent tissues specially breast. According to some studies, genetic polymorphism in genes related to the biosynthesis and metabolism of hormones is associated with an altered risk of breast cancer and other types of cancer. It is assumed that, among ethnic groups, genetic differences exist that affect steroid hormone metabolism and transport. The action of steroids is mediated by steroid hormone receptors, which are members of nuclear receptor family of ligand-activated transcription factors. The activities of receptors are modulated by various factors viz co-activators/ co-suppressors including growth factors and their receptors and components of various cell signaling pathways. Studies have been undertaken to assess the possible association of low penetrance genes mainly CYP 17, VDR gene and AR-CAG repeat polymorphisms with breast cancer risk in North Indian women.

CYP17 Gene Polymorphism and its Association with High-Risk North Indian Breast Cancer Patients

Hormonal risk is one of the non-genetic factors that may contribute to the disease etiology. Several clinical, epidemiological and experimental studies have proved that estrogen and progesterone play a major role in the growth of normal breast tissue, and breast cancer risk is strongly related to exposure to these endogenous steroid hormone levels, specially estrogen. Although the exact mechanisms remain to be fully elucidated, the alkylation of cellular molecules and the generation of active radicals can damage DNA together with the potential genotoxicity of estrogen and some of its metabolites (e.g. the catechol estrogens) have been implicated in breast carcinogenesis. Because cumulative exposure to circulating estrogen has been linked to increased risk of breast cancer, interest has been focused on the enzymes regulating the biosynthetic and metabolic pathways of these hormones. The estrogen-forming ovary expresses the CYP17 gene, which codes for the cytochrome P450c17a enzyme and mediates both steroid activities and functions at key branch points in human steroid genesis. A single polymorphism, T-> C, in the CYP17 gene creates an additional Sp I- type promoter site (CCACC) which is associated with increased estrogen level. The base pair change also creates a recognition site for the MspA1 digestion of a polymerase chain reaction (PCR) product that has been used to designate two alleles, A1 and A2, arbitrarily. It has been reported that women who are heterozygous or homozygous for the CYP 17 A2 allele show comparatively higher serum levels of estradiol than do women with an A1/A1 genotype.

To investigate the influence of genetic polymorphism of CYP17 on breast cancer susceptibility mainly

in high-risk Indian breast cancer patients, targeting early onset breast cancer and familial cases, a total of 242 histopathologically confirmed breast cancer (106 early onset, 80 late onset and 56 familial) and 212 age-matched controls were investigated for CYP 17 gene polymorphism by PCR-RFLP method at National Institute of Pathology, New Delhi. The genotypes identified were assigned as homozygous wild type (A1A1), heterozygous variant (A1A2), and homozygous variant (A2A2) (Fig. 3.2). The association of CYP17 polymorphisms with expression of estrogen receptor was also investigated. The results showed significantly increased ($p \leq 0.001$) frequency of heterozygous (A1A2) and homozygous (A2A2) CYP17 genotypes in breast cancer patients (94.3%) than in controls (80.3%). A highly statistically significant increased risk in carriers of homozygous A2 allele was found in young patients ($P < 0.001$) in comparison with patients having late onset condition ($P = 0.260$). However, no significant association between the genotype and breast cancer risk was observed among women with strong family history. Further, data had showed that patients (80.6%) with at least one A2 allele exhibit ER-independent cell proliferation, although statistical significance could not be established ($P = 0.160$).

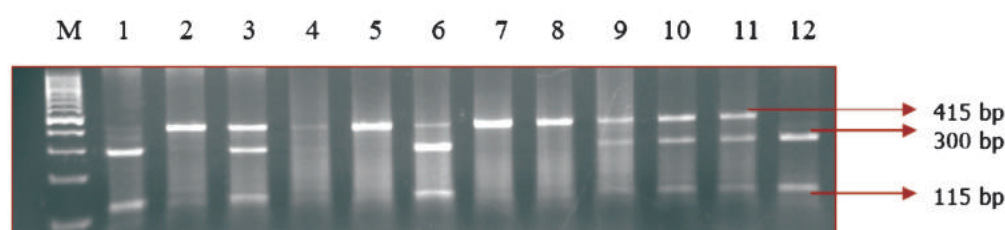


Fig. 3.2: MspAI restriction fragments of CYP17 alleles separated by agarose gel electrophoresis. Lane 1: 100 bp ladder, 1-12 are samples; Lane 2, 7, & 8: homozygous A1A1 (415bp); Lane 3, 9, 10, & 11: heterozygous A1A2 (415bp, 300bp, 115bp); Lane 1, 6, & 12: homozygous A2A2 (300 bp).

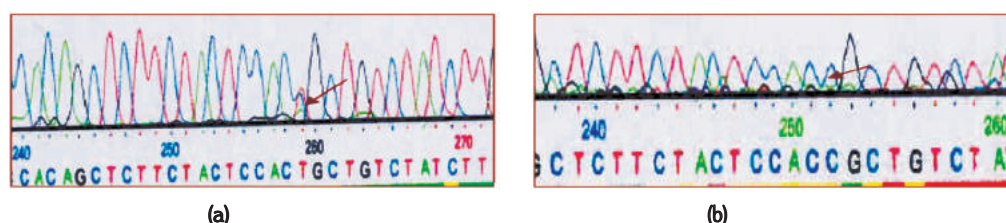


Fig. 3.3: Electropherogram showing heterozygous (a) and homozygous (b) A2 alleles of CYP17 T>C polymorphism.

CYP17 A2 allele gene polymorphism plays a significant role for development of breast cancer Indian women at young age and can also serve as a target for therapy. Due to association of A2 allele of CYP17 gene with increased estrogen level, the steroid hydroxylases are turning out to be potential new targets for drug development. Instead of inhibiting the action of steroid hormones at the level of their receptors by using antihormones, CYPs may help to develop new inhibitors of hydroxylase enzymes.

VDR Polymorphism(s) and Breast Cancer Risk in North Indians

Identification of $1, 25(\text{OH})_2\text{-D}_3$ and VDR as components of a signaling network that affects breast tissue proliferation, differentiation and apoptosis raises the possibility that vitamin D may play a protective role against mammary transformation and common VDR gene variants may be associated with the risk of breast cancer. This study examined the association between VDR (ApaI/TaqI/Poly A) genotype and breast cancer risk in North Indian population. Genotypes for polymorphism at 3' end of VDR gene were studied in 160 cases and 140 control subjects using PCR-RFLP method to determine Apa-I and Taq-I polymorphisms and fragment analysis to determine variable length Poly-A micro satellite repeats.

Patients having long poly-A repeats ($\chi^2 = 87.03$, $\text{df} = 2$, $p < 0.001$) and homozygous L allele (odds ratio

26.97, 95% CI, 12.27, 59.17) showed a significant association with disease while no significant association was found with VDR Apa-I ($\chi^2=1.00$, df2, $p=0.60$) and Taq-I polymorphism ($\chi^2=0.35$, df2, $p=0.83$). The results indicated that VDR Poly-A genotypic variants confer susceptibility to breast cancer with increasing number of Long (L) *PolyA* allele while, Apa-I polymorphism may modify the individual susceptibility to disease. Analysis of the VDR genotype in relation to covariates had revealed that the LL genotype was significantly associated with high grade breast cancer suggesting that this VDR variant might be associated with disease progression.

The genotype also mediated an increased risk of breast cancer in women with early onset compared to late onset. The associations between disease and genotype AATTLL (OR; 5.09, 95% CI 1.23 – 20.80; $p=0.02$) was significant. The decrease in risk in terms of odds ratio from 5.09 to 2.08 even with single (a) in genotype AaTTLL may indicate the protective role of the same in the presence of susceptible (L) allele.

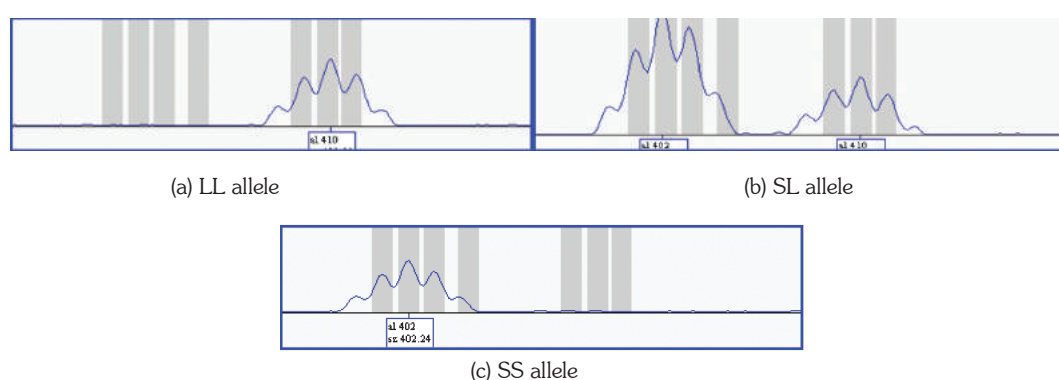


Fig. 3.4: Fragment analysis of VDR (Poly A repeat) gene showing (a) homozygous long alleles (LL) heterozygous long and short allele (SL) and homozygous short alleles (SS) in breast cancer cases.

AR-CAG repeat Polymorphism and Breast Cancer Risk in North Indians:

To determine the role of AR-CAG repeats in risk of breast cancer and their association with response to neoadjuvant chemotherapy, genotyping of the AR CAG repeat region was done in 70 patients and 80 healthy aged matched female controls. Genotype carriers with 20 CAGn showed decrease of AR mRNA expression however no association could be established with the risk ($p = 0.47$) (Fig. 5). In patients with locally advanced breast cancer, higher AR mRNA expression was seen prior to neo-adjuvant chemotherapy in responders ($p < 0.02$) which decreased significantly after neo-adjuvant chemotherapy ($p = 0.014$). Although, expansion of the CAGn in the AR gene did not show any association with breast cancer risk, however, significant correlation was found with response to neoadjuvant chemotherapy in LABC patients.

Multigenic Model of Breast Cancer Susceptibility

Considering CYP17 A2 and VDR Poly-A L as high risk alleles, a multigenic model of breast cancer susceptibility was developed to identify women carrying a combination of alleles which put them at relatively higher risk to develop breast cancer. The risk among women carrying three high-risk alleles was 4.68 [95% confidence interval (CI), 0.77–28.0; P for trend = 0.10] compared with those who carried none. The conditional logistic regression analysis revealed that the heterozygous TC genotype for CYP17 and AR1AR2 of AR, imparted significantly four fold risk for the breast cancer risk, in comparison to the referent genotype TT and AR1AR1 [adjusted ORs: 3.705(1.236, 11.106), $P=0.019$] and [4.391(1.324, 14.557), $P=0.016$ respectively]. Gene-gene interaction showed that the combinations TC*AA, TC*Aa, TC*aa and CC*Aa imparted significantly four to fifteen fold more risk for the breast cancer [(4.377 (1.159, 16.520), $P=0.029$); 4.041(1.092, 14.956), $P=0.036$); (15.071(0.975, 232.81), $P=0.052$); (4.151(1.053,

16.371), $P=0.042$) respectively]]. These findings suggest that breast cancer risk has a strong genetic component and support the theory that the underlying mechanism of “complex trait” can be understood using a multigenic model of low penetrant genes. Genes involved in hormone synthesis and metabolizing pathway play significant role in breast cancer development.

Study of Behavior of Human Mammary Epithelial Cancer Cells *in vitro*.

The studies were undertaken to define the characteristics of human mammary epithelial cancer cells (HMEC) which govern their ability to grow *in vitro* as primary culture as well as continuous or established cell lineages.

Molecular Events Leading to Immortalization/Senescence of Human Mammary Epithelial Cancer (HMEC) Cells *in vitro*.

Role of phenotypic expression of estrogen receptor (ER), progesterone receptor (PgR), c-erb B2 oncoprotein and epidermal growth factor receptor (EGFR) have been studied on capacity of HMEC cells to grow *in vitro* as monolayer. Expression of ER retarded the growth of HMEC cells, while expression of EGFR resulted in increased cell proliferation. Expression of c-erb B2 oncoprotein which correlates with actively proliferating and dedifferentiated breast cancer cells with high metastatic potential *in vivo* did not exert significant control over proliferation of tumor cells *in vitro*. Cultured HMEC cells in subsequent passages showed a decrease in their proliferative capacity, alteration in phenotypic characterization and development of morphologic features of terminal differentiation, resulting in senescence (Fig. 3.5). It was also observed that the ER –ve cells in the primary culture outgrow ER +ve cell population during propagation because of their proliferative advantage suggesting that primary cultures may not represent the heterogeneous population of primary tumors, and may not be an ideal tool for conducting drug sensitivity studies.

Study on role of various growth regulatory factors in inducing senescence in cultured HMEC cells showed that the histological grade of tumour played major role in controlling the proliferation and phenotypic characteristics of cultured HMEC cells during primary culture. However, on subsequent passages the tumor cells developed senescence irrespective of any tumor grade or phenotype. To evaluate the role of various autocrine growth factors produced by HMEC cells, conditioned media (CM) was collected during primary and senescent cultures and their effect had been studied on proliferation of autologous HMEC cells used as target cells. Significant increase in proliferation of target cells was noticed on their exposure to CM from primary cultures, while cessation of their proliferation was found on their exposure to CM from senescent cells, suggesting that HMEC cells produce growth promoting factors during primary culture and growth inhibitory factors on subsequent passages, inducing features of cellular senescence in them. The role of epidermal growth factors (EGF) and transforming growth factors (TGF) and as autocrine factors in inducing proliferation/senescence of cultured HMEC cells was also investigated. Deletion of EGF from growth media initially caused decreased proliferation to target HMEC cells, followed by improvement in their proliferation. Supplementation of growth media by TGF α induced significant increase in proliferation of target cells. Addition of epidermal growth factors receptor (EGFR) antibody to cells exposed to media devoid of EGF and media supplemented with TGF α showed marked suppression of proliferation of target cells. The morphologic and phenotypic characteristics of target HMEC cells exposed to TGF α were also found similar to those HMEC cells grown during primary culture, suggesting autocrine production of EGF and TGF α by cultured HMEC cells during primary culture. Supplementation of TGF β to growth media induced marked suppression of proliferation to target cells along with morphologic and phenotypic features of terminal differentiation or senescence. Exposure of senescent cells to media supplemented with EGF and TGF α could not induce their proliferation. This suggested that during primary culture; HMEC cells

produce TGF α while on subsequent passages they underwent some genetic and phenotypic alterations resulting in secretion of growth inhibitory factors like TGF β which induced cessation of proliferation and development of senescence.

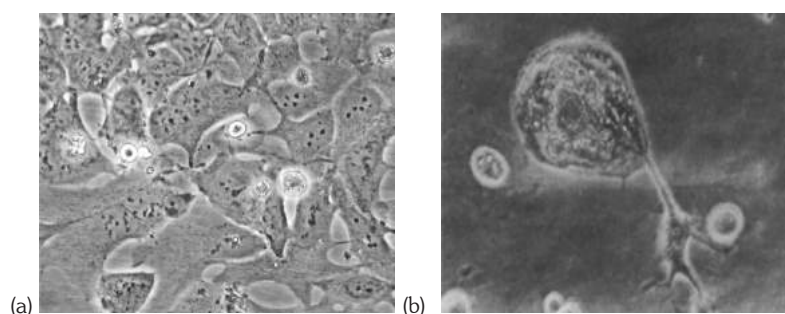


Fig. 3.5: Phase contrast photomicrograph of a) primary culture and b) senescent cell showing a marked increase in size and bizarre shape with formation of cytoplasmic vacuoles x 490.

Establishment and Characterization of Breast Cancer Cell lines from Primary Breast Tumors in Indian Women

Cell lines provide an unlimited, self-replicating source of cells that can be widely distributed to facilitate comparative studies. Hence, well-characterized cell lines are powerful research resources for studying cancer cell biology and for the development of new strategies against cancer. In India, the onset of breast cancer is a decade earlier compared to western countries. There are no cell lines available from early onset breast cancer patients in India.

Two novel triple negative breast cancer cell lines, NIPBC-1 and NIPBC-2 were successfully established from primary tumors (infiltrating duct carcinoma) of two young breast cancer patients aged 39 and 38 years respectively (Fig. 3.6 & 3.7). Characterization of these cell lines confirmed its epithelial origin with expression of epithelial specific antigen and luminal origin with cytokeratin 18 expression and also showed presence of microfilaments, secretory vesicles, microvilli, tight junctions and desmosomes on ultra-structural examination. Both the cell lines showed anchorage independent growth and invasion of matrigel coated membranes. Karyotype analysis showed aneuploidy, deletions and multiple rearrangements in chromosomes 7, 9, X and 11 and isochromosomes 17q in both the cell lines. P53 mutational analysis revealed no mutation in the coding region in both the cell lines; however, NIPBC-2 cell line showed presence of heterozygous C/G polymorphism, g.417 C>G (NM_000546.5) resulting in Arg/Pro allele at codon 72 of exon 4. Screening for mutations in BRCA 1&2 genes revealed presence of three heterozygous polymorphisms in exon 11 of BRCA1 and 2 polymorphisms in exons 11 and 14 of BRCA2 gene in both the cell lines. Both the cell lines showed presence of CD 44⁺/24⁻ breast cancer stem cells and capability of producing mammosphere on culture.

Although a number of normal and breast cancer cell lines established from western patients are available, there is a paucity of cell lines established from Indian breast cancer patients. So far only two breast cancer cell lines have been established from late onset breast cancer patients (>60 years) from India using primary tumors, hence the two established cell lines are unique with respect to the ethnicity and also are the only breast cancer cell lines established from Indian early onset breast cancer patients.

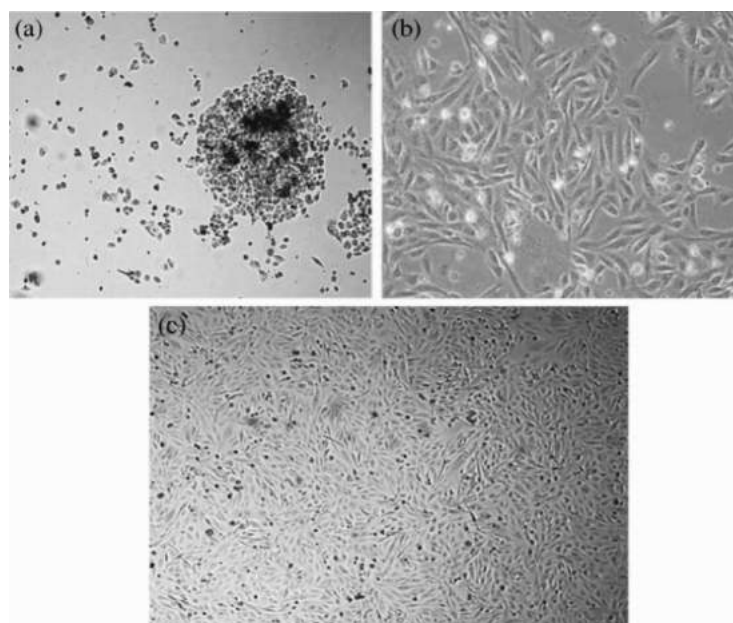


Fig. 3.6: Establishment of NIPBC-1 cell line showing stages of (a) Initiation (b) purification and (c) completely purified confluent NIPBC-1 cell line.

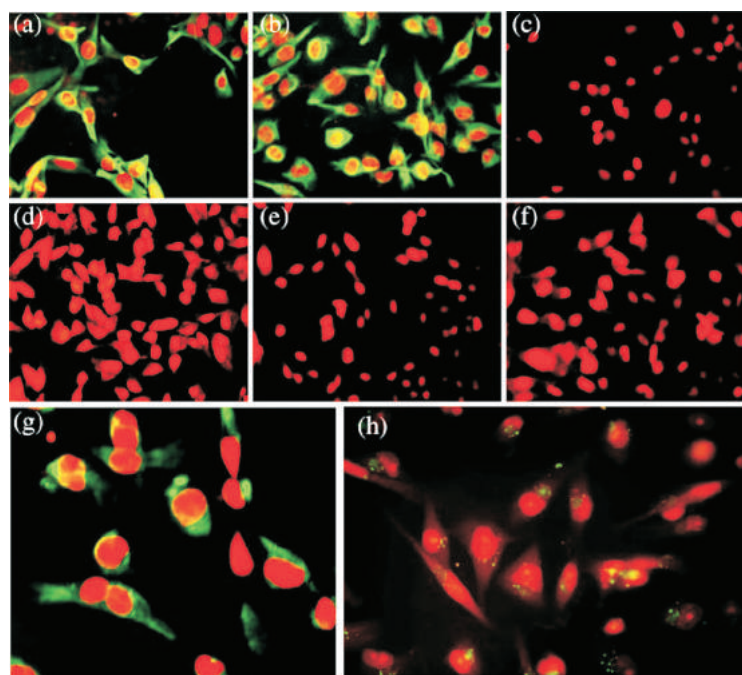


Fig. 3.7: Expression of biological markers in the established breast cancer cell line NIPBC-1. (a) Epithelial membrane antigen, (b) Cytokeratin 18, (c) Cytokeratin 5/6, (d) Estrogen receptor, (e) Progesterone receptor, (f) HER2/neu(g) P53 and (h) Vimentin.

Two triple negative breast cancer cell lines, NIPBC1 and NIPBC2 were established from primary tumors of two early onset breast cancer patients (<40 years), which can serve as novel *in vitro* model to study mechanism underlying breast carcinogenesis in younger age group patients and also for the identification of new therapeutic modalities.

Drugs Effects on ALDH1A1-Expressing Breast Cancer Stem Cells

Tumor recurrence and treatment failure are two well-known factors in cancer therapy associated with poor survival, which have recently been linked with cancer stem cells (CSCs). The CSCs may not respond to conventional chemotherapeutic drugs similar to bulk cancer cells, resulting in resistance, relapse and metastasis. Breast Cancer Stem Cells (BCSCs) are characterized by the combined expression of cell surface markers CD44⁺/CD24⁻, presence of aldehyde dehydrogenase (ALDH) enzyme and by their ability to pump out dyes like Hoechst (side population cells). The antineoplastic alkaloid ellipticine (5,11-dimethyl-6Hpyrido [4,3-b] carbazole), found in Apocynaceae plants, e.g., *Ochrosiaborbonica*, *Ochrosiaelliptica*, is a prodrug. Cytochrome P450 enzymes oxidize ellipticine, forming up to five metabolites, viz., 7-hydroxyellipticine, 9-hydroxyellipticine, 12-hydroxyellipticine, 13-hydroxyellipticine, and ellipticine N2-oxide. Out of these, 9-hydroxyellipticine and 7-hydroxyellipticine are the major detoxification products of ellipticine oxidation *in vivo*. In human breast adenocarcinoma MCF7 cells, ellipticine causes arrest of the G2/M phase of the cell cycle associated with an increased expression of p53, p21/WAF1, and KIP1/p27 proteins and growth inhibition induced by mitochondrial proapoptotic pathways. In addition, several derivatives of ellipticine have shown promising results in the treatment of metastatic breast cancer. Effect of ellipticine on BCSCs expressing the ALDH1A1 has been studied on mammosphere formation and self-renewal ability *in vitro* and compared with paclitaxel which is administered as conventional chemotherapy. Further, the efficacy of ellipticine was also evaluated on the activity of human ALDH1A1 by molecular docking of ellipticine and its derivatives with the energy-minimized model of *Homo sapiens* ALDH1A1 preceded by *in silico* homology modeling of the enzyme.

Human breast cancer cell lines, MCF7, SUM159 and normal breast cell line, HBL100 were grown as adherent cultures, cells were then stained with aldefluor and ALDH⁺ cells were then cultured in non-adherent, non-serum conditions to form primary mammospheres (termed M1 mammospheres), followed by serial passaging every 10th day leading to the generation of M2, M3, M4 mammospheres demonstrating the self-renewal of stem cells. A dynamic increase both in the number of mammosphere forming units and sphere forming efficiency was seen until the second passage followed by a dramatic reduction (Fig. 3.8).

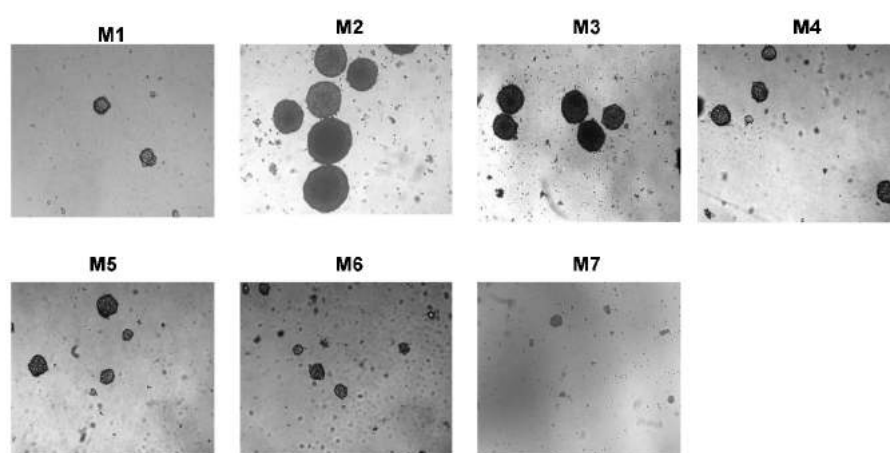


Fig. 3.8: Representative pictures of mammospheres formed by MCF7 after serial passaging.

Study of efficacy of ellipticine and paclitaxel to inhibit the aldehyde dehydrogenase 1 class A1 (ALDH1A1) positive BCSCs *in vitro* showed that at 3 mM concentration, ellipticine decreased the expression of ALDH1A1-positive BCSCs by 62 % ($p=0.073$) in the MCF7 cell line, and by 53 % ($p=0.024$) in the SUM159 cell line compared to vehicle-treated cultures. Ellipticine significantly reduced the formation of mammospheres, whereas paclitaxel enhanced mammosphere formation in both the treated cell lines.

Interestingly, when treated with a combination of ellipticine and paclitaxel, the percentage of ALDH1A1-positive BCSCs dropped by several fold *in vitro* (Fig. 3.9). A homology model of Homo sapiens ALDH1A1 was built using the crystal structure of NAD-bound sheep liver class I aldehyde dehydrogenase [PDB ID: 1BXS] as a template. Molecular simulation and docking studies revealed that the amino acids Asn-117 and Asn-121, Glu-249, Cys-302, and Gln-350, present in the active site of human ALDH1A1, played a vital role in interacting with the drug.



Fig. 3.9. Paclitaxel increases while the combination of ellipticine plus paclitaxel decreases both the number and size of mammospheres. Representative pictures of primary mammospheres formed in the presence of varying concentrations of ellipticine and paclitaxel and their subsequent second and third generation mammospheres after drug withdrawal by a MCF7 and b SUM159 cell lines.

Study showed that the cancer stem cells did not respond to conventional chemotherapy effective for bulk cancer cells and require additional drugs to arrest their self-renewal capacity. Ellipticine arrested the self-renewal ability of ALDH1A1-positive BCSCs and could be used in combination with conventional cytotoxic drug like paclitaxel for effective targeting of BCSCs.

Predictive Tumor Markers in Breast Cancer in Indian Women

Majority of breast cancer patients in India present at locally advanced stage (IIIB, IIIA and IIB) and standard of care for these patients is neoadjuvant chemotherapy (NACT) followed by surgery in the form of modified radical mastectomy and three more cycles of adjuvant chemotherapy. NACT facilitates local as well as distant control of the disease and provides an *in vivo* chemo sensitivity test for a particular regime. It is vital to predict response to chemotherapy in order to tailor the regime in a particular patient for an optimum response and to avoid chemotoxicity in a non-responder. The prediction would help in avoiding the toxicity induced by an ineffective chemotherapeutic regime in a non-responder and would also help in the planning of an alternate regime. Studies had been undertaken to assess the predictive role of various tumor markers viz p-glycoprotein, p53, apoptotic markers, mismatch repair genes, androgen receptor gene and type 1 growth factor receptor family genes in search for an ideal predictor.

Role of p-Glycoprotein Expression in Predicting Response to Neoadjuvant Chemotherapy (NACT) in Breast Cancer

Development of resistance to chemotherapeutic agents is a major problem and one of the mechanisms considered responsible is the expression of 170-kDa membrane glycoprotein (usually referred to as p-170 or p-glycoprotein), which is encoded by multidrug resistance (MDR1) gene. This glycoprotein acts as an energy dependent pump, which actively extrudes certain families of chemotherapeutic agents from the cells. The expression of p-glycoprotein at initial presentation has been found to be associated with refractoriness to chemotherapy and poor outcome. Against this background, a prospective study was conducted by studying expression of p-glycoprotein in pre-treatment biopsy of tumor tissue to ascertain whether pretreatment detection of p-glycoprotein expression could be utilized as a reliable predictor of response to neoadjuvant chemotherapy in patients of locally advanced breast cancer (LABC). Fifty cases of locally advanced breast cancer were subjected to trucut biopsy and the tissue samples were evaluated for expression of p-glycoprotein, ER and PR by immunohistochemistry. The response to neoadjuvant chemotherapy was assessed clinically and by using ultrasound after three cycles of FAC (fluorouracil, adriamycin and cyclophosphamide) regime. The clinical response was correlated with both the pre and post chemotherapy p-glycoprotein expression. A significant correlation was observed between positive p-glycoprotein expression and poor clinical response rates ($p < 0.05$). Significant increase in expression of p-glycoprotein was seen after three cycles of NACT, 52% before initiation of NACT increased to 73.5% after NACT. The chemotherapy induced p-glycoprotein positivity observed in the study could possibly explain the phenomenon of acquired chemoresistance and may also serve as an intermediate end point in evaluating drug response particularly if the adjuvant therapy is planned with the same regime.

The study concluded that p-glycoprotein expression indicated a poor clinical response to NACT.

Role of p53 and Apoptotic Markers in Predicting Response to Neoadjuvant Chemotherapy (NACT) in Breast Cancer

The pathway associated with tumor cell death is mainly apoptosis or programmed cell death and chemotherapeutic drugs like DNA-damaging agents act on rapidly multiplying cells including both the tumor and the normal cells by following this pathway. This could account for both the response and toxic effects. Absence or decreased apoptosis has been found to be associated with chemo resistance. The change in expression of apoptotic markers (Bcl-2 and Bax proteins) brought about by various chemotherapeutic regimens has been used to identify drug resistance in the tumor cells. A prospective clinical study was conducted to assess whether chemotherapy induced toxic effects could serve as reliable predictors of apoptosis or response to neo-adjuvant chemotherapy in patients with locally advanced breast cancer. Fifty cases of locally advanced breast cancer after complete routine and metastatic work up were subjected to trucut biopsy and the tissue evaluated immunohistochemically for apoptotic markers (bcl-2/bax ratio). Three cycles of neoadjuvant chemotherapy using FAC regime (5-fluorouracil, adriamycin, cyclophosphamide) were given at three weekly intervals and patients assessed for clinical response as well as toxicity after each cycle. Modified radical mastectomy was performed in all patients three weeks after the last cycle and the specimens were re-evaluated for any change in the Bcl-2/Bax ratio. The clinical response, immunohistochemical response and the drug-induced toxicity were correlated and compared. Immunohistochemical response was defined as decrease in the Bcl-2/Bax ratio.

Clinical response including the reduction in the tumor size and axillary lymph node status was observed in 70% of patients and was found to be statistically significant ($p < 0.0001$). There were no patients in the N0

group and 29.4% of the N1 patients were down staged to N0 while 70.6% remained N1. In patients with N2 disease 7.7% were down staged to N0 status while 46.2% were down staged to N1 status and 46.2% did not show any response. Change in immune-histochemical expression (IHC response) was observed in 60% patients which was statistically significant ($p = 0.008$). Correlation between immunohistochemistry and clinical response was also found to be statistically significant ($p < 0.0001$).

Acute vomiting was observed in 63.3% patients. Vomiting was found in 81% clinical responders ($p = 0.002$) and 78% IHC responders which was statistically significant ($p=0.04$). Alopecia was observed in 86% clinical responders ($p = 0.000$) and 94% IHC responders ($p=0.000$). Leucopenia was observed in only 14% and 17% of clinical and IHC responders respectively and was found to be an insignificant predictor of response in the present study. When multiple toxicities were correlated with the clinical and IHC response, 46.7% of patients had both acute vomiting and alopecia. 67% clinical responders ($p=0.001$) had both vomiting and alopecia. 72% immune-histochemical responders ($p=0.001$) had both vomiting and alopecia.

A significant positive correlation was observed between the presence of vomiting ($r = +0.558$), alopecia($r = +0.802$) and response to neoadjuvant chemotherapy. A significant negative correlation was observed between the absence of side effects and poor response to neoadjuvant chemotherapy. The chemotherapy-induced toxicity was observed to be a cost effective and reliable predictor of response to neo-adjuvant chemotherapy.

Correlation of Mismatched Repair Genes with Response to Neo-adjuvant Chemotherapy in Breast Cancer

The DNA mismatch repair (MMR) pathway is an important post-replicative repair process. It is involved in the maintenance of genomic stability and MMR genes have therefore been named the *proof readers* of replicating DNA. These genes repair the replicative errors of DNA and are thus imperative for genomic stability.

The MMR genes have been found to be involved in promoting cytotoxicity, apoptosis, p53 phosphorylation and cell cycle arrest following exposure to exogenous DNA damaging agents. Loss of MMR function prevents the correction of replicative errors leading to instability of the genome, and can be detected by polymorphisms in microsatellites nucleotide repeat sequences scattered in whole genome. This phenomenon, known as microsatellite instability (MSI), is a hallmark of MMR dysfunction and can be used as a marker of MMR dysfunction in various malignancies. An alternative method for detection of MMR dysfunction is to test the expression of protein products of the MMR genes by immunohistochemistry (IHC), as mutations in these genes lead to reduced or absent expression of their gene products. Correlation between loss of MMR function, histopathological, behavioral parameters of the tumor with response to chemotherapy in breast cancers may be of value in predicting response to neoadjuvant chemotherapy (NACT).

Thirty-one cases of locally advanced breast carcinoma (LABC) were studied to assess the correlation between MMR dysfunction, clinicopathological parameters and clinical response to neoadjuvant chemotherapy. Analysis for immunohistochemical expression for four MMR protein products MLH1, MSH2, MSH6 and PMS2 was done in the pre NACT trucut biopsy specimen and after three cycles of NACT with CAF (cyclophosphamide, adriamycin, 5-fluorouracil) regimen, in the modified radical mastectomy specimen. 54.83% patients showed down staging of the tumor size and axillary lymph node status in response to NACT including 2 patients with complete pathological response ($p=0.000$).

A significant correlation was observed between expression of MLH1 and MSH2 with histological grade ($p=0.048$ and 0.038 respectively). Tumors with decreased expression of MLH1 and MSH2 showed poor

differentiation. Combination of loss of expression of MMR proteins with histological grade may serve as a predictor of aggressive behavior and poor outcome. Cases with high post NACT expression of PMS2 were poor responders to chemotherapy. MSH6 was the most frequently altered MMR gene, with loss of expression in 48% patients; high expression was associated with poor response to NACT.

Expression of p-glycoprotein in pre-treatment biopsy predicted a poor clinical response to NACT in patients having LABC. The chemotherapy-induced toxicity (vomiting and alopecia) correlated significantly with clinical and immunohistochemical response (reduction in bcl2/bax ratio) and were found to be a cost effective and reliable predictor of response to neo-adjuvant chemotherapy. Alteration in expression of MMR proteins was found mainly in high histological grades and predicted aggressive behavior and poor response to neoadjuvant chemotherapy in patients with LABC.

Role of Androgen Receptor (AR) in Breast Cancer

Androgen receptor (AR) belongs to a family of intracellular steroid hormone receptors that function as ligand dependent transcription factor which regulates target gene expression. The full length AR protein is a 110 kDa phosphoprotein, which mediates its physiological functions by binding to its ligand testosterone or after its conversion to 5 -dihydrotestosterone (DHT) by 5 -reductases. AR has an N-terminal transactivation domain, which contains the poly-Glutamine (CAG) repeat sequence, a DNA-binding domain (DBD) having two C4 type Zinc fingers, a hinge region and a C-terminal ligand-binding domain (LBD) which gets activated upon binding to androgens. Androgen binding to C-terminal of AR leads to the dissociation of chaperone proteins and dimerization of AR leading to a conformational change whereby its nuclear localization signal (NLS) is exposed. Exposed NLS then aids in the translocation of AR to the nucleus, where it binds to androgen-response elements (AREs) present in the promoters of several target genes in a tissue-specific manner. In the nucleus, AR recruits many other proteins, such as general transcription factors and RNA polymerase to activate androgen-responsive genes. Traditionally, estrogen receptor (ER) and progesterone receptor (PR) are known to be the prominent players in the progression and development of breast cancer. Recent evidence suggests that AR antagonizes ER function and plays an anti-proliferative role in ER+ve breast cancers whereas it plays a significant role in facilitating tumor cell growth in an androgen-dependent manner in an ER-/AR+ breast cancers.

Role of Androgen Receptor as Independent Predictor for Response to NACT in Locally Advanced Breast Cancers (LABC)

The clinical significance of AR in breast cancer has been studied by correlating its expression with clinic-pathological parameters, other steroid receptors (ER and PR) and growth factor receptors (EGFR and CD105) in 100 cases by immunohistochemistry (IHC). Risk Ratio (RR) along with 95% confidence interval (CI) was estimated to assess the strength of association between the markers and clinicopathological characteristics. Categorical Principal Component Analysis (CATPCA) was applied to obtain new sets of linearly combined expression, for their further evaluation with clinicopathological characteristics. In 31 cases presenting with locally advanced breast cancer (LABC), the expression of AR, ER, PR, EGFR and CD105 correlated with response to neoadjuvant chemotherapy (NACT). The results indicated the association of AR+ ($p=0.001$) and AR+/EGFR- ($p=0.001$) with the therapeutic response to NACT in LABC patients. The AR expression exhibited maximum sensitivity, specificity and likelihood ratio of positive and negative test for response to NACT.

Correlation of expression of steroid (AR) and type 1 growth receptor genes (EGFR, ERBB2, ERBB3) and gene associated with metabolism of chemotherapeutic drugs (MDR1), before and after therapy,

was studied in LABC patients with response to treatment. Significant correlation was found with high levels of pre-NACT AR gene expression ($p = 0.016$) with responders, which decreased following NACT ($p = 0.008$). Hence, AR upregulation can serve as a useful predictive marker for response to NACT. A significant post-therapeutic increase in the expression levels of EGFR and MDR1 gene in responders ($p = 0.026$ and $p < 0.001$) as well as in non-responders ($p = 0.055$, $p = 0.001$) suggested that expression of these genes changes during therapy but they did not have any impact on tumor response, whereas a post-therapeutic reduction was observed in AR in responders (Fig. 3.10). The study indicated an independent predictive role of AR with response to NACT.

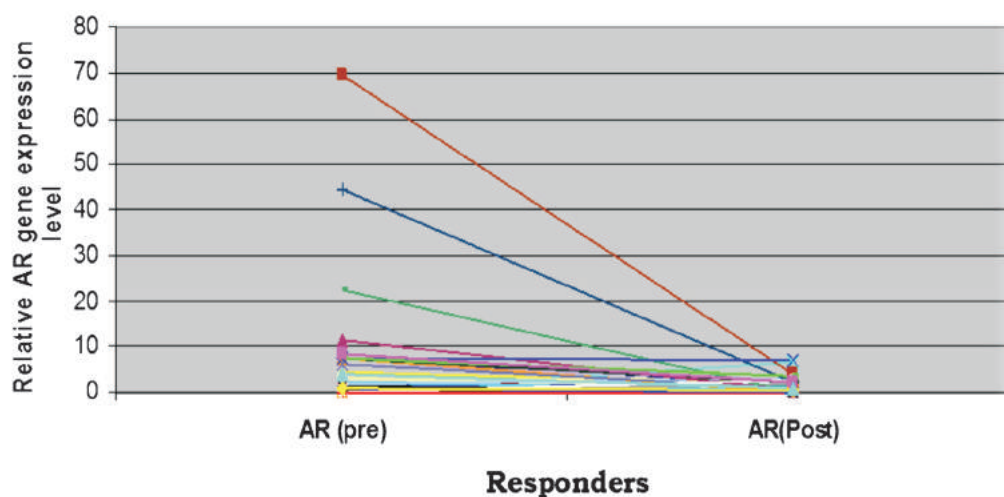


Fig. 3.10: Expression levels of AR measured in pre NACT tumor biopsy specimens and post-therapeutic tumor resection samples in responders.

Study of Androgen Receptor (AR) Signaling in Breast Cancer

AR has been implicated in the development and progression of breast and prostate cancers. Therefore, understanding the regulatory mechanisms of the functioning of AR in ER-/AR+ breast cancer will provide many novel targets for the purpose of therapeutic intervention.

To get the molecular insights into androgen receptor signaling in breast cancer AR targets having prominent roles in cell cycle, apoptosis and metabolism by using bioinformatics tools were identified and 10 genes as AR targets by studying the regulation of these genes in MDA-MB-453 cell line on stimulation by androgens like 5 -dihydrotestosterone (DHT) using RT-qPCR and ChIP assay were validated. The experimental findings showed that treatment of breast cancer cells with DHT promoted cell proliferation and decreased apoptosis while treatment with anti-androgen like bicalutamide was able to reverse the effect of DHT thereby demonstrating the application of bicalutamide along with taxanes as novel therapy for the treatment of TNBCs, which were positive for downstream AR signaling. In order to understand how androgen receptor influences DNA binding, an exhaustive, rigid-body docking between individual ARE and DNA binding domain (DBD) of the AR protein was performed. To decipher the DNA-protein contact points between the AR-DBD and ARE(s), 5 genes namely, AATK, CDT1, KLF6, LIPH and SGOL2 were selected on the basis of TF search score and fold enrichment in ChIP-qPCR experiments for the purpose of interaction studies (Fig. 3.11). The results showed that novel residues K567, K588, R591 and K592 were involved in the process of DNA binding and make important contributions in increasing the stability of DNA-protein complexes. To verify these specific DNA-protein interactions electrostatic energy term calculations for each residue were determined using the linearized Poisson-Boltzmann equation.

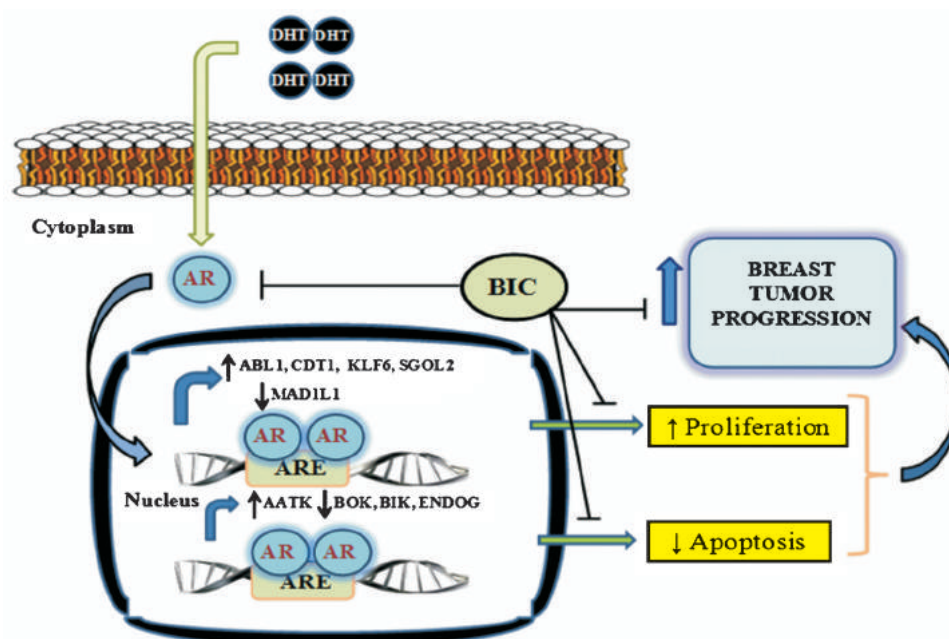


Fig. 3.11: Proposed model shows AR regulation of the breast tumour progression. On DHT stimulation Androgen Receptor (AR) translocates into the nucleus and bind to its cognate androgen response elements (AREs). Inside the nucleus, AR upregulates the expression of ABL1, CDT1, KLF6 and SGOL2, while it represses MAD1L1 expression to induce cell proliferation. At the same it induces AATK expression and down regulates the expression of BOK, BIK and ENDOG to decrease apoptosis and promote breast cancer progression. Bicalutamide (BIC) reverses the effect of AR on cell cycle and apoptosis by binding and preventing its activation. Due to its ability to negate the effects of AR, bicalutamide can be used to block breast cancer progression.

Androgen receptor has been identified as independent predictive marker for response to neoadjuvant chemotherapy in locally advanced breast cancer cases. Androgen receptor can serve as novel therapeutic target for Triple Negative Breast Cancers. Bicalutamide (anti androgen) can be used along with other chemotherapeutic drugs as novel therapy for the treatment of TNBCs, which are positive for downstream AR signaling.

Breast Cancer in North Eastern India

Study of Environmental and Genetic Risk Factors for Breast Cancer in North East India

Northeast region of India shows high incidence of tobacco-related cancer with widespread consumption of betel quid and tobacco in different forms. There is an increasing incidence of breast cancer with extensive use of tobacco in females in this region. Thus, the role of tobacco exposure and polymorphisms in detoxification enzymes in breast cancer risk was analyzed using samples collected from patients registered at Dr B Borooah Cancer Institute, Guwahati. Polymorphisms in five gene variants, GSTT1, GSTM1, GSTP1, TP53 and CYP17, and four environmental exposure variables (tobacco smoking, tobacco chewing, betel quid chewing, and alcohol) were analyzed in 117 breast cancer cases and 174 cancer free controls. Multifactor dimensionality reduction (MDR) identified betel quid chewing as the single main risk factor and women with betel quid chewing history had five times the risk of developing breast cancer [OR: 4.78 (2.87-8.00) $p=0.001$]. In logistic regression analysis, GSTT1 null and GSTM1 null genotypes conferred 41% [0.59 (0.34-1.03) 0.06] and 55% [0.58 (0.30-1.02) 0.05] reduced risk to breast cancer respectively. However, the risk increased in women with GSTP1 variant G allele which conferred 1.43 times [(0.96-2.11) 0.07] more risk to breast cancer. MDR analysis revealed betel quid chewing to be the single factor imparting the main effect [testing accuracy of 0.6851 and cross validation consistency 10/10, $p=0.05$]. Fig. 3.12 depicts the interactions between nine attributes from the MDR analysis via a graphical representation of a 'dendrogram'. It showed betelquid chewing, GSTT1 and GSTM1 on a

separate branch imparting their independent effects to breast cancer risk. The study highlighted betel quid chewing as a significant and independent risk factor for developing breast cancer in NE region. Moreover, the absence of functional detoxification enzymes GSTT1 and GSTM1 were associated with reduced breast cancer risk.

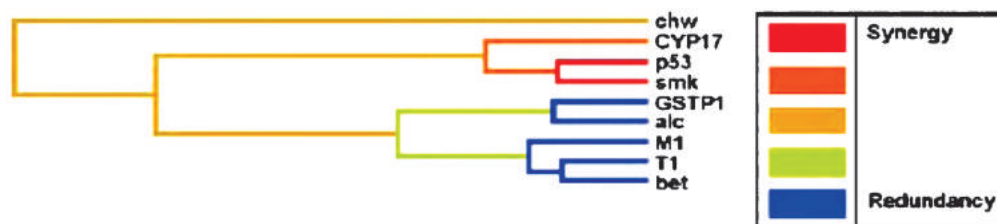


Fig. 3.12: Interaction dendrogram for the breast cancer dataset: graphical representation of interactions between nine attributes (GST1 (T1), GSTM1 (M1), GSTP1, CYP17, TP53 (p53), tobacco smoking (smk), tobacco chewing (chw), betel quid chewers (bet) and alcohol consumption (alc)) from the multifactor dimensionality reduction analysis using an 'interaction dendrogram'.

Association of DNA Repair and Cell Cycle Gene Variations with Breast Cancer Risk

Polymorphisms in DNA repair and cell cycle genes contribute to increased breast cancer risk. Their association and interaction in relation to betel quid and tobacco chewing habits needs exhaustive multi-analytical investigation to explain breast cancer predisposition due to DNA damage.

Polymorphism in TP53-72Arg>Pro, RAD51-135G>C, BRCA2, and CCND1-G870A were examined in 204 Breast Cancer (BC) cases and 217 controls from Northeast Indian population. Multifaceted analytic approaches were used to explore relationships between polymorphisms, tobacco history and BC susceptibility. Betel quid chewing was identified as the predominant risk factor. CCND-AA and dominant model showed protection towards BC in BQC (betel-quid chewer) and NBQC (non betel-quid chewer). TP53-Pro/Pro genotype showed protection towards BC in NBQC. RAD51-C allele was associated with BC risk in BQC. Two BQC cases had BRCA2 8415G >T:K2729N mutation in exon18.

The hierarchical interaction graph showed (Fig. 3.13), large independent effect of betel quid chewing (9.38%) among environmental factors for total sample set. A strong interaction (1.32%) was seen between RAD51 and TP53. MDR analysis showed best 4 locus model in NBQC. Interaction diagram concurred the interactions between TP53 and RAD51 (1.32%) with independent effect (1.89%) of CCND1 in NBQC (Fig. 3.14). In CART analysis, BQC with CCND1 GG genotype were at risk followed by combination of BQC, CCND1, No-Smk, Alc. Risk was also observed in BQC, CCND1, No-Smk, Non-Alc, TP53 combination and BQC, CCND1, No-Smk, Non-Alc, TP53 (Fig. 3.15). NBQC group showed risk with combination of NBQC and TP53.

These data indicated that common genetic variations in DNA repair and cell cycle genes contributed towards breast cancer risk. In addition, different predisposition was observed amongst BQC and NBQC breast cancer patients rendering dissimilar susceptibility towards breast cancer. The elevated risk for breast cancer in BQC may be attributable to betel quid carcinogens and minor roles of *BRCA2* mutation and C allele of *RAD51*. Whereas NBQC could be at slightly lower risk for breast cancer due to the protection offered by the Pro/Pro-TP53 form. CCND1 polymorphism conferred protection irrespective of the betel quid chewing status.

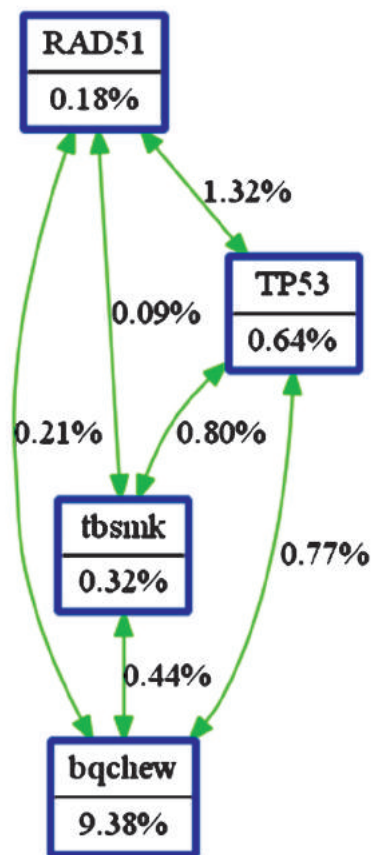


Fig. 3.13: Interaction dendrogram using orange software for the total data set.

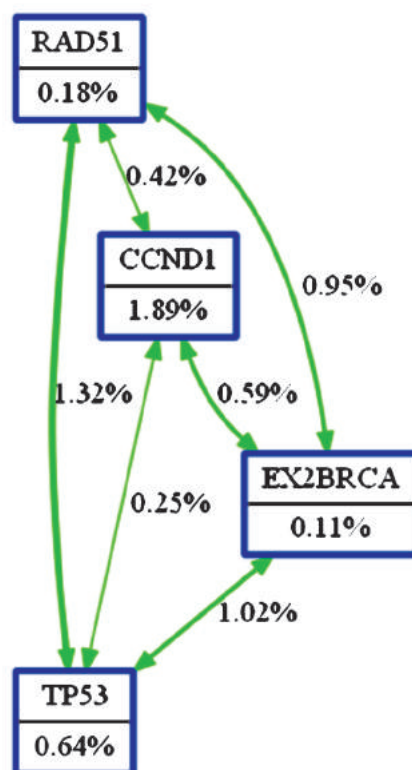


Fig. 3.14: Interaction dendrogram using orange software for the NBQC data set.

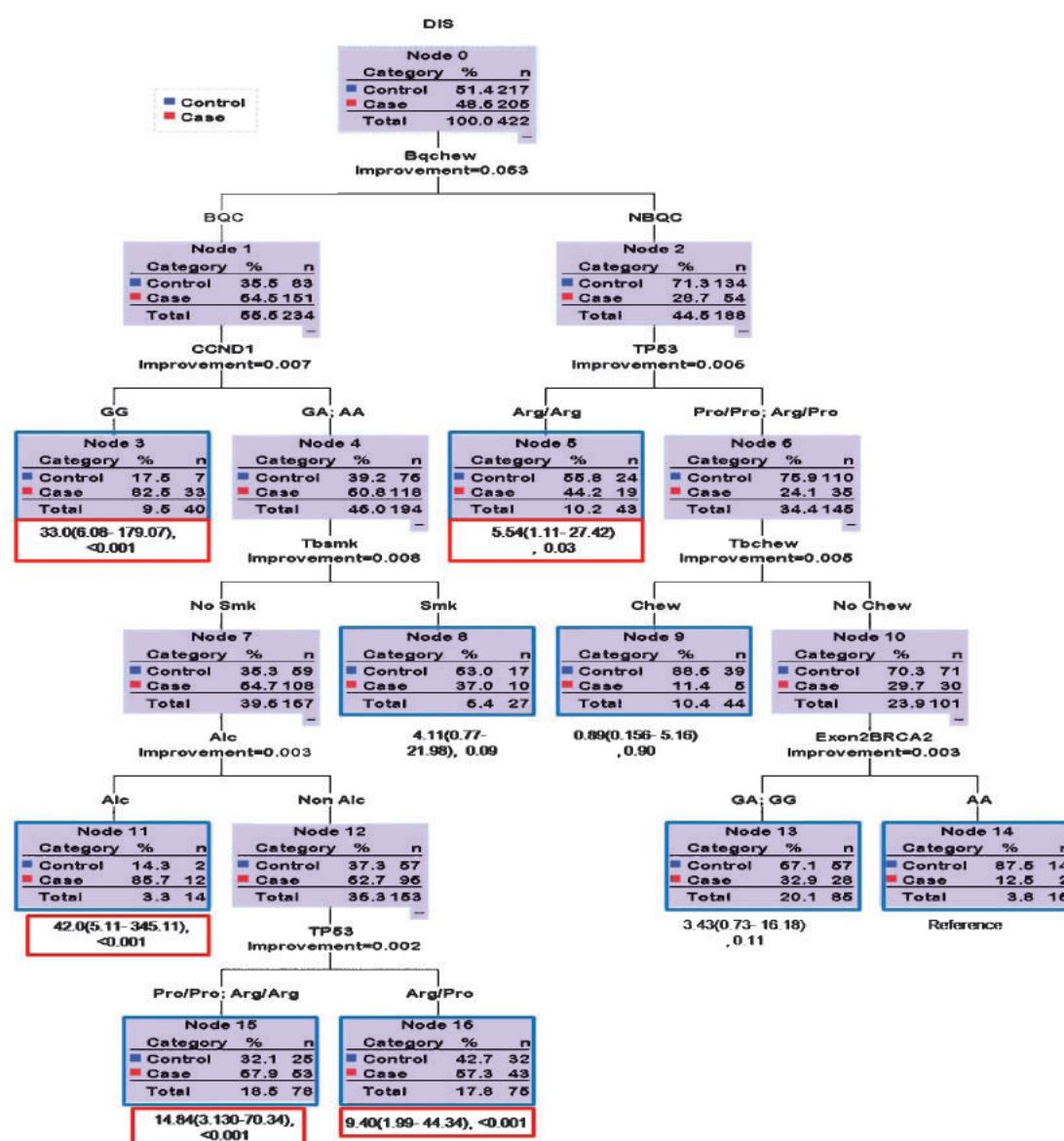


Fig. 3.15: Classification and regression tree (CART) analysis for the DNA repair and cell cycle genes and environmental risk factors. Terminal nodes are bordered thick blue. Red bordered odd ratio boxes values are <0.05.

Genomic Alterations in Breast Cancer Patients in Betel Quid and Non Betel Quid Chewers

To elucidate the mechanism of betel quid carcinogens in North East Indian population; role of genomic alterations in breast cancer patients with and without BQ chewing history was investigated. Study of copy number variations and pathway networks using SNP array and Ingenuity Pathway Analysis (IPA) was done in BQ chewers (BQC) and non BQ chewer (NBQC) breast cancer patients. BQC tumors showed significantly ($p < 0.01$) higher total number of alterations, as compared with NBQC tumors. Incidence of gain in fragile sites in BQC tumors were significantly higher ($p < 0.001$) as compared with NBQC tumors. The chromosomal regions 7q33 and 21q22.13 were significantly ($p < 0.05$) associated with BQC tumors while regions 19p13.3–19p12 and 20q11.22 were significantly associated with NBQC tumors. Gene ontology (GO) and network analysis showed that genes associated with BQC regions, 7q33 and 21q22.1 were enriched for oxidoreductase ($p < 0.001$) and aldo-keto reductase activity ($p = 0.015$) in contrast to G-protein coupled receptor protein signaling pathway ($p = 0.005$) and cell surface receptor linked signal transduction ($p = 0.012$).

for 19p13.3–19p12 and 20q11.22 regions associated with NBQC. IPA analysis for BQC associated regions revealed one top network (score=20) “Drug Metabolism, Molecular Transport, Nucleic Acid Metabolism” encompassing genes like AKR1B1, AKR1B10, AKR1B15, ERG, ETS2 (Fig. 3.16). IPA analysis for NBQC genes revealed two top networks (score=29) “Molecular Transport, Nucleic Acid Metabolism, Small Molecule Biochemistry” and “Cellular Development, Embryonic Development, Organismal Development” (Fig. 3.17) encompassing genes like RPN2, EMR3, BLCAP and VAV1, NNAT and MUC16 respectively.

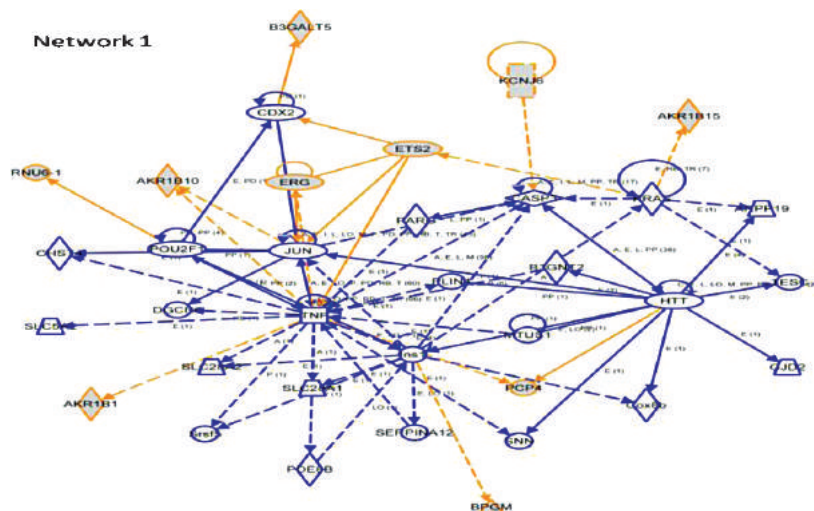


Fig. 3.16. BQC Network 1 Drug Metabolism, Molecular Transport, Nucleic Acid Metabolism.

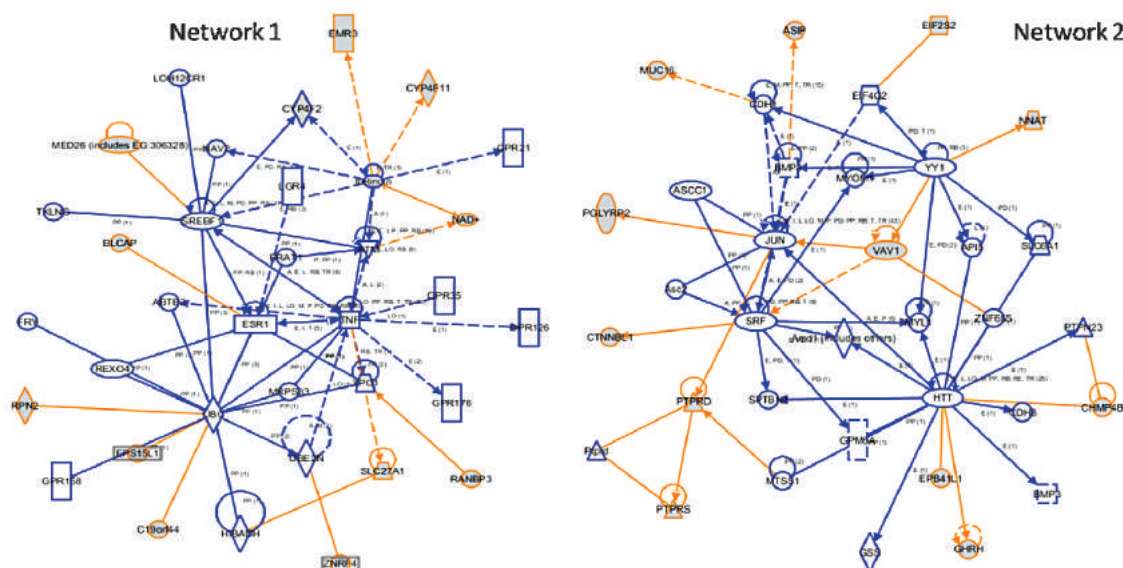


Fig. 3.17. NBQC Networks: Molecular Transport, Nucleic Acid Metabolism, Small Molecule Biochemistry (Network 1) and Cellular Development, Embryonic Development, Organismal Development (Network 2).

Twenty seven common regions of gain were illustrated between BQC and NBQC tumors. Both groups exhibited gain on chromosomes 1q, 5p, 7p, 8q, 12q, 16p, 17q and 20q. Enrichment and IPA of genes associated with these regions show activation of protein kinase activity ($p=0.009$) and cell junction ($p=0.01$). IPA analysis revealed three top networks, (Fig. 3.18). Network 1 functions in cellular movement, connective tissue development and function, cellular assembly and organization (score=43) with key role played by PTK2. Network 2 functions in cell-to-cell signaling and interaction, tissue development, organismal injury and abnormalities (score=43) with RPN2, EMR3, VAV1, NNAT and MUC16 important

genes. Network 3 functions in cell morphology, cellular assembly and organization, cellular compromise (score=32) with key roles played by MYC and YWHAZ. Among all the tumor associated canonical pathways enriched were GNRH signaling (p=2.92E 04), cAMP-mediated signaling (p=3.60E 04), protein kinase A signaling (p=3.77E 04), CXCR4 signaling (p=4.99E 03), molecular mechanisms of cancer (p=8.58E 03), phospholipase C signaling (p=1.01E 02), RAR activation (p=3.16E 02), ILK signaling (p=4.21E 02).

The study showed that breast cancer associated with betel quid chewing exemplifies genetic alterations differing from those observed in the non betel quid chewer. A number of genetic changes are shared in both tumor groups considered as crucial in breast cancer progression.

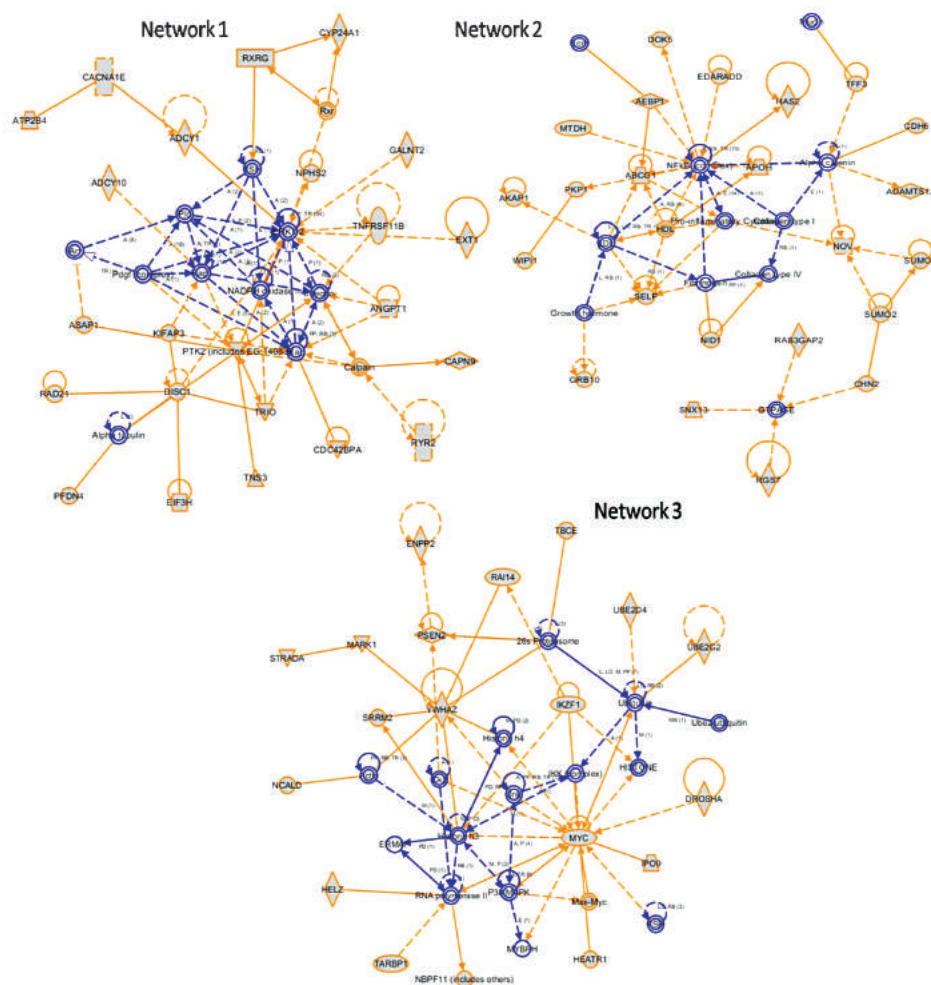


Fig. 3.18: Common Networks: cellular movement, connective tissue development and function, cellular assembly and organization (Network 1), cell-to-cell signaling and interaction, tissue development, organismal injury and abnormalities (Network 2), cell morphology, cellular assembly and organization, cellular compromise (Network 3).

Betel quid chewing has been shown as a significant and independent risk factor for developing breast cancer in NE region. Although common genetic variations in DNA repair and cell cycle genes contributed towards breast cancer risk, BQC were at an elevated risk for breast cancer attributable to betel quid carcinogens with minor roles of BRCA2 mutation and C allele of RAD51. Breast carcinogenesis associated with betel quid chewing showed genetic alterations different from those observed in the non betel quid chewers. A number of genetic changes were shared in both tumor groups considered as crucial in breast cancer progression.

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Magnitude of Cervix Cancer in India

Cancer of the cervix has been the most important cancer in women in India over the past two decades. Very high age adjusted incidence rates, varying from 22.5 per 1,00,000 to 30.2 per 1,00,000, has been found in Pasighat district, Mizoram district, Aizawl district and Pampumpare district, Barshi and Chennai. Earlier Chennai used to register the highest incidence for cervical cancer in India. After the initiation of Cancer Atlas Project, at least 5 districts with even higher incidence have been recorded. Cancer of the cervix accounted for 16% of all cancers in women in the urban registries in 2005. However, it constitutes approximately 30% of the cancers in females in Barshi.

All the urban population based cancer registries (PBCRs) at Bangalore, Bhopal, Chennai, Delhi and Mumbai have shown a statistical significant decline in the AARs of this site of cancer. This decline is despite the absence of any organized screening programme or any other intervention. As of today in the urban PBCRs the cervical cancer ranks at number two following breast cancer. However, in the rural areas cervical cancer still is a leading site of cancer among females. Since over 70% of the Indian population resides in the rural areas, cervix cancer still constitutes the number one cancer among females. Based on the data of the PBCRs, the estimated number of new cancers during 2007 in India was 90,708. The projected number of cervical cancers in India for the years 2020 will be 1,23,291.

Most of the Indian women with cervical cancer present at a late stage because of the lack of awareness about the disease and its treatment. Over 63-89% of all cervical cancers had regional disease at the time of presentation in different hospital cancer registries of ICMR.

Cervical Cancer Screening through Para-medical Health Workers under Re-orientation of Medical Education Programme (1985-1990)

Important aspects of cancer screening are high quality, high coverage and close monitoring of various activities. In 1980s, the role of para-medical personnel was considered to be minimal. However, it was not possible to cover a substantial population unless para-medical personnel are involved in such programmes. The ICMR initiated a project to assess the feasibility of involving health services through performance of several cancer screening activities by para-medical workers. The study was carried out in Alipur block of Delhi through Institute of Cytology and Preventive Oncology.

The project aimed at assessing the feasibility of utilizing Auxiliary Nurse Mid wives (ANMs), Village Health Guides, Anganwadi workers, Interns and Medical Officers of primary health center, after appropriate training in the detection of precancerous and cancerous lesions of the uterine cervix, by means of clinical downstaging for all married women and by Pap smear of women above 35 years and those symptomatic even below 35 years of age. The project was also aimed to establish a system of referral for patients detected through the Pap smear screening programme. The 5-year study was initiated in 1985. The preparatory

phase included baseline survey, finalization of strategy, development of various questionnaires & record keeping mechanisms, referral procedures, development of health education material, etc; medical interns, medical officers of the PHC and anganwadi workers were also involved. The main activities carried out during the intervention phase included (i) orientation and re-orientation training; (ii) health education & motivation; (iii) Pap smear collection & follow up; (iv) clinical observations & Pap smear analysis; (v) management and follow up of cases; and (vi) concurrent assessment of all health personnel. Coverage in the age group above 35, during the effective intervention period of 38 months was 36.8%. Smears could not be collected in 11.9% of women covered under the project. The Pap smears were adequate in 85.5% cases. The study registered 19 cervical dysplasia cases and no case of malignancy was encountered. Comparison of results from surveys on knowledge, attitude and practices before and after intervention revealed 40% increase in knowledge about cervical cancer.

In summary, the project provided confidence that cancer screening activities can be carried out through para-medical health workers after adequate training, providing optimal facilities, and close monitoring. This was encouraging and helped in designing projects on involving health services at a larger level.

District Level Cervical Cancer Control Projects

The lessons learnt by ICMR in 1980s were clear that para-medical personnel can be involved in screening for cervical cancer. It was also clear that starting a vertical programme for cancer control would not be possible. Thus the next logical step was to start involving health infrastructure motivated by experts from Regional Cancer Centres (RCCs). The first step was to reaffirm the lessons learnt during conduct of the block level cervical cancer screening programme in Delhi. A pilot project in Tamil Nadu (in 1989 & 90), studied the feasibility of training female MPWs in clinical examination of the uterine cervix of the eligible population and collection of the Pap smear. The study demonstrated that the village health nurse can satisfactorily be trained to conduct regular cervical screening programme, both in identifying an abnormal cervix and in taking adequate Pap smear.

The operational research task force project on down staging carcinoma cervix envisaged utilizing the existing health infrastructure to take the initiative for the early detection of cancer cervix. The twin center project (in Gujarat and Karnataka) aimed at assessing the efficacy of clinical down staging through cervical examination and selective cytology for control of cervical cancer. The specific objectives were to assess if the para-medical were able to educate women to participate in screening programme; identify abnormal cervixes; able to collect adequate Pap smears; and refer the women as per established criteria. The project started through Bangalore & Ahmedabad RCCs in 1991. The female para-medical workers of health services were already performing insertions of the intra-uterine contraceptive device, which also involved first ensuring that cervix was normal in appearance. These workers were provided adequate training to impart health education, perform visual inspection of the cervix and identify abnormal cervixes and suspicious of malignancy and accordingly refer; take a Pap smear. The anganwadi workers were trained to impart health education. The referral was for the Lady Medical Officer at the Primary Health Center, who was also trained to verify their observation and treat or appropriately refer to the concerned RCC. The compliance to this referral was also to be assessed. The ultimate outcome parameter was to assess if the intervention would result in change in the stage of the disease at the time of presentation for treatment. A substantial increase in the knowledge of the workers was observed after their training (Fig. 4.1). The Pap smears collected by para-medical workers in the field were considered to be satisfactory, in view of the adequacy varying between 91% and 95% (Fig. 4.2). The performance of para-medical workers in differentiating abnormal & normal cervixes was also good, the agreement rate of their findings with PHC medical officer being 89.7%, and with the project gynaecologist being 95.2%. After an 18 months

intervention, the proportion of women covered for health education at Karnataka was 8.3% in the area with clinical examination in the field, and 22.0% in the area with only health education in the field. The coverage for health education at Gujarat was near total. The coverage for clinical examination of cervix was more in Gujarat, if the examination was carried out in the field (28.3% vs 0.8%).

The coverage for clinical examination in Karnataka was 8.3% when the examination was done in the field, and 9.3% in the PHC approach. A total of 154 dysplasia cases were detected out of total of 2,110 women screened in the area with clinical examination in the field, in Gujarat. In Karnataka, 86 dysplasia were detected out of total of 889 screened women. The sensitivity and specificity rates of the procedure of clinical down staging were low (Table 4.3 (a to d) and were not considered to be enough for the purpose of the procedure independently qualifying as a screening tool. The sensitivity and specificity were observed to be low, whether inflammatory lesions were classified as normal or as abnormal.

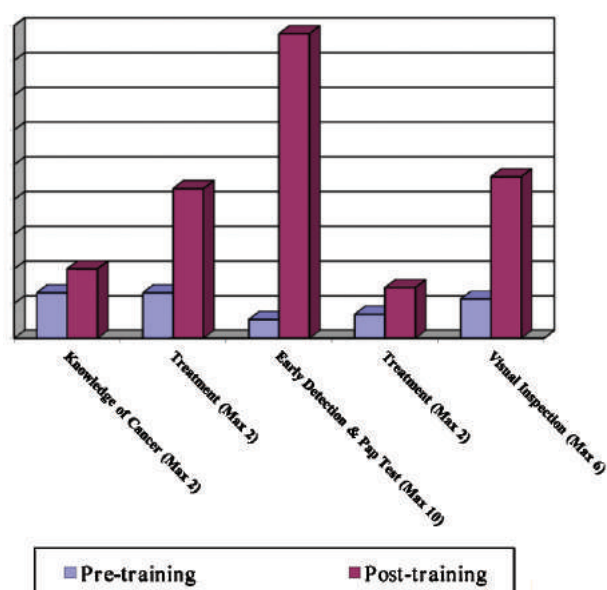


Fig. 4.1: Pre & Post training Knowledge Score of Paramedical Workers, Kheda Centre.

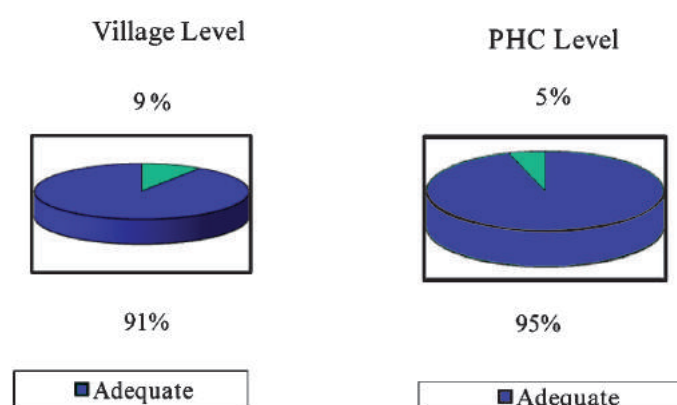


Fig. 4.2: Adequacy of Pap Smear (Karnataka).

Table 4.1: Comparative Results of Visual Examination Between Para-medical Workers and PHC Medical Officer (Gujarat).

PHC Medical Officer	Para-Medical Worker		
	Normal	Abnormal	Total
Normal	372 91.4%	35 8.6%	407 100%
Abnormal	56 11.8%	419 88.2%	475 100%
Total	428	454	882

Mcnamar •²Test p = 0.0277
Agreement Rate = 89.7%

Table 4.2: Comparative Results of Visual Examination Between Para-medical Workers and Project Medical Officer (Gujarat).

Project Medical Officer	Para-Medical Worker		
	Normal	Abnormal	Total
Normal	550 97.2%	16 2.8%	566 100%
Abnormal	22 10%%	198 90%	220 100%
Total	572	214	786

Mcnamar •²Test p = 0.3304 Agreement Rate = 95.2%

Table 4.3 (a-d)

(a)

Comparison of Findings of Visual
Inspection and Pap Smear (Gujarat)
Dysplasia Vs Others as Normal

Visual Inspection	Cytological Examination		Total
	Normal + Inflammatory	Dysplasia	
Normal	1186 60.6%	73 47.4%	1259
Abnormal	770 39.4%	81 52.6%	851
Total	1956 100%	154 100%	2110

Sensitivity= 52.6%; Specificity= 60.6%

(b)

Comparison of Findings of Visual
Inspection and Pap Smear (Gujarat)
Normal Vs Abnormal (Dys+Inf)

Visual Inspection	Cytological Examination		Total
	Normal	Dysplasia + Inflammatory	
Normal	469 81%	790 51.6%	1259
Abnormal	110 19%	741 48.4%	851
Total	579 100%	1531 100%	2110

Sensitivity= 48.4%; Specificity= 81%

(c)

Comparison of Findings of Visual
Inspection and Pap Smear (Karnatka)
Dysplasia+Ca Vs Others as Normal

Visual Inspection	Cytological Examination		Total
	Normal + Inflammatory	Dysplasia + Ca	
Normal	634 79%	49 57%	683
Abnormal	169 21%	37 43%	206
Total	803 100%	86 100%	889

Sensitivity= 43%; Specificity= 79%

(d)

Comparison of Findings of Visual
Inspection and Pap Smear (Karnatka)
Normal Vs Abnormal (Dys+Inf+Ca)

Visual Inspection	Cytological Examination		Total
	Normal	Dysplasia + Ca Inflammatory	
Normal	282 81.7%	401 73.7%	683
Abnormal	63 18.3%	143 26.3%	206
Total	345 100%	544 100%	889

Sensitivity= 26.3%; Specificity= 81.7%

An important lesson learnt was that screening activities need to be provided to women as close to their home as possible, as the asymptomatic women may not be motivated enough to visit far away clinics which sometimes may be at the expense of losing their daily wages or spending their savings. The study also indicated that with adequate training the Pap smears collected in smears were satisfactory and the procedure can be adopted for the purpose of screening. Their performance in terms of differentiating abnormal cervix from normal was also good as shown by a high agreement rate of their examination findings with the medical officers. The higher rate of agreement with the project medical officer perhaps can be explained by the fact that their training was carried out by the project medical officer. The study showed that the operational aspects may be satisfactory; suggesting that with careful planning the health services may be gainfully utilized in carrying out cervical cancer screening activities. The strategy may however, require additional aspects to be added in view of the low sensitivity and specificity rates for clinical examination as compared to cytology as a standard.

Consultation on Cervical Cancer Control in India

A national consultation was organized to review the existing infrastructure, manpower and fiscal resources to start a cervical cancer control programme; to take the opinion of the experts at the national level from different cancer centres as well as from the professional organizations to evolve the best strategy of action in cervical cancer control; to consult the international experts in implementation of cervical cancer control programs; to review and update the existing Indian guidelines and managerial strategies for such a program at the national level; and to help health administrators to adopt an appropriate policy and to integrate the program in the existing health infrastructure.

The consultation was organized by ICMR on 18th to 20th March 2002, at New Delhi, in collaboration with the Ministry of Health & Family Welfare, International Network on Control of Gynaecological Cancer (INCGC), and WHO-India. The group noted that in recent years there seemed to be a substantial increase in cytological facilities and manpower. However, the number of smears examined in these laboratories was quite low, and none of the laboratories was fulfilling the criteria of WHO of 20,000 smears/year to maintain diagnostic skill. Quality assurance was not mandatory. Costs of Pap smear test were variable, from free of charge to the patient to Rs. 200 in private laboratories. Screening was mainly opportunistic, although there were some attempts at organized screening. Facilities for cytology were inadequate. In some centres pilot studies for alternative methods were ongoing (e.g. visual inspection with acetic acid, visual inspection with Lugol's Iodine, HPV, colposcopy). There was no general consensus concerning the results. In some states the efficacy of health workers' detection of abnormalities matches with that of medical officers. Mobile detection camps offer the screening facility to the patient at the periphery. However, cost-effectiveness was reduced and follow-up was poor when aiming at a target-population. This approach can be useful as a component of an organized screening program. Pap test was being used for screening in most centres.

The issues related to the implementation of a cervical cancer control program that would achieve the reduction in morbidity and mortality from the disease in a phased manner were deliberated. The experts stressed on the need for awareness about the various risk factors and the necessity of early detection, which should be imparted not only to the women but also to their husbands and other members of the family. Societal attitudes need to change with emphasis on education of school children. Screening for cervical cancer was considered to be ideal, the strategies for which should be evolved in a phased manner so that optimum benefit can be achieved utilising the available resources and taking into consideration the ground realities. A stepwise approach was suggested. In view of the available expertise with the regional cancer centres, it would be useful if in the first phase, these institutions could collaborate with

the state health services to undertake cervical cancer screening programme in one district. The areas where the Modified District Cancer Control Program was already running under the National Cancer Control Program by the different Regional Cancer Centres (RCCs) can be taken up to develop a model of such organized screening program. Subsequent expansion of programs to other districts may be through involvement of medical colleges/institutions and non-governmental organisations in the area.

For a short-term measure the women attending the primary health centre/hospital outpatients or in-patients for consultation for gynaecological or other problems, antenatal clinics, immunization and sterilization camps can be motivated to undergo tests for opportunistic detection of cervical pre-cancer and cancer. These programs concentrate resources in low risk populations with limited effect on death rates. These current efforts may continue, whilst efforts were made to initiate a population based organized screening program. To have such an organized cervical screening program emphasis had to be given to improve the existing health infrastructure, to train the manpower resources and to set up a workable call-recall system. Such programmes need to be undertaken on a continuous basis.

The screening programmes would be more successful if a high number of women (>80%) within a specified age group can be covered rather than using the test repeatedly on a smaller number of women. All women should be offered at least one cervical screening test in their lifetime between ages 40-45 years age. This was based on data from India in relation to peak incidence for pre-invasive and invasive disease. For a long-term strategy, depending upon the availability of resources, a step up approach can be followed, the direction of stepping up being: once in a lifetime screen at age 40-45 yr.; 10 yearly screen at age 35, 45, & 55 years of age; 5 yearly screen 35-64 years of age. When coverage of target population is more than 80% and if resources are available, one may start screening at age 25 years onwards.

Cancer Screening in Himachal Pradesh

The study was initiated on the backdrop of the knowledge that health services should be able to undertake a cancer screening programme if it is projected as one of their work element aimed at reducing mortality and morbidity in the community. The lesson necessitated that one should look forward for a state willing to initiate cancer control activities by integrating the activities in their routine primary health care operations. This study was initiated when the Government of Himachal Pradesh showed their inclination to initiate a cancer screening project. Like other state governments, Himachal Pradesh (HP) was also undertaking some programmes for early cancer detection through popularization of warning symptoms. Another reason for choice of HP was its terrain for coverage being difficult, suggesting that a system may be more likely to be replicable for rest of the country.

One of the important decisions was to decide on what cancer sites should be covered. The ICMR had already carried out projects on cervical cancer screening, and it remained one of the priority cancers for India. Breast cancer was another site where intervention elsewhere has shown to be effective. Oral cancer was agreed to be another site for intervention, taking the analogy of cervical cancer in terms of a pre-cancerous state, report of possibility of success on such intervention at Kerala, and the public health importance of the disease in India. Lung cancer was also an important cancer site for India, but in view of non-availability of an evidence based screening tool, this component was limited to anti-tobacco education only.

Strategy for screening was another major aspect to be finalized. It was realized that the most important aspect in cancer screening would be to establish a system for cancer screening in the health infrastructure which should be able to take care any change in strategy in future if a need was observed. At the initiation of the programme, only the evidence based strategy should be incorporated. For cervical cancer, Pap smear

examination was the best known successful strategy, and some studies on visual inspection after acetic acid/ Lugol's iodine had been reported. On a wider consideration, it was realized that long term effects of these alternate strategies were not known and the strategy would not result in complete avoidance of cytology services. Thus, a network for cytology smear, their transport, examination and reporting had to be built in the programme. Thus, in order to minimize confusion among the health care personnel, it was decided to initiate the study with cytology based strategy. Mammography was known as the best approach for screening for breast cancer. However, as the programme was to be carried out at a big level (minimum one district), the cost for mammography machines with guided biopsy facility was considered to be prohibitive. Therefore, mammography was decided to be included as a confirmatory support on the lines of WHO recommendations. For oral cancer, the approach of anti-tobacco education coupled with follow up of oral pre-cancerous lesions was taken as the appropriate approach. The extent of the intervention was another decision. The state suggested covering as much area as practical. However, the immediate intention was to develop a system appropriate to the state, it was decided to begin with three districts which should be geographically away from each other. Thus, districts of Mandi, Hamirpur and Bilaspur were selected for intervention.

The operational research study aimed at establishing a system for screening for cancers of cervix, breast and oral cavity in three districts of Himachal Pradesh. The major components of the study were anti-tobacco community education; education regarding risk factors of cervical and breast cancers; screening for early diagnosis of cancers of oral cavity, cervix and breast, by the paramedical workers.

For this; an Expert Committee and a Monitoring Committee (for close and frequent interaction) were constituted. The decision on geographic areas for intervention, cancer sites, and strategies, etc. (as described under background), was taken by these committees after consultation with state health professional. The Monitoring Committee provided useful suggestions.

It took several discussions and visits to finalize and complete the activities related to preparatory phase. The salient activities included finalization of strategy for the project; enumeration of target population; development of health education aids; preparation of training manuals and proformae; and conduct of a baseline survey on knowledge, attitude & practices. Each of the activities required careful planning without interrupting the routine activities of the health system. These activities were carried out with the help of the Monitoring Committee. A training programme for establishment of a population based cancer registry for the entire State was carried out. Quality control plan for connected activities was also finalized. There were some delays in training of medical officers, paramedical workers and subsequent initiation of the intervention. The training of the health staff was carried out by the state government themselves. In early 2007, it was decided that clear directions and necessary technical inputs have been provided to the state governments by ICMR, and the state government may carry the intervention on their own.

In 2010, the state government requested ICMR to get involved in the project again and provide technical and financial support for running of cancer screening activities in these three districts or even the whole of state. Despite assurance from the health department report submitted by investigators lacked important details and was accepted by monitoring committee with limitation. It was decided to consolidate the work carried out in the original three districts to begin with and then consider expansion of the project.

In summary, it was clear from the study that the state governments were interested in initiating a cancer screening programme. However, they would like to carefully develop the needed elements for integration of cancer screening with the existing activities without disrupting the routine activities. This aspect was achieved under the project, but the time taken was more than what had been planned at the outset. The exercise also suggested that a one-time training may not be sufficient for a major programme like cancer

screening, but a continuing education plan (re-training of existing and training of the newly transferred staff) would also be essential. The study identified the activities that may be necessary for initiation of a cancer screening activity by the state governments. The experience also showed the need for integrated development of all the activities at correct timing. It may be necessary for ICMR to remain connected with undertaking of intervention activities for some time after development of the intervention package.

Human Papilloma Virus and Cervical Cancer

A study supported by ICMR in early 1990s in Delhi, tried to determine the humoral and cellular immune response against human papilloma virus (HPV) 16, in patients with benign and malignant lesions of the uterine cervix and to correlate the response with the clinical status of the patients. HPV 16 E7 and L1 proteins as well as synthetic peptides of two B cell epitopes of HPV 16 E7 gene product was used for this purpose. However, no correlation could be established between the severity of the disease and T cell responses. The study suggested that peptides PI and PII are the two major immunodominant B cell epitopes of the HPV16E7. This study was considered to be an initial aspect before embarking on development of immunological tests for diagnosis of HPV and also to develop a vaccine against HPV. However, failure to identify immunogenic response or tests resulted in keeping the subject on backburner for some time.

By the year 2000, four studies were reported on development of vaccine against HPV 16 and/or 18. Thus the focus was shifted to understand and assess the efficacy of these vaccines and to check if this vaccine would be useful through health services network. With this background a WHO-ICMR consultation on “HPV vaccines as a tool for prevention of cervical cancer” was organized at New Delhi on 18th & 19th September 2002. This helped in consolidating the thoughts on assessing the role of HPV in control of cervical cancer. It was also noted that the only non-profit organization working on HPV vaccine at that time was the National Cancer Institute of USA. The Secretary, Ministry of Health & Family Welfare was requested to write to NIH, for possibility of provision of HPV vaccine developed by them to India, for studying feasibility of its use through health services and also assess its efficacy in Indian population. However, the NCI, USA, informed that the vaccine technology has already been handed over to private sector in USA and suggested contacting Merck & Co. for any further studies.

In late 2006, Bill and Melinda Gates Foundation awarded a multi-country project to PATH, for studying the feasibility of introducing HPV vaccine for purposes of control of cervical cancer in less developed countries. The four-country study also included India. The PATH contacted India to provide advisory support for carrying out this study in India.

WHO-ICMR Consultation on HPV Vaccines as a Tool for Prevention of Cervical Cancer

Cervical cancer is an important public health problem for India. Infection with human papilloma virus (HPV) is considered to be an important aetiological factor in its causation. In early 2000s, it had been possible to develop vaccines against certain high-risk as well as low-risk strains of HPV. The consultation was organized to consider whether a cervical cancer vaccine would be a solution for cervical cancer disease control in India. Over the years India had collected many data on the prevalence of HPV 16 and 18 in cervical cancer, though other HPV types have not been studied in detail. In a few population based studies and several hospital based studies on HPV in India, it was shown that HPV 16 comprised about 70 % of infections associated with cervix cancer while HPV 18 comprised 12 %. In addition molecular mechanisms of oncogenicity had also been studied. The group reviewed the existing literature from India and from other parts of the world to decide on the role of HPV vaccine in cervical cancer control activities in India.

The group agreed with the need for testing the efficacy of vaccines(s) against HPV in order to assess its impact on cervical cancer. The group discussed the type of vaccines that would be ideal for India, the difficulties in their testing; the age & sex groups on which the trial should be carried out; dosage schedule; existing organizations through which the trial/ intervention programme could possibly be carried out; ethical issues associated with a vaccine trial; costing and cost-effectiveness of a HPV vaccine programme; strategies for prevention of cervical cancer; end-points for a vaccine trial, possible sites for a vaccine trial; etc. Questions were put forward to the meeting participants and upon extensive discussions and debate based on scientific evidence the panel members agreed on the following recommendations:

1. Recognizing that cervical cancer is the most common cancer of women in India, that it is responsible for a heavy load of morbidity and mortality, and that there are grossly inadequate measures for its early detection and treatment, there is a need for clinical evaluation of vaccines against HPV for prevention of cervical cancer.
2. It was noted that HPV 16 was predominantly associated with cervical cancer cases in various parts of the country. However, there was meagre data on incidence and transmission dynamics of HPV in the community, especially young (including interaction between HPV types and other RTI). Such data needs to be generated and a carefully designed vaccine trial could generate it cost-effectively.
3. Several laboratories in the country have the capability of detecting and typing HPV and undertaking cytological and histological procedures. However, inter-laboratory procedures and standardization as well as internal and external quality control procedures need to be strengthened in preparation for the trials. In addition therapeutic facilities for management of diseases detected during the trial need to be optimized.
4. Social and behavioural research should be undertaken to understand the acceptability or resistance of the population towards vaccination. Suitable and optimum health education measures should be taken to sensitize the population in the entire trial area. Care should be taken to ensure that there is no stigmatization or discrimination of the participants.
5. Selection of study population should be based on the principle of selecting those with low probability of HPV infection at enrolment but high probability of getting infected within a short period after vaccination. Current data suggests that priority should be given to trials in women but recognizing trials of vaccination among men may be necessary at a later stage.
6. Since vaccination programmes are not imminent at this stage, only ethical and gender issues that are relevant to the conduct of clinical trials need to be considered at this stage.
7. Cost-effectiveness and financial sustainability are extremely important. Currently data is inadequate to optimally address the issue. Trials may provide an excellent opportunity to address certain economic aspects, which may help in this kind of analysis.
8. The trial should preferably be multi-centre and the selection of trial sites should be based on objective assessment and site visits. The workshop helped in consolidating the thoughts on undertaking further research on HPV vaccine and also on the role of HPV virus in aetiology of cervical cancer in India.

Demonstration Project on HPV Vaccination in Andhra Pradesh and Gujarat

HPV infection is an important cause of cervical cancer. PATH in partnership with State Governments of Gujarat and Andhra Pradesh implemented an operational research study related to cancer of the cervix prevention in India. ICMR provided technical support & consultation for development of protocol and plan of monitoring. The study was a part of the four-country project awarded to PATH by the Bill

& Melinda Gates Foundation, to assess the feasibility of use of HPV vaccine under the public health programmes of less developed countries, as a part of broad cervical cancer control programme.

The objectives were to generate critical data and experience for evidence-based decision making about public sector immunization programs as part of a broader cancer-of-the-cervix prevention and control strategy. This was an operational research project, on assessing the health services needs and preparedness in case HPV vaccine is introduced to health services in future.

The study was carried out in three blocks of Andhra Pradesh and Gujarat, through the existing health infrastructure of respective states. The study was carried out with the help of an expert group (National Technical Advisory Committee) after obtaining appropriate clearances from the Institutional Ethics Committees and the Health Ministry's Screening Committee on Indo-Foreign projects. Study was a part of the four-country project (Peru, Uganda, Vietnam and India) sanctioned by the Bill and Melinda Gates Foundation to PATH (Program for Appropriate Technologies in Health). None of those countries made any commitment to ongoing vaccination after the research was completed. The vaccine for this purpose was donated by the manufacturers of the vaccines (Merck & Co. for Andhra Pradesh, and GSK for Gujarat). Thus, financial support for the project was not provided by any government agency in India. If the utility of HPV vaccine was agreed, the participating countries were expected to wait for the day when these vaccines become affordable, so that an evidence-based decision can be made on what to do. It was also understood that for effective cervical cancer control, countries must work to develop and maintain screening programs with treatment for adult women and possibly vaccination for adolescent girls.

Phase I of the study was in the form of a formative study carried out by PATH in collaboration with the Department of Social and Behavioural Sciences at the National AIDS Research Institute (ICMR), Pune, in 3 blocks each of Khammam district (Andhra Pradesh) and Vadodara district (Gujarat). Through expert consultation, focus group discussions (FGD) and in-depth interviews (IDI), supplemented by systematic health systems study (HSS), data was collected from adolescent girls; parents, guardians or decision makers for adolescent girls; community (geographic and affinity groups) leaders; health professionals at different levels; policy makers and representatives of NGOs; and experts on immunization-related issues at the national level. The demonstration project was designed on these findings. Study was started after receiving requisite clearances from the local Ethics Committees, Health Ministry's Screening Committee, and Drug Controller General of India.

The demonstration project to study the feasibility of HPV vaccination among girls aged 10 to 14 years was carried out by the respective state health infrastructure in 3 blocks each of Khammam district (Andhra Pradesh) and Vadodara district (Gujarat). PATH received the grant for the project from the Bill and Melinda Gates Foundation. For the purpose of this project Merck and GSK agreed to provide the vaccine to PATH, free of cost in one state each.

The study utilized both the brands of HPV vaccine available in the market (Gardasil by Merck in Andhra Pradesh; and Cervarix by GSK in Gujarat). The study was initiated after these vaccines had been approved by the Drug Controller General (India) (DCG(I)) for marketing and sale in Indian market. The technical component was finalized in consultation with ICMR, including plan for appropriate monitoring and clearance from appropriate authorities. The technical expertise on implementation and monitoring to the health services was provided by PATH. Cold chain for the vaccine was maintained and monitored as per standard protocol. The vaccination started in Gujarat from 17th August 2009 with GSK vaccine; and in Andhra Pradesh from 16th July 2009 with Merck vaccine. Only those for which consent and assent had been obtained received the vaccine.

Gujarat: In Gujarat, three 0.5ml doses of Cervarix™ (HPV vaccine) were administered intramuscularly according to the standard schedule of 0, 1 and 6 months. A total of 10,686 girls between the ages 10-14 years were vaccinated with 1st dose of HPV vaccine manufactured by GSK, out of which 97 girls (0.90%) were reported to have adverse events, none of which was serious. 10,259 girls were administered the 2nd dose of the vaccine, out of which 13 (0.13%) reported adverse events, none of which was serious. 9,637 girls were administered the 3rd dose of the vaccine, out of which 14 (0.15%) reported adverse events, none of which was serious.

Andhra Pradesh: In Andhra Pradesh, three 0.5 ml doses of Gardasil® (HPV vaccine) were administered intramuscularly according to the standard schedule of 0, 2, and 6 months. A total of 14,091 girls between the ages 10-14 years were vaccinated with 1st dose of HPV vaccine manufactured by Merck & Co., out of which 38 girls (0.27%) were reported to have adverse events, none of which was serious. 13,930 girls were administered the 2nd dose of the vaccine, out of which 52 (0.37%) reported adverse events, none of which was serious. 13,791 girls were administered the 3rd dose of the vaccine, out of which 239 (1.73%) reported adverse events, none of which was serious.

Deaths among girls vaccinated under the project in Andhra Pradesh and Gujarat: Death sometime after vaccination may not mean death due to vaccination for any diseases. Details of deaths observed among the girls vaccinated with HPV vaccine under the project are given below. The deaths were observed by the health workers during their routine house-to-house visits or as an effort for call for the next dose of vaccination.

Gujarat: No deaths were reported from Gujarat after 1st dose. Two deaths were reported from Gujarat after 19 & 28 days after 2nd dose (54 & 64 days after 1st dose). These girls had not reported any serious adverse event immediately after vaccination. Causes of deaths among these girls were suspected snake bite; and severe anaemia with malaria. No deaths were reported after 3rd dose.

Andhra Pradesh: A total of five deaths were reported from the group of girls vaccinated with HPV vaccine in A.P., Three deaths were reported 24 days to 49 days after 1st dose of HPV vaccination. These girls had not reported any serious adverse event immediately after vaccination. Causes of deaths among these girls were suicide, viral fever and drowning in open well. One more death was reported 99 days after 2nd dose (196 days after 1st dose), cause of death being suicide. One death was reported 95 days after 3rd dose, cause of death being heat stroke.

The causes of deaths among the girls vaccinated under the project in Andhra Pradesh and Gujarat did not relate to HPV vaccination.

Following media reports on the deaths in trial sites (which were later not found to be associated with the vaccine, as per report of Dr. S.S. Aggarwal Committee), the Government of India suspended all HPV vaccine related trials in the country in April 2010 and Committee submitted its report to Ministry of Health & Family Welfare, New Delhi, 2011.

Natural History of Cervical Cancer and Biological Behavior of Dysplasia

In order to carry out secondary prevention for cervical cancer, natural history of the disease is most essential. In the western countries, the cervical cancer has been brought down by regular cytology screening programmes and natural history of the disease has been well studied. This information however, was lacking in developing countries including in India. In order to bridge this gap, Institute of Cytology and Preventive Oncology (ICPO), one of the premier institutes of ICMR initiated a long term prospective study in six major hospitals of Delhi. About 1,20,000 women were screened from the gynaecology department and over 1000 women who were resident of Delhi with different grades of dysplasias were registered for

long term follow up along with equal number of age and parity matched controls. The study was initiated in 1979 till 1989. The end point of the study was carcinoma *in situ* till seventy two months of follow up, a cumulative incidence of mild dysplasia progressing to carcinoma *in situ* was 4.89, for moderate dysplasias 36.7 and for severe dysplasias 65.4. In short, one in every 25 mild dysplasias, one in every third moderate dysplasias and one in every alternate severe dysplasias progress to carcinoma *in situ*. This information highlighted the importance of degree of dysplasias and its further management for the first time for Indian women. Further, it was noticed that any dysplasia which has to progress to carcinoma *in situ* would do so in about three years time. Further from carcinoma *in situ* to invasive cervical cancer would take ten years. This fact informed about a long drawn out natural history of the disease where ample time was available for women who underwent intervention.

Risk factors of cervical cancer

Factors influencing cancer of the uterine cervix in North India:

This study was conducted over the 3-year period from 1960-63 which comprised of 693 cases of cervix carcinoma diagnosed from a population study of 39,587 women. The findings of this study suggest the importance of the socioeconomic factors of poverty, rapid sexual maturation, early marriage and consequent early sexual experience, and early pregnancy and multiparity in producing the high incidence of cancer of uterine cervix in Indian women.

Analysis of the age distribution by 5-year groups showed that the cervical carcinoma rate was significantly higher in the groups 45-54 years of age, in both hindu and muslim females. It was noted that in hindu women, the highest rate of cervical carcinoma was found in the group who married at the age of 10 years or earlier, whereas in muslim women, the rate of carcinoma was highest in those women who married between 11 and 13 years of age. The incidence of cancer of the cervix was significantly higher in those hindu women who gave birth to the first child at or before the age of 17, whereas in the muslim women, it was highest in those who had the first childbirth between 18 and 21 years of age. A direct correlation was shown between number of pregnancies and incidence of cervical carcinoma, i.e., as the number of pregnancies increase, the rate of carcinoma increases. The cancer rate was markedly high in both groups among women who had more than 4 full-term deliveries.

From the present findings, it was apparent that the cancer rate was definitely higher in women who married at an early age and had a larger family than in those who married late and had few children. The younger age at marriage and higher number of pregnancies might be taken as indicative of prolonged exposure to active sexual life. The majority of the patients with carcinoma of the cervix belonged to the low socioeconomic group.

Risk Factors Related to Biological Behaviour of Precancerous Lesions of the Uterine Cervix.

In this study of factors related to cervical carcinogenesis, a cohort of 1,107 cervical dysplasia along with 1,077 controls matched for age and parity were followed up prospectively. During the follow up 75 dysplasia cases progressed to carcinoma *in situ*. The overall rate of progression of dysplasia to malignancy was observed to be 15.7% at the end of 108 months of follow-up. The analysis of progression rates in relation to various factors revealed significantly higher progression rates for initially higher grade of dysplastic lesions, and early age at consummation of marriage. The other factors, such as religion, literacy status of the patient, number of pregnancies, presence of cervical erosion, history of fetal loss and positivity to HSV-II antibodies, did not reveal statistical significance. The case-control comparison for detection of HPV 16/18 by *in situ* hybridisation revealed the presence of HPV 16/18 sequences

in 67.3% of the dysplasia subjects progressed to carcinoma *in situ* while 27.3% of precancerous cases regressed to normalcy ($P < 0.001$).

Early Detection of Cervical Cancer

Prevention of cervical cancer is now possible through early detection of cervical cancer and its precursors. Organized cytology screening programmes in the west have brought down morbidity and mortality from cervical cancer to a large extent. The success of cervical cancer is dependent on several factors such as awareness of early detection methods among general physicians and the lay public, availability of Pap's screening network throughout the country, concomitant quality control for cytology, availability of adequate trained manpower, call and recall mechanism, referral system from periphery to the tertiary level of health care, availability of evaluation facilities for cytologic abnormalities through colposcopy, availability of colposcopists, availability for histologic and treatment services for both cervical cancer and its precursor lesions. Most of the developing countries lack these facilities and therefore cytology screening on a large scale is not feasible. ICMR through its intramural and extramural research projects is trying to develop different models for early detection facilities.

Hospital Based Cytology Screening

A study carried out at ICPO demonstrated that a hospital based cytology screening programme was successful in detecting cervical cancer in 55% of the cases as compared to only 6% reporting in a cancer clinic of Lok Nayak Hospital, New Delhi. In the same study, it was revealed that 76% of cases in carcinoma *in situ* stage were detected among dysplasia cases under observation. This clearly demonstrated the potential of early detection of cervical cancer in a single cytology screening among women reporting to the gynaecology department.

In order to bring down mortality from cervical cancer, community based screening is mandatory. Opportunistic screening may not be successful in bringing down the mortality from cervical cancer. However, it is also noticed that women in India, though not avail preventive services, they report to gynaecologic services whenever symptoms appear. This provided an opportunity to screen women for cervical cancer. It is hoped that such a programme will be the starting point of creating awareness among women.

Community Based Cytology Screening Programmes – a Feasibility Study

ICPO initiated a project for community based cytology screening programme involving the existing health infrastructure and the reorientation of medical education (ROME) scheme of Maulana Azad Medical College at Alipur PHC, Delhi. It was possible to train the existing ANMs to clinically diagnose cervical conditions. The overall agreement between ANMs and the doctors was as high as 89% after 2 weeks training programme. The adequacy rate of taking Pap smear by the trained ANMs was more than 95%. However, it was not possible to utilize the existing manpower at the PHC for routine Pap smear screening programme. The main reason for the failure was due to involvement of ANMs in achieving MCH and family planning targets.

Similar experiences were gained in one of the ICMR funded projects in South India at Chennai. This was a feasibility study involving Village Health Nurses (VHNs) in the diagnosis of cervical lesions. The agreement between the gynaecologists and the VHNs in identifying cancer among women with abnormal cervix was 95% and 80% of Pap smears taken by VHNs were adequate by WHO criteria. Though the training programmes for diagnosing cervical conditions and taking a Pap smear by trained para-medical staff was highly successful, their involvement in the screening programme was not successful due to their preoccupation in achieving given targets in the MCH and family planning programmes. This clearly shows that additional manpower is required for any cancer control activities at the PHC level.

Another study carried out at All India Institute of Medical Sciences, Delhi showed that the clinical examination had a sensitivity of 81.7% and specificity of 97.3% in identifying cancerous lesions.

Alternative Modalities for Cancer Detection

Considering the problems for undertaking mass scale cytology screening in developing countries research efforts were directed towards finding alternative means for early detection for cervical cancer. ICPO worked extensively on a strategy of clinical down-staging using direct visual inspection of cervix among women attending Gynaecology Department of six major hospitals of Delhi. The study was carried out in two different settings: i) in the setting of a longitudinal follow up of a cohort of 1000 women with different grades of dysplasias and ii) cross-sectional study of women attending gynaecology Out Patient Department (OPD) of six major hospitals of Delhi. In the prospective follow up study of dysplasias, it was observed that only 1.6% of dysplasias had high risk clinical signs (bleeding ectopies, unhealthy cervix and a growth) at the stage of recruitment. At the stage of progression to carcinoma *in situ*, 52.4% had displayed these high risk signs. Likewise the cross-sectional studies of about 45000 women revealed that the women with high risk signs had 19 fold higher risk of detecting carcinoma *in situ* and early invasive cancers compared to women having normal looking cervix. This study also revealed that about 63% of women with carcinoma *in situ* / early invasive cancer (stage 0-II A) could be detected by referring about 10% of women with high risk signs attending gynaecology departments. These strategies can downstage women with cancer with an attendant reduction in mortality from cervical cancer. However this strategy is sub-optimal as it could not detect precancerous lesions and thus was not in a position to bring reduction in incidence of cervical cancer. WHO recommended evaluating visual inspection to identify early curable cancers (1996).

In order to improve the test characteristics such as sensitivity and specificity of visual inspection of cervix, a new strategy of application of 3-5% acetic acid was initiated which is called Visual Inspection using Acetic Acid (VIA).

AV MAGNIVISUALIZER

AV Magnivisualizer is low cost self-illuminated and magnifying device for visualizing the cervix, which can be used where there are no cytologic and colposcopy facilities available. This device can also be used even where there is no electricity. It can also be operated on 12 volt portable rechargeable battery. This unique device has following features:

White light of AV Magnivisualizer has color temperature of 5500K – 6000K (Tungsten bulb has the color temperature ranging from 3000 K to 3400 K) with full visible spectrum of light (wave length 4000 Å to 7800 Å). Human eyes are most sensitive for 5550 Å in yellow-green spectrum of visible light. In the AVMagnivisualizer, photographic quality of halogen bulb was used and the yellow light (halogen bulb also have little amount of yellow light) was corrected by adding special blue filter to make the light source optically white, equal to the day light.

AV Magnivisualizer could detect aceto-white lesion more clearly, which was difficult to detect in yellow light of tungsten bulb. This instrument was used and tested in hospital based population. The Magnivisualizer improved the detection rate of early cancerous lesions from 60% for unaided visual inspection to 95%. It also permitted detection of 58% of cases of low-grade dysplasia and 83% of cases of high-grade dysplasia; none of these cases were detectable by unaided visual inspection. For low-grade dysplasia the sensitivity of detection by means of the Magnivisualizer was 57.5%, in contrast with 75.3% for cytological examination. However, the two methodologies had similar sensitivities for higher grades of lesions. The specificity of screening with the Magnivisualizer was 94.3%, while that of cytology was 99%. First prototype of Magnivisualizer was prepared in 1998 and was used as a tool for visual screening in research

setting. An improved version and commercial viable model was prepared in 2005. The simple technique of visual inspection by means of the AV Magnivisualizer can be undertaken even in a primary health care center; the only additional requirement is an examination table. A biopsy can be performed at the same time, whereas with the result of the cytological investigation is not known for some time afterwards, “see and treat” approach can easily be adopted.

Cancer Screening Strategies Workshop at ICPO:

Recognizing the key role that effective early detection and screening programmes can have in reducing the cancer burden in India, ICPO in collaboration with the US National Cancer Institute Center for Global Health, held a workshop in September, 2013, to summarise feasible options and relevant evidence for screening and early detection of common cancers. The review of these evidence-based recommendations has been published in *Lancet Oncology* July 2015, and is intended to act as a guide for policy makers, clinicians, and public health practitioners who are developing and implementing strategies in cancer control for the three most common cancers in India.

Summary of Recommendations for Cervical Screening in India:

Primary screening with HPV DNA testing is the most reproducible and sensitive approach, even in a resource-limited setting such as in India, since a highly sensitive test such as HPV detection needs less frequent screening and could save programme costs. However, issues such as cost per test, protocols for transportation of cervical cell specimens, testing, reporting, treatment, and follow-up care of HPV-positive women still need to be addressed. Effective translation of evidence relies on a range of implementation issues. Development of an affordable, simple, rapid, and accurate test for HPV DNA detection would be ideal, and research funding for such a device should be a priority. Meanwhile, India should follow the stepwise implementation plan advocated by WHO and initiate the screening programme with visual inspection with acetic acid linked to treatment as the core intervention. The HPV detection test could be introduced as an expanded intervention if resources are available to introduce and sustain the technology.

Secondly, self-collection of vaginal samples for HPV DNA testing for detection of cervical cancer would reduce the need for human resources, and has the advantage of self-empowerment and privacy for these women. The key to successful implementation of self-sampling HPV testing was to promote awareness among women in the community in terms of screening, prevention, early detection, and treatment. This could be achieved, in part, by community health workers. The National Health Mission is a large community-oriented, multifaceted basic health-care delivery programme in India that employs many female community health workers (known as Accredited Social Health Activists). These women can be trained to perform door-to-door counselling and motivate women to undergo cervical cancer screening. With additional remuneration, these workers could obtain self-collected samples from women and transport them to the HPV testing centres.

Thirdly, the number of screening programmes based on visual inspection with acetic acid has been increasing in India. Visual inspection with acetic acid screening is likely to be used in national programmes following WHO guidelines until resources allow the introduction of HPV tests. The expert group recommended the inclusion of quality assurance and evaluation in these programmes, following the latest WHO guidelines on quality assurance of a visual inspection with acetic acid based programmes. The development of strategies for enhancing the implementation of visual inspection with acetic acid screening in India would help to introduce HPV testing as and when a more affordable and point-of-care test is available in the future.

Fourthly, if HPV testing is introduced and resources for high-quality cytological tests are available, women who are HPV positive can be triaged on the basis of this test before referral to colposcopy.

Fifthly, the task group consensus recommendation for the appropriate age and way to screen HPV is to include visual inspection with acetic acid screening for women aged between 30 and 49 years and HPV testing as it becomes affordable for women aged 30 years and older. On the basis of WHO guidelines, a woman should be screened for cervical cancer at least once in her lifetime in a resource-poor setting. Once this is achieved, screening women for cervical cancer at more regular intervals (5 years or 10 years) can be considered.

Finally, screening of eligible women who visit healthcare systems for other reasons (opportunistic screening) will increase the level of awareness of cervical cancer in women, and help to integrate cervical cancer screening with other non-communicable disease services.

Circulating Immune Complexes (CIC) in Cervical Cancer

Presence of circulating immune complexes was studied in cervical cancer patients by polyethylene glycol (PEG) precipitation method and compared with CIC levels from healthy blood donors. 88% samples from cervical cancer patients showed significantly elevated CIC levels as compared to controls. Tumour associated antigen could be detected in CIC.

Expression of Cytokeratins 14, 18 and 19 Among Tumour Progressing to Uterine Cervix

This study was carried out at Regional Cancer Centre, Thiruvananthapuram. The expression of cytokeratins (CKs) in normal cervical epithelium, low grade squamous intra epithelial lesions (SIL), high grade SILs and squamous cell carcinoma (SCC) were analyzed using four different monoclonal antikeratin antibodies. In normal cervical epithelium, CK18 showed strong immunoreactivity in basal and parabasal layers. CK19 and 14 were expressed only in the basal layer while CK13 was found selectively in the spinal cells. As the lesions progressed from the LSILs to HSILs, immunoreactivity of CK18, 19 and 14 in the basal cell compartment increased while the expression of CK13 decreased. In SCC, well-differentiated tumours showed decreased immunoreactivity for CK18, 19 and 14 with CK13 showing a strong and focal (localized) immunoreactivity. Undifferentiated carcinomas totally lacked CK13 reactivity. These findings suggested that expression of CK 18, 19 and 14 may be directly related to tumour grade and CK 13 may be a marker of differentiation in cervical lesions

Single Nucleotide Polymorphisms of Cytokine Genes in Human Papillomavirus (HPV)-mediated Cervical Cancer

Only a fraction of high risk HPV (HR-HPV), with persistent infection develop cervical cancer after a long latent period showing thereby that in addition to HPV infection immunological, environmental or genetic co factors may mediate cervical carcinogenesis. Among the host genetic factors, variations in effective host immune response may be an important determiner and susceptibility to cervical cancer. The role of single nucleotide polymorphisms (SNPs) in immunomodulatory genes involved in the development of HPV-mediated cervical carcinogens was studied at ICPO, Noida. The potential association of SNPs at -308 G/A and -238 G/A of TNF with susceptibility to HPV associated cervical cancer was investigated. The frequency of -308 A allele in TNF was significantly higher in cancer cases compared with control subjects with an odds ratio of 2.7 (95% CI = 1.41-5.51). Women carrying A allele for this locus presented 3 times increased susceptibility to HPV 16 infection as evident from carrier genotype distribution between HPV positive cases and control subjects (OR= 3.1; 95% CI 1.60 – 6.03). No such association was found for TNF -238 (G/A) polymorphism with the risk of development of cancer.

Further an association of HLA II (DRB1 and DQB1) alleles with the susceptibility to HPV associated cervical precancer and cancer cases in a hospital based case control study in north Indian population was investigated at ICPO. Results showed that DRB1*15 allele / DRB1*15 – DQB1*06 haplotype may have predisposition for HPV infection or cervical cancer / precancerous lesions. While DRB1*04 allele/ DRB1*04-DQB1*03 haplotype might exhibit susceptibility to cervical precancerous lesions. DRB1*13 allele/DRB1*13-DQB1*06 haplotype was found to be strongly protective against risk to HPV infection as well as cervical cancer among north Indian population.

Glutathione S-transferase M1 and T1 Polymorphisms, Cigarette Smoking and HPV Infection in Precancerous and Cancerous Lesions of the Uterine Cervix.

Glutathione S-transferases (GSTs) play an important role in detoxification of carcinogenic electrophiles. The null genotypes in GSTM1 and GSTT1 have been implicated in carcinogenesis. This study evaluated the influence of genetic polymorphisms of GSTM1 and GSTT1 gene loci in cervical carcinogenesis. DNA from clinical scrapes of women with minor gynaecologic complaints attending Gynaecology OPD, LokNayak hospital, New Delhi and tumor biopsies of cervical cancer cases attending the cancer clinic was extracted. HPV DNA was detected by standard polymerase chain reaction (PCR) using L1 consensus primer pair. Polymorphisms of GSTM1 and GSTT1 were analysed by multiplex PCR procedures. Differences in proportions were tested using Pearson's Chi-square test with Odds ratio (OR) and 95% confidence interval (CI). The risk of cervical cancer was almost three times in women with GSTM1 homozygous null genotype (OR-2.62, 95%CI, 1.77-3.88; $p < 0.0001$). No association of GSTM1 or GSTT1 homozygous null genotypes was observed in women with normal, precancerous and cervical cancerous lesions among 35 or >35 years of age groups. Smokers with null GSTT1 genotype had a higher risk of cervical cancer as compared to non-smokers (OR-3.01, 95% CI, 1.10-8.23; $p = 0.03$). The results further showed that a significant increased risk of cervical cancer was observed in HPV positive smoker women with GSTT1 (OR-4.36, 95% CI, 1.27-15.03; $p = 0.02$) and GSTM1T1 (OR-3.87, 95% CI, 1.05-14.23; $p = 0.04$) homozygous null genotypes as compared to HPV positive non smokers. The results demonstrated that the GST null genotypes were alone not associated with the development of cervical cancer, but interacted with smoking and HPV to exert effects in population.

Cyclin D1 Gene Polymorphism in Cervical Precancer and Cancer

The primary cause in the development and progression of cervical neoplasia has been shown to be dependent on the integration of HPV DNA into the host genome. The two early genes of high-risk HPV types E6 and E7 encode the main transforming proteins. The products of these early genes interfere with the normal function of tumor suppressor genes p53 and Rb, leading to deregulation of G1-S phase of cell cycle which is regulated by cyclins, cyclin-dependent kinases, and their inhibitors. Evaluation of single nucleotide polymorphism of the cell cycle regulatory genes Cyclin D1, p16 and RB1 in the development of cervical cancer was studied at ICPO.

G870A polymorphism at codon 242 in exon 4 of cyclin D1 gene was investigated and found to be strongly associated ($p = 0.0002$, OR=2.46; using recessive genetic model for association) with increased risk of cervical cancer in Indian population.

The second common G/C polymorphism at nucleotide 1722 within CCND1 3'UTR, was also studied. The frequency of polymorphic 'C' allele was significantly higher in controls (50.7%) compared to cases (42.3%) and strongly associated with reduced risk of cervical cancer ($p = 0.019$; OR=0.71; 95%CI=0.54-0.94.). Therefore, G870A association in Cyclin D1 may play as risk factor whereas G1722C variant may

play protective role in Indian HPV associated cervical cancer.

The SNPs in p16 gene at C540G (rs11515), C580T (rs3088440) in the 3'-UTR of exon 3 and RB1 gene at A153104G (rs4151580) located in the intron 18 were genotyped and confirmed by direct sequencing. SNP at C580T of p16 gene was found to be negatively associated with the risk of cervical cancer ($p=0.0004$, OR=0.04, 95% CI=0.002-0.63). Contrary to this, SNP at A153104G of RB1 gene showed statistically significant association ($p=0.035$, OR=1.69, 95% CI=1.06-2.68) with increased susceptibility for the development of cervical cancer. In addition, women carrying p16 (rs11515) carrier genotypes 540 (CG/GG) may have protective effect for the development of cervical cancer ($p=0.0001$, OR=0.31, 95% CI=0.17-0.56). p16 (540C/580T) has emerged as a major risk haplotype ($p=0.033$, OR=1.47, 95%CI=1.05-2.07) whereas p16 (540G/580T) as a chief protective haplotype ($p=0.014$, OR=0.39, 95%CI=0.18-0.83) for the development of cervical cancer among Indian women. These results suggested that single nucleotide polymorphisms in p16, RB1 genes may affect the susceptibility to cervical cancer collectively.

Genetic Basis of HPV mediated Cervical Cancer in Indian Women

The primary cause in the development and progression of cervical neoplasia has been shown to be dependent on a series of cellular, genetic and epigenetic events including mutation, deletion, polymorphism and/or methylation of various tumor suppressor genes, checkpoint genes of cell cycle machinery etc. Little is known about whether genetic variation in tumor suppressor genes and cell cycle regulatory genes among Indian women associate with susceptibility to cervical cancer. The ICPO's study results showed that differential promoter methylation of FHIT & RASSF1A genes correlates well when comparing various clinical stages & histological grading of cervical carcinoma in Indians.

The study identified a novel mutation at nucleotide position 655, at codon 98 from CAT CGT with ultimate replacement of amino acid histidine by arginine in cervical cancer cases. Molecular modelling was performed to predict the effect of this mutation in disease pathology. It is predicted that this change, his to arg substitution in substrate binding domain may generate catalytically inactive protein with loss of tumor suppressor activity. This would give important insights into genetic predisposition to cervical cancer in Indian population.

Homocysteine levels and Methylenetetrahydrofolate Reductase (MTHFR) Gene Polymorphisms as Cervical Cancer Marker

Human papillomavirus is considered to be a major etiological factor but is not sufficient for development of cervical cancer. Other host factors including altered homocysteine levels, a functional marker of folate inadequacy, might contribute to the carcinogenic process. The potential association of homocysteine levels and *MTHFR* polymorphisms with cervical cancer in histologically confirmed cases including 39 precancer cases and 231 healthy controls with normal cervical cytology were investigated. Both patients and controls were screened for Human Papillomavirus infection. The homocysteine and consequently cysteine levels were significantly higher in cases-both cancer and precancer ($p<0.001$) as compared to controls. However, polymorphisms in *MTHFR* gene (677 C/T and 1298 A/C) that are reported to modulate homocysteine levels were not associated with disease. This study at ICPO established an association of total homocysteine levels with the risk of developing carcinoma of the uterine cervix. Therefore, homocysteine levels were associated with cervical cancer independent of *MTHFR* gene polymorphisms in Indian population.

Allelic Variations CYP2D6 Gene and Susceptibility to Cervical Cancer

Tobacco smoking is an independent risk-factor for cervical neoplasia. It has been found that polymorphism at loci that encode carcinogen-metabolizing enzyme such as cytochrome P450 2D6 (CYP2D6) catalyzing the detoxification of carcinogens may determine susceptibility to cervical cancer. It is likely that an understanding of these allelic differences is important for determining an individual's risk of cancer and susceptibility to potentially toxic agents.

A study at ICPO was initiated to elucidate the role of CYP2D6 polymorphism and susceptibility to squamous cell carcinoma of uterine cervix in Indian population. The results showed that the frequency distribution for the combination of CYP2D6 EM genotype and smoking between mild/moderate and severe dysplasia was statistically significant ($p=0.047$) suggesting that women with cervical intraepithelial neoplasia I/II (CIN I/CINII) and CYP2D6 EM genotype who smoke appeared to have more chances for the lesions to progress to CIN III. Whereas frequency distribution for the same combination between severe dysplasia and squamous cell carcinoma failed to attain any statistical significance suggesting that CINIII with CYP2D6 EM genotype had less chance to progress to cervical cancer.

HPV Diagnostics

Development of Multiplex PCR using Paper Smear for Simultaneous Detection of all Mucosal Human Papillomaviruses and typing of HR HPV Types 16 and 18

Apart from Hybrid Capture II which is a generic test for HPV infection; HPV detection and typing is most commonly done by PCR-based assays which are well established and constantly being improved to enhance the sensitivity and specificity for detecting HPV infection. Diagnosis of HPV genome by PCR relies on assay of cellular DNA extracted from either cervical scrapes or tumor biopsy specimens. However, these methods are cumbersome and suffer from several technical limitations associated with collection, transport and storage of biological specimens. Scraped cervical cells or tumor biopsies are generally collected in cold PBS or kit specific solutions, transported on ice, and require constant refrigeration or it can be stored in liquid-based preservation media (Cytoc or ThinPrep solution) that allow storage at room temperature. These methods are, however, hazardous and cost-intensive for developing countries like India. The procedures of DNA isolation by standard proteinase K digestion and chloroform-phenol extraction or by using commercially available DNA extraction kits are complex, hazardous, time consuming and neither cost-effective nor suitable for screening large number of specimens at a time. To overcome these difficulties, ICPO had earlier developed an easy 'Paper Smear (PS)' method for dry collection of cervical scrapes and biopsy imprints from field stations or clinics for screening of HPV infection. This method has been used by several other researchers for collecting variety of biological specimens to detect various infectious agents, gene sequences and metabolic or genetic diseases. This is an extremely sought after method for easy sample collection, transport and storage without liquid medium or refrigeration specifically in resource-poor settings. However, the yield of DNA per disc obtained from PS was found to be relatively low as compared to that in liquid-based collection and was a limiting factor for doing multiple reactions and experiments. Moreover, conventional uniplex PCR is time consuming, laborious and not suited for large scale screening programs.

In order to make the HPV testing simple and cost-effective with cervical scrapes and applicable for the robust mass screening, the HPV test needs to detect all mucosal or genital HPV genotypes and identify at least the two most common and clinically important HR-HPV types 16 and 18 that are responsible for more than 80% of cervical cancer. Therefore, ICPO developed a simple multiplex PCR (MPX-PCR) using a single paper smear punch to perform an easy and cost-effective method for screening a broad spectrum

genital HPV types and typing of HR-HPV types 16 and 18 along with β -globin as internal control in clinical specimens in a single tube reaction. The efficacy of this test was validated in clinical settings by collecting PS samples at four different national centres located at different regions of the country.

A total of 100 cervical cancer tissue biopsies and corresponding cervical scrape samples on 'Paper Smear' from women diagnosed at different clinical stages (III & IV) were obtained from patients attending the Cancer Clinic of Gynecology Out-Patient Department (OPD) of Lok Nayak Hospital, New Delhi and of Jawaharlal Nehru Medical College and Hospital, Aligarh (India). Each biopsy specimen was bisected and one-half was subjected to histopathologic examination and the other half was immediately frozen in liquid nitrogen for molecular investigation. The cervical scrapes from these cancer patients were collected as 'Paper Smears' (PS) before taking the punch biopsies, with the help of Ayre's spatula by rotating 360° in the endocervix and smearing it within 0.5-1.0 cm diameter on the 3mm Whatman filter paper cut to the size of a microscopy glass slide. Informed consent was obtained from each patient prior to collection of the specimens. In addition, 100 cervical scrapes were collected from the women with or without abnormal cytology attending the OPD for routine checkup or problems other than malignant lesions. Sixty histopathologically proven cancer biopsies with good quality extracted DNA and LI consensus primers were selected for the standardization of multiplex PCR and for comparison of performance of the multiplex technique on DNA extracted by boiling using corresponding Paper Smears. For multicentric evaluation of PS-based sample collection and their influence on the rate of detection of HPV DNA sequences, a total of 248 samples were collected from Tata Memorial Cancer Hospital, Mumbai (n=55), Kidwai Memorial Hospital, Bangalore (n=35), Regional Cancer Centre, Trivandrum (n=58), LokNayak Hospital, New Delhi and Jawaharlal Nehru Medical College Hospital, Aligarh (n=100). The samples on paper smear were collected, dried and transported to ICPO at room temperature for subsequent HPV analysis.

Development and Standardization of Multiplex PCR Assay

The multiplex PCR assay for simultaneous amplification of HPV LI region, HR-HPV type 16, and HPV type 18 alongwith the β -globin gene as an internal control was developed by using equimolar mix of HPV 16 and HPV 18 plasmids in a background of human genomic DNA for detection of HPV infection. The developed assay was then applied on genomic DNA of tumor tissues. The primers employed were MY09/MY11 targeting HPV LI consensus, HPV 16 URR, HPV 18 E6 and β -globin gene. The primers were custom synthesized and HPLC purified by Microsynth GmbH (Switzerland). The primer sequences alongwith final concentration and other characteristics are given in Table 4.4. A common annealing temperature for all four primer pairs was standardized by performing gradient uniplex PCR for each primer set and was worked out to be 55 °C. The PCR was carried out in 25 μ l reaction volume containing either about 100 ng of genomic DNA from biopsies or known copies of HPV 16/18 plasmid DNA, 10 mM Tris-HCl (pH 8.4), 50 mM KCl, 2.5 mM MgCl₂, 200 μ M of each dNTP's (dATP, dTTP, dCTP, dGTP) plus indicated concentration of each oligonucleotide primers (see Table 4.4) and 0.5 units of Taq DNA Polymerase (AmpliTaq Gold, Applied Biosystems, USA). PCR was performed in DNA Engine Tetrad with initial denaturation at 95 °C for 9 min was followed by 35 cycles with denaturation at 95 °C for 30 sec, annealing at 55 °C for 30 sec, and extension at 72 °C for 60 sec, which was extended for 7 min at the last cycle. Ten μ l of PCR products was electrophoresed in a 3% agarose gel, and stained with ethidium bromide. Transilluminated gel images were digitized and analyzed by AlphaDigiDoc™ RT Gel Documentation System (Alpha Innotech Corporation, USA). The sensitivity of the Multiplex PCR assay was determined by using 10 fold dilution series of HPV 16 and HPV 18 plasmid mix in the background of C33a DNA and the assay was calibrated using WHO International Standards for HPV 16 and HPV 18 developed under WHO HPV LabNet program.

Results & Summary

To evaluate the test performance of cervical scrapes as PS, 60 histopathologically confirmed cervical cancer cases with known HPV positivity were tested by MPX-PCR on the DNA extracted from PS by boiling method and the results obtained were compared with results of uniplex and also with multiplex PCR amplification of DNA extracted from corresponding punch biopsies by standard protocol. (Fig. 4.3). Out of 60 PS samples, 58 were positive for HPV 16, 4 for HPV 18 and 2 had co-infection of HPV 16 and HPV 18 as revealed by MPX-PCR. As indicated in Table 4.4 DNA extracted from punch biopsies by standard procedure also revealed similar positivity in cases by uniplex PCR that matched with multiplex results except in 6 PS cases where the β -globin amplification also failed and thus indicating sample inadequacy. Therefore, overall adequacy using a single 4 mm disc of PS was almost 92%. PS-MPX PCR was also found to capture the co-infection similar to the one detected by uniplex PCR in corresponding tissue biopsy DNA. Therefore, similar to standard protocol the PS technique can effectively detect HPV 16 and HPV 18 individually as well as in co-infections along-with other HPVs. Use of β -globin in MPX-PCR provides an obvious advantage by distinguishing an HPV negative sample from an inadequate sample. The results obtained show a good correlation of PS-MPX PCR with that of conventional methods employed for sample collection, DNA extraction or uniplex/multiplex PCR for detection of HPV infection. The use of single tube MPX PCR enabled to detect HPV infection not only with same sensitivity and specificity as of uniplex PCR but also minimizes the carryover of PCR products, thus reducing the chances of cross contamination.

Furthermore, to validate the PS protocol for its utility for large scale screening programs or vaccine monitoring, the performance of PS-MPX PCR was also determined on specimens collected from normal population. A total of 100 samples were collected from normal asymptomatic women for cytological diagnosis and HPV detection by both standard protocol as well as PS method. The standardized multiplex protocol was employed to detect HPV in the genomic DNA extracted from PBS collected samples by conventional method or from PS samples by boiling method. As described in Table 4.5; by using standard protocol on cervical scrapes collected in PBS, the adequacy as measured by β -globin amplification was 92% whereas use of a single 4 mm disc from PS for multiplex analysis demonstrated an adequacy of 83%. However, the HPV positivity as determined by PS-MPX PCR was comparable to the positivity determined by standard procedure. Therefore, the technique was also found suitable for screening of women with pre-cancer lesions or normal cytology where the number of scraped cervical epithelial cells as well as DNA per sample is relatively lesser as compared to that in cancer cases.

The sample adequacy played a critical role in success of PS-MPX methodology. Considering the non-standardized procedure of cervical scrape collection, variability in expertise of sample collectors and preparation of paper smears by the healthcare professionals of variable expertise, the variations in test results were quite obvious. It was therefore determined the magnitude of this variation in a national multi-centric evaluation in clinical settings of 4 different cancer hospitals having no prior exposure to collection of cervical scrapes on PS. For this multicentric validation study, each centre collected cervical scrapes on PS from women diagnosed with cervical pre-cancer or cancer, transported the PS samples to laboratory where the PS-MPX PCR was performed on all samples. Table 4.6 describes overall analysis of samples by PS-MPX PCR. Sample adequacy as evidenced by β -globin amplification was found variable with respect to each centre and ranged from as low as 77% (27/35) to as high as 82% (82/100). Majority of these samples were positive for HPV infection and demonstrated variable HPV 16 and HPV 18 positivity that could be ascribed to variable prevalence of HPV 16 and HPV 18 infection in these populations. It is, however, important to note that the present study was focused mainly on the validation of PS-MPX PCR assay for HPV genotyping rather than to determine the prevalence of HPV infection in different populations in Indian subcontinent, therefore, the prevalence data for HPV infection as indicated in tables

should be used with caution. To test the effect of multiple numbers of PS discs on PCR adequacy, 60 PS samples collected from four different national laboratories that were found to be negative by β -globin amplification (See Table 4.7) were randomly assigned to four groups of 15 samples each. From each group 2 to 5 PS discs were punched out and DNA was isolated as described for single PS disc and analyzed for their amplification adequacy. As observed, by increasing the number of PS discs per case from two to five the positivity for β -globin increased gradually from 26% to 74%. The positivity was found maximum (74%) when 5 discs were utilized simultaneously for analysis. No further increase in the detection rate could be achieved if the number of PS discs increased beyond 5. Therefore the variation due to non-standardized procedures of scrape collection could be taken care of by increasing the amount of scraped cervical cells by increasing the number of PS discs as starting materials for DNA extraction.

The study thus described a simple single tube multiplex PCR-based assay for detection of genomic and clinically relevant HR-HPV genotypes which utilizes a small amount of DNA extracted by boiling a disc or two of paper smear which has been employed for collection of cytological samples in dry form, storage and transportation at room temperature and validated its use in mass cervical screening programs. In view of recent reports showing immense importance of one time HPV screening in the control of cervical cancer present technique offers a suitable, simple and economically viable alternative not only for HPV screening but may also serve as an effective solution for large epidemiological studies directed towards understanding the prevalence of two most clinically relevant carcinogenic HPV types HPV 16 and HPV 18, as well as evaluating impact of HPV vaccination programs, particularly in remote areas and resource-poor countries like India.

Table 4.4: Oligonucleotide primers used for establishing MPX-PCR.

Primer	Nucleotide position	Amplicon size (bp)	Sequence (5'-3')	Final concentration (mM)	Reference
MY09 (forward)	L1	450	CGTCCMARRGGAWACTGATC	20	Manos et al. (1989)
MY11 (reverse)	L1		GCMCAGGGWCATAAYAATGC		
HPV-16 forward	7763-7781	217	AAGGCCAACTAAATGTCAC	20	Das et al. (1992)
HPV-16 reverse	57-75		CTGCTTTTATACAACCGG		
HPV-18 forward	463-482	100	ACCTTAATGAAAAACACGA	20	Das et al. (1992)
HPV-18 reverse	534-562		CGTCGTTTAGAGTCGTTCTG		
b-Globin forward		268	GAAGAGCCAAGGACAGGTAC	10	Das et al. (1992)
b-Globin reverse			CAACTTCATCCACGTTACACC		
M5A or C; W5A or T; Y5C or T; R5A or G.					

Table 4.5: Comparison of HPV detection in cervical biopsies by uniplex PCR with the corresponding PS by MPX-PCR (n560).

Histopathological grade	Total	HPV positivity (n) in biopsy DNA			Total	HPV positivity (n) in PS DNA		
		HPV-16	HPV-18	Co-infection		HPV-16	HPV-18	Co-infection
HSIL (n 57)	7	7	0	0	5	5 (2*)	0	0
WDSCC (n 518)	18	17	1	0	17	16 (1*)	1	0
MDSCC (n 522)	22	21	1	0	21	20 (1*)	1	0
PDSCC (n 510)	10	10	1	1	10	10	1	1
Adenocarcinoma (n 53)	3	3	1	1	2	2 (1*)	1	1
Total	60	58 (97 %)	4 (7 %)	2 (3 %)	55 (92 %)	53 (88 %)	4 (7 %)	2 (3 %)

*Number of PS specimens that were inadequate (i.e. negative for b-globin amplification).

HSIL, High-grade squamous intraepithelial lesion; MDSCC, moderately differentiated squamous cell carcinoma; PDSCC, poorly differentiated squamous cell carcinoma; WDSCC, well-differentiated squamous cell carcinoma.

Table 4.6: Detection of HPV infection by MPX-PCR from PS specimens collected from a normal asymptomatic population (n5100).

Cytological Diagnosis	Samples analysed (n)	HPV detection by conventional in-house methods				HPV detection by PS-MPX-PCR			
		Adequate samples (n)	HPV-positive samples (n)	HPV-16 (n)	HPV-18 (n)	Adequate samples (n)	HPV-positive samples (n)	HPV-16 (n)	HPV-18 (n)
Normal	58	55	6	6	0	50	6	6	0
Inflammation	32	28	3	3	0	25	3	3	0
ASCUS	3	3	1	1	0	2	1	1	0
LSIL	5	4	2	2	0	4	2	2	0
HSIL	2	2	2	2	0	2	2	2	0
Total	100	92	14	14	0	83	14	14	0

ASCUS, Atypical squamous cells of undetermined significance; HSIL, high-grade squamous intraepithelial lesion; LSIL, low-grade squamous intraepithelial lesion.

Table 4.7: Adequacy and rate of HPV detection in a single 4 mm punch from a PS of cervical scrapes with cervical pre-cancer and cancer collected from four different centres.

Sample attribute	Sample collection site				Total
	LH, New Delhi/JNMCH, Aligarh	RCC, Trivandrum	TMH, Mumbai	NCBS, Bangalore	
Samples analysed (n)	100	58	55	35	248
Inadequate samples (n)	18	13	15	8	54
Adequacy (%)	82	78	73	77	78
Samples analysable for HPV infection (n)	82	45	40	27	194
HPV-positive samples [n (%)]	70 (85)	38 (84)	34 (85)	22 (81)	164 (85)
HPV-16 positive [n (%)]	57 (81)	28 (74)	26 (76)	16 (73)	127 (77)
HPV-18 positive [n (%)]	5 (7)	2 (5 %)	2 (6)	1 (5)	10 (6)
HPV-16/HPV-18 co-infection [n (%)]	1 (1.5)	1 (3)	0	0	2 (1)

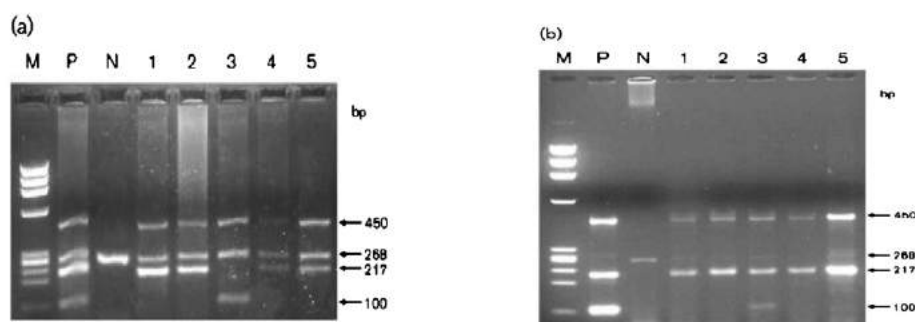


Fig. 4.3: Consensus and type-specific identification of HPV by MPX-PCR using (a) genomic DNA from cervical cancer biopsies collected in PBS and extracted using a standard chloroform/ phenol method (b) genomic DNA from the corresponding cervical scrapes collected on Paper Smears and isolated by a rapid boiling method. Lanes: 1–5, genomic DNA from cervical cancer biopsies; P, cocktail of HPV-16 and -18 plasmids; N, negative control with genomic DNA from C33a cells.

HPV Therapeutics

Elimination of High-Risk Human Papillomavirus Type HPV 16 Infection by ‘Praneem’ Polyherbal Tablet in Women with Early Cervical Intraepithelial Lesions

Recently two virus-like particle (VLP)-based vaccines, Gardasil and Cervarix, have been developed and these vaccines are found to be highly immunogenic and generate protective antibody response in over 99% of recipients preventing new infection with these viruses. Recently, there have been reports of *in vitro* as well as animal studies showing potential anti-HPV activity of various microbial and herbal derivatives. However, these anti-HPV molecules are yet to be tested clinically for their safety and efficacy in human subjects.

India has one fourth of world’s cervical cancer burden and annually about 96,922 new cases appear and more than 60,078 deaths occur as reported by Globocan 2018. Of almost 100% of cervical cancer cases infected with HPV, the prevalence of high risk HPV type 16 is predominantly high. In the absence of organized screening program in India, majority of the asymptomatic infections remain undetected. In addition, absence of suitable low-cost and non-ablative, anti-HPV therapeutics also contribute to high incidence of cervical cancer in India as well as other developing countries.

Praneem, a polyherbal derivative dispensed as pessary or tablet have successfully completed Phase-I safety and Phase-II efficacy studies for treatment of abnormal vaginal discharge due to a variety of reproductive tract infections (RTIs). After intra-vaginal use of Praneem tablet for 7 days, 97.2% of the 141 women suffering from abnormal vaginal discharge reported complete or partial regression of the symptoms. On speculum examination, 74% of the women were found to be objectively relieved of their symptoms. Trichomonas was eliminated in 100% of women affected; the cure rate of vaginosis was 68% while it was 77.4% for *C. albicans*.

The study was designed to evaluate the effect of intra-vaginal, topical application of Praneem for clearance of HPV infection in women infected with high risk HPV type 16.

Study Participants and Clinical Specimen

A total of 772 women who attended the Cancer Clinic of the Department of Gynecology and Obstetrics, Lok Nayak Hospital, New Delhi for routine gynecological examination were screened for presence of HPV infection. Out of these 772 patients, 57 women were tested positive for HPV infection by consensus primer, while 51 women were found positive for HPV type 16. In each case, clinical history was recorded regarding their age and clinical symptoms. Cervical scrapes were collected using Ayre’s

Spatula for both Pap test and HPV DNA testing. Colposcopic examination was performed on case to case basis as determined by the clinical investigator. Pregnant and lactating women, patient with menstrual irregularities, or any other tumor or having any concomitant therapy were excluded from the study. Twenty women with molecularly-confirmed HPV 16 infection with or without low grade squamous intraepithelial lesions (LSIL) were enrolled for an open-label, placebo-controlled study in two groups receiving either Praneem or a look-alike placebo using non-random, deterministic method based on alternation. Mean age of subjects enrolled in Praneem arm was 34.9 years and the placebo arm was 33.3 years. Cytological diagnosis of Pap smear of patients in Praneem arm (n = 10) showed 2 subjects with normal cytology, 5 LSIL with HPV infection, 2 with benign cellular changes (BCC) and 1 with BCC and metaplasia whereas in placebo group (n = 10), 3 subjects with normal cytology, 4 with LSIL and HPV infection, 2 with BCC and 1 had BCC with metaplasia.

Study Design

The details of study design and protocol are presented in Fig. 4.4. It was conducted as a pilot study and was carried out in 20 women molecularly diagnosed to be positive for HPV type 16 infection with or without LSIL as determined by cytological assessment of the Pap smear by two independent cytopathologists. Treatment compliance was checked and confirmed by the clinical investigators. HPV DNA test was performed as primary outcome by generic and type-specific HPV PCR. Pap smear, visual inspection and colposcopic examination of the cervix were carried out before and after the treatment to study effect of treatment on lesion or on cytology as secondary outcome.

DNA Extraction and Diagnosis of HPV Infection:

Genomic DNA was extracted from cervical scrapes of patients prior to and after completion of the treatment by the standard method of proteinase K digestion and phenol chloroform extraction procedure. PCR amplification was performed using the consensus degenerate primers (MY09:5'-GCMCAGGGWCATAA Y A ATGG-3', MY11:5'-CGTCCMARRGGAWACTGATC-3' where M=A/C; W=A/T; Y=C/T; R=A/G) as described earlier. Further typing of high risk HPV types 16 was done by type-specific primers [HPV16-(1)5AAGGCCAACTAATAGTCAC-3', (2)5'CTGCTTTTATACTAACC GG-34]. -globin gene was used as internal control (1)5'-CAACTTCATCCACGTTCCACC-3' \ (2)5'-GAAGAGCCAAGGACAGGTAC-3').

Cytological Diagnosis

For cytomorphological analysis, cervical smears were collected using Ayre's spatula by rotating it at 360° around the circumference of the ecto-cervical region and endo-cervical canal, and Pap smears were prepared. Each smear was examined by two independent cytopathologists and Bethesda System of classification was used for reporting.

Results

Of 772 women initially screened for HPV positivity, 57 were found HPV L1 positive and 51 (89%) of them were positive for high risk HPV 16. Out of these, 26 women were enrolled in the trial and received either Praneem or placebo tablet. Of 25 women not enrolled in the study, 10 (40%) showed high grade lesions and were subjected to treatment, 12 (48%) were lost to follow-up as they did not return after first screening and 3 (12%) did not give their consent to enroll for the study. Among those who were placed on trial, 4 women in Placebo arm and 2 in Praneem arm did not return for follow up evaluation of treatment. During and after completion of treatment, patients were interviewed regarding any adverse effect after vaginal application of Praneem. None of the subjects either in treatment or in placebo arm experienced any adverse effect of the vaginal application of Praneem.

Upon completion of the treatment course of 30 days, subjects in both treatment and placebo groups were evaluated for HPV infection by PCR. 6 out of 10 (60%) subjects on Praneem-treated group did not have any detectable HPV DNA in both LI consensus and HPV 16- specific PCR (Fig. 4.5) whereas in 4 subjects HPV infection however, still persisted. In contrast, 9 out of 10 (90%) subjects in placebo arm showed persistence of HPV infection and only in one subject HPV DNA could not be detected. The subjects who still showed persistent HPV infection following Praneem treatment were further treated with another course of Praneem for 30 days and evaluated thereafter. HPV DNA sequences got cleared in only two of these patients while other two patients still showed presence of HPV infection. Thus in all, 8 out of 10 patients (80%) with HPV infection showed clearance of HPV 16 infection following intravaginal Praneem treatment, whereas in the placebo group the spontaneous HPV clearance rate was found to be only 10% and was found to be highly significant ($p < 0.005$) as determined by Fishers Exact Test. Patients who had undergone treatment with Praneem showed very good improvements in their clinical symptoms and cytological abnormalities. Patients who had detectable cervical lesions were subjected to colposcopic examination before and after the treatment. Pre-neoplastic lesions characterized by aceto- white epithelium and abnormal transformation zone showed a conspicuous regression along with remission of condylomatous changes in women treated with Praneem polyherbal tablets. Improvements in clinical findings were also corroborated with cellular changes in cervical epithelial cells in Pap smears following treatment with Praneem tablets. The Pap smears prior to initiation of Praneem treatment showed small rounded cells appearing in chain with unevenly distributed and heterogeneous cytoplasm, small pyknotic eccentric nuclei indicating early koilocytotic changes suggestive of HPV infection whereas the follow-up Pap smear after Praneem treatment showed clearance of HPV-associated changes and reversal of cellular abnormalities 'within normal limits'. None of the women in Placebo group, except one who had spontaneous clearance of HPV infection, showed any improvement in clinical or cytological abnormalities.

Summary

In the present study, anti-HPV activity of a polyherbal formulation, Praneem, derived from *Azadiractaindica* and other medicinal plants, the anti-viral, anti-carcinogenic and anti-inflammatory activities of which are known for ages in Indian traditional medicine was evaluated. The results showed for the first time, that a 30 day application of the Praneem polyherbal tablet can result in elimination of highly carcinogenic HPV type 16 from the uterine cervix. One course of Praneem treatment resulted in elimination of HPV in 6 out of 10 (60%) cases. A repeat treatment resulted in an overall 80% clearance of HPV 16 against a spontaneous clearance of HPV in 1 out of 10 (10%) cases in the Placebo arm. The elimination of HPV DNA was found to be accompanied by marked improvements in clinical symptoms as well as cytological abnormalities in Praneem-treated subjects. Though 8 cases among 10 (80%) in the Praneem treatment arm showed clearance of HPV DNA, the remaining 2 cases did not respond to the treatment. Apart from having high viral load and possibility of insufficient treatment duration, the likelihood of viral integration into the host cell genome or re-infection in these cases cannot be ruled out. Since more than 80% of HPV infections resolve spontaneously within 12 to 36 months, it can be argued that the cases who had cleared their HPV infection could be those which might have shown spontaneous clearance even if they were not treated. This possibility cannot be ruled out.

Most of the therapeutics reported to have efficacy against HPV infection were performed in high grade lesions (HSIL; CIN2/CIN3) where the lesions are prominent and there is high probability of malignant progression or recurrence after treatment. Apart from outcome, these treatments are either ablative procedures which require specialized infrastructure or are expensive (e.g. recombinant IFN γ). The findings

demonstrated a simple, safe and cost-effective, polyherbal formulation that possesses potent anti-HPV properties and can be easily applied topically for elimination of HPV infection in women and can be well-employed in resource poor countries. Moreover, detection and non-invasive curing of HPV infection in CIN1 or inflammatory lesions using simple herbal preparations such as Praneem can prove extremely useful in prevention of cervical cancer progression at an early stage and its control.

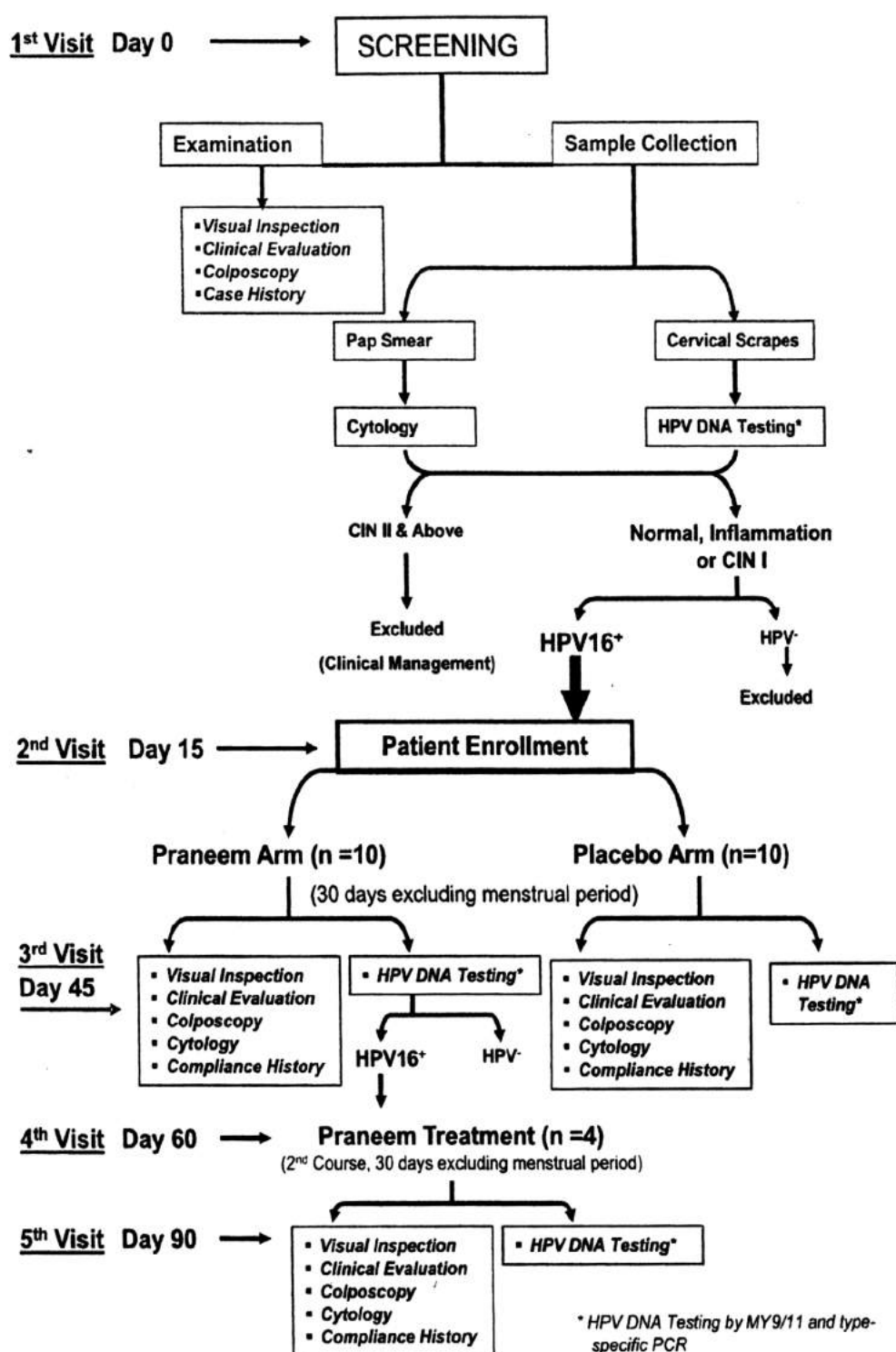


Fig. 4.4: Schematic presentation of Praneem clinical trial protocol.

Twenty women molecularly diagnosed of HPV 16 infection without or with low grade squamous intraepithelial lesion (LSIL) were randomly assigned to receive intra-vaginal, topical application of either Praneem tablet or placebo for 30 days excluding the days of menstrual period and were evaluated for persistence of HPV infection within 3 days of completion of the treatment.

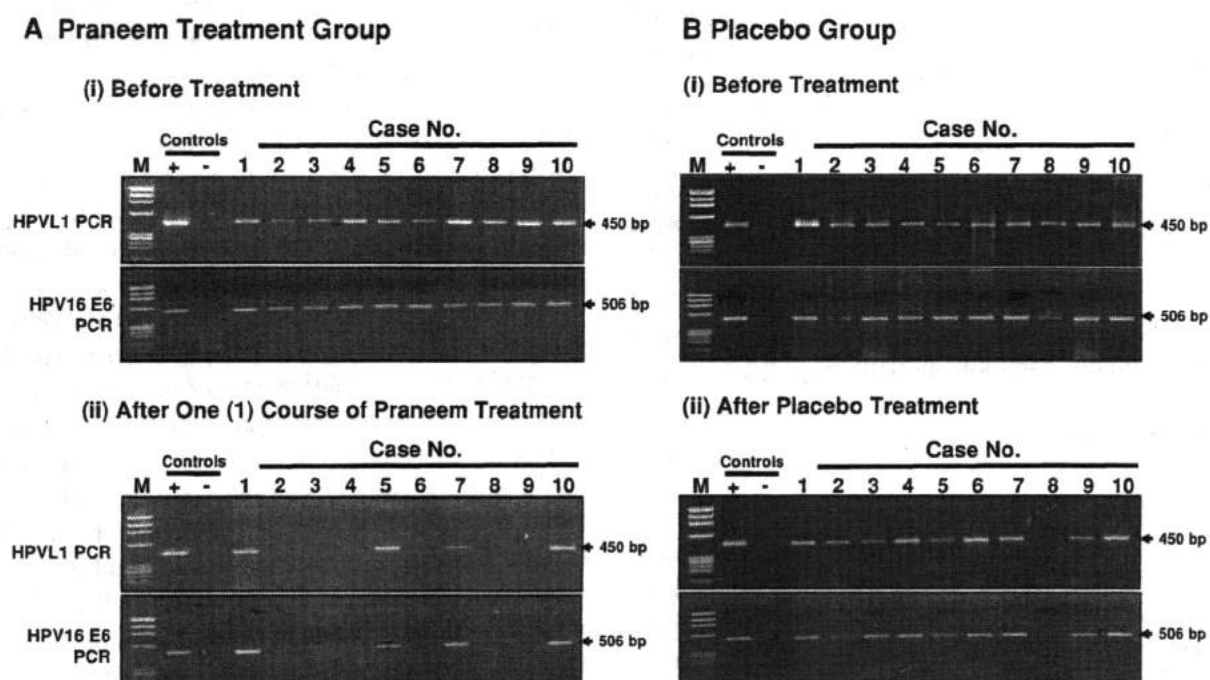


Fig. 4.5: Effect of Praneem (A) and placebo (B) treatment on HPV infection in HPV 16 positive study subjects. Upper pane) shows presence of HPV DNA as detected by HPV L1 consensus PCR and amplification of 450 bp DNA, whereas lower panels show HPV 16 positivity by type-specific PCR for E6 region. Composite panels in A show PCR amplifiable sequences before (i) and after (ii) treatment with Praneem (A) or Placebo (B). M marker.

HPV Prognosis

Aberrant Expression and Constitutive Activation of STAT3 in Cervical Carcinogenesis: Implications in High-Risk Human Papillomavirus Infection

STAT3 has been observed in a wide number of human cancer cell lines and primary tumors including blood cancers and solid tumors and has been shown to be associated with the poor prognosis in various types of malignancies. Though some studies demonstrate presence of STAT3 in a subset of cervical lesions, not much is known about the expression and activation of STAT3 during cervical carcinogenesis in general, and its relation with HPV infection, in particular. Recently, a potential STAT3 binding site has been mapped on to 5' region of LCR of HPV 16 that controls expression of viral oncogenes, thus suggesting a plausible productive interaction between HPV infection and STAT3 signaling.

The cervical cancer provides a unique window of opportunity for studying the expression of important markers of disease progression as the transformation of cervical epithelial takes 10-15 years to occur. Moreover, cervical carcinogenesis progresses through histopathologically well-characterized precursor stages. Keeping in view the above advantages, present study was designed to evaluate the expression pattern of STAT3, its phosphorylation and cellular distribution, and DNA-binding activity in different grades of cervical pre-cancer and cancer lesions in association with HPV 16 infection to determine the role of STAT3 in HPV 16-induced cervical carcinogenesis.

Cell lines and Clinical Specimens

Established cervical cancer cell lines C33a (HPV negative), SiHa and Ca Ski (HPV 16 positive) and HeLa (HPV 18 positive) cells free of intra/inter species cross-contamination were procured from ATCC and were maintained in prescribed culture conditions. A total of 120 fresh cervical tissue biopsies were collected comprising 70 malignant, 30 premalignant and 20 normal (control) cervical tissues prior to any chemo/radio therapy from the Cancer Clinic, Gynaecology Out Patient Department of LokNayak Hospital, New Delhi, India.

Results:

STAT3 Expression and Phosphorylation in Cervical Cancer Cell Lines and Tumor Biopsies

Immunoblot analysis of cellular proteins demonstrated constitutive STAT3 expression in all cervical cancer cell lines although the degree of expression was variable. In contrast to HPV⁻ C33a cells, all HPV positive cells had higher expression of STAT3. Interestingly, all four cell line irrespective of HPV infection expressed STAT3 phosphorylated at Y705 & S727 residues, though HPV16⁺ cells expressed a comparatively higher level of phosphorylated forms in comparison to HPV⁻ negative (C33a) or HPV18⁺ positive (HeLa) cells.

Further, the expression and phosphorylation of STAT3 in cervical pre-cancer and cancer tissues was examined and compared with normal controls. Only trace amounts of STAT3 were expressed in normal cervical tissues whereas both low grade as well as high grade precancer lesions (LSIL and HSIL) expressed either moderate or high levels of STAT3 respectively. STAT3 was consistently over-expressed in cancerous tissues. Immunoblotting for pSTAT3(Y705) and pSTAT3(S727) revealed a concordant increase in pSTAT3 levels in precancer and cancer lesions indicating that STAT3 expressed in these lesions was phosphorylated both at Y705 and S727 residues. The expression of STAT3 and its phosphorylated forms was found to increase as a function of severity of the cervical lesions from pre-cancer to cancer stages.

To assess whether STAT3 is also elevated at mRNA level, total RNA isolated from cervical cancer cell lines or from a subset of cervical tissues comprising normal, pre-cancer (LSIL and HSIL) and cancer lesions were subjected to cDNA synthesis and STAT3-specific RT-PCR. Both HPV positive cervical cancer cell lines as well as HSIL and cancer tissues expressed high levels of STAT3 transcripts. On the other hand, level of STAT3 transcript was moderate in HPV negative cell line C33a and LSILs and was completely absent in normal tissues.

Increased Expression and Nuclear Localization of STAT3 and Phosphorylated STAT3 in Cervical Pre-cancer and Cancer Lesions

Immunohistochemical analysis demonstrated complete absence of phosphorylated (Y705 or S727) or non-phosphorylated STAT3 in normal tissues, however, a few control tissues with inflammatory cytology demonstrated low or moderate immunopositivity for STAT3 and pSTAT3. Majority of the low grade precancer lesions (LSIL) had low or undetectable STAT3 or pSTAT3 expression, though in some LSIL, STAT3 and pSTAT3 showed focal positivity in both basal and suprabasal layers that was found to be equally distributed among nuclei and cytoplasm. In contrast, high grade precancer lesions (HSIL) had variable level of STAT3 and pSTAT3 expression that frequently localized to the nuclei. Of 50 cancer biopsies examined for immune-histochemical analysis of STAT3, 42 were found positive for STAT3 and out of these, 71% had either moderate or high STAT3 expression and had variable degree of nuclear positivity. Moreover, low levels of STAT3, pSTAT3 (Y705) and pSTAT3(S727) were also detectable in the cytoplasm.

Constitutive Activation of STAT3 in Cervical Cancer Tissues and Cell lines

EMSA analysis of nuclear proteins derived from human cervical cancer cell lines, C33a, SiHa and Caski and a subset of freshly collected biopsies from normal, precancer and cancer lesions showed presence of active STAT3 complexes in both HPV negative and HPV 16 positive cervical cancer cell lines. Compared to HPV C33a cells, HPV16+ SiHa and Caski cells showed marginally higher DNA-binding activity. Binding of hSIE probe to STAT3 was found to be specific as antibody to pSTAT3 could specifically bind and supershift retarded DNA-protein complex, confirming that the activity corresponds to STAT3. On the other hand, normal cervical tissues expressed low or undetectable STAT3 DNA binding activity except in a few cases that had inflammatory cervix. In comparison, precancer lesions (both LSIL and HSIL) showed moderate STAT3 DNA binding, whereas, cancer biopsies revealed a prominent STAT3 activity.

Localized Expression of Activated STAT3 in HPV16+ Cervical Pre-cancerous Lesions

LSILs which showed focal positivity of STAT3 in some cases but not others were stratified with respect to HPV 16 positivity to perform a comparative immunohistochemical analysis of STAT3 expression and pSTAT3 localization. HPV negative LSIL sections had a low background staining with no nuclear positivity for STAT3 as well as pSTAT3. In contrast, HPV 16 positive LSIL sections revealed a strong focal positivity and nuclear localization of both STAT3 and pSTAT3 (Y705 & S727) in basal and suprabasal cell layers of cervical epithelium.

Differential Expression and Activation of STAT3 in Various Histopathological Grades of the HPV 16 Positive Lesions

The cancer biopsies with confirmed histopathology and HPV 16 positivity were re-evaluated for STAT3 and pSTAT3 expression. A comparative immunoblotting and immune-histochemical analysis revealed a lower expression of STAT3 and pSTAT3 (both Y705 and S727) in WDSCC cases in comparison to MDSCC and PDSCC that had high STAT3 and pSTAT3 expression. Elevated level of pSTAT3 in MDSCC and PDSCC were also corroborated with intense nuclear positivity of STAT3 in histologically-advanced cancer tissues and was found in as high as 78% and 88% of cancer cells in MDSCC and PDSCC respectively. In contrast, only in 53% of cells in WDSCC sections showed nuclear localization of STAT3 proteins. Together, these findings indicated that constitutive activation of STAT3 was a frequent occurrence in high grade malignant cervical carcinomas and positively correlated with poorer histopathological grades.

Summary

The study demonstrated aberrantly expressed and constitutively active STAT3 both in cervical cancer cell lines and in cervical precancer and cancer lesions. Expression of STAT3 was elevated at transcript level and was found associated with simultaneous increase in phosphorylation at both, Tyr705 and Ser727, that are known to regulate STAT3 dimerization, nuclear transport, DNA-binding and transactivation. Dually phosphorylated STAT3 present in cervical precancer and cancer lesions was found to localize to the nuclei and possessed a functional DNA-binding activity. Immunoblotting, IHC and DNA binding assays revealed that aberrant STAT3 activity increases as a function of severity of the disease from pre-cancer to cancer during cervical carcinogenesis and was found associated with HPV 16 positive lesions. Moreover, STAT3 expression and activation correlated well with HPV 16 positivity in cervical precancer and cancer lesions which indicated its possible involvement in establishment of HPV infection and persistence. In addition, when correlated with different histopathological grades in HPV 16 positive cancer lesions, cases with more advanced histopathological grades had significantly higher expression of active STAT3.

Apart from its tumorigenic properties, STAT3 has been proposed to promote HPV oncogene expression through binding to its upstream regulatory region that controls viral transcription. However, the interaction of HPV 16 or its oncogenes with STAT3 signaling in cervical cancer and the mechanism of HPV 16-mediated STAT3 activation is yet to be elucidated. Never-the-less, the study demonstrated presence of active STAT3 with functional DNA binding activity that accumulates gradually during the process of cervical carcinogenesis and understanding mechanism of disease pathogenesis may help in developing novel STAT3-based approaches for therapeutic interventions against HPV infection and cervical cancer.

Humoral and Cellular Immune Response Against Human Papilloma Virus (HPV) 16

Serum samples from histopathologically confirmed patients with carcinoma cervix and nine healthy asymptomatic women were analysed for detection of antibody response to HPV 16 E7 and L1 fusion proteins in a western blot assay. Antibodies to HPV 16 E7 fusion protein were detected in 69.2% carcinoma patients and 11.1% healthy women, whereas all the patients and healthy women revealed the presence of antibodies to HPV 16 L1 fusion protein. ELISA was carried out with the synthetic peptides of two B cell epitopes of HPV 16 E7 gene product for detecting the presence of antibodies in patients cytopathologically confirmed to have cervical intraepithelial neoplasia CIN-1, patients of CIN2, patients of CIN3, patients of histopathologically confirmed carcinoma cervix and healthy asymptomatic women. Two synthetic peptides of HPV 16 E7 comprising the amino acids sequences QAEPDRAHYNIVTF (a.a. 44-57) and QLNDSEEEDEI (a.a.27-38) were designated as peptides P1 and PII respectively. 43.3%, 19%, 53.8% and 26.6% patients with CIN-1, CIN-2, CIN-3 and carcinoma cervix respectively had antibodies to P1 peptide, whereas 30%, 76%, 46% and 60% patients with CIN1, CIN2, CIN3 and carcinoma cervix respectively had antibodies to PII peptides. None of the healthy women had antibodies to P1 and PII peptides of HPV 16 E7.

Lymphoproliferative assays (LPA) were carried out with synthetic peptide (EPDRAHYNIVTFCC: a.a. 46-59) of HPV 16 E7 T cell epitope for detection of T cell responses. 6.6% of the controls and 63.3%, 42.8%, 30.7% and 36.6% patients with CIN1, CIN2, CIN3 and carcinoma cervix respectively showed positive response to T cell epitope. No correlation could be established between the severity of the disease and T cell response. The study thus suggested peptides P1 and P2 are two major immune-dominant B cell epitopes of the HPV16 E7 and PII is more immune-dominant compared to P1.

DNA samples of biopsy material from carcinoma cervix patients were tested for the presence of HPV DNA in dot blot hybridization experiments. 76.6% DNA showed the presence of HPV DNA under low stringency conditions and 43.4% of these samples showed the presence of HPV 16 DNA. ELISA showed the presence of antibodies to HPV 16 E7 in 70% of the 10 patients with HPV 16 DNA, 46% of the patients with HPV DNA and 85.7% of the patients with no evidence of HPV DNA by dot blot hybridization. LPA reactivity to HPV 16 E7 T cell epitope was observed in 30% HPV 16 positive, 53.8% HPV 16 negative but other HPV positive and 14.3% HPV negative cervical cancer patients. These findings suggest that LPA reactivity to HPV 16 E7 T cell epitope was depressed in most HPV 16 positive cervical cancer patients.

The ICMR's National Institute on Research on Reproductive Health (NIRRH), Mumbai has undertaken research in reproductive cancers. Emphasis of the studies was to assess prevalence of different HPV types, the causative agent of cervical cancer in clinic based populations, to detect cervical cancer using Papanicolaou stain (Pap) smear in urban slums of Mumbai, and also to enhance knowledge and promoting health seeking behavior of couples on cervical cancer in these urban slums. In addition, exploratory studies are being carried out to elucidate the mechanisms associated with pathogenesis of prostate and cervical cancers. Besides the focus on reproductive cancers, the Institute also has interest in fertility issues of cancer survivors. Because of advanced treatment, more and more cancer patients are surviving and

leading a normal life. Survival rate of childhood cancer is almost 80-85%. Cancer is no more a disease of aged. On the contrary 1% of cancer patients are adolescents who have yet not completed their family. Majority of the survivors have compromised fertility as a side effect to their life saving cancer treatment. As a result, a new group of patients have emerged who are rendered infertile viz. azoospermia in men and premature ovarian failure in women. In order to improve the quality of life of these survivors and to enable them to attain biological parenthood, a new field, 'oncofertility' has emerged with a focus on fertility preservation prior to therapy.

Highlights/Achievements;

- An indigenous PCR based test was developed for detection of HPV and its types (HPV 16,18,6 & 11) in biological samples such as cervical sample, tissue or first void urine.
- Screening of HPV in women (n=470) with normal cervical cytology, revealed HPV infection in 38 (8.1%) women. HPV 16, 18, 6, 11 and mixed infection was observed in 26.3%,10.5%, 36.8%, 5.2% and 15.8% of these infected women, respectively, while 36.8% had other HPV types. High incidence (36.8%) of high-risk HPV types 16/18 indicated a clear evidence of the limitations of Pap smear test in cervical cancer screening.
- Study in 104 males, indicated 12(11.5%) had HPV infection, 50% of these were below 30 years. Nine of them were married and three were unmarried, suggesting that these males though not married, were sexually active and should be included in screening programs.
- Genes such as HLA have been identified to be associated with development of cervical cancers in HPV infected women. Studies are in progress to confirm these observations using larger sample size.
- Studies directed to understand the significance of PSP94 (Prostate secretory protein 94), first identified at the institute, in the pathogenesis of prostate cancer revealed that PSP 94 and its interacting partner - CRISP-3 independently cause growth inhibition in cell line specific manner. This was suggestive of CRISP-3 having some PSP94-independent role during prostate tumorigenesis.
- Studies carried out to identify the androgen dependent regulators of telomerase, a RNA-dependent DNA polymerase, in androgen dependent prostate cancer cell line, revealed lesser expression of androgen receptor, p53 (a negative regulator of telomerase) and c-Myc (a positive regulator of telomerase). No change in the expression of Mad1 (an inhibitor of telomerase) after silencing of androgen receptor in prostate cancer cell line was observed.
- Various cryopreservation protocols were established for gonadal tissues so as to ensure maximum post thaw viability in cancer survivors, rendered infertility due to cancer treatment.
- Novel populations of pluripotent stem cells in adult human testis and ovary were identified. These stem cells were found to be resistant to busulphan treatment in mice possibly because of their quiescent nature. These results opened up an exciting new perspective for fertility restoration in cancer survivors.

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North East Region of India comprising of eight states viz., Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura (Fig. 5.1). This region covers an area of about 2,55,082 sq km, surrounded by China, Myanmar, Bangladesh and Nepal with mountains, hills and valleys of varying heights and depth with difficult surface communication. The unique diversity of ethnic groups and sub-groups with different customs, socio-economic status, dietary habit and life styles covers the entire north eastern region. These states have diverse population structure, ethnic background and linguistic groups with their unique life styles, customs, cultures, dietary habits and consumption of peculiar food items/food processing. Major ethnic groups who migrated several centuries ago from South China, Thailand, Myanmar and other South East Asian countries are now living as indigenous people of these states.



Fig. 5.1: North East Reigion of India.

People of this region have peculiar food habits and consume different items. This region of the country have also reported a very high incidence of oesophageal cancer in Assam, stomach cancer in Mizoram, lung cancer in Mizoram & Manipur, nasopharyngeal cancer in Nagaland and hepatocellular carcinoma (liver cancer) in Sikkim and Arunachal Pradesh.

1. An Atlas of Cancer in India (2001-02)

To have a preliminary idea about the cancer burden in Northeast India the World Health Organization funded project on “Cancer Atlas” and in the region involving the pathologists located in different parts of Northeast India. The results showed an overall high incidence rates of cancer were observed in both males and females in the districts of Mizoram. Comparable high incidence rates in the districts of other states especially in the North East (Manipur, Sikkim, Assam) were also seen. The cancers of the stomach were particularly high in all the districts of the Mizoram, Nagaland, Sikkim and Manipur in both males and females. The cancers of the oesophagus in both males and females were high in Assam. Cancer of the nasopharynx was high in Nagaland and Meghalaya. Lung cancer in females and males were several fold higher in the districts of Mizoram and Manipur.

2. Cancer Registry in Northeast:

A hospital based cancer registry was established in Assam Medical College along with rest of the country to record cancer cases coming to Assam Medical College, Dibrugarh. Initial data from Cancer Atlas project showed very interesting and diverse cancer incidence from this project. Therefore expert committee (steering committee) suggested starting population based cancer registry in different parts of northeast region.

Initially six population based cancer registries (PBCRs) were started in 2003 in Northeastern region. Assam being mostly populated state of the region, 3 PBCRs were started in Assam. One at Assam Medical College, Dibrugarh, to cover Dibrugarh District and to get a comprehensive population based cancer incidences in upper Assam, second in B Borooah Cancer Institute, Guwahati, to cover Kamrup Urban district and to get an estimate of cancer in lower Assam and the third one was started at Silchar Medical College, Schar to cover southern Assam. Rest three were started in Aizawl Civil Hospital to cover the state of Mizoram, Regional Institute of Medical Sciences of Imphal to cover Manipur. (Though this registry initially envisaged covering entire state of Manipur, but subsequently it was restricted to Imphal district only) and another started in STNM hospital, Gangtok to cover Sikkim state. Subsequently 5 more PBCRs were added; one at Shillong, Meghalaya, two in Arunachal Pradesh (one at Pashighat Civil Hospital, other at Naharlagun Civil Hospital), one in Kohima for Nagaland and other one at Agartala, Tripura.

Cancer incidence rate expressed as age adjusted or age standardized rate (AAR) (according to world standard population) per 100,000 persons showed highest AAR from PBCR Aizawl, Mizoram in both males and females. In males, Aizawl district is followed by East Khasi Hills district of Meghalaya state and in females by Kamrup Urban district not only in North East but in India. Among males, cancer oesophagus was the leading site in the registries of Assam and Meghalaya. Stomach cancer is the leading site in Sikkim and Mizoram while cancer of the nasopharynx was highest in Nagaland. Among females, cancer of the gallbladder and oesophagus followed by cancer of breast are the leading sites in Dibrugarh and Kamrup Urban respectively. Lung is the leading site in Manipur and Mizoram (particularly among females). East Khasi Hills district of Meghalaya had the highest relative proportion of tobacco associated cancers for males and females at 69.3% and 43.0% respectively when compared to all sites of cancers. Among the new PBCRs, males in Nagaland showed highest AAR of 21/100,000 for nasopharyngeal cancer. East Khasi Hills district of Meghalaya showed the highest AAR in cancers of the oesophagus (in both males and females), cancers of the hypopharynx and larynx in males. The states of the North East have reported high incidence rates of cancers of the upper aero digestive tract that would include anatomical sites such as oral cavity, pharynx, hypopharynx, nasopharynx, larynx, oesophagus and even

stomach. The international comparisons of the incidence rates of these sites of cancer revealed that the PBCRs at Meghalaya (especially East Khasi Hills district) and Mizoram (Aizawl district) and Kamrup Urban in Assam have the highest AARs of cancers of these sites, particularly oesophagus, hypopharynx and larynx.

ANATOMICAL SITE WISE STUDIES:

1. Oesophageal Cancer:

North East Regional Population Base Cancer Registry (NERPBCR) had identified Assam as one of the high incidence state for oesophageal cancer in comparison to other areas.

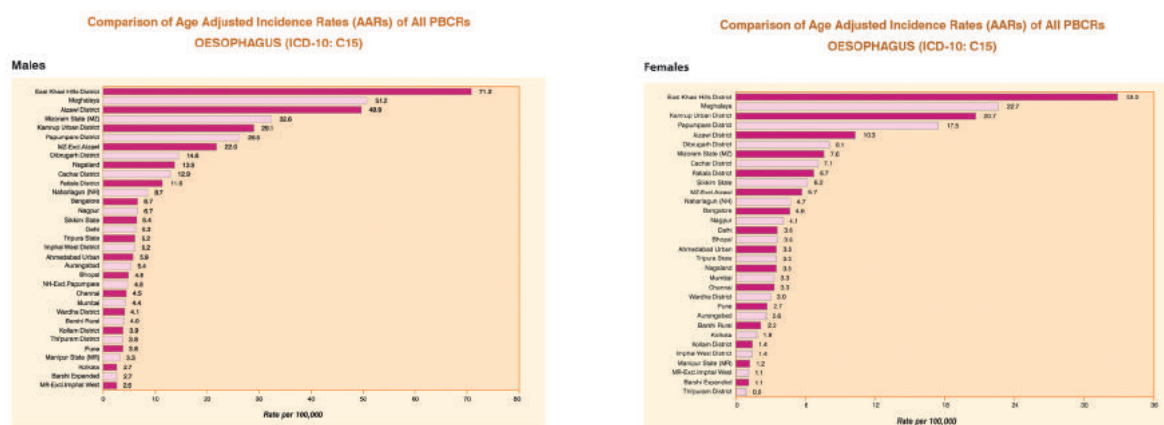


Fig. 5.2: Comparison of Age Adjusted incidence rates (AARs) of All PBCRs OESOPHAGUS (ICD-10:C15)

Case control study on oesophageal cancer carried out during 2014 in Kamrup Urban district showed 65.8% of the cases were males while 34.2% females. Majority of male cases were of 60 - 70 yrs and cultivators (44.3%); while majority of females were housewives (65.8%). Topography indicated more tumours in middle-third of esophagus (C15.4) in both sexes. Histologically most of the cancers were squamous cell carcinoma (90.7%) and moderately differentiated (G2) (81%) irrespective of gender. Chewing of tobacco was significantly associated with increased risk of esophageal cancer (OR=2.17; $p=0.040$). Significant risk association was also observed for those who consume betel nut, betel leaf with lime, *katha* and *zarda* (OR=7.30; $p<0.001$). Smoking of bidi (OR=2.18; $p=0.018$) and cigarette (OR=3.21; $p=0.020$) were significantly associated with esophageal cancer. Significant risk was also observed for those who consumed alcohol (OR=2.23; $p=0.001$). whisky (OR=3.41; $p=0.002$) and rice beer (OR=2.75; $p=0.011$). Frequent and high consumption of a unique alkali preparation (mostly used in Assam) called *kalakhar* (extract made from charred banana stem and false leaf used as food additives) showed significant association with oesophageal cancer (OR=2.77; $p=0.029$). The significantly protective effect was observed for the intake of curd among the study subjects (OR=0.40; $p=0.018$).

Burden of different cancers in the region was reported by 6 (six) population based cancer registries is located at Dibrugarh, Guwahati, Silchar, Imphal, Aizawl and Gangtok. To cover more population number of PBCR in the north east was later on increased to 11 (eleven) i.e., Shillong, Agartala, Pasighat, Naharlagun and Kohima.

Regional Medical Research Centre had been conducting studies on cancers in the north eastern region since 1990s and completed the epidemiological studies on nasopharyngeal cancer in Nagaland, oesophageal cancer in Assam, stomach cancer in Mizoram and the studies on the tobacco and pesticide related cancer in all the population based cancer registry areas of north east.

Centre has initiated molecular epidemiological studies on lung cancer in Manipur and Mizoram, oral cancer in Assam & Meghalaya, breast cancer in Assam, Meghalaya, Mizoram and Tripura and stomach cancer in Mizoram.

Peculiar Food Habits and Dietary Ingredients Use in Assam:

i. Betel Nuts & Tobacco:

Chewing of betel nut with or without tobacco in Assam is totally different from rest of India. In Assam 'raw' (green), 'ripe' (red) and 'fermented' (underground, processed) betel nuts are chewed (Fig. 5.3 & Fig. 5.4). The later (underground), locally known as '*Bura Tamul*', is prepared by keeping 'ripe' betel nuts for 3-4 months covered with bark from the betel tree, cow dung and soil in a 4-5 feet deep hole in the ground. During this period, the outer fibrous shell of the nuts decays. However, people also take chopped or crushed nuts at the different stages of ripening. These betel nuts in pieces are wrapped in betel leaf and chewed with or without tobacco. The proportion of betel nut is more than betel leaves.



Fig. 5.3: Betel nut plant.



Fig. 5.4: : Betel Leaf.



Fig. 5.5: Traditional way of storing tobacco.

Tobacco leaves packed in bamboo stem are kept long time for subsequent uses (Fig. 5.5), locally known as "Dhapat". People chew this tobacco preparation with betel nut. A mixture of finely cut and dried, 'raw' or 'ripe' betel nut ('*supari*') and finely cut, scented tobacco ('*Zarda*') is also chewed in Assam.



Fig. 5.6: Mixing of tobacco with slake lime (calcium carbonate).



Fig. 5.7: Putting the tobacco mixture in mandibular groove.

Some people use dried tobacco mixed with lime processed with tips of thumb on the palm of other hand (Fig. 5.6). The mixture is kept in between gum and lip. This preparation is locally known as “*Khaini*” (Fig. 5.7). Irrespective of the composition, the quid is retained in the mouth for about 20-25 minutes. Tobacco is also processed with molasses and stored in bamboo stump locally known as “*Mala Tankhu*” and is also used for smoking in “*Hukka*” or “*Silim*”.

ii. Peculiar Foods:

The indigenous population of Assam very frequently uses a unique food additive commonly known as ‘*kalakhar*’. *Kalakhar* is a highly alkaline (pH 11-12) substance made from charred false of banana stem and sun dried sliced stem of banana or peels of a particular variety of banana called “*Bhim Kol*”. It is preserved for use in the form of decoction or in dark grey powder form and is used as an additive during the preparation of *curry* or “*dal*”. Mature banana stems are chopped or false stem is peeled, cut and dried in the sun. After they are completely dry, the false stems are charred in a special furnace or in an open fire. The product is stored in earthen pots for variable duration (months to years). The watery extract is highly alkaline brown liquid that is filtered through cloth or a coarse filter (Fig. 5.8 (a-f)). *Kalakhar* has been in use in Assam since ancient times, and probably the practice began as a herbal remedy for sore throat, gastric acidity, antipyretic and gradually become an important and popular food additive and household remedy.

Preparation of Kalakhar



Fig. 5.8(a)



Fig. 5.8(b)



Fig. 5.8(c)



Fig. 5.8(d)



Fig. 5.8(e)



Fig. 5.8(f)

2. Peculiar Food Habits and Dietary Ingredients Use in Mizoram:

Tobacco smoking in Mizoram is very high among adults including adolescents irrespective of sex. A peculiar habit of using “tobacco smoke-infused” water locally known as “*tuibur*” has also been observed in Mizoram. The habit of chewing betel quid, containing ‘*raw*’ (green), ‘*ripe*’ (red) betel nut, slaked lime wrapped in betel leaf is also wide spread in Mizoram. Tobacco is often used. Dried tobacco mixed with lime processed with tips of thumb on the palm of other hand into a powder that is placed near the gum known locally as “*Khaini*” also chewed in Mizoram like in Assam.

i. Tobacco and its Product:

a. *Tuibur*:

Unlike other smokeless tobacco products, a unique tobacco smoke-infused water is used in Mizoram and is locally known as *tuibur*. This product is made locally by passing smoke generated by burning tobacco, through water until the preparation turns cognac in colour and has a pungent smell (Fig. 5.9). *In vitro* studies showed the geno toxic nature of *tuibur*. Indigenous crude devices are used for the production of *tuibur* on small scale. Users take about 5-10 ml. *tuibur* orally and keep it in the mouth for some time and then spit it out. Most of the users take it several times a day. *Tuibur* is available in open market throughout the state (Fig. 5.10).

Fig. 5.9: *Tuibur* plant (smokeless tobacco).Fig. 5.10: *Tuibur* in the market of Mizoram.

b. Meiziol or Zoial:

It is a handmade local cigarette/bidi made from vaihlo (*Nicotiana dadacum*) tobacco. After plucking, the tobacco leaves are thrashed by feet until the leaves become soft. Then they are dried in the sun or sometimes in a warm place like over the fireplace without applying direct heat. These are cut into small pieces and rolled directly using a thin paper or leaf of a plant. The tobacco content of each *meiziol* is about 0.8 -1.0 gm. The length of each *Meiziol or Zoial* is 6-7 cm.

i. Peculiar Foods:

Pork, beef and other meats and fish are popular non-vegetarian foods in Mizoram. They preserve them by smoke-drying and salting for future consumption. Sa-um (fermented pork fat) and bekang (fermented soya bean) are two other unusual foods in Mizoram. Commonly used food items and their methods of preparation in Mizoram are as follows:

a) Smoke Dried Salted Fish:

Locally available fresh fish are salted, dried over the fire for few days, and stored in containers for months for future use (Fig. 5.11). They are consumed after boiling or frying.



Fig. 5.11: Smoke dried salted fish.

b) Smoke Dried Salted Meat:

Fresh red meat (pork or beef) are cut into small pieces, mixed with salt and dried over fire for varying periods and stored at room temperature. This stored meat is boiled before consumption (Fig. 5.12). Smoke drying is also done at commercial scale (Fig. 5.13).



Fig. 5.12: Smoke dried salted meat



Fig. 5.13: Smoke drying of meat in a market area.

c) SA-UM (Fermented Pork Fat):

Pork fat is boiled in water and stored in a special container called *UM* (container made from a gourd shell) for 3-4 days or until it gives off a peculiar smell. The final product is called *SA-UM*. *SA-UM* is mixed with vegetables or chillies for consumption (Fig. 5.14).



Fig. 5.14: SA-UM (fermented pork fat) preserved in UM (container made from Gourd)

d) Bekang (fermented soya bean):

Soya beans are boiled and kept in a warm, air-tight environment under the sun or near fire for about 3 days to ferment. They are eaten directly or with chillies.

e) Typical kitchen of the rural areas of Nagaland



Fig. 5.15

(iv) Family History and Oesophageal Cancer

Studies at RMRC, Dibrugarh indicated possible familial & geographical aggregation of oesophageal cancer in Assam. There is an aggregation of oesophageal cancer in lower Assam indicated by red spot (Fig. 5.16).

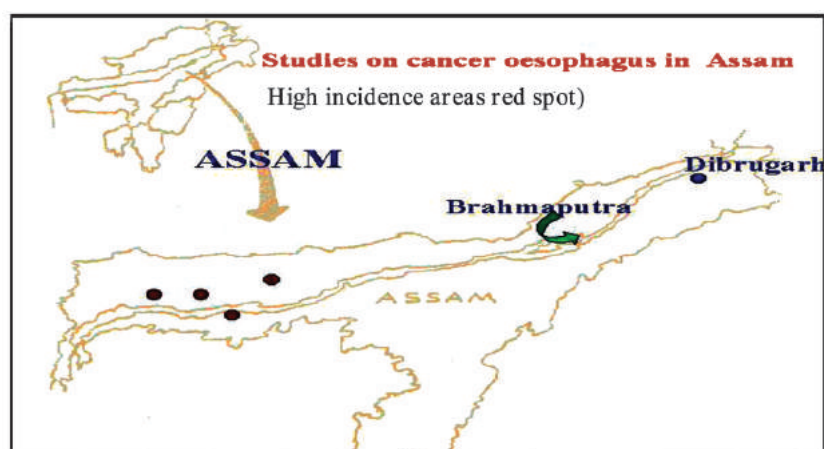


Fig. 5.16: Studies on cancer oesophagus in Assam.

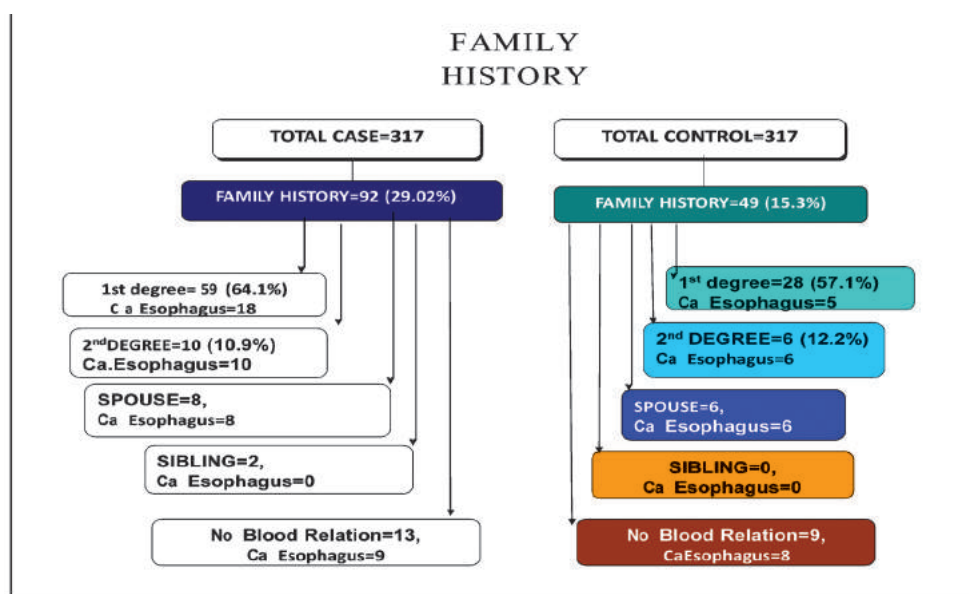


Fig. 5.17

About 36.9% (117 out of 317 cases) of the cases had the history of cancer in their families and this history were first identified by verbal autopsy and then confirmed by personal visit to the affected family whereas only 15.5% (49 out of 317) of control subjects had the history of cancer in their families. Out of 117 cases, 80 families (68.4%) of the cases had oesophageal cancer. Oesophageal and other cancer had occurred in 72 number of 1st degree relatives (61.5%) of the index cases and out of 72 first degree relatives, 66.7% of them suffered from carcinoma of oesophagus. But among the 2nd degree relatives though the occurrence of cancer was only 12.8% the carcinoma oesophagus occurred in 91.7% among the 2nd degree relatives. Among the spouses 15. were suffered from cancer and out of them 53.3% were suffered from carcinoma of oesophagus. But no siblings were found to be suffered from cancer. 15.4% of the relatives among no blood relation with the index cases had suffered from cancer and among them 13 were suffered from carcinoma of oesophagus (Fig. 5.17).

This centre carried out a preliminary study to find out the survival of oesophageal cancer patients and revealed that only 4% of the esophagus cancer patients survive upto 3 years in Assam after diagnosis and cancer directed treatment. The study reveals that patients undergoing surgery live significantly longer than other mode of treatments.

V) Genome-Wide Analysis of Genetic Alternations in Patient's Esophageal Cancer Using Single Nucleotide Polymorphism Arrays (2011 - 2014)

The epidemiological part of the project was carried out at RMRC, Dibrugarh. On completion of project, the information of esophageal cancer cases and controls was collected. The 65.8% of the cases were males while 34.2% were females. Majority of male cases belong to the age group of 60 - 70 years. Male cases are mostly cultivators i.e. 44.3%; while among the female cases, majority of them are housewives i.e. 65.8%. Topography inclination indicates more towards C15.4 (middle-third of esophagus) among both the sexes. Squamous cell carcinoma is mostly predominant (90.7%). The pathological grade of most of the cases is moderately differentiated (G2) showing 81% of the total cases irrespective of gender. Chewing of tobacco is significantly associated with increased risk of esophageal cancer among the study subjects (OR=2.17; $p=0.040$). Significant risk association was also observed for those who consume betel nut leaf with lime, *katha* and *zarda* (OR=7.30; $p<0.001$). Smoking of bidi (OR=2.18; $p=0.018$) and

those who smoke both bidi and cigarette (OR=3.21; $p=0.020$) were also significantly prone to develop esophageal cancer. Significant risk was also observed for those who consumed alcohol (OR=2.23; $p=0.001$). While among the type of alcohol consumption significant risk was observed for those who consumed whisky (OR=3.41; $p=0.002$) and rice beer (OR=2.75; $p=0.011$). From the dietary habits, significant risk association for esophageal cancer was observed for the intake of *kalakhar* (extract made from banana leaf used as food additives) (OR=2.77; $p=0.029$). In the study, significantly protective effect was observed for the intake of curd among the study subject (OR=0.40; $p=0.018$).

Gene expression profile and copy number variation has been studied by genome-wide approaches in esophageal cancer patients from Assam and Kamrup districts showing high incidence (AAR of 33/100,000) and familial clustering (29%) with aim to identify the molecular functional pathways and link between tobacco use and genetic variation. Up-regulation of B-cell receptor signaling pathway genes (PRKCB1) and down regulation of Natural killer cell mediated cytotoxicity genes (NFAT5, KLRC1) in familial ESCC suggests that immune response contribute significantly to familial esophageal cancer. Down regulation of genes associated with Xenobiotic metabolism viz. GST & EPHX1 and mutations in BRCA2 gene have been found associated with risk for familial ESCC and can help in screening. Up-regulated GPCR activity gene NPY and MAPK activity gene FGF12 and down-regulated ribosome proteins S4, L32 and cytokeratin genes KRT4 are found important prognostic factors for both familial and non-familial tobacco and betel quid associated esophageal cancer. DHFR gene has been found important drug target for this cancer. A high through put tissue micro array containing 120 esophageal squamous cell carcinoma from the patient pool has been constructed to validate the gene expression profile.

vi) Epigenetic Studies in Esophageal Cancer

The epidemiological component of the study was carried out at RMRC, Dibrugarh. A total of 75 esophageal cancer samples and 75 controls were collected. Among the cases 74.6% were males and out of which majority of them belong to the age group of 51-60 years. On the other hand, female percentage is 25.3% and majority of the cases belong to the age group of 51-60 yrs. Most of the cases have the family income in the range of Rs.1000-5000 and among controls majority have their income in the range of Rs.5000-10000. Male cases were mostly cultivators (42.85%) followed by business (19.64%), service (16.07%), unskilled (12.5%) and skilled (8.92%). Among the female cases majority were housewives (78.9%) followed by cultivators (10.52%) and service (10.52%). Squamous cell carcinoma type which was mostly predominant (94.7%) followed by adenocarcinoma (5.3%). Most of the tumors are found to be moderately differentiated (66.7%) compared to well differentiated (20%) and poorly differentiated (13.3%). Tobacco smoking and tobacco and betel nut chewing was higher among the cases. 70.7% cases were smokers and 93.3% cases were chewers. Alcohol drinking habit was also more among the cases (about 45.3%).

2. Stomach cancer

High incidence of stomach cancer in Mizoram (2nd highest in the world) (Fig. 5.18) was documented and the use of “*Meizilor Zoial*” (alocalcigarette), “*tuibur*” (smokeless tobacco), “*rakzu*” (Locally made alcoholic drink) and the consumption of “*smoked salted fish and meat*”, “*SA-UM*” (fermented pork fat) and “*Soda*” (alkali), was found to be very high in this state.

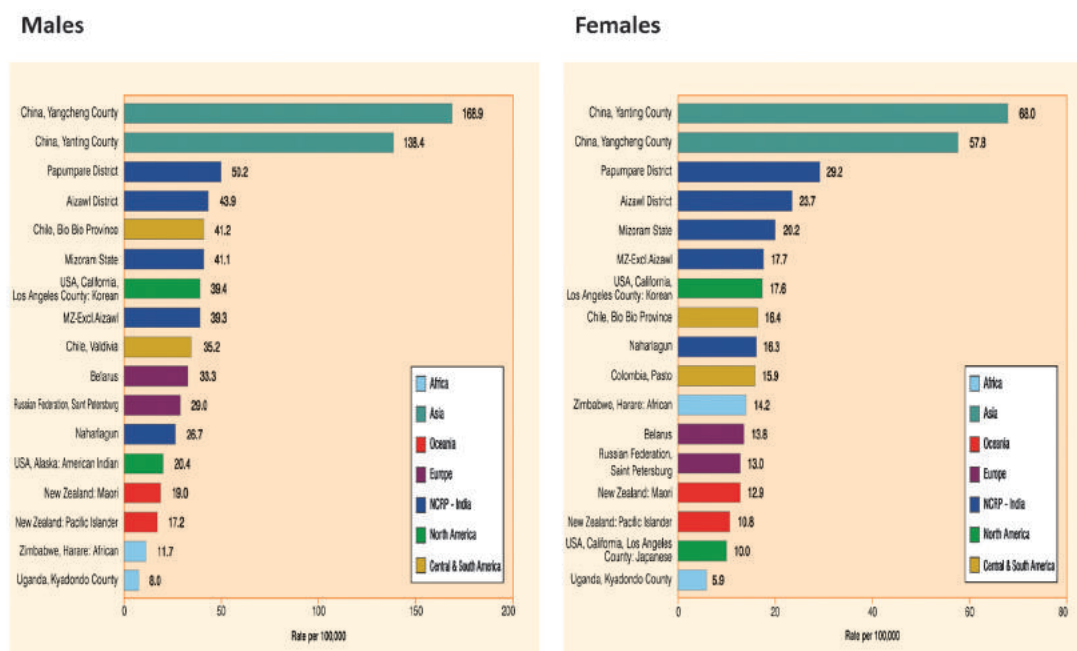


Fig. 5.18: International Comparison of AAR with that of PBCRs in India STOMACH (ICD-10: C16).

1. Genetic Studies in Stomach Cancer

Although CYP and GST enzymes are involved in the activation and detoxification of N-nitrosamines which are some of the most potent carcinogens present in smokeless tobacco, snuff and tobacco smoke and related compound, studies on the relationship between genetic polymorphisms of CYP1A1, GSTT1, and GSTM1 and the risk of gastric carcinoma (GC) have not been carried out in this high risk region of India so far and the previous reported studies from other part of the world are conflicting.

To evaluate the association between polymorphic detoxifying genes and GC risk, a case-control study including histologically confirmed 134 gastric cancer patients and 282 healthy controls was conducted in a gastric cancer high-risk area in North east region of India to screen the GST, CYP1A1(2A,2C) genes and p53 codon 72 polymorphisms.

The frequency of *GSTT1* and *GSTM1* null genotypes did not show significant differences between cases and controls. Variant genotypes (*Ile/Val* and *Val/Val*) of *GSTP1* were found more frequent in cases of gastric cancer as compared to controls but the difference was not statistically significant (OR=1.25, 95% CI 0.77-2.03, p=0.37). No significant difference in frequency distribution of CYP1A1 2A genotypes (ww, wv, vv) and CYP1A1 2C genotypes (II, IV, VV) was found in cases and controls, suggesting no associated risk for gastric cancers with GST and CYP1A1 genes. Conditional logistic regression analysis of frequency distribution of p53 codon 72 Arg/Arg, Arg/Pro and Pro/Pro in the cases and controls showed higher risk associated with Arg/Pro genotype whereas the Pro/Pro appeared to be a protective genotype. These observations were represented both by OR1 and OR2 but lacked statistical power. No significant interactions of risk habits were found with *P53* genotypes, however, interaction of betel quid chewing with Arg/Pro genotype reached a near significance level with a 2-fold risk for gastric cancer (OR2=2.40, 95% CI=0.91-6.26; p=0.07).

Risk estimates for *H. pylori* status, tobacco, betel quid and alcohol consumption habits in 112 cases and 66 healthy age matched controls showed significant association of tobacco chewing with risk of development of GC, while there was no significant association with *H. pylori* as reported earlier. However,

tobacco chewers who were also *H. pylori* positive showed a seven-fold risk (OR=7.12, 95% CI=5.02-36.31, $p=0.0001$) whereas betel quid chewers who were also *H. pylori* positive showed a four fold risk (OR=3.52, 95% CI=1.16-10.86, $p=0.026$) for developing gastric cancer.

To understand the molecular mechanism underlying the development and progression of GC in this high-risk region in NE India, the gene expression profiles of tumor tissue from 5 patients having history of tobacco/ betel quid chewing, alcohol consumption and sero positivity for IgG antibody against *H. pylori* were studied and compared with normal appearing pooled gastric mucosal tissues. A total of 170 genes were found to be differentially expressed. Of the 108 significantly up-regulated genes, genes involved in apoptosis (RYK, SH2D2A, CSNK1E, LRP12, HSPE1, FGB, PXN, CHP), pro-inflammatory response (IL9), angiogenesis (SH2D2A, ANG), cell adhesion (CD151, PXN, WASF1), cell proliferation (CDC2L1, JAG2, IL9), regulators of I-kappa B kinase/NF-kappa B cascade (CXXC5), Wnt signaling (CHP, CSNK1E, RYK), Notch signaling (JAG2), Hedgehog signaling (CSNK1E, BMP7), metabolic pathway (LDHB), thyroid cancer (RET) and epithelial cell signaling in *H. pylori* infection (ATP6V1G2) were found to be biologically relevant in tumorigenesis. Of the 62 significantly down-regulated genes, genes involved in base-excision repair (RAD51L3), inhibition of cellular proliferation (INSM1), immune response (RFX1, CXCL10, C3AR1, MALT1), viral response (ATP6V1G2, HBXIP, ACE2), epithelial cell signaling in *H. pylori* infection (JAM3) were found to be relevant in tumorigenesis. Analysis of the molecular functional pathways by Genowiz™ identified four different potential relevant molecular functional pathways that were deregulated in these patients including mainly Hedgehog signaling (CSNK1E, BMP7), Wnt signalling (CHP, CSNK1E, RYK), Notch signaling (JAG2) and epithelial cell signaling in *H. pylori* infection pathways (JAM3, ATP6V1G2). It is reported that a balance between Wnt-FGF-Notch and BMP-Hedgehog signaling network was found to be important to regulate the homeostasis among stem and progenitor cells and disruption of the signaling network results in cancer.

Validation of LDHB, PXN, RYK, RET, ANG, BMP7, RAD51L3 and CXCL10 genes by real time PCR in 37 cases of gastric cancer including 26 *H. pylori* positive and 11 *H. pylori* negative cases showed significant down-regulation in expression of RAD51L3 ($p=0.0013$), CXCL10 ($p=0.0038$) and BMP7 ($p=0.0002$) genes whereas significant up-regulation in expression of PXN ($p=0.0077$), ANG ($p=0.0015$), LDHB ($p=0.019$), RYK ($p=0.044$) and RET ($p<0.0001$) genes in *H. pylori* positive cases compared to adjacent normal tissue. The *H. pylori* negative cases showed a significant up-regulation of expression of RET ($p=0.024$) and a significant down-regulation of expression of BMP7 ($p=0.018$) as compared to its adjacent normal tissue. The gene expression of PXN ($p=0.039$) was alone found to be significantly up-regulated in *H. pylori* positive gastric cancer cases on comparison to *H. pylori* negative gastric cancer cases. A significant direct correlation of expression of PXN with RET (0.523, $p=0.05$) was observed in *H. pylori* positive patients while *H. pylori* negative cases did not show any correlation in the expression of any of the genes studied.

The diverse patterns of expression of tumorigenesis related genes in *H. pylori* positive and negative patients suggest the probability for the existence of different and distinct molecular pathways in the pathogenesis of gastric cancer. It is possible that factors other than *H. pylori* infection including high tobacco consumption and food habits such as high salt intake interact with and contribute significantly to the initiation of inflammation in gastric cancer. This is a novel pro-tumorigenic mechanism of inflammation in gastric cancer in which tumor-related pathways such as Wnt (CHP, CSNK1E, RYK), Notch (JAG2), Hedgehog (CSNK1E, BMP7) and epithelial signaling pathway in *H. pylori* (ATP6V1G2, JAM3) may play an important role in the development of gastric cancer in *H. pylori* infected and non-infected individuals via distinct molecular pathways.

Studies at Regional Medical Research Centre showed GSTM1 and GSTT1 gene, p53 codon 72 polymorphism and dietary and tobacco habit interactions influence stomach cancer development in Mizoram, India. Although no direct association between CYP2E1RsaI polymorphism and stomach cancer was observed, relations between CYP2E1 RsaI polymorphisms and different tobacco and dietary risk habits in terms of developing stomach cancer exist in this high risk population of north-eastern part of India. GSTM1 and GSTT1 genotypes were determined by PCR and H. pylori infections were determined by ELISA. Polymorphisms of GSTM1 and GSTT1 genes appeared to be effect modifiers. Persons habituated with tobacco smoking and/or tuibur habit had increased risk of stomach cancer if they carried the GSTM1 null genotype and GSTT1 non-null genotype.

Study revealed that p53 codon 72 polymorphism, the Pro/ Pro genotype was significantly associated with a higher risk of stomach cancer (OR, 2.54; 95%CI, 1.01-6.40) as compared to the Arg/Arg genotype. In gender stratified analysis, the Pro/Pro genotype showed higher risk (OR=7.50; 95% CI=1.20-47.0) than the Arg/Arg genotype among females. Similarly, the Pro/Pro genotype demonstrated higher risk of stomach cancer (OR=6.30; 95% CI=1.41-28.2) among older people (>60 years). However, no such associations were observed in males and in individuals <60 years of age. Smoke dried fish and preserved meat (smoke dried/sun dried) consumers were at increased risk of stomach cancer (OR, 4.85; 95%CI, 1.91-12.3 and OR, 4.22; 95% CI=1.46-12.2 respectively) as compared to non-consumers. Significant gene-environment interactions exist in terms of p53 codon 72 polymorphism and stomach cancer in Mizoram. Tobacco smokers with Pro/Pro and Arg/Pro genotypes were at higher risk of stomach cancer (OR, 16.2; 95% CI=1.72-153.4 and OR, 9.45; 95% CI=1.09-81.7 respectively) than the non-smokers Arg/Arg genotype carriers. The combination of tuibur use and Arg/Pro genotype also demonstrated an elevated risk association (OR, 4.76; 95% CI=1.40-16.21).

1. Oral Cancer in Assam & Meghalaya

India has the highest rate of oral cancer in the world. It comprises 30-40% of total malignancies in India and ranks as the leading cancer site in males and the third leading site in females. A retrospective study (from 1993 to 2004) on the prevalence of head and neck cancers in NE India showed significantly high prevalence 55%, affecting males more than females in the age group of 40-69 years. Moreover, a very high AAR of carcinoma of the tonsil has been reported from at least five districts in Assam State with Kamrup district recording the second highest AAR in India. Approximately 30% of oral cancers are attributed to betel/tobacco chewing alone and an additional 50% to the combined habits of chewing and smoking. Ninety-two percent of oral squamous cell carcinoma (OSCC) in men and 61% of OSCC in women are attributable to tobacco usage. The combined use of peculiar betel/tobacco quid chewing habits, widely prevalent in this region, increases the relative risk of developing oral cancer. The pathogenesis and molecular mechanism implicated in smokeless tobacco consumption is unknown. Currently there are no genetic studies available to explain the alarmingly high prevalence of oral cancers from North east region of India

The quantitative absorption, distribution, metabolism, and excretion of carcinogenic tobacco constituents depend on the activity and efficiency of metabolic and enzymatic detoxification pathways. The enzymatic detoxification process is mainly divided into three phases. Phase I enzymes involved in activation of toxic compounds predominantly by oxidation into more reactive intermediates that are neutralized and conjugated by phase II family of enzymes such as glutathione-S-transferase (GST), N-acetyltransferase (NAT). The resultant water-soluble and less-toxic conjugated product can easily be eliminated from the cell by phase III transport mechanisms for the elimination of glutathione conjugates. The detoxification efficiency of GST enzymes is determined by the presence and nature of the isoenzymes coded by GSTT1,

GSTM1, and GSTP1 genes. The prevalence of tobacco and betel quid chewing habits as well as the occurrence of tobacco-associated cancers is high in the northeast (NE) region of India. However, the prevalence of polymorphism in GST genes in tobacco-associated cancer patients from this region is not well known.

Frequencies for GST enzyme genes (GSTT1, GSTM1, and GSTP1) polymorphism was analyzed in 160 oral cancer cases and 169 age and sex matched controls from the same ethnic group. The conditional logistic regression analysis revealed that the risk of developing oral cancer significantly increased in tobacco chewers (OR =2.44, 95% CI=1.47–4.05, $p=0.001$), tobacco smokers (OR=1.72, 95% CI=1.08–2.73, $p=0.02$), and betel quid chewers (OR=2.20, 95% CI=1.29–3.76, $p=0.004$). When adjusted for other variables under consideration, no significant association was found for GSTM1 and GSTT1 null genotype independently or in combination with oral cancer risk. Although no significant independent association of oral cancer with null genotypes of GSTT1, GSTM1 and the variant alleles of GSTP1 was found, individuals with both GSTM1 null genotype and variant alleles of GSTP1 were found to have marginal increased risk for developing oral cancer (OR=1.84, 95% CI=0.91–3.72, $p=0.08$). Study of polymorphisms in three genes [CYP1A1 (Msp1 and Nco1), NAT2 and NQO1] in two hundred and thirty five oral cancer cases and 289 healthy controls from this region showed significantly higher frequency of both heterozygous Pro/Ser (AOR=1.64, 95% CI=1.04–2.58) as well as homozygous Ser/Ser (AOR=1.81, 95% CI=0.98–3.32) NQO1 genotypes in cases than controls suggesting polymorphic variant genotypes of NQO1 are significant risk for oral cancer with homozygous variants genotypes showing a higher risk compared to heterozygous genotypes in univariate analysis. Gene-environment interaction analysis however showed that the variant genotypes of NQO1 did not interact with tobacco consumption habits. Although variant genotypes of NQO1 may play an important role in the genetic susceptibility to oral cancer, its pathway appears to be unrelated to the detoxification mechanism of tobacco constituents. No significant association was found between polymorphisms of CYP1A1 (Msp1 and Nco1) and NAT2 genes with oral cancer risk.

The human TP53 tumor suppressor gene plays a central role in many cellular processes, regulating cell growth, DNA maintenance and apoptosis. It is an important component of DNA repair machinery in response to DNA damage induced by radiation or adduct formation. This might explain the occurrence of the *P53* gene mutation and alteration in about 50% of all cancers, particularly tobacco related cancers. Studies have shown a relationship between tobacco smoke exposures, carcinogen-DNA adduct formation, tumor specific mutation of TP53 gene and cancer risk.

Numerous polymorphism in the wild type *P53* have been reported both in coding and non-coding regions. Out of the five polymorphisms described in the coding region, polymorphisms in codon 47 and 72 in exon 4 are functionally well characterized. More common of the two, codon 72 polymorphism is a single base substitution of cytosine for guanine, leading to arginine (A72) being replaced by proline (P72) that has been reported to be associated with the risk of several cancers. Studies on codon 72 polymorphism have revealed striking ethnic differences in frequency of *P53* variant allele that varies with latitude, increasing in a linear as populations near the equator. Thus ethnicity might be related to allelic distribution of the gene and its determinacy in disease involvement; however, some studies do refute the ethnicity-risk confounding relationship.

A case control study was undertaken in cases of oral cancer and normal age matched controls collected from Dr. B. Borooah Cancer Institute, Guwahati, Civil Hospital, Aizawl, and Sir T.N.M. Hospital, Gangtok in three north east states to investigate the role of p53 codon 72 polymorphism and its interaction with tobacco, betel quid and alcohol. Tobacco chewing was the strongest risk factor for oral cancer with a

3 fold increased risk ($OR_2 = 3.05$, 95% $CI = 1.79-5.20$; $p < 0.001$). Both tobacco smoking ($OR_2 = 1.68$, 95% $CI = 1.00-2.81$; $p = 0.04$) and betel quid chewing ($OR_2 = 1.85$, 95% $CI = 1.02-3.33$; $p = 0.04$) also increased risk for oral cancer. A strong association of risk of oral cancer was seen in patients carrying Arg/Arg and Arg/Pro genotypes ($OR_2 = 8.62$ 95% $CI = 2.19-33.93$; $p = 0.002$ and $OR_2 = 5.22$ 95% $CI = 1.38-19.75$; $p = 0.01$ respectively) irrespective of smoking habit. Although tobacco and betel quid chewing emerged as a risk factor yet no interaction of *p53* genotypes with the habit showed statistical significance.

Case control study on oral cancer in Assam and Meghalaya, where betel nut chewing with or without tobacco is very high, was conducted. Major cancer sites were gum (39%) followed by buccal mucosa of cheek (31%). Major clinical presentation of the cases was ulceration in the mouth (39%) followed by swelling (19%). The cases were predominantly from illiterate and lower income group from rural areas. Betel nut chewer has synergistic effect on oral cancer. Betel quid and the saliva were shown to be mutagenic in studies conducted at the Centre. Among the genotypes studied GSTM1 null genotype and MSP1 mutant genotype of CYP1A1 gene were showing independent risk to oral cancer. Risk of oral cancer in subject with GSTM1 null genotype is enhanced by habit of tobacco chewing and smoking. Risk of oral cancer in subjects with MSP1 mutant genotype is increased by GSTT1 null genotype.

Toxic Effect of Tender and Fermented Ripe Betel Nuts

The toxic effect of aqueous extract of fresh (BNE1) and fermented (BNE2) betel nuts was studied employing modified Allium test. Toxic effect of both the extracts on the root growth of Allium bulbs was observed. EC_{50} values were 0.024% and 0.030% for BNE1 and BNE2 respectively. Brown coloration of roots was observed. Crochet hooks were distinct in BNE1. Mitotic indices were lower than the control except in 0.01% concentration of BNE2. Sticky chromosomes, C-mitosis, vagrant chromosomes and fragment/micronuclei were also noted. The study indicated higher toxicity of fresh betel nuts than that of fermented one.

Toxic Properties of Saliva of Betel Nut Chewers

Saliva of betel nut (fermented by keeping underground) chewers from Assam (India) have been found to induce significant toxic effect on both macroscopic and microscopic parameters in modified Allium test. The Allium bulbs were treated with four concentrations (0.25, 0.50, 0.75 and 1.00%) of saliva samples collected after 2-3 minutes of betel nut chewing without tobacco (BQ) and with tobacco (BQT). Saliva from both BQ and BQT chewer showed significant inhibition of root length as compared to control. Twisted and colored roots were also observed in the treated groups. Reduction of mitotic index (MI) was recorded more along with increase of c-mitosis, vagrant chromosome, sticky chromosome and micronuclei formation in both the treated group.

Breast Cancer

Population Based Cancer Registry data from North East showed, breast cancer is occupying number one position among females in many states of Northeast region. Younger breast cancer victim and late reporting have been observed in the region. Betel nut chewing with or without tobacco in all the states of the region is rampant. There is an increasing incidence of breast cancer and eminent use of tobacco in females in this region. Like all other high incidence cancer of the region breast cancer also showed high association with betel nut chewing. Betel nut chewing with or without tobacco was found to be an important risk factor in Tripura but not in Mizoram. Smoking habit was found to be associated in both Tripura and Mizoram.

1. Contribution of Environmental Factors and Low Penetrance Genes in Breast Cancer

(i) Low Penetrance Detoxifying Genes and Environment in Breast Cancer

Northeast region of India shows high incidence of tobacco-related cancer with widespread consumption of betel quid and tobacco in different forms. There is an increasing incidence of breast cancer and eminent use of tobacco in females in this region. Thus, the role of tobacco exposure and polymorphisms in detoxification enzymes in breast cancer risk was analysed in this study. Polymorphisms in five gene variants, GSTT1, GSTM1, GSTP1, TP53 and CYP17, and four environmental exposure variables (tobacco smoking, tobacco chewing, betel quid chewing, and alcohol) were analysed in 117 breast cancer cases and 174 cancer free controls. Multifactor dimensionality reduction identified betel quid chewing as the single main risk factor and women with betel quid chewing history had five times the risk of developing breast cancer [4.78 (2.87-8.00) 0.001]. In logistic regression analysis, GSTT1 null and GSTM1 null genotypes conferred 41% less [0.59 (0.34-1.03) 0.06] and 55% less [0.58 (0.30-1.02) 0.05] reduced risk to breast cancer, respectively. However, the risk increased in women with GSTP1 variant G allele which conferred 1.43 times [(0.96-2.11) 0.07] more risk to breast cancer. The Multifactor Dimensionality Reduction (MDR) analysis revealed, betel quid chewing to be the single factor imparting the main effect [testing accuracy of 0.6851 and cross validation consistency 10/10, $p=0.05$]. Figure below depicts the interactions between nine attributes from the MDR analysis via a graphical representation of a 'dendrogram' (Fig. 5.19). It shows betel quid chewing, GSTT1 and GSTM1 on a separate branch imparting their independent effects to breast cancer risk. In conclusion, this study suggests betel quid chewing as a significant risk factor for developing breast cancer.

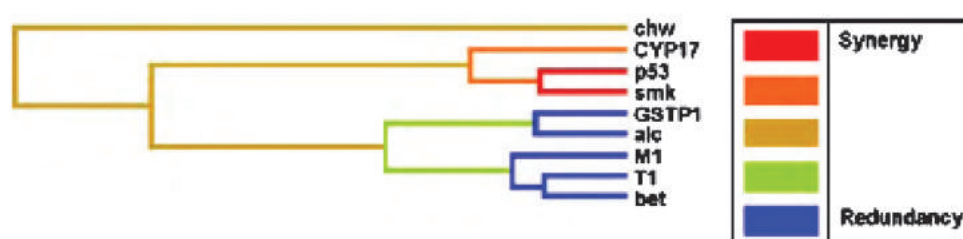


Fig. 5.19: Interaction dendrogram for the breast cancer dataset: graphical representation of interactions between nine attributes (GST1 (T1), GSTM1 (M1), GSTP1, CYP17, TP53 (p53), tobacco smoking (smk), tobacco chewing (chw), betel quid chewers (bet) and alcohol consumption (alc)) from the multifactor dimensionality reduction analysis using an 'interaction dendrogram'.

2. Contribution of High Penetrance Genes and Breast Cancer:

Genetic mutations in BRCA1 gene was found in 12% cases of Mizoram but only 2.44% of cases from Tripura. Ten cases had family history of breast cancer and out of 10 cases; 1 has mutations in BRCA1. The BRCA2 gene was analyzed for mutations and polymorphisms in Northeast India population. The cases and controls were analyzed by PCR- DHPLC analysis. Variants in four samples was observed by DHPLC and confirmed by sequencing. 10204T>C: silent mutation in exon 27 was seen in two samples. 8415G > T: K2729N mutation in exon 18 was observed in two samples. None of the controls showed the above two mutations. Two variants suspected to be polymorphisms were seen in exon 2 and exon 27. -26G>A variant in exon 2 confirmed as polymorphism by sequencing was analyzed in 204 samples and 240 controls for risk analysis (Fig. 5.20). Wild type GG genotype was present in 35% of cases and 31% controls, heterozygous AG genotype was present in 46% cases and 52% controls whereas variant homozygous AA genotype was present in 17% cases and 15% controls. Women with AA [0.83 (0.56-1.24, 0.78)] genotype and GA [0.78 (0.51-1.18, 0.24)] genotype confer 17% and 49 % lower risk towards breast cancer which was not significant. 10462A > G: I3412V variant in exon 27 also confirmed as a

polymorphism by sequencing was analyzed in 204 samples and 240 controls. Wild type AA genotype was present in 55% of cases and 97% controls whereas heterozygous AG genotype was present in 4.8% cases and 2.7% controls. No homozygous variant was observed. Women with the AG genotype conferred a 1.81 times higher risk [1.81 (0.67 – 4.88), 0.25] towards breast cancer which was insignificant.

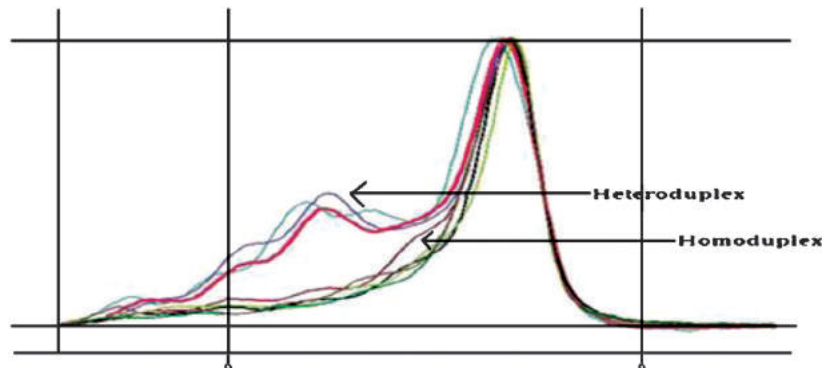


Fig. 5.20: A variant G>A in Exon2 of BRCA2 gene.

3. Contribution of Gene Copy Number Variations- a Genome Wide Approach

In order to have insight into the genetic changes that are taking place during breast tumorigenesis in this population, the copy number analyses of breast tumors has been done using SNP 10K array. DNA copy number gains were obtained as determined by dChip using analysis of signal intensity values based on the Hidden Markov Model. Arrays with >93% call rates were included in the analysis as per Affymetrix manual. Both the raw copy number and log₂ ratio (signal/mean signal of normal samples at each SNP) were computed to estimate copy number changes in chromosome view. For each SNP, the signal values of all of the normal sample were averaged to obtain the mean signal of 2 copy (male X chromosomes were multiplied by 2 before averaging) and the observed copy number was defined as (observed signal/mean signal of two copy) * 2 and visualized either log₂ ratio displayed in blue to white, then to red color scale or white (0 copy) to red color scales. In total, 116 regions had genomic aberration, out of which 20 regions had copy number (> 3.5) gain and 90 regions showed copy number (> 2.8) amplification. Losses were less as compared to gains and only 6 regions showed copy number loss (Figs. 5.21 & 5.22).

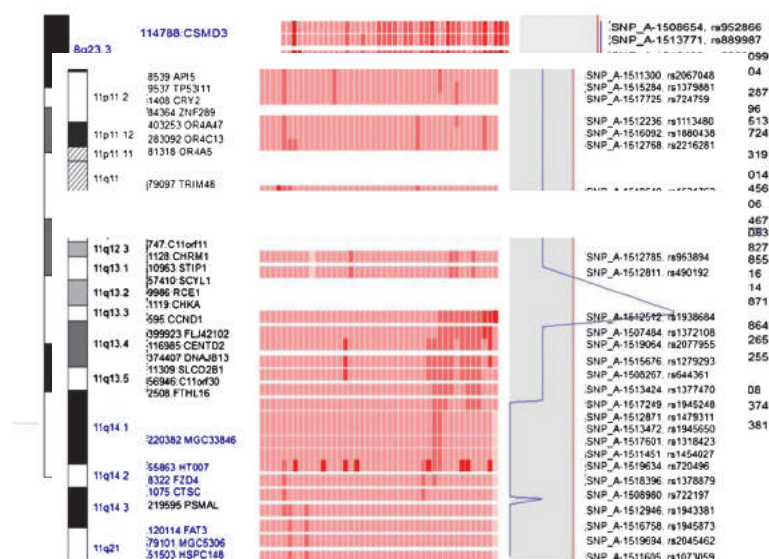


Fig. 5.22: Showing samples with gain of the *CCND1* gene on locus 11q13.

(ii) Copy Number and LOH Analysis in Breast Cancer

Genomic alterations amongst tobacco and non-tobacco users in many cancers using genome-wide approaches have been documented. Betel quid which is another form of tobacco consumption is widely used in the Northeastern part of India. The previous report showed a strong association of breast cancer risk with betel quid chewing in northeast Indian population which led to examine the genomic alterations in betel quid chewers (BQC) in breast cancer patients. The analysis of 26 betel quid chewer and 17 non betel quid chewer (NBQC) breast cancer patients from Northeast India using SNP array was undertaken. On average, betel quid chewers showed more alteration than the non-chewers 45.57% vs. 31.64 % respectively. Seven chromosomal regions (3p26.3, 3q26.1-3q27.2, 4p16.1, 5q11.2-5q12.1, 6q25.3, 7q33 and 21q22.13) were significantly ($p < 0.05$) associated with BQC tumors while five regions (16p13.12-16p11.2, 17q11.2, 19p13.3-19p12, 19q13.32-19q13.43, 20q11.22) were significantly associated with NBQC tumors. Incidence of gain in fragile sites observed was more prominent in BQC tumors (average 33.85%) than in NBQC tumors (average 23.52%). Common frequent alterations ($>30\%$) were also seen in 27 regions suggesting changes irrespective of any etiological factors. These data show that betel quid exposure by chewing can cause early changes in breast leading to carcinogenesis. Alterations seen in non-betel quid chewers indicate additional mechanism of breast tumorigenesis. The findings in the present study provide the first evidence, indicating that betel quid exposure may produce a specific DNA damage profile leading to breast carcinogenesis.

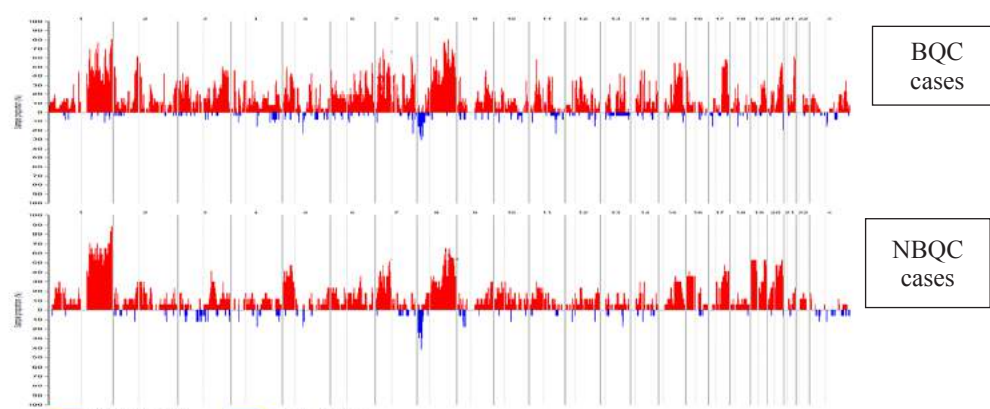


Fig. 5.23: Prevalence (%) of patients with 3 copies (red) and 1 copies (blue) in BQC and NBQC tumors, respectively. The x-axis represents the positions in genome/chromosomes, and the y-axis represents the prevalence.

(iii) Study of DNA Repair Genes in Breast Cancer

The northeast region of India reports an increase in incidence of breast cancer and eminent use of tobacco in various forms by females. Environmental genotoxic stress like tobacco smoke and smokeless tobacco contain polycyclic aromatic hydrocarbons (PAHs), tobacco-specific nitrosamines, nitrosamino acids, aldehydes, metals, aromatic and heterocyclic amines and other genotoxic carcinogens which are related with DNA adduct formation which causes DNA damage, acting as initiators of cellular alterations. DNA repair and cell cycle control mechanisms maintain genomic stability. When DNA damage occurs, DNA repair pathways, cell cycle arrest, and apoptosis are activated. Single nucleotide polymorphisms (SNPs) in genes involved in DNA repair and cell cycle control can affect repair efficiency and increase cancer risk. Therefore the relationship between breast cancer risk and SNPs in DNA repair and cell cycle genes in a high risk population which was exposed to high levels of environmental genotoxic stress was assessed. The analysis of breast cancer cases and age and sex matched controls from the northeast region of India for polymorphisms in BRCA2 (10462A > G: I3412V), RAD51 (5'UTR G135C) and CCND1

(G870A) genes by dHPLC and PCR-RFLP respectively. The exons for mutations in the BRCA2 gene by DHPLC was also analyzed. The heterozygous *CCND1* GA genotype significantly reduced the breast cancer risk in women [OR = 0.39, (95% CI 0.22-0.70), 0.001] when compared with the GG genotype. Women with high risk allele of RAD51 gene had an increased risk to breast cancer but the result were not significant [1.31, (0.54-3.20), 0.65]. Polymorphism analyzed in the BRCA2 gene also did not infer a significant risk to breast cancer [1.65, (0.59-4.65), 0.44]. Two samples showed a T>G mutation in exon 2, two samples showed missense variant (8415GT : K2729N) in exon 18. 3 samples showed a silent C>T mutation in exon 27. Therefore, mutation analysis revealed that BRCA2 mutations do not contribute significantly for the breast cancer risk in the Northeast Indian population.

Studies in Mizoram and Tripura showed Genetic polymorphism of tumor suppressor gene p53 and XRCC1_Arg399Gln were associated with increased risk of breast cancer in this study. Case control study showed carriers of mutations in XRCC1 and XRCC3 and TP53 genes significantly increased the risk of breast cancer subjects. Next Generation Sequencing data along with TaqMan allelic discrimination facilitated in identification of a novel mutation in PKHD1 gene which predisposed carriers of the variant allele to increased risk of breast cancer (OR = 3.1; CI = 1.4 – 7.2; p = 0.007). As a proof of concept the present study based on Next Generation Sequencing (NGS) technology has revealed that NGS have a significant utility for healthcare diagnostic especially by quickly screening mutomes which will be specific for different ethnic groups.

4. Lung Cancer:

North-eastern states in India have reported a very high prevalence of aero-digestive tract cancers when compared with other regions of India. Prevalence of lung cancer (AAR: 37.9 in males and 40.8 in females) has been reported highest in the Aizawl district of Mizoram. The use of tobacco is very high in Aizawl district, where locally prepared tobacco products such as Tuibur, a unique tobacco smoke infused water and Mieziol, a local cigarette made from vaihlo leaf are widely used. The habit of chewing betel quid, containing fresh betel nut and slaked lime wrapped in betel leaf is also widespread in this region. The state of Assam reports widespread chewing habit of tobacco with peculiarly fermented betel nut, which contains arecoline, that produces 3-methyl nitrosaminepropio-nitrile, a potent carcinogen and safrole-like DNA adducts that are shown to be genotoxic and mutagenic. In addition when fermented, betel nut is contaminated by fungi, leading to production of carcinogenic aflatoxins which further adds to the risk of this cancer in the region.

The lung cancer patients and normal healthy controls were investigated for GST polymorphisms. Of these, 81 (80%) of the cases and 104 (47%) controls were smokers, 55 (54%) cases and 123 (56%) controls were tobacco chewers and 79 (78%) cases and 130 (59%) controls were betel quid chewers. Lung cancer was significantly associated with smoking OR= 4.23 95% CI=2.24-7.96, p<0.001) and betel quid chewing (OR=2.16, 95% CI: 1.05-4.43, p=0.04). The frequency of GSTT1 and GSTM1 null genotype was 19% and 33% in lung cancer cases and 31% and 47% in controls, respectively. Variant genotypes of GSTP1 (Ile/Val and Val/Val) were found more frequently in lung cancer cases (46%) when compared with controls (40%). But the difference was not statistically significant. GSTT1 null (OR=0.47, 95% CI=0.24-0.93, p=0.03) and GSTM1 null (OR=0.52, 95% CI=0.28-0.96, p=0.04) 0.96, p=0.04) genotypes appeared to be protective factors for lung cancer. The geographical and cancer specific estimates revealed that chances of developing lung cancer were significantly higher for smoking individuals in Assam (OR=3.12, 95% CI: 1.39-6.97, p=0.006). The association of gene dosage effects by tri-modal phenotype distribution of the three genotypes of *GSTM1* and *GSTT1* with lung cancer on 154 lung cancer cases and 154 controls. No association was observed between the *GSTM1* genotypes alone or in combination with

environmental risk factors. The homozygous and null genotype of *GSTT1* conferred 34% (OR=0.66; 95% CI=0.36-1.22; $p=0.19$) and 68% (OR=0.32; 95% CI=0.15-0.71; $p=0.005$) reduced risk when compared to the wild genotype. Combined *GSTM1* and *GSTT1* null genotypes conferred significantly reduced risk (OR=0.23; 95% CI=0.06-0.80; $p=0.02$). Tobacco smoking was a major environmental risk factor to lung cancer (OR=3.03; 95% CI=1.73-5.31; $p<0.001$). However, its interaction with the homozygous and null genotype of *GSTT1* conferred reduced risk (OR=0.59; 95% CI=0.24-1.43; $p=0.25$ and OR=0.30; 95% CI=0.10-0.91; $p=0.03$ respectively). The results suggested that, true effects of the GST genes can be estimated when accounting for the actual copy numbers present, which clearly demarcates the associations of the three genotypes.

Multi-analytical approach including logistic regression (LR), classification and regression tree (CART) and multifactor dimensionality reduction (MDR) was applied in 188 lung cancer cases and 290 controls to explore high order interactions among xenobiotic metabolizing genes and environmental risk factors. Smoking was identified as the predominant risk factor by all the three analytical approaches. Individually, CYP1A1*2A polymorphism was significantly associated with increased lung cancer risk (OR=1.69; 95% CI=1.11-2.59, $p=0.01$) whereas EPHX1 Tyr113His and SULT1A1 Arg213His conferred reduced risk (OR=0.40; 95% CI=0.25-0.65, $p<0.001$ and OR=0.51; 95% CI=0.33-0.78, $p=0.002$ respectively). In smokers, EPHX1 Tyr113His and SULT1A1 Arg213His polymorphisms reduced the risk of lung cancer whereas CYP1A1*2A, CYP1A1*2C and GSTP1 Ile105Val imparted increased risk in non-smokers only. While exploring non-linear interactions through CART analysis, smokers carrying the combination of EPHX1 113TC (Tyr/His), SULT1A1 213GG (Arg/Arg) or AA (His/His) and *GSTM1* null genotypes showed highest risk for lung cancer (OR=3.73; 95% CI=1.33-10.55, $p=0.006$) whereas combined effect of CYP1A1*2A 6235CC or TC, SULT1A1 213GG (Arg/Arg) and betel quid chewing showed maximum risk in non-smokers (OR=2.93; 95% CI=1.15-7.51, $p=0.01$). The MDR analysis identified two distinct predictor models for the risk of lung cancer in smokers (tobacco chewing, EPHX1 Tyr113His, and SULT1A1 Arg213His) and non-smokers (CYP1A1*2A, GSTP1 Ile105Val and SULT1A1 Arg213His). These results identified distinct gene-gene and gene environment interactions in smokers and non-smokers.

Study of p53 codon 72 polymorphisms in lung cancer patients and controls showed Pro/Pro genotype more frequent in controls (25.2%) than cases (23.0%), however this genotypic distribution was not significant. Risk of cancer was higher for individuals carrying Arg/Pro than Pro/Pro genotypes but the results were not statistically significant. Frequency of smokers was higher in cases (68.3%) than controls (50.7%) ($\chi^2=12.824$, $p<0.0001$) and smoking conferred a significant risk (OR=1.88, 95% CI=1.11-3.19, $p=0.018$). Distribution of betel quid chewers differed significantly between cases and controls ($\chi^2=21.15$, $p<0.0001$) and conferred a significant risk of developing lung cancer (OR=3.54, 95% CI=2.01-6.25, $p>0.001$). Interaction combinations between betel quid chewing and P53 gene variants were highly significant and increased lung cancer risk upto six fold for all the three genotypes (OR=5.90, 95% CI=1.67-20.81; $p=0.006$, OR=5.44, 95% CI=1.67-17.75; $p=0.005$, OR=5.84, 95% CI=1.70-19.97; $p=0.005$ for Arg/Arg, Arg/Pro and Pro/Pro respectively). Interaction of Arg/Arg genotype with tobacco chewing and alcohol use conferred 64% and 74% less chance of developing the cancer (OR=0.36, 95% CI=0.13-0.97; $p=0.04$ and OR=0.26, 95% CI=0.07-0.95; $p=0.042$ respectively). No interaction between smoking and P53 genotypes rendered significant risk. No significant effect of the polymorphism on susceptibility to lung cancer was seen. Tobacco smoking and betel quid was found to be major risk factors.

Microarray and real-time validation was carried on 35 samples of lung cancer. Gene ontology and pathway analyses were performed on 734 differentially expressed (1.5 fold, $p<0.05$) genes. Gene enrichment

terms related to epidermal growth factor, homeobox related transcription activity terms and MAPK signaling. Most important biological function related to cell death, cancer, cell cycle and cellular assembly and organization. Furthermore, a signature of 24 differentially expressed zinc finger proteins (10 up-regulated and 14 down-regulated) including a zinc transporter SLC30A1 was identified. The novel genes (TMSB10, RPS, PPFIA1, TNS3, NGFR, CLK3 and PFDN6) were not associated to lung cancer before, which could act as putative bio-markers.

Risk factors of lung cancer in Mizoram were similar to the stomach cancer in relation to tobacco or tobacco product use. A case-control study revealed significantly higher risk for environmental exposure like users of Meiziol, a locally made cigarette (OR=2.25; $p=0.006$) as compared to general cigarette users (OR=1.68; $p=0.171$) among women in Mizoram. Current users of *Tuibur*, a liquid tobacco (OR=1.59; $p=0.039$) was also significantly associated with increased risk for lung cancer among women in Mizoram. Traditional dietary practices in Mizoram such as consumption of smoked fish (OR=2.57; $p=0.006$), smoked meat (OR=3.03; $p=0.001$) and Soda (OR=7.96; $p<0.001$) were more prone to develop lung cancer among women in Mizoram. Exposure of cooking oil fumes (OR=2.56; $p<0.003$), wood as heating source for cooking (OR=1.50; $p=0.044$), kitchen inside living room (OR=1.79; $p=0.001$), improper ventilated house (OR=2.10; $p=0.003$) were also significantly prone to develop lung cancer among women in Mizoram. GSTM1 null genotype and CYP1A1 homozygous mutant were significantly associated with the increase risk of lung cancer in both Manipur (GSTM1 OR=2.59; $p<0.001$ & CYP1A1 OR=4.12; $p<0.001$) and Mizoram (GSTM1 OR=2.95; $p<0.001$ & CYP1A1 OR=2.23; $p=0.003$). Gln/Gln alleles of both XRCC1 and XPD genes appear to amplify the effects of household exposure, smoking and betel quid chewing on lung cancer risk in the study population of both the states. Gln/Gln genotype was higher among cases than control groups in both XRCC1 (8.5% vs 6.3%) and XPD genes (12.9% vs 10.5%). Significant protective effect for lung cancer was observed for intake of bamboo shoots (OR=0.15; $p<0.001$) and egg (OR=0.05; $p<0.001$) in both Manipur and Mizoram. In addition to these among the traditional dietary habits of Manipur consumption of citrus fruits (OR=0.07; $p<0.006$) and garlic (OR=0.01; $p<0.001$) also plays an inverse association for lung cancer risk specifically in Manipur.

5. Hepatocellular Carcinoma (HCC) in Sikkim and Arunachal Pradesh

Hepatocellular cancer in Arunachal Pradesh and Sikkim was high in comparison to other states of India. Incidentally helicobacter B virus (HBV) infection among some tribes in Arunachal Pradesh was also very high. AAR reported was second highest in males both in Arunachal Pradesh and Sikkim (38 and 7.4) as compared to females (15.4 and 6.1) respectively. Majority of the cases were in the age group of 51 – 70 years, illiterate cultivators. More than half of the cases were from poor family (income range of Rs.500-1750). Most of the tumors were moderately differentiated (85.7%) followed by well differentiated (14.3%) in Arunachal Pradesh while in Sikkim most of the tumors are poorly differentiated (50%) followed by moderately differentiated (30%). Common presentation was abdominal pain (32%), inability to eat (30%) and weight loss (27%). Hepatitis B positive is found to be the common history of illness in both the states. Chewing of tobacco (Khaini) was significantly associated with increased risk of Hepatocellular Carcinoma (HCC) among the study subjects. People in Arunachal, a peculiar alcoholic drink is prepared called *Chai Mod* and is consumed by many. The process of preparation is peculiar and unique. Alcohol consumption was also significantly associated with increased risk of HCC among the study subject especially those who consume local drinks. Diet mostly includes rice, pulses and green leafy vegetables along with non-vegetarian foods which comprises of eggs, meat, and fish (fresh, dried and smoked). Most (89.5%) of the people consume non-vegetarian diet. Consumption of fermented and non-fermented bamboo shoot is widely preferred by the people of Arunachal Pradesh. Fruits are occasionally consumed which includes

citrus fruits, apples, jackfruit, etc. Risk was also observed in case of intake of spices though it was not significant.

Nasopharyngeal Carcinoma:

Nasopharyngeal Carcinoma (NPC) is a rare disease but incidence is high in southern part of China. Incidence of NPC vary across the world and is about 0.5/100,000 per year in the U.K. Earlier PBCR data from Mizoram showed the incidence is as high as 8.4/100,000 in men in Aizawl district of Mizoram to 2.5/100,000 in Sikkim (Fig. 5.24). In India also; though NPC is a rare yet in states like Nagaland, Mizoram, Sikkim, Manipur it is high. In view of high occurrence of NPC in Nagaland (Hospital record), RMRC, Dibrugarh conducted studies during 1992-94 and 1996-97, to evaluate the risk factor of the disease and reported that the consumption of smoked dried meat to be a risk factors for NPC in Nagaland. Epstein Barr Virus (EBV) antibody positivity was higher among NPC patients as compared to healthy controls. Significant mutagenic effect was observed in smoked dried and preserved meat. The following schematic model was proposed for the nasopharyngeal carcinoma in the north east region of India (Fig. 5.26).

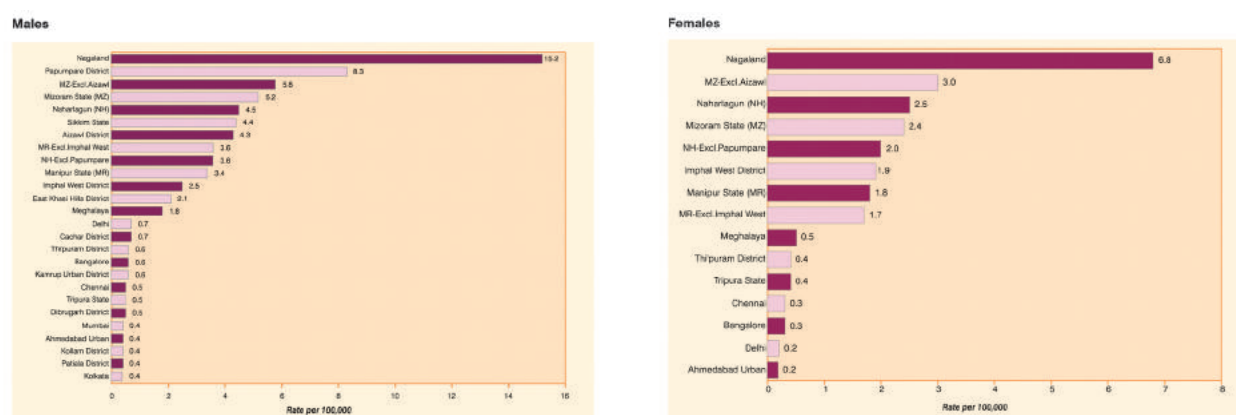


Fig. 5.24: Comparison of age adjusted incidence rates (AARs) of all PBCRs NASOPHARYNX (ICD-10: C11) in males and females.

Typical kitchen of the rural areas of Nagaland



Fig. 5.25: Typical kitchen of the rural areas of Nagaland.

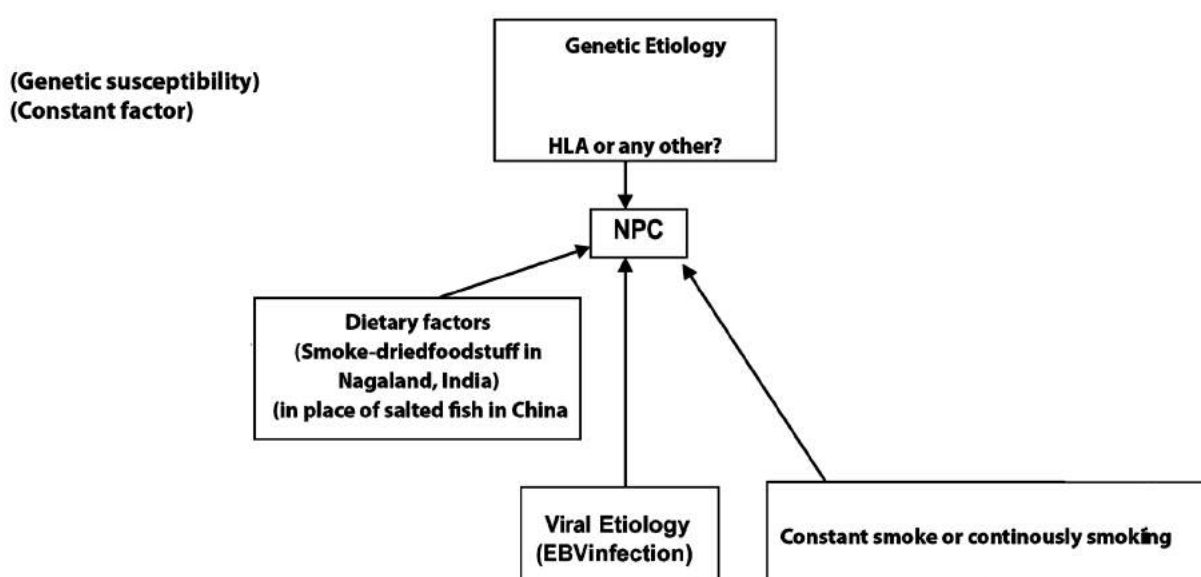


Fig. 5.26: Possible etiological factors associated with Nasopharyngeal Cracinoma in N.E. Region of India.

8. Pattern of Survival and Quality of Life of Esophageal and Stomach Cancer Patient in North Eastern Region

The epidemiological information has been collected from a total of 100 cases of Esophageal Cancer which include 18 cases from PBCR Dibrugarh, 42 cases from PBCR Guwahati, 24 cases from PBCR Silchar, 16 cases from PBCR Aizawl and 55 cases of Stomach Cancer which include 5 cases from PBCR Dibrugarh, 37 cases from PBCR Guwahati, 7 cases from PBCR Silchar and 6 cases from PBCR Aizawl. Out of the total 100 case of esophageal cancer male are preponderance with an estimated rate of 68% and maximum of the cases belongs to the age group of 50-59 yrs, whereas in case of stomach cancer female are preponderance with the estimation rate of 53% out of total 55 cases belonging to the age group of 50-59 yrs. Regarding caste/ethnic group is concerned, the non tribe is the main victims of both esophageal and stomach cancer. It is also being found out that the married people suffer the most than the unmarried one. Geographically the esophageal cancer is more prominent towards urban area with an estimation of 48% whereas stomach cancer is more prominent in rural area with 52.7% out of total cases. Maximum of the esophageal cases are found to be service holder with education upto middle level whereas in stomach cancer house wife is the most victim and they were found to be generally illiterate. The nuclear families were the main sufferer of both esophageal and stomach cancer. It was also observed that the maximum of the esophageal cases were from the middle class group and the maximum of the stomach patient were from the lower category. In case of anatomical stage, the cancer has spread towards stage IIIA and in both esophageal and stomach cancer. In esophageal cancer Radiotherapy is used for treatment by most and in stomach cancer chemotherapy has been used by many for treatment. 60% of the esophagus cancer and 50.9% of the stomach cancer patients were given treatment prior to the Reporting Institution (RI). 90% of the esophagus cancer patient and 53% of the stomach cancer patient were continuing the prescribed treatment. Maximum number of the esophagus and stomach cancer patient have chosen to take treatment in their own state rather than going out of state for treatment. It was found out that 78% of the esophageal patient was alive and 22% of them are dead, whereas in stomach cancer 65% of the patient was alive and 34.5% of them are dead. In context to clinical information most of the esophageal cases are Squamous cell carcinoma with pathological grade G1: Well differentiated

and the topography inclination indicates more towards C15.5: Lower-third of esophagus. In case of stomach cancer adenocarcinoma is the most common type of cancer with pathological grade G1: Well differentiated and the topography inclination indicates more towards C16.3: Gastric antrum. The overall average survival probability of esophageal cancer patient is above 10-11 months and stomach cancer patient is 8-9 months are found. It was also observed that EORTC QLQ C30 questionnaire gives reliable results on quality of life of both esophageal and stomach cancer patients. Results showed that almost all the multi-item scales of the questionnaire registered high value of ($\alpha > 0.70$) signifying that the scales are internal consistent.

Study also revealed that only 4% of the esophagus cancer patients survive upto 3 years in Assam after diagnosis and cancer directed treatment. The study reveals that patients undergoing surgery live significantly longer than other mode of treatments.

Publications

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An estimated 250 million persons in India use tobacco in one form or the other. Most common form of tobacco smoking is bidis followed by cigarettes. Smokeless tobacco is used in the form of khaini, gutka, mawa, etc. The morbidity and mortality caused by tobacco is extremely high, which necessitates strict measures to control its use. India is one of the earliest signatory of the Framework Convention on Tobacco Control (FCTC) of World Health Organization. The ICMR helped the government in identifying and consolidating its stance during various stages of negotiations, especially during the initial stage of identification of strategies by the Working Group on FCTC.

Indian Council of Medical Research (ICMR) considers tobacco use as an important public health issue. Tobacco is known to be an important risk factor in causation of 25 disease entities, including cancer of several body organs and tissues. The role of tobacco in cancers in India had been reported as early as 1902. But the need for legislative and public health action was considered important only during the second half of 20th century, especially after the publication of first report of the US Surgeon General in 1954. Several Indian studies, especially from Mumbai and Uttar Pradesh, during 1950s and 1960s highlighted the important aetiological role of tobacco use (both in smokeless and smoking form) in causation of various cancers, like oral cavity, pharynx, lung, oesophagus, etc.

The Cigarette Act of 1975 was the first legislative attempt by the Government of India aimed at tobacco control in India, which mandated cigarette packets to very clearly carry a specified warning. The limitations of this warning were recognized by Indian scientists very soon after its implementation. ICMR was proactive in advocacy for stronger tobacco control measures. However, the economic and social aspects of tobacco were considered compelling and delayed major action against tobacco. Under such a scenario, ICMR identified in 1970s, that generation of scientific evidence on tobacco related diseases may be an important direction for research. Such research was initially started on cancer; subsequently important projects were also carried out on role of tobacco in cardiovascular diseases, chronic obstructive lung diseases, stroke, etc. Important studies which changed the perspective included: assessment of magnitude of tobacco use, magnitude of tobacco related diseases, feasibility of reduction of tobacco use through existing infrastructures, role of pan masala (with or without tobacco) in oral cancers, cost of major tobacco related diseases to the country, feasibility of quickly carrying out large community based surveys on tobacco use patterns, etc. An important role performed by the ICMR was to provide impartial analysis of the existing situation to the Government along with possible control measures. This support on scientific data analysis, undertaking research studies and identifying control measures helped the government to initiate action on tobacco control, including promulgation of comprehensive tobacco control legislation, anti-tobacco education activities, tobacco cessation measures, etc.

The following portion provides an account of various research studies by ICMR and how it changed the scientific knowledge and control activities in India.

Tobacco: A Major Public Health Problem in India

Based on the scientific literature on tobacco, ICMR noted in 1970s that the tobacco being a major public health problem had not been projected adequately. Thus, it was decided to collect and analyze scientific data on the subject and assess the need for public health action, as well as decide on the research priorities. The research findings from Mainpuri (Uttar Pradesh) and Mumbai on oral cancers highlighted the importance of smokeless tobacco use which was peculiar to Indian sub-continent. Tobacco was considered as an important revenue as well as foreign exchange earner. The problem of population control and control of communicable diseases was taken as the paramount priority. On the other hand, the magnitude of non-communicable diseases had not been adequately documented.

Therefore, the first logical step seemed to be to prepare a document on tobacco, providing both its role in disease causation as well as a contributor to Indian economy. The document printed in 1984, provided the data on prevalence rates of tobacco use in India, diseases caused by tobacco, tobacco crop production figures and exports, revenue & employment by tobacco, legislation for tobacco control, need for tobacco control, etc. This document attracted a lot of attention including the need to review the then existing Cigarette Act 1975 which mandated writing a warning on all cigarette packages. ICMR advocacy in this direction succeeded, when a decision to develop a new legislation for all tobacco products was taken in a meeting of Ministry of Health & Family Welfare in 1986.

Prevalence of Tobacco Use in India

Information on prevalence of tobacco use in India before 1980s was through sporadic community surveys, carried out as a part of specific research projects. It may not have been sufficient to assess the overall magnitude of the tobacco problem, leave alone understand the finer nuances and trends of tobacco use pattern. Surveys conducted prior to 1970 showed that the prevalence of tobacco usage varied from 61% to 86% among males and from 15% to 67% among females. Prior to 1960, the major tobacco habit among males was chewing. However, the trend seemed to have reversed perhaps between 1960 and 1970. Population based surveys conducted in seven rural areas in 1960s indicated that more men smoked rather than chewed tobacco, the exception being Pune and Dharbhanga. A substantial proportion had mixed habit. Among the females, the major habit was tobacco chewing, except in Srikakulam and Dharbhanga where smoking was more popular. It would be logical to conclude that the changeover occurred at different time periods in different areas.

Under an ICMR collaborative study on the effect of intervention programme on nonmedical use of drugs in the community, baseline surveys on the prevalence of drug usage, including tobacco, were conducted. The intervention was carried out in Delhi only. The prevalence of tobacco usage in the baseline survey varied from 46 to 74% among males and 2 to 50% among females. Smoking was the commonest habit among males in Bangalore (urban and rural both) and Delhi. Chewing was the predominant habit among males in Dibrugarh & Ranchi and among females in all areas.

In surveys conducted in 1980s in three rural areas, under an ICMR project on anti-tobacco community education, the main habit among males was smoking and among females was chewing. However, in Uttar Pradesh, nearly all the users (males or females) chewed tobacco. The overall prevalence of tobacco usage among the males over the age of 5 years was 34% (range 31.9 to 36.7%) in Uttar Pradesh (Mainpuri and Shekohabad) and 32% (range 33 to 36%) in Karnataka. The prevalence among females above the age of 5 years was 26% (range 20.1% to 29.1%) in Uttar Pradesh and 36% (32 to 43%) in Karnataka. In Goa, the survey on tobacco usage was conducted on population over the age of 15 years and showed an overall prevalence of 33% (range 23.4% to 40.8%) among males and 20% (range 15.3% to 26.4%) among females.

Overall, various studies in India in the 1980s showed that the prevalence of tobacco use among men over 15 years of age varied between 46% and 63% in urban areas and between 32% and 74% in rural areas. Among women, it varied between 2% and 16% in urban areas and between 20% and 50% in rural areas.

Thus, the wide variations in tobacco use prevalence did not permit a valid conclusion on prevalence of tobacco use. The use of tobacco crop pattern seemed to be the only assessment method for study of trends, which may have its inherent limitations. The data also suggested that prevalence of tobacco habit among males in 1980s was lower than those found in earlier surveys. There was no explanation for such a decrease, as no tobacco control measures were available and the surveys also did not substantiate population of quitters. Therefore, the possible reasons for this change were differences in methodologies, definitions, analysis, and possible geographic variations. The situation necessitated large community based surveys on tobacco use prevalence.

Nationwide Surveys by National Sample Survey Organization (NSSO) of India

The ICMR noted the possibility of undertaking large surveys on tobacco use, as a component of the ongoing surveys conducted by the National Sample Survey Organization (NSSO) of India, which is an institution under Government of India. NSSO surveys are on consumer expenditure and are carried on yearly basis from July to June of next year. At the request of Indian Council of Medical Research, the tobacco use data was first included in the fourth quinquennial NSS survey (1987-88). The NSSO informed that the detailed questionnaire suggested by ICMR may not be possible but a small number of questions on regular or casual use for major habits were included. The questions were repeated during the fifth quinquennial survey (1993-94). Under these surveys, households were selected on a random basis, in about 12,000 to 14,000 villages and blocks in the central sample (covered by NSSO) and an independent sample of about 14,000 to 16,000 villages and blocks in the state sample (covered by the governments of various states and union territories). The entire area of the country – rural and urban – was covered, with the exception of some interior areas of certain states. The surveys adopted a stratified two-stage sampling design with census villages/urban frame survey blocks at first stage and households at the second stage units. Consumption data was collected with a reference period of “last 30 days”. Being a continuous survey, the reference period was a moving one, varying from household to household.

Availability of the nationwide surveys through NSSO provided a mechanism of assessing the total number of tobacco users in India both in smoking as well as smokeless forms. Application of age wise prevalence rates of 1987-88 data of NSSO to the 1991 census data indicated the number of tobacco users in India to be approximately 214 million. Based on the data from the 1993-94 survey data, the number of tobacco users in India was approximately 239 million in 1996. The details of tobacco habit prevalence from the 1993-94 survey data is given in the table below (Table 6.1 and 6.2):

Percentage of Persons Aged 10 Years and above Reporting Habit of Consumption of Tobacco in Various Forms per 1000 Persons by Sex and Age Group. National Sample Survey, 50th Round (1993-94).

Table 6.1: All India Urban													
Form of Tobacco Consumption	Regularity	Age (in Years)											
		10-14		15-24		25-44		45-59		60 & Above		All Ages	
		Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
Smoking	Regularly	0.1	0	5.5	0.2	30.5	0.5	38.1	1.8	26.6	1.8	20.2	0.7
	Casually	0.1	0.1	0.9	0	2.8	0.1	2.5	0.2	1.9	0.2	1.8	0.1
Chewing, Zarda, Dokta, etc.	Regularly	0.1	0	3.1	0.4	11.4	2.6	14.1	6.6	13.3	7.2	8.0	2.6
	Casually	0	0.1	0.6	0.1	1.7	0.4	1.7	0.6	1.3	0.6	1.2	0.3
Snuff	Regularly	0	0	0.1	0.1	0.3	0.3	0.9	0.8	1.4	1.9	0.4	0.4
	Casually	0	0.1	0.1	0	0.1	0.1	0.1	0.1	0.2	0	0.1	0.1
Burnt Tobacco/ Powder/ Paste	Regularly	0.2	0.1	0.6	0.6	1.9	1.4	2.8	2.9	2.5	2.8	1.5	1.5
	Casually	0	0.1	0.1	0	0.2	0.1	0.2	0.2	0.1	0.1	0.1	0.1
Consumption of Tobacco in Any Form	Regularly	0.4	0.2	8.7	1.2	40.7	4.5	50.9	11.4	39.5	13.0	27.7	4.7
	Casually	0.1	0.1	1.3	0.2	3.2	0.5	2.7	0.8	2.7	0.7	2.1	0.4

Table 6.2: All India Rural													
Form of Tobacco Consumption	Regularity	Age (in Years)											
		10-14		15-24		25-44		45-59		60 & Above		All Ages	
		Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
Smoking	Regularly	0.5	0.1	12.0	0.7	43.4	2.5	49.8	4.6	39.8	4.8	29.3	2.3
	Casually	0.1	0.1	1.3	0.1	3.1	0.4	3.0	0.5	2.8	0.5	2.1	0.3
Chewing, Zarda, Dokta, etc.	Regularly	0.4	0.3	6.6	1.9	20.5	5.9	25.9	10.3	26.4	10.7	15.1	5.3
	Casually	0.2	0.1	1.3	0.4	2.6	0.8	2.9	1.1	2.2	0.9	1.9	0.7
Snuff	Regularly	0	0	0.4	0.3	0.7	0.7	1.2	1.6	1.3	1.9	0.7	0.7
	Casually	0	0.1	0	0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Burnt Tobacco/ Powder/ Paste	Regularly	0.4	0.5	1.8	2.0	4.4	3.9	6.0	5.5	6.8	5.2	3.5	3.3
	Casually	0.1	0.1	0.2	0.1	0.5	0.2	0.5	0.3	0.4	0.3	0.3	0.2
Consumption of Tobacco in Any Form	Regularly	1.3	0.9	19.1	4.6	61.3	12.2	72.3	20.4	65.0	21.2	43.0	10.9
	Casually	0.3	0.2	1.9	0.5	3.0	1.1	2.8	1.5	2.9	1.3	2.2	0.9

Source: Sarvekshana, January-March 1998; p76.

National Family Health Survey

The National Family Health Survey (NFHS) is a large-scale, multi-round survey conducted in representative sample of households throughout India. The survey is organized by the Ministry of Health and Family Welfare of India. Three rounds of the survey have been conducted since the first survey in 1992-93. The survey provides state and national information for India on fertility, infant and child mortality, the practice of family planning, maternal and child health, reproductive health, nutrition, anemia, utilization and quality of health and family planning services. The survey is providing important data for policy purposes. Data on tobacco use (in smoking or smokeless form) was collected during second (1998-99)

and third round (2005-06) of the survey. While data on tobacco use during second round (like the NSSO survey data) was collected from the household respondent, the data for tobacco use during the third round was collected from individuals who were interviewed with individual questionnaire, by asking four specific questions on current use of tobacco (smoke and non-smoke variants). The data was obtained by asking about use of tobacco during past 24 hours.

Table 6.3: Tobacco use by women and men

Percentage of women and men age 15-49 and men age 15-54 by their use of tobacco, and percent distribution of those who smoke cigarettes or <i>bidis</i> by number of cigarettes/ <i>bidis</i> smoked in the 24 hours preceding the survey, India, 2015-16							
	Women			Men			
Tobacco use	Urban	Rural	Total 15-49	Urban	Rural	Total 15-49	Total 15-54
Use of tobacco							
Smokes cigarettes	0.1	0.1	0.1	16.2	12.0	13.6	13.7
Smokes bidis	0.2	0.6	0.5	8.9	15.8	13.2	14.3
Smokes cigars	0.1	0.1	0.1	0.4	0.5	0.4	0.5
Smokes a pipe	0.0	0.0	0.0	0.1	0.2	0.2	0.2
Smokes a hookah	0.1	0.2	0.2	0.4	0.7	0.6	0.6
Chews paan masala or gutkha	1.7	2.5	2.2	13.6	16.3	15.3	14.9
Uses khaini	0.6	1.6	1.3	7.7	14.9	12.1	12.4
Chews paan with tobacco	1.2	2.2	1.8	4.1	6.3	5.4	5.6
Other chewing tobacco	0.5	0.8	0.7	1.7	2.6	2.3	2.3
Uses snuff	0.1	0.2	0.1	0.0	0.1	0.1	0.1
Other	0.3	0.8	0.6	0.2	0.6	0.4	0.5
Does not use tobacco	95.6	91.9	93.2	61.1	52.0	55.5	54.5
Number of respondents	242,296	457,390	699,686	39,549	63,862	103,411	112,122
Number of cigarettes/bidis smoked in the past 24 hours							
0	1.4	0.2	0.4	5.4	6.1	5.9	5.5
1-4	61.2	50.6	52.5	41.7	33.9	36.7	35.1
5-9	13.1	24.1	22.2	21.2	19.5	20.1	20.2
10 or more	24.1	25.1	24.9	31.0	39.7	36.5	38.5
Missing	0.1	0.0	0.0	0.7	0.8	0.8	0.8
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Number of cigarette/ bidi smokers	625	2,985	3,610	8,682	15,104	23,785	26,963

Source: All India Report of 4th NFHS (2015-16) of India. <http://rchiips.org/NFHS/NFHS-4Report.shtml>. Accessed on 5th January 2018.

ICMR- WHO Survey in Uttar Pradesh and Karnataka

Tobacco use prevalence from the NSSO or NFHS surveys is available for smoking and smokeless forms of tobacco for specified age groups. However, several details about tobacco use patterns according to different tobacco types and use characteristics including intentions to use and quit, etc., are not available. Such data is also not available to develop region or area specific intervention strategies. The data also does not provide information on changes in specific habit types at different time periods. Therefore ICMR organized tobacco use prevalence surveys in Uttar Pradesh and Karnataka, with financial support & technical discussions with WHO SEARO in 2001. The surveys also demonstrated the feasibility of

quick survey on large population using cluster sampling approach. The study on about 60,000 population in 2 states was completed in 7 months. The study also provided detailed data on tobacco consumption patterns, dynamics of tobacco use during life time of respondents, quantum and duration of each habit, quitting patterns and opinion of community on tobacco control.

Prevalence of Tobacco Use in Uttar Pradesh and Karnataka in India (2001)

The study was a part of the three-nation project by the World Health Organization, South East Asian Regional Office (WHO SEARO), conducted in India, Indonesia and Bangladesh using a standard survey tool. The standard survey tool was to use cluster sampling approach to undertake a fast survey on tobacco use patterns. The timeframe for the entire survey was 6 months. Through this survey, WHO intended to help member countries to provide a baseline data on tobacco use which when repeated at periodic intervals may provide trends in tobacco use. ICMR was requested to carry out the Indian component of the survey. The overall survey was to assess the prevalence of use among different sexes, age groups, and social classes. A small sample was proposed for the survey to collect data on presence or absence of tobacco use. However, during the discussions the need for a higher sample size was realized in India due to presence of multiple tobacco habits and known differences in several demographic and social classes. Therefore, each survey in India was on a sample of approximately 30,000 population. The initial objective was to cover the entire country, but later on the survey was limited to two Indian states (Karnataka and Uttar Pradesh).

The survey was carried out with the objectives of building a database on prevalence of tobacco use in the total population and among specific population subgroups, for the purposes of advocacy for tobacco control and planning tobacco control interventions and evaluation.

Three districts from each state were selected through simple random sampling method. To cover a population of about 30,000 above 10 years of age, 600 clusters in each of the state were selected by simple random method from the list of villages or urban wards, obtained from the Registrar General of India. The proportion of urban & rural population as observed from the census data from the specific district, was used for allocation of clusters to urban and rural population. Within a cluster, approximately 17 houses were visited by trained workers according to a pre-defined movement plan to provide approximately 50 persons above 10 years of age. The cluster size was decided based on the fact that each interviewer would be able to cover approximately 50 persons during each round. The data collection took about 5 months and the entire study encompassing 60,000+ persons was completed in 7 months after receipt of funds, indicating that fast survey on a large community is feasible. The survey activity started early in the morning. The low refusal rate was achieved by revisiting the missing persons later on during the same day. The proportion of persons missed or refusal for interview was small, being 3.1% in Karnataka and 0.2% in Uttar Pradesh.

The overall prevalence of ever-use of any kind of tobacco products among the population above 10 years of age was observed to be 29.6% in Karnataka and 34.6% in Uttar Pradesh. The overall prevalence of current use of tobacco in the same population group was observed to be 28.4% in Karnataka and 34.4% in Uttar Pradesh. The difference in ever-usage and current usage prevalence was observed to be very small, suggesting that tobacco quitters were very few in numbers (being 1.2% in Karnataka and 0.2% in Uttar Pradesh). The figures below (Fig. 6.1 and 6.2) provide the prevalence of ever-use and current-use of tobacco according to sex and urban/rural residence status

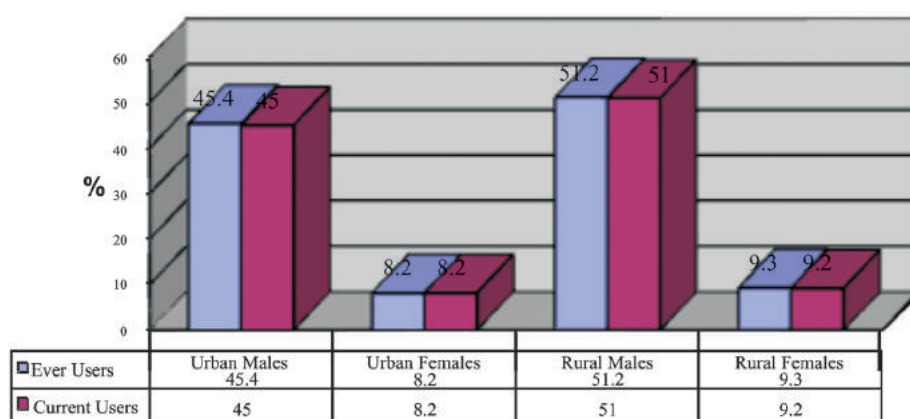


Fig. 6.1: Prevalence of Tobacco use Uttar Pradesh (2001).

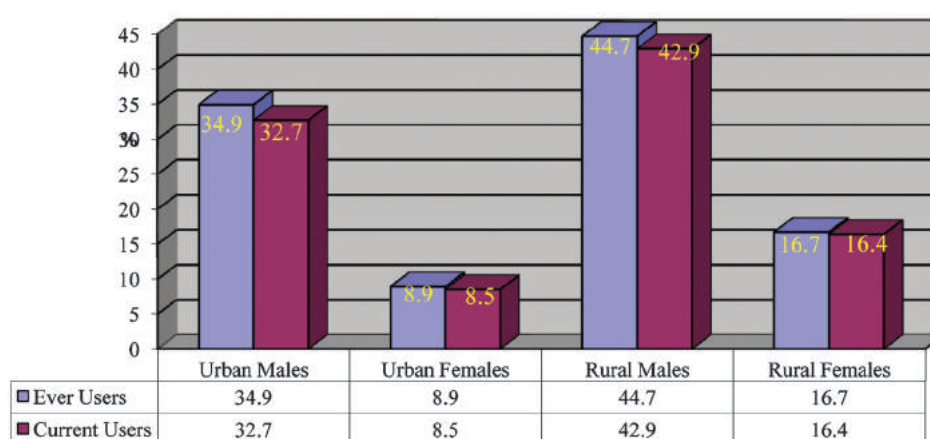


Fig. 6.2: Prevalence of tobacco use Karnataka (2001).

The prevalence of tobacco use increased with increasing age. The peak values were observed after 55 years of age or even continued to rise till the 70+ year age group in certain categories. The prevalence rates of tobacco use were higher in rural areas as compared to urban areas in most age-groups, the exception being the teenagers, where prevalence was comparable or higher in urban areas (Fig. 6.3).

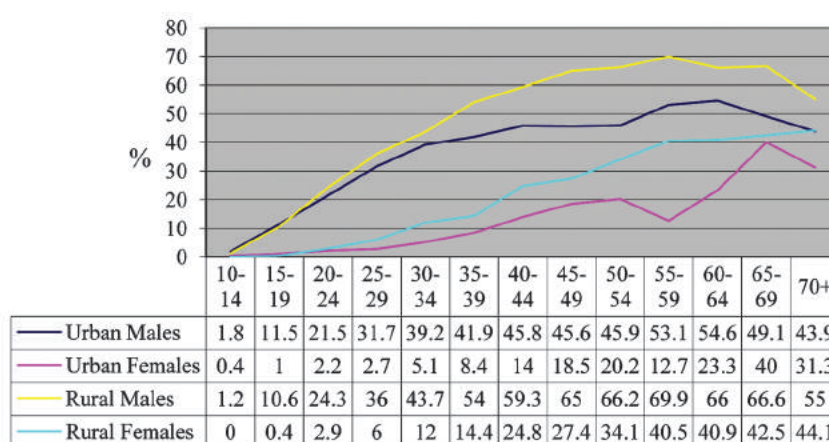


Fig. 6.3: Age-wise prevalence (%) of Tobacco use in Karnataka according to sex and rural/urban status.

Negative association between education and prevalence of tobacco use was observed overall and in most of the age-groups in both the states. Similarly, higher family income levels were associated with a lesser prevalence of current tobacco use, although no difference in prevalence rates was observed between monthly family income categories below Rs. 2,000 per month. Religion did not seem to have any association with prevalence of current use of tobacco.

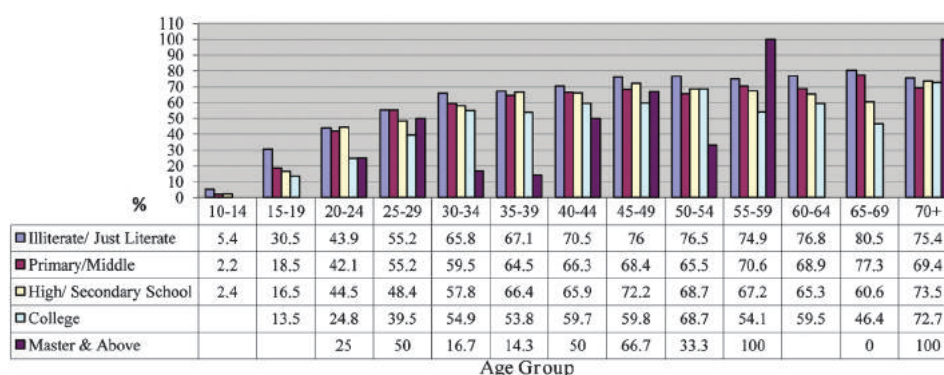


Fig. 6.4: Age-wise prevalence (%) of tobacco use in Uttar Pradesh males according to education.

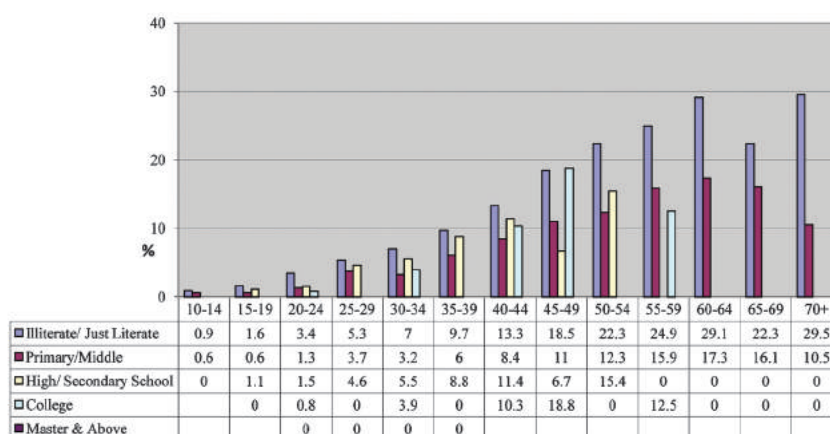


Fig. 6.5: Age-wise prevalence (%) of tobacco use in Uttar Pradesh females according to education.

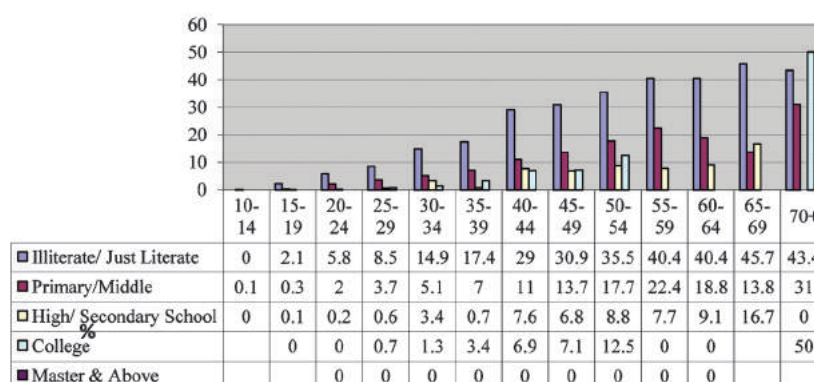


Fig. 6.6: Age-wise prevalence (%) of tobacco use in Karnataka females according to education.

Tobacco use in smokeless form was predominant among women and among men below 30 years of age, both in urban and rural areas. However, smoking was the predominant form of tobacco use among men above 30 years of age. The overall prevalence of current smoking was 15.5% (29.3% among men & 0.6% among women) in Karnataka and 18.0% (27.5% among men & 2.6% among women) in Uttar Pradesh with gradual increase with age. The overall prevalence of current use of smokeless tobacco was observed to be 13.9% in Karnataka (13.4% among men and 14.4% among women) and 17.5% in Uttar Pradesh (24.3% among men and 6.6% among women).

Bidi was observed to be the most popular modality of tobacco smoking accounting for 91.7% and 84.5% of smoking habit (with or without other tobacco habits) in Karnataka and Uttar Pradesh, respectively. Cigarette smoking was practiced by 5.7% and 4.1% current tobacco users (with or without other tobacco products) in Karnataka and Uttar Pradesh respectively. Cigarette smoking was more popular in urban areas as compared to villages.

Only a limited number of persons started their tobacco habit before 10 years of age. Among those starting tobacco use before 10 years of age, 66.7% men (12 out of 18) in Karnataka and 98.4% men (62 out of 63) in Uttar Pradesh had also started using tobacco on a regular basis before the age of 10 years.

The mean daily frequency of tobacco use was 13.0 ± 9.7 in Karnataka (14.9 times/ day among men & 7.5 times/ day among women) and 9.1 ± 6.5 in Uttar Pradesh (10.0 times/ day among men & 7.9 times/ day among women). Cigarette smoking was mainly limited to men with 36.6% and 55.2% of them using it 6 or more times a day in Karnataka and Uttar Pradesh, respectively. About 89% persons in Karnataka and 87.2% men & 74.8% women in Uttar Pradesh smoked bidis 6 or more times a day.

Duration of use of tobacco products increased with increasing age. In Karnataka, 18.4% of men and 9.3% of women had ever considered tobacco cessation, but only 4.0% men and 2.2% of women could quit the habit. In Uttar Pradesh, 3.5% of men and 1.4% of women ever contemplated cessation, but only 5 men and 1 women (<0.1%) could quit the habit. The data also suggested that several of the persons reporting quitting may actually have switched to other forms of tobacco use (say from smoking to smokeless tobacco). Among those ever considering cessation, 59.4% in Karnataka and 63.0% in Uttar Pradesh were currently interested in quitting. Common reasons for quitting the habit included long-term health effects, economic impact, & moral/ religious reasons. Self-determination was the chosen method in most of the cases.

In Karnataka, at-least one of the parents was reported to be a tobacco user among 68.3% men and 74.3% women tobacco users. Tobacco use by a parent was reported by 49.8% men and 45.2% women never-users of tobacco. In Uttar Pradesh, parental tobacco use was reported by 73.8% men and 60.3% women tobacco users, in contrast with 66.4% men and 59.9% women never-users of tobacco.

In Karnataka, 61.7% of tobacco users and 78.0% of never-users reported knowledge of at least one harmful effect of tobacco. In Uttar Pradesh, 90.8% of tobacco users and 88.7% of never-users reported knowledge of harmful effect of tobacco. Cancers was the commonest response in all age groups in both the states, both amongst users as well as non-users of tobacco, followed by respiratory diseases, heart diseases and stroke. Second-hand smoke was considered to be harmful for adults by 57.1% men tobacco users and 26.6% women tobacco users in Karnataka, while in Uttar Pradesh, 95.2% men and 92.7% women tobacco users considered it harmful. Second hand smoke was considered to be harmful for children by 56.4% men & 26.4% women tobacco users in Karnataka and 95.3% men & 93.7% women tobacco users in Uttar Pradesh.

Overall, 61.3% men & 38.0% women tobacco users and 75.0% men & 69.6% never-users of tobacco in Karnataka, felt that spending money on tobacco products makes people poorer. In Uttar Pradesh, 80.2% men & 74.3% women tobacco users and 79.3% men & 80.1% women tobacco never users felt that expenditure on tobacco makes people poorer. Reasons for tobacco use included relaxation, sophistication, or fun. Some users also found tobacco to be repulsive or immoral.

The common responses in decreasing order about perception about tobacco industry included revenue generation, harm to economy, provision of jobs and killing citizens in both the States. Among tobacco using respondents in Karnataka, 77.9% supported discontinuation of advertising and sponsorship by tobacco industry; 75.8% supported increase in prices of tobacco products; 79.9% supported banning of smoking in public places; and 80.5% supported banning of sale to minors. About one-fifth (between 19.0% and 21.5% for different questions) of respondents had no opinion on various control measures. About 3% of tobacco users opposed increase in prices of tobacco products, while opposition for other measures was less than 1%. Support for control measures by tobacco users of Uttar Pradesh was 91.8% for discontinuation of advertising; 74.5% for increase in prices; 90.9% for banning smoking in public places; and 90.5% for banning sale to minors. Opposition for above measures was voiced by 5.0%, 11.2%, 3.8% and 3.6% of tobacco users, respectively.

The study demonstrated the feasibility of undertaking large community based survey to collect detailed data on tobacco use. The earlier large studies on the subject were carried out over a long time, sometimes over 2 to 3 years, which may have difficulty in related to a particular time period.

The earliest attempt by ICMR on collection of nationwide data on tobacco use prevalence was to motivate National Sample Survey Organization (NSSO) to include information on tobacco survey in their surveys on quinquennial basis in early 1980s. The NSSO had agreed to only a small number of questions due to the length of their existing questionnaire. While the NSSO data was able to provide indication of the overall prevalence of tobacco use data both in smokeless and smoking forms, it was not optimal to provide information on specific forms of tobacco use, trends in change of specific habits, knowledge of harmful effects of different tobacco types, perceptions about tobacco use, reasons for initiation and quitting of tobacco habit, community view on control measures, etc. Such information would be useful not only about designing intervention programmes and materials, but would also help in assessing the effect of intervention strategies and plan legislative and administrative actions.

The study also demonstrated the feasibility of detailed data collection about different types of tobacco habits in a short period of time. The analysis also suggested the need for continuing such methodology, due to the suggestion that several persons who earlier could have been categorized as tobacco quitter, may actually have been persons who changed from one type of tobacco habit to another, say cigarettes to bidis or from smoking to smokeless form of tobacco. The questionnaire was able to find out the tobacco habits of a person at any time, thus, providing the opportunity of knowing when the person changed tobacco habit. This may explain the small number of quitters observed under the study. The finding would be important for expansion of tobacco cessation activities in India.

The study showed the limited knowledge on tobacco hazards in the community and the fact that level of this knowledge may be different in different areas. This highlighted the need for developing specific intervention programmes suited to particular region, besides intervention programmes common to the country or a larger area. The data also demonstrated that a large number of people were willing to quit, but they did not have the will power and the social support to quit. Most of the people including tobacco users also do support legislative and administrative measures for tobacco control, including even ban on tobacco.

The questionnaire was specifically designed to obtain detailed information on different forms of tobacco use, based on the experience from earlier ICMR projects on the subject. Some of the limitations of the questionnaire observed during the study were corrected during the conduct of another study needing tobacco usage information during on about 70,000+ persons in a community survey in New Delhi. This study indeed showed that there were several people who as per the earlier studies could have been identified as tobacco quitter, actually might have changed their form of tobacco use.

Age at Initiation and Prevalence of Tobacco Use among School Children

Tobacco use among school children is becoming a serious problem in developing countries. The early age at initiation highlights the imperative need to intervene and protect this group.

At ICMR's National Institute of Cancer Prevention (erstwhile Institute of Cytology and Preventive Oncology) at Noida; the study was undertaken with an aim to assess the prevalence of tobacco use among school children and to determine the age at initiation of this habit. The study included 4786 students of class 7-12 (age range 11-19 years) studying in various schools in Noida, U.P., India. Cluster and random sampling methods were used for selecting the students and a self-administered questionnaire was prepared. Any kind of tobacco use was seen in 537 (11.2%) of the students. Of these, 419 (8.8% of total) were 'ever smokers' (including current smokers) and 219 (4.6%) were 'ever tobacco chewers'. Tobacco use exclusively by smoking was elicited in 179 students (3.7%) while 118 (2.5%) were 'exclusive chewers'. The mean age at initiation of tobacco use was 12.4 years. Nearly 70% boys and 80% girls 15 year of age initiated the habit before the age of 11 years. Private school students reported a significantly early uptake of tobacco chewing as a habit. More than 50% of tobacco chewers reported use of khaini at least once prior to the study. This study highlighted the emerging threat of tobacco use among children. In addition, a downward shift in the age at initiation of tobacco uptake and rising prevalence among girls was also noted.

Risk factors Associated with Tobacco Use among Adolescents: a Cross-Sectional Study

Subsequent to elucidation of the prevalence of tobacco use among school children, a study was undertaken to determine the factors associated with tobacco use in this group. Knowledge of the factors leading to tobacco use at this young age might be helpful in devising ways to curb this problem. The same study group utilized in prevalence study was administered questionnaire regarding factors associated with their tobacco habit. The various factors associated with tobacco use in the study group included: father in blue collar jobs, less educated mothers and tobacco use by father/ mother/ siblings/ friends. The habit was manifold higher if the students bought tobacco for teachers (10.6-fold), brother (6.4-fold) or father/ relatives (3.1-fold). The reasons for tobacco habit included: refreshing themselves (31.5%) and influence of actors smoking in films (61%). About half of the students preferred smoking outside their home. Socio-economic status of the family, education level of the parents and influence of family and peer habits were some of the most important factors in tobacco habits in children. Hence, the methods of information, education and communication (IEC) should be directed at these to have a major impact on curbing this habit in children and adolescents.

Economics of Tobacco Use in India

Computation of economics of tobacco was one of the identified research area for ICMR. In order to achieve this goal, it was identified that cost of management of tobacco related diseases was an important missing link. ICMR initiated studies on this aspect in 1989. However, by mid 1990s, the importance of information on tobacco economics was a matter of concern for the Government of India. The Ministry of Health & Family Welfare constituted an Expert Committee on Economics of Tobacco in India vide a notification issued on 26th July 1995. The ICMR acted as the secretariat for this Committee.

The Expert Committee was set up by the Department of Health, Ministry of Health and Family Welfare, Government of India, to undertake a comprehensive study on economics of tobacco use inter alia examining the tax revenue and foreign exchange earnings, employment and consumer expenditure on the one hand, and the cost of tertiary level medical care facilities for treatment of tobacco related diseases, losses due to fire hazard, ecological damage due to deforestation and disposal of tobacco related waste on the other hand, with a view to making an economic study of the impact of tobacco consumption.

The Expert Committee initially had 12 members. Subsequently by means of further notifications (the last being dated 13th September 1996), the Committee was expanded to include 21 members. Most of the additions to the Committee were from tobacco industry. The Committee held a number of meetings. During the first meeting of the Committee, the components for computing the economics of tobacco as developed by ICMR representative were discussed and agreed to. Several members of the Committee were requested to prepare specific chapters. The final report of the Committee was submitted to the government on 13th February 2001. The long time taken in submitting the report was attributed to, lack of literature, fairly difficult and delayed access to lots of data, particularly regarding the cost of treatment of tobacco related diseases, no more than nominal research and other facilities available to the Committee, the gaps in literature and data". Besides review of the scientific literature, the Committee collected data from major institutions like Department of Oncology and Cardiology, Ministries of Finance, Agriculture, Commerce, Labour, Industry Environment and Forests.

The Committee report had noted that there were hardly any benefits, which accrue either at the individual or social level to the consumers of tobacco products, the demand for which is, by and large, supplier-induced and determined. The indirect, incidental, short-run presumed benefits in terms of employment, output, foreign exchange earnings, tax revenue, etc., were there because the consumers are in various ways induced and addicted to spend a part of their incomes on this commodity. The Committee had 21 members with 14 of them directly or indirectly associated with tobacco promotion. The report highlighted the losses to the society due to tobacco use as well as economic gains accrued through its commercial trade. The benefits of its use were for different people/ groups whereas the losses were for different set of people, thereby the two components were not directly comparable. The report concluded that tobacco was a demerit product. The costs of medical treatment and other external costs incurred by the patients and society, even in their underestimated form, were considered to be so staggering as to dwarf the putative indirect financial benefits. The cost to the society due to three major disease entities in India during the year 1999 was Rs. 27,761 crores (Details in the Table 6.4). The value of all the tobacco products in India at the same had been assessed at Rs. 24,000 crores. The report had tremendous implications on the till then perceived economic contribution to the economy.

Table 6.4: Cost of Tobacco to India (1999)

	Tobacco Related Disease		
	Cancers	Coronary Artery Disease	Chronic Obstructive Lung Disease
No. due to Tobacco			
1996	1,54,300	4,200,000	3,700,000
1999	163,500	4,450,000	3,920,000
Average Cost (1999)	350,000	29,000	23,300
Total Cost India (1999) in Billion Rupees	57.225	129.05	91.336
Total Cost (1999) = Rupees 277.611 Billion or US\$ 6.5 Billion			

The issues addressed here were mainly the impressions provided by the chairman and the member-secretary as given in the report and the preface written by them. The report was not scrutinized by ICMR and thus may not provide the official stance of ICMR.

The literature on tobacco economics at that time brought it out clearly that there was a good deal of work which has been in various ways inspired by the tobacco interests, more particularly by the organized cigarette manufacturing industry. There seemed to be a great need for and large scope for many studies in this largely unexplored area.

The report categorically stated that “after the initial formation of the Committee, after a few months another group of members, representing various tobacco interests were added to the Committee. It appeared that the Committee had, for this reason, acquired a character, which created its own special problems like presentation of representation to various agencies about the thrust, etc., of the Report prior to the submission and publication of the Report”.

Some of the members belonging to tobacco promotion group through letters wanted, *interalia*, better coverage to employment, alternative crops and investments by farmers. However, despite of repeated requests for providing chapters, no material was received from them. The report included everything available at that time on those issues, including that supplied by the Tobacco Board and CTRI. The Report included a detailed section on employment which analyses the available scientific macro and other data on these activities.

Some members of the Committee belonging to tobacco promotion suggested inclusion of some elements like effect of any legislation on employment, or add figures on employment as projected by tobacco industry (but not supported by available scientific literature). This was not done as the terms of reference did not require to make any policy/ legislative suggestions or to go into the impact, harmful or otherwise, of any hypothetical legislative action. However, a report on the impact of a hypothetical legislation was sponsored by the Ministry of Commerce around the same time.

During every meeting, the members belonging to tobacco promotion groups made sweepingly subjective statements that possibly arose from their fears and apprehensions about some possible, hypothetical policy moves. The report however, incorporated and analyzed every specific point raised by the members. Out of twenty, ten members had communicated their concurrence to the report; five members expressed their non-concurrence; and five members either did not respond or their response reflected neither concurrence nor non-concurrence. It actually was clear that many members were driven by the policies of the groups they belonged to and wanted the report to be written as per the priority of their group, and not according to the critical review of the available literature.

Cost of Management of Tobacco Related Diseases

During early 1980s, need for tobacco control was increasingly being recognized by the scientific community of India. However, the suggestions were not being translated into control measures. One of the major stumbling block was the economic contribution of tobacco in the form of revenue and employment generated by tobacco industry. Some studies elsewhere had suggested that the revenue provided by tobacco industry to the governments was less than the costs posed by diseases caused by tobacco. However, no such data was available from India. An ICMR advisory committee on cancer research suggested that ICMR should carry out a study on the subject.

The preliminary work at ICMR was in the form of an inventory for comparing the benefits and costs of tobacco for India. Tobacco industry had generated a lot of data on advantages and contribution of tobacco to Indian economy, but the data on costs was only in the form of estimates. The available

estimates were on costs of treatment of some tobacco related cancers through adding the cost of drugs during the treatment period. However, in reality, the cost of any disease includes direct as well as indirect medical costs for treatment incurred by the patients, their families and their friends. It was also important to consider the costs borne by the government institutions as well as out of pocket expenditure by the patients/ families/ friends. The societal cost of any disease also included secondary & tertiary costs, due to absenteeism, premature death, indirect economic costs as a result of tobacco production, occupational diseases, etc. Assessment of the diseases burden caused by tobacco use (not the magnitude of tobacco related diseases) was also important. A look at the inventory suggested that it may be possible from the scientific literature to make a fair assessment of the disease burden of cancers, coronary artery disease and chronic obstructive lung diseases, caused by tobacco. Stroke was considered to be an important tobacco related disease category for India, but the literature was not sufficient at that time to assess the disease burden caused by tobacco. However, no data existed on comprehensive cost of any tobacco related diseases. The western studies had calculated the data based on the hospital costs for treating patients. But, Indian hospital had never calculated such costs and there was absolutely no information on the money spent or lost by the patients, which was likely to be more than the money spent by hospitals. Thus, it was decided to undertake a study on cost of management of tobacco related diseases as a first step.

As the magnitude of the disease, and studies on attributable risks of disease due to tobacco was likely to be available for three disease entities (cancers, coronary artery disease and chronic obstructive lung disease), the ICMR study was also restricted to costs of management of these diseases. No distinction on disease caused by tobacco was necessary for this study, as the management principle was (and is) the same for all cases regardless of aetiology. However, the study design was to be adjusted as per the availability of data on magnitude of these disease entities. Thus, a longitudinal study design was finalized for study on tobacco related cancers (as the data was available on incidence rates of cancer); and a cross-sectional design was considered adequate for coronary artery disease and chronic obstructive diseases (as the data on magnitude was available through prevalence studies).

1. To estimate the average cost of diagnosis and treatment of tobacco related diseases by the patients and their relatives/friends.
2. To determine variables which influence the expenditure by patients on treatment of their diseases.
3. To estimate the average cost of diagnosis and treatment of tobacco related diseases by the institution.
4. To estimate the loss of productivity due to absenteeism as a result of the illness, for the patients and their relatives/ friends.
5. To estimate the loss of productivity due to death and disability due to tobacco related cancers.

The study collected data from patients of the identified diseases and their relatives/ friends on the expenditure on diagnosis & treatment of their disease; travel for treatment/ diagnosis; additional expenses for lodging & food during the treatment period; and loss of wages because of the disease. The expenditure incurred by the treating institution on management of these patients was also collected. In case of premature death, the cost imposed upon the society (due to premature death) was also calculated.

A cohort approach was adopted for assessment of the cost involved in management of tobacco related cancers. The cohort included cases of cancers of the oral cavity (including gums), pharynx (excluding nasopharynx), larynx, and lungs. The patients were followed up for three years after registration at the

hospital or till death, whichever occurred earlier. At the time of first contact, the patients were enquired about demographic details, the duration of the illness, the health agencies contacted by them for diagnosis and treatment of their illness (specific or non-specific). The item wise expenditure made by the patients, their relatives/ friends, was recorded, under various headings, namely, consultation, investigations, treatment with different modalities, travel for the purpose, and any additional cost incurred for lodging and boarding. The information was also collected on any loss of wages for treatment of the disease, or if the disease resulted in loss of job. Specially trained medico-social workers collected the information on a pre-tested questionnaire. Intangible costs caused by the disease, and tertiary costs of tobacco use, were not attempted in this study. The follow up information was attempted every 3 months either during the scheduled visit of the patients to the hospital or by visiting the homes of the patients. All the expenses or losses by the patients incurred during the study duration were combined to provide the total expenses by the patients and their relatives/ friends. The procedure of discounting was adopted for the expenditure incurred by the patients (or their relatives/ friends) during different years, the rate of discounting being 10%. The expenditure given in the report pertained to the year 1990, the first year of the study. The information sought from the patients was on recall basis. The medico-social workers engaged in the study were trained to know the prevalent charges for various services by the private hospital in the city, approximate transport charges by various modalities of travel, charges of food in various facilities around the hospital, etc., for the purpose of further probing, to assess the reasons for variations. The costs on coronary artery diseases and chronic obstructive diseases were collected on similar lines and items (as for cancers), but on a cross-sectional basis. The reference period for collection of data was last one year.

Expenditure by various departments of the hospital was determined by the investigations rather than the diagnosis of the patients. Thus, the data collection included, identification of various investigations and service activities undergone by the patients; the determination of unit cost of various investigations and other services needed by patients of tobacco related cancers; the charges paid by the patients for undertaking the investigations, etc.; and calculation of the excess expenses incurred by the institution in treating these patients. The details of investigations & other hospital services, and charges paid by them, were collected from the patients during interview. Data was collected from various concerned departments of hospital, on the staff and the equipment available with them to perform the functions needed for treatment and investigations of tobacco related cancer cases. The hospital costs pertaining to tobacco were calculated through analysis on optimum number of investigations per day per concerned personnel, number of personnel engaged on work, salaries of the staff, the proportion of time spent for carrying out that investigation/ service, the purchase value & annual maintenance of equipment, and cost of reagents/ consumables used for undertaking the investigation/ services, and per patient costs of general maintenance of the hospital. The cost of the equipment was also obtained. The data collected from patients was on recall basis and it was accepted that the costs thus collected would be lower than actual due to expected recall bias. In case of any estimation, the lower expected value was used for calculation, thus, sticking to the principle of underestimation (in case of doubt) followed throughout the study.

The difference between the actual age at death and expectation of life at that age (as available from the Registrar General of India) was used to compute the salary loss, savings of pension to the government or the organization (in case the patient was entitled to pension), loss of family pension. The retirement age in India at that time was 58 years, which was considered as the productive age for those in job, whereas for those engaged in business the remaining life expectancy was considered as the productive age. As the age of the spouse of the deceased person was not collected, the age of the deceased was used for calculation of the family pension (which again was considered to be an under-estimate).

The average cost due to a case of tobacco related cancer was observed to be Rupees 1,34,449 (discounted to 1990 level). The patients in the cohort, spent an average of Rupees 17,965 (including loss of income

due to absenteeism), with another Rupees 4,009 being contributed by the institution in the form of various services. The loss due to premature deaths of patients of tobacco related cancers amounted to Rupees 1,12,475. Annual per capita direct expenditure by patients of CHD and COPD was Rs. 8,520.3 and Rs. 2,257.6, respectively. The annual indirect losses by the patients & state/ employer for patients of CHD and COPD amounted to Rs. 6,388.4 & Rs. 9,694.1, respectively. Thus the total average annual expenditure for a patient of coronary artery disease was Rs. 14,909 and for a patient of chronic obstructive lung disease was Rs. 11,952.

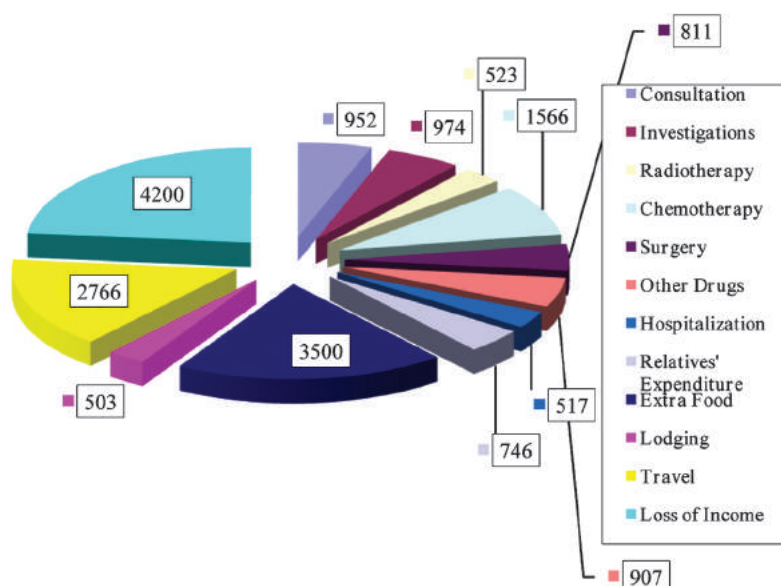


Fig. 6.7: Expenditure by patients of Tobacco related cancers (1990 prices).

Total Average Expenditure by a Patient of Tobacco Related Cancer = Rs. 17,965

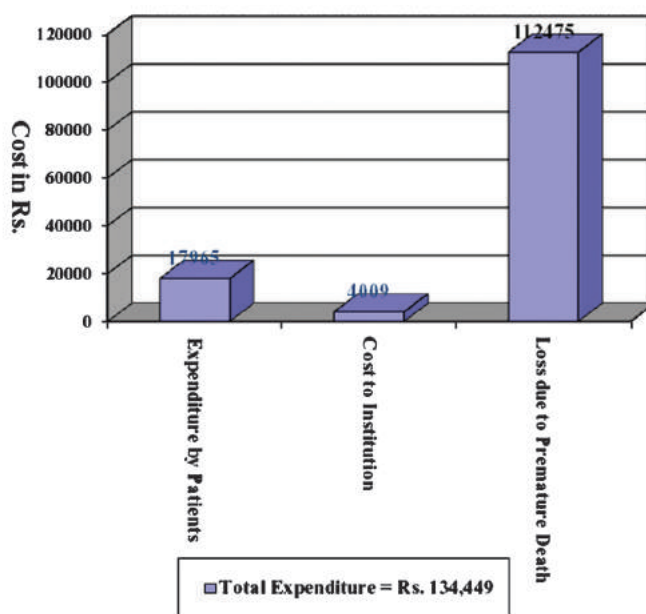


Fig. 6.8: Average per patients cost of Tobacco related cancers (1990 prices).

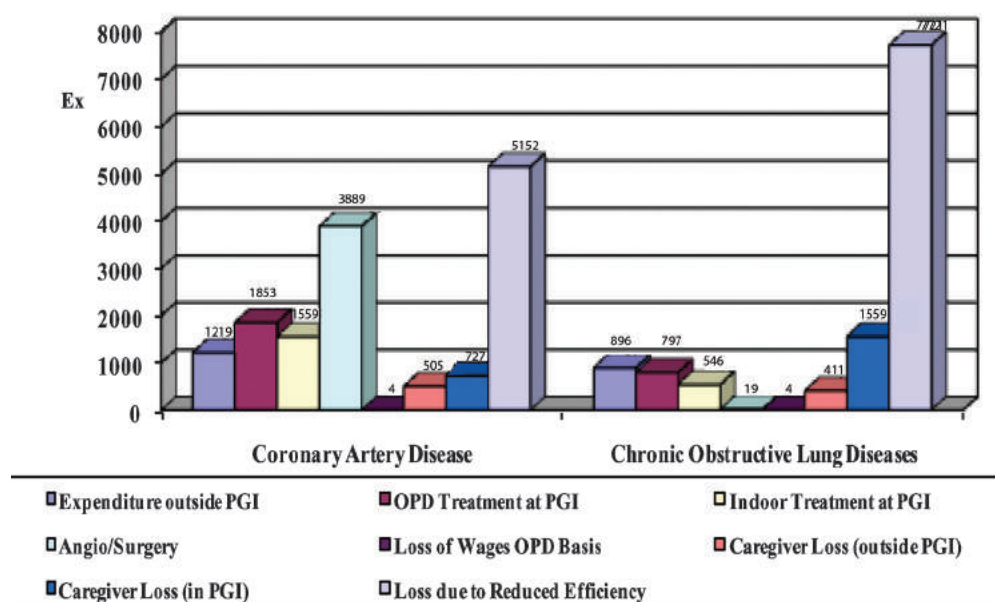


Fig. 6.9: Average expenditure by patients of Coronary Artery Disease and Chronic Obstructive Lung Diseases (1992 prices).

Total Average Expenditure by Patients of Coronary Artery Disease =Rs. 14,909

Total Average Expenditure by Patients of Chronic Obstructive Lung Disease = Rs. 11,952

There was very little difference in expenditure by the patients on items related to direct medical treatment, according to different demographic attributes of the patients. The few exceptions where such differences were noted included a lower expenditure on chemotherapy among old patients; a higher expenditure by residents of Delhi on consultation and surgery; and higher expenses on radiotherapy on patients where the intent of treatment was curative. The direct non-medical expenditure (on travel, lodging, etc.) on treatment was influenced by personal characteristics of the patients, suggesting a variation in expenditure due to their paying capacities. Better occupation, greater distance of the hospital from their place of residence, younger age of the patient, and curative intent of treatment (probably influenced by longevity and higher degree of follow up), was associated with higher expenditure.

The study provided important inputs for computing the overall cost of three major disease entities to the country. The earlier estimates on the subject related only incomplete estimates on direct medical costs of these diseases. Thus, the study was a turning point for country's policy.

The study did require meticulous planning in terms of identification of components which may be required while analyzing the data. It was clear that if some component is missed at the time of planning, the study may need to be conducted all over again to get information on that aspect. Therefore, the phase of pre-testing of the questionnaire was extremely critical. The questionnaire for this study was finalized after three rounds of pre-testing. This also provided an opportunity to identify the diagnostic tests and therapies the patients preferred from outside the base hospital and thus prepare the interviewers on probable involved issues.

The study also required very close day-to-day monitoring, especially in the beginning when the interviewers required very frequent guidance. Although not anticipated during planning, it became clear very soon that there would be no option but to adopt the principle of underestimation of costs in case of doubt.

The study very clearly showed an important role of socio-economic factors in determining the overall costs of the diseases. The investigations and management practices had also changed since that time.

There has been major increase in availability of cancer treatment facilities in private sector in India. The GDP and economic status of Indian population has also increased since the conduct of the study. Data on magnitude of more tobacco related diseases like stroke may also be available. The above highlights that if a similar study is carried out today, the cost of tobacco related diseases is likely to be higher than those observed by the study.

The above data was converted into the total cost for the country due to three major tobacco related disease entities. The data based on the final data on cancer related study and preliminary results from the coronary artery disease and chronic obstructive lung diseases was presented during a WHO- Government of India conference in January 2000. The cost to the country due to three major disease entities for the year 1999 was estimated at Rs. 27,761 crores. The data changed the strategy of advocacy by the tobacco promotion groups in the country, who used to harp on the economic advantage of tobacco at any indication of government taking a major step on tobacco control. Obviously, the study helped in making a paradigm shift in the policy related decision-making of the government.

Support for Tobacco Control

The rich experience in research related to tobacco was utilized in helping and guiding decision makers in matters related to tobacco control. Some of the specific areas where major inputs were provided included, economics of tobacco in India (through an expert committee constituted by the Ministry of Health & Family Welfare), health hazards of pan masala containing tobacco (through expert committee constituted by the Director General of Health Services), preparation of health education messages to be broadcast through television (at request of Ministry of Health & Family Welfare), provision of facts and information to the Parliament's Committee on Sub-ordinate Legislation on the Cigarette Act, organization of International conference on Global Tobacco Law: Towards a WHO Framework Convention on Tobacco Control, and organization of an Inter-Ministerial Committee meeting to discuss issue related to framework convention on tobacco control. ICMR was also represented in the meeting of the WHO's Working Group on Framework Convention on Tobacco Control.

Development of a Comprehensive Tobacco Control Legislation

The need for a comprehensive legislation on tobacco control was accepted in 1986 itself, wherein the gravity of tobacco problem was noted along with the fact that ineffective of the existing Cigarette Act. It has to be realized that any legislation is enacted after consultation and agreement with all ministries and departments of the government. While the Ministry of Health & Family Welfare was convinced, it took sometime before other ministries could be sensitized about the need for major action on tobacco control. The Ministry of Health and Welfare started consultations in early 1990s with other ministries, which suggested that an expert group on study of tobacco economics in India be constituted. A national workshop on tobacco control held in 1991 identified major tobacco control activities (both for smoking and smokeless tobacco use) in the country. In the meantime, a Parliament Committee on Cigarette Act of 1975 was constituted in 1995. It was prudent that the recommendations of this Committee are considered by the government. In fact, the report of this Parliament Committee helped a lot in not only proposing a strong legislation for tobacco control, but also took initiative in requesting various departments to initiate activities in their sphere for tobacco control. The observations of the Committee were based on the consultations carried out with various concerned departments, parties, companies, NGOs, etc. In fact several inter-ministerial meetings were held during the next few years to suggest and monitor actions taken on the recommendations of the Committee. ICMR played a very active role during these consultations. The details of the observations and recommendations of the Parliamentary Committee are briefly given here.

Observations in the 22nd report of the Committee on Sub-ordinate Legislation:

The Committee observed that the statutory warning on cigarette packs laid down by the Cigarette Act 1975 has not proved effective. Limitations of the warning identified by the Committee included: its language, being only English; the monotony of the message; small lettering and its presence on only one side of the pack; its size, being smaller than the brand name; the absence from the purview of the Act of *bidis*, the most prevalent form of tobacco smoked in India; the absence of a maximum permissible limit for tar and nicotine content and the absence of tar and nicotine concentrations on the packs. The Committee observed that the existing provisions on tobacco advertising were ineffective & insufficient; glamorous tobacco advertisements create illusion about smoking being pleasurable, sophisticated and helpful for achieving success; sponsorship of sports and cultural events gives the tobacco industry high visibility; tobacco industry's claim that advertising is for brand switching and does not increase tobacco consumption was wrong and grossly misleading; risk for disease also exists for those who passively inhale tobacco smoke; and ban on smoking in public places in other countries has resulted in widespread change in attitudes, leading to smoking cessation. The Committee also noted that parental practices at home and peer habits at school affect initiation of tobacco habit, which makes it essential that a coordinated, rational educational strategy is carried out at both these levels. The Committee also observed the success of a nationwide radio programme, 'Radio Date', carried out jointly by AIR and ICMR, as well as the anti-tobacco community education projects of ICMR. The Committee also felt that increasing the production of tobacco on the one hand and framing comprehensive legislation to curb its use cannot be justified, and the need for taking action on substitution of tobacco crop by some other crop and addressing the issues of livelihood of millions of workers engaged in the manufacture of *bidis* and cigarettes.

Recommendations in the 22nd report of the Committee on Sub-ordinate Legislation

The Committee provided detailed recommendations on various facets related to tobacco control which related to all concerned sectors, like health, agriculture, commerce, sports, etc. Salient aspects for various actions are given below.

Statutory Warning on Tobacco Products: The Committee recommended that the health warning(s) should appear on all tobacco products; it should be worded strongly, be rotated periodically and be supplemented by symbols or pictures; should be in English and regional language(s); lettering of the warning should be as large as that of the brand name, and be displayed on both sides of the pack; concentrations of tar and nicotine should be printed on packs and cartons of all tobacco products and maximum permissible limits be fixed; individual cigarettes should also bear the health warning; health warnings should also be displayed prominently at every shop where tobacco products are sold; and imported cigarettes and other tobacco products must meet the statutory warning requirements of India.

Tobacco Advertising: Government should make provision in the proposed legislation for a total ban on all forms of advertisement for tobacco, with stringent penal provisions for violation of the law. There should be a total ban on the sponsorship of major sports events by cigarette companies.

Prohibition of Tobacco Smoking in Public Places: The Committee recommended a complete ban on smoking at least in all public places where large numbers of people are expected to be present for long periods, such as hospitals, dispensaries, other health-care establishments, educational institutions, conference halls, cinemas, theatres, offices, all workplaces and railway waiting rooms. There should also be a complete ban on smoking in public transport systems, domestic flights and Government vehicles.

Social Awareness about Tobacco: People should be educated about the effects of smoking and passive smoking; non-smokers who object to smoking and other forms of tobacco use should have legal backing;

anti-tobacco education should be compulsory in schools and colleges; teachers should not smoke within school premises; sale of tobacco and tobacco products should be banned in the vicinity of schools and colleges, with provision for punishment of vendors for violation; persons below a minimum age (say, 18 years) should not be sold cigarettes; social awareness should be created through electronic and print media; as far as possible, scenes in which characters smoke in an obtrusive manner should not be included in programmes shown on television; and the Government should allocate adequate resources and personnel to carry out effective anti-smoking education.

Alternative Cash Crops to Replace Tobacco: Initiative should be taken by the Ministry of Agriculture to persuade farmers to switch over to alternative crops, first on an experimental basis, and the results then widely publicized to convince farmers of its viability; farmers be given anti-tobacco education; government might consider giving monetary assistance to farmers to help them change to alternative crops; Indian Council of Agricultural Research develop new techniques for high-yield varieties and for fertilizers for alternative cash crops; research be continued to explore alternative uses for tobacco. Tobacco Board should not promote internal consumption of tobacco, although the Committee had no objection to production of tobacco for export; a gradual approach be adopted wherein efforts were made to phase out cultivation of tobacco for human consumption over time; and conduct a study of the resources required for rehabilitating workers in tobacco production and the areas in which they could be absorbed, to formulate a concrete proposal in this regard.

The Committee report provided a lot of support for tobacco control activities and helped in further expanding and widening the scope of the comprehensive tobacco control legislation. Various ministries did in fact start experiments to achieve the recommendations of the Committee. ICMR strongly recommended the initiation of the Framework Convention on Tobacco Control (FCTC), during the initial consultation of WHO with various countries. The ICMR very actively participated in developing a country stance for FCTC, and represented India in the two meetings of the Working Group of FCTC which laid down the provisions of FCTC based on scientific knowledge. ICMR also took very active part in suggesting unbiased approach for tobacco control, developing stance on various provisions, likelihood of considering FCTC provisions under the country's legislation, organizing meetings to discuss various issues, preparing background notes for various Ministry meetings including inter-ministerial meetings on FCTC, tobacco control legislation, consideration of some legislations like ban on smokeless tobacco use, warnings on smokeless tobacco products, tobacco cessation clinics, educational strategy, alternatives for tobacco farmers, work related to health hazards of pan masala, economics of tobacco use in India, etc. A paper on multi-sectoral approach for tobacco control was also prepared.

Anti-tobacco Community Education

The study was to assess if anti-tobacco education can effectively be imparted to community, through use of existing government infrastructure. Such a possibility was expected to be cost-effective, long-lasting, and more acceptable to the community. The study was started in 1985, at a time when quitting tobacco was considered to be extremely difficult (which still remains to be the prevailing opinion among tobacco users) and the health services had not perceived the need for an active role in this direction. No tobacco cessation guidelines or services existed, neither in the government nor in private sector. At the time of initiation of the project health services and schools were considered to be two existing infrastructure for the purpose of testing their response to carry out an organized anti-tobacco education activity.

The multicentre task force project of ICMR was to study the feasibility of involving existing infrastructures in anti-tobacco community education. The outcome measures were to find any change (reduction as a positive outcome) in tobacco use prevalence brought about by (i) reduction in tobacco use among tobacco habituees; and (ii) reduction in uptake of tobacco by non-users.

The intervention by health services was directly on the community by para-medical health personnel. The direct intervention by school network was on the children attending schools but the indirect and secondary effect was expected on the community and the change in tobacco use was measured in the overall community.

The study was carried out in Karnataka (near Bangalore), Kerala (Trivandrum district), Uttar Pradesh (Mainpuri district) and Goa (entire Goa state). At the time of initiation, health services were proposed to be involved in all the four areas. In Goa one section was to carry out intervention through schools alone, the second through health services and schools, with the third to act as control. The willingness to carry out the intervention by health services was high in Karnataka, whereas it was low in Kerala and minimal in Goa and Uttar Pradesh. The workers considered it as an additional burden and did not carry out intervention. The study in Kerala was truncated to see the work of some para-medical workers who had opted to participate. The study in Uttar Pradesh at the advice of expert group was modified to raise a group of community volunteers to carry out intervention for a short period. The Goa study compared two areas with school based intervention with one area acting as control. During the preparatory phase, the first control area in Karnataka was considered to be too close to the city (which may make analysis difficult due to possible overall education through other sources) and thus a second control area was added.

The study involved one year of preparatory phase, wherein the concerned workers were trained, requisite educational material was prepared and a baseline survey on knowledge, attitude and practice (KAP) was carried out on a representative sample of the community. The intervention was for a period of 3 years in Karnataka & Goa, and for a period of 1 year in Kerala and Uttar Pradesh. The KAP survey was repeated after intervention. The data analysis was also carried out during the fifth year of the study. The trained primary health workers in Karnataka and community health volunteers in Uttar Pradesh also examined the oral cavity to identify and classify lesions as normal and abnormal. All the abnormal lesions identified by the workers were advised to attend a clinic specially arranged by the investigators for this purpose.

Pre-tested health education material was prepared by the project staff and used by the existing infrastructure personnel. The main health education aid available with the health workers in Karnataka was a photograph of oral cancer patient, which helped in initiating the discussion with individuals. The educational sessions were carried out individually or in groups by the health workers, during their routine visits to the community which happened approximately once in 15 days. This educational procedure was aided by organization of mobile exhibitions of posters in the community. Similar procedure was followed by the community volunteers in Uttar Pradesh. The health education material in Goa was more extensive and included pamphlets, posters, booklets, models, exhibitions and dramas. The school children were provided educational material for taking to their homes and read them out for their parents and neighbors in case they were illiterate.

The overall reduction in the prevalence of tobacco usage in Goa was 11.8% among men and 9.1% among women in intervention zone 1; 13.4% among men and 13.3% among women in intervention zone 2; and 2.0% for men and 10.2% for women in control zone. The proportional reduction in the rate of tobacco habit was 33.5% in men and 45.5% in women of intervention zone 1; 32.6% in men and 50.4% in women of intervention zone 2; and 8.5% in men and 33.3% in women of control zone (Fig. 6.10). Thus, the school based intervention was considered to be effective and feasible for the community. Based on the experience of this project, Ministry of Education, state of Goa, included an 8 hour course on tobacco as a part of co-curricular activities for standard five and above.

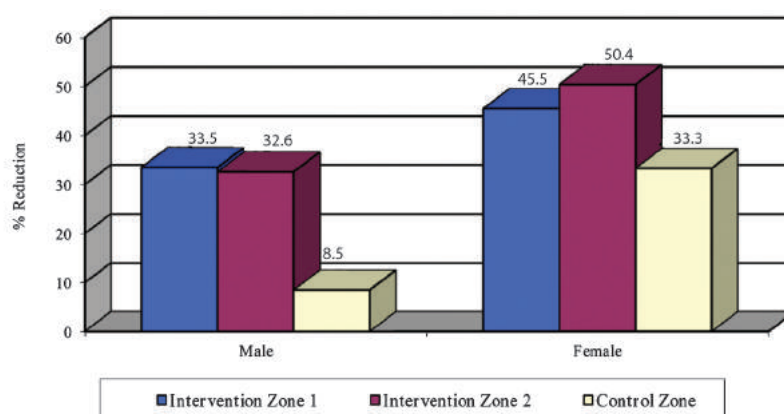


Fig. 6.10: Proportionate reduction in Tobacco use in Goa.

Intervention at Bangalore centre achieved a reduction of tobacco habit in experimental area, amounting to 5.7% in the males and 6.9% in the females. The control area I showed an increase of 3.8% among male and 7.8% among female, while in control area II, among men there was a 2.9% increase in habit and 4.6% decrease among females (Fig. 6.11).

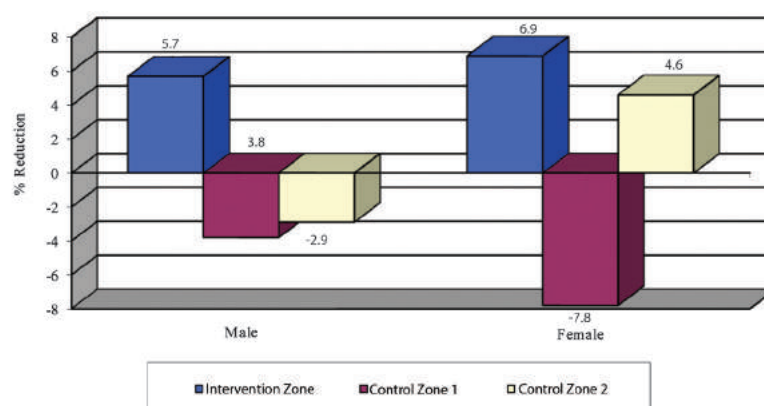


Fig. 6.11: Proportionate reduction in Tobacco use in Karnataka.

The intervention through community volunteers at Agra centre showed that 26.3% males and 10.5% females left tobacco and another 10.1% males and 4.3% females as likely quitters (6 months have not passed after leaving tobacco), after an intervention of about one year. The project at Trivandrum centre could not achieve optimum participation of health care workers. The nine workers, who worked on the project, referred 408 patients out of which 258 reported, giving a compliance of 63.2%. About 59% of these were found to have cancers (10) or pre-cancers. Of the 10 cancer cases five were in stage I & II. 29 old cases were also examined, out of which four recurrences were detected.

The study demonstrated the feasibility of undertaking anti-tobacco education by health services, if receptive. A major lesson learnt from the study was that local health services should be approached through the administrators at the central/ regional levels. The Karnataka centre started the project by contacting the state health administrators at Bangalore, who in turn issued a circular for the concerned area to carry out the intervention as per the instructions by the investigators. The limited data from Kerala also suggested that if all the workers were receptive, the gains could have been expected. The Uttar Pradesh centre showed that community was willing to work for this purpose, if guided adequately.

The Goa centre results were unique, by showing that children could be powerful tool to bring about a social change in the community. This happened through careful planning for an organized plan supported under their curriculum. The schools did require some health education aids (like video players and screens/ TVs), which were provided to them. The cooperation provided by the health infrastructure was also extraordinary.

While the study showed the efficacy of simple as well as multiple health education aids, the study did not plan efficacy of individual health education strategies or packages. The study reiterated that well conducted educational activities add to the learning of individuals at any age, which actually changed the impression of the subject as an interventional activity and not under the research domain. The study also helped in learning and hypothesizing that health education aids should be aimed at desired change for the target community and the intervention should be considered as a package of activities relevant to the local area.

Radio DATE (Drug, Alcohol & Tobacco Education)

Background

The project Radio DATE was a collaborative effort of Indian Council of Medical Research (ICMR) and All India Radio (AIR). The acronym DATE stood for Drugs, Alcohol and Tobacco Education. The project was conceived and agreed when a team of high officials from AIR visited ICMR with the request for help for a small educational activity. At that time India had only government controlled radio system. Since the officials belonged to the central set up, ICMR suggested conduct of a nationwide programme on education on tobacco, alcohol and drugs. The suggestion was readily agreed by AIR and the name of the programme (with acronyms suggested by ICMR) was agreed during the same meeting. The discussions also finalized that the first episode would be aired from World Health Day of 1990 (7th April 1990). ICMR agreed to provide the subject areas for different episodes along with the scientific contents for that purpose. AIR agreed to develop the episodes with the help of provided scientific contents and translate them to local languages. The suggestions on experts to feature in the programme were also provided by ICMR, who were contacted directly by AIR. The coordination for the broadcast of episodes was done by AIR, whereas ICMR helped in pre-testing and analysis of the data. The surveys to assess the reach and efficacy of the programme were carried out by ICMR.

Overall Objectives

To assess the effect of Radio as an instrument of social change in regard to Drugs, Alcohol and Tobacco consumption in the community.

Specific Objectives

Developmental Objectives

1. To design and develop a series of radio programmes aimed at generating awareness in the community on the holistic approach to health hazards of drugs, alcohol and tobacco and their management.
2. To inform and educate the community in promoting healthy life styles, which are impaired with the consumption of drugs, alcohol and tobacco.

Evaluation Objectives

3. To assess the immediate effect of such radio programmes in a defined group of radio listeners (registered audience).
4. To assess the reach of the programme in general population.

Methodology

The radio programme was in the form of 30 weekly episodes of 20 minutes each. Ten episodes focused on tobacco, eight each on alcohol and drugs, and two episodes on legal aspects. The introductory and concluding episodes touched all the three themes. The episodes were broadcast from 84 stations of All India Radio (out of 104 existing at that time) at prime time, simultaneously in sixteen languages.

The major content areas for different episodes were finalized by ICMR with the help of an expert panel, members of which also provided background scientific material to be provided to the team of AIR producers. The AIR developed the storyline combining the scientific contents and also featured experts. The prototype was developed in Hindi and was sent to selected radio stations of All India Radio for translation in the concerned regional language, as per the specified guidelines. While developing regional language episodes, experts from the local area with ability to communicate in the regional language were invited. Thus, the contents and delivery modality was closely controlled by AIR, although each episode was finally translated in local language. The translated episodes were sent to all the radio stations requiring the concerned language episode. The broadcast was during a specified time with the exact time to be decided by the radio station but definitely between 8.00 A.M. and 9.00 A.M. on Sundays, with a repeat broadcast during the week, generally in the evening. The name Radio DATE was used by radio stations all over the country, but some radio stations also coined and used names in local languages.

Evaluation of the reach and effect of the tobacco component of the programme was carried out through two community based surveys, carried out after the broadcast of tobacco episodes (which was the first topic to be covered). These surveys were among randomly selected persons above 15 years of age in selected rural Goa and Karnataka, where no organized anti-tobacco programmes were being conducted. The control areas of an ICMR project on anti-tobacco community education were used for this survey. The intervention phase on the earlier project had just finished in that area and the only intervention that had happened in these area was through the project Radio DATE.

A call for sending posters was given to audience by AIR, during the last episode of the programme. These posters were screened by an expert group to select three best posters for each subject. The number of entries also helped AIR in assessing the reach of programme.

Observations

The ICMR surveys carried out immediately after the broadcast of tobacco episodes, showed that the potential listeners of radio comprised 80.4% of the population in Goa and 59.1% of the population in Karnataka. In Karnataka 31.6% of the potential listeners and in Goa 26.7% of the potential listeners, had heard at least one of the first eleven episodes (on tobacco). Education level was observed to be the determinant of reach in Goa; while education and occupation influenced the reach in Karnataka.

The mean number of episodes heard by the listeners was 2.6 ± 1.46 in Goa and 2.57 ± 1.13 in Karnataka. The factors associated with listening higher number of episodes included higher education, male sex, non-usage of tobacco and radio ownership in Goa; and, higher education, radio ownership and caste in Karnataka. Most of the listeners considered the programme to be very good or good, and felt that it would have effect on the tobacco users to quit their habit as well as on children to prevent the initiation of habit. About 4% tobacco users in Goa and about 6% users in Karnataka quit their habit after hearing the programme (Fig. 6.13). About 98% to 99% of the listeners expressed that such programmes should continue.

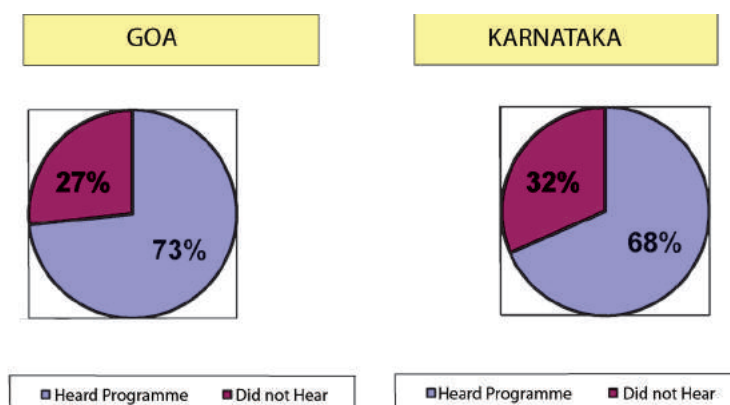


Fig. 6.12: Proportion of persons hearing the programme in rural Goa and Karnataka.

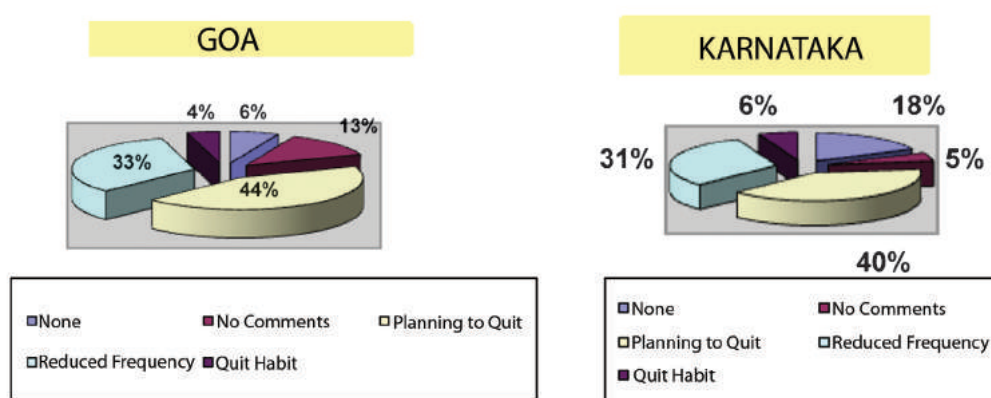


Fig. 6.13: Effect of Radio DATE in rural Goa and Karnataka.

Summary

All major tobacco control programmes have tended to use mass media, either as a major plank or as a mechanism to inform people about the intervention. Scientists generally favoured television over radio, believing that it can provide behavioural skills by actors or by programme participants. The radio at the time of carrying out the study, was considered to have a bigger reach in the Indian community (97% vs 15%) than television. The Indian radio network had capability of translating the programmes in all 17 official languages of India and the interest shown by the radio managers for the programme was exemplary. Thus, the study group considered that radio in India was expected to have the same advantages as reported for television. At the time of planning this project, there was only one scientific study in a small area of USA, which reported useful role of radio in reducing tobacco use in the community. This ICMR project was the largest project of its kind wherein the entire India was covered.

Quick planning and development of the episodes was the hallmark of the study. The broadcast started about 14 weeks after this mega project was conceived. The episodes had been prepared in about 4 months after conception of the project.

While the programme was on air, print media extensively reviewed it. These comments were encouraging and at times also provided suggestions on the contents for future episodes. Most of these suggestions had been incorporated and thus required no change in the episodes which had already been prepared. However, two additional episodes on legal aspects were prepared based on suggestions of mass media and internal discussions.

The project helped in generating interest on developing stories on the three chosen subjects (tobacco, alcohol & drugs) on their own. After the programme, the sole existing television channel at that time in India (Doordarshan), made a 20 minute film on the making of "Radio DATE", and televised it at prime time. One year after the project, ICMR made one and a half minute spot on tobacco and provided it to Doordarshan, which used it as a time filler, between the main programmes (at prime times as well as at other times). The film was run for more than 500 times (about 300 times during the first year itself) but ICMR did not have to pay for it. After Radio DATE, requests from print media for background material on tobacco also increased. The radio stations also prepared and aired programmes on their own. An evaluation of number of programmes and duration accorded to the subject of tobacco on Delhi station of All India Radio, showed that both the annual number of programmes and time to the subject increased after the programme Radio DATE, sustaining for the next two years and decreased slightly thereafter. All this publicity was achieved without any expenditure.

The programme also received appreciation from scientific community as well as politicians. The biggest political accolade was the appreciation from the Union Health Minister of India and the request that more such programmes should be developed. Audio cassettes prepared and distributed to all ICMR institutions for further duplication and distribution, as well as for use during their educational activities.

The cost of such a mega programme would have been massive or even prohibitive. The project was possible only because of the active participation of the radio with the scientific group of ICMR, highlighting the importance of both media and scientific managers. The broadcast was during the regular broadcast hours of AIR and was considered as an opportunity availed. Thus, the cost of broadcasting was not considered by AIR or ICMR. The additional expenditure for preparation of the programme was towards arranging the participation of experts, for travel funds for the team, and for consumables. The additional expenditure for the programme was Rs. 6,35,000 in 1990, which amounted to Rs. 0.58 per 100 listeners (US \$ 0.03 per 100 listeners, at 1990 conversion rate). The cost per 100 population was Rs. 0.07 (or US \$ 0.004).

The project also provided a rich experience on collaborating with media (both electronic and print media). Most media personnel would not have time and/or aptitude for extensive research or collection of information. While anti-tobacco groups wish for wide dissemination of information, the media also looks for material, which should be new (to them), non-repetitive, crisp & in simple language, to be used by them as a base to the total story. Both scientists and media personnel are looking for gains from such stories, but the type of credits to be gained by the two groups are different. Thus, preparation of such stories is a mutually beneficial exercise. The interaction also highlighted the need for considering the aptitude of target group rather than considering mainly the sensitized group of audience. Thus, the active role of both media managers and scientific managers can result in execution of projects which in isolation may be an extremely difficult proposition.

Oral Cancer

Oral cancer accounts for around 30% of all cancers in India and is the most common cancer among Indian men. India has one third of oral cancer cases in the world. As per the Globocan, 2018 data, there have been 119,992 new cases and 72,616 deaths due to oral cancer (including lip cancer) in India. In general, more men suffer and die from oral cancer than women.

Risk Factors for Oral Cancer

- Tobacco consumption

All forms of tobacco, including cigarettes, beedi, pipes, cigars, and chewing (smokeless) tobacco, can cause oral cancer. Keeping tobacco quid inside mouth may lead to cancer of cheek, gums and inner cavity of mouth. Paan and betel nut are also causal agents.

- Alcohol consumption

Alcohol increases the risk of oral cancer. The risk is about twice as high in people who have 3 to 4 alcoholic drinks per day compared to those who don't drink alcohol. The risk of oral cancer is even higher in people who use both alcohol and tobacco.

- Sharp teeth or ill fitting dentures
- Diet: Lack of proper nutrition and diet low in fruits and vegetables: linked with an increased risk of cancers of the oral cavity and oropharynx (*World Cancer Research Fund/American Institute for Cancer Research. Food, Nutrition, Physical Activity, and the Prevention of Cancer-a Global Perspective. Washington DC:AIICR;2007*).
- Human Papillomavirus (HPV)

Infection with certain high risk HPV types increases the risk of oral cancer, especially in younger people.

- Weak immune system

People with weakened immunity are more prone to suffer from oral cancers. Certain immune deficiency diseases at birth, radiotherapy and chemotherapy, medicines given to organ transplant recipients and the Acquired immunodeficiency syndrome (AIDS) may be responsible for weakened immune system.

- Exposure to sun's ultra violet rays: may cause lip cancer.

ORAL-PRE-CANCERS: USE OF PAN –MASALA:

Study on Association of Oral Pre-cancers with Use of Plain Pan Masala

In addition to tobacco use (both smoking as well as smokeless forms), plain pan masala use is of public health importance to India. While studies are available on effect of smokeless tobacco on oral cavity, the effect of plain pan masala and areca nut are not well proven. It is also known that oral cancer is often preceded by pre-cancerous conditions of the oral cavity. Some precancerous conditions like oral submucous fibrosis have increased in recent times, which coincide with the increase in the habit of use of gutka or pan masala (with or without tobacco). Areca nut is used in India as a component of betel quid (with or without tobacco), pan masala (with or without tobacco) or gutka. Areca nut is also used as such (plain or scented supari). Any confidence of pan masala being carcinogenic is derived mainly from IARC's recommendation about betel quid being carcinogenic. These studies have not tried to differentiate the different forms of areca nut usage. Moreover, epidemiological studies on carcinogenicity of pan masala mixture were not available. The substance has however, been shown to be mutagenic. Thus, an epidemiologic study was needed to fill the gap in knowledge on carcinogenicity of plain pan masala.

The study objectives were to conduct a survey to obtain the (i) prevalence of use of plain pan masala; (ii) prevalence of various oral precancerous lesions among users of plain pan masala, as well as among and

non-users of pan masala / tobacco/ areca nut; and (iii) to calculate the odds ratio of development of oral pre-cancerous lesions due to use of plain pan masala chewing.

Major limitation for a case control study on the subject was the non-availability of sufficient number of oral cancer or pre-cancer cases with history of use of plain pan masala alone. Therefore, the study is in the form of a survey to be carried out among 425,000 persons above 15 years of age to provide about 1,700 persons using plain pan masala. The study was carried out in population residing within the area encircled by the Ring Road of Lucknow city. The data on use of plain pan masala, tobacco and areca nut was collected through personal interviews of all persons by trained social workers after a house-to-house visit. The examination of the oral cavity was carried out simultaneously by trained doctors/dental surgeons. Regular supervisory visits were made to address any field level problem. The persons with oral pre-cancerous conditions were advised appropriate treatment.

The study pre-tested and finalized a questionnaire, and procured the necessary material for survey and examination of the oral cavity. In summary, the study tested the feasibility of using the ICMR questionnaire on collecting detailed information on tobacco habits.

Carcinogenicity of Pan Masala containing Tobacco and Consideration of Ban on Use of Tobacco in Smokeless Form in India

A major change in modality of use of smokeless tobacco occurred in early 1970s, when pan masala was introduced in Indian market. Pan masala contains the ingredients of traditional betel quid, except the betel leaf. Pan masala removed the inconvenience of frequent preparation of pan due to its ready to use nature and longer shelf life. The increasing popularity of pan masala resulted in introduction of pan masala-containing tobacco and gutka. Gutka has been defined by manufactures as tobacco with ingredients of pan masala, and thus essentially would be similar to pan masala-containing tobacco but with higher proportion of tobacco. Availability of this product in small sachets further increased its popularity. In fact, the ICMR survey on tobacco use in 2001 showed that pan masala containing tobacco was the commonest tobacco modality among population below 40 years of age.

The Indian scientists in 1980s pointed out that certain toothpastes and toothpowders being sold in India contained tobacco. This product produces dependence among its users forcing them to brush their teeth several times a day with this preparation. Therefore, the Directorate General of Health Services and Ministry of Health and Family Welfare decided to ban this formulation, which was contested by the manufacturers in High Court and later in Supreme Court. In 1994, while upholding the validity of the government of India notification for banning the addition of tobacco to toothpaste and tooth powder, the Hon'ble High Court of Rajasthan directed the Central Government to appoint a committee of experts to assess the public health impact of use of tobacco in pan masala, gutka, and to prohibit the manufacture of these products if found to be injurious to health. The Directorate General of Health Services constituted an Expert Committee on use of tobacco in pan masala, gutka, etc., in August 1994.

The commonest modality of tobacco use in smokeless form in India till mid 1980s had been the betel quid containing tobacco, whose causative role in oral cancer had been reviewed and accepted by International Agency for Research on Cancer, in its monograph on smokeless tobacco use. The expert committee of the Directorate General of Health Services on pan masala-containing tobacco and gutka discussed the available literature on the subject during its first two meetings. Based on the published literature, most scientists believed that tobacco use in any form was harmful to health. However, some scientists pointed out that this assumption may hold good for programmes on awareness on tobacco, but in the court of law specific evidence would be necessary if there was a move to ban these products. These scientists

believed that sufficient evidence was not available to prove the association of pan masala-containing tobacco with oral cancers and it would not be appropriate to suggest a ban on these products. The ICMR was requested to prepare a review paper for consideration of the expert committee. The background paper was published after the committee had provided its recommendations. The paper considered all the studies on pan masala-containing tobacco, related mixtures, and its individual components. The interpretation on carcinogenicity of pan masala containing tobacco was made based on (i) studies on pan masala containing tobacco; (ii) studies on similar mixtures; and (iii) studies on effect of individual ingredients of the mixture and the likely effect of their combination. The literature showed a limited evidence of pan masala containing tobacco being carcinogenic to animals. Proportion of areca nut and tobacco in pan masala containing tobacco is between the proportion of these substances in two known tobacco-areca nut mixtures of India (Mainpuri tobacco and Mawa). Studies on Mainpuri tobacco indicate it to be carcinogenic, while literature suggests an association between mawa use and oral sub-mucous fibrosis. The review concluded that based on human evidence on similar mixtures and limited studies on pan masala containing tobacco, pan masala containing tobacco is expected to be carcinogenic. Direct evidence of its carcinogenicity in human beings is not likely to be available at present, even with well designed studies. This is due to the fact that the incubation period of oral cancer is 15 to 20 years, and the habit of chewing of pan masala containing tobacco is less than the incubation period.

The conclusions of the background paper were accepted unanimously by the committee during its third meeting in 1996. The committee however, debated if indirect evidence would be enough in the court of law. The chairman of the committee felt that direct epidemiological evidence was must if a drastic action like banning the product was to be taken. Therefore, preparation of epidemiological research projects to assess the association of the product with pre-cancerous conditions was suggested. The issue was debated again during the fourth meeting of the expert committee in September 1997 under the chairmanship of the Director General of Health Services. By this time there were many reports from ENT and dental specialists about increase in number of cases of sub-mucous fibrosis. It was pointed out that even an epidemiological study on pre-cancerous conditions may not tremendously change the scenario. The review had tried to assess the health hazards of the product in many ways and a direct question was posed to the committee if they could think of any possible reason to conclude that the product may not be carcinogenic. It was also argued that it would be against the principles of preventive medicine if a decision was not taken and people are allowed to fall prey to diseases for the next 15 to 20 years, which was expected to be the minimum period for availability of direct evidence. The argument was accepted and the expert committee unanimously concluded that pan masala-containing tobacco and gutka, are hazardous to health, and the hazards are likely to be similar to those of conventionally chewed tobacco, mainly sub-mucous fibrosis and oral cancers. As suggested by the paper, the committee also felt that the incubation period of the disease caused by pan masala containing tobacco is likely to be shorter than after the use of conventionally used betel quid with tobacco. The committee felt that the background paper has used the principles of preventive medicine in pre-empting the health hazards of newer mixtures. It was felt that at present, modified tobacco mixtures can be introduced in the market without any license, and it would not be appropriate to wait for the next 20 years to carry out human studies on its carcinogenicity and then recommend its ban.

The opinion of the expert committee was considered by the Central Committee on Food Standards (CCFS), which is a statutory committee under the provisions of Prevention of Food Adulteration Act 1954, to advise the Government on matters relating to food safety and quality. The CCFS examined the report in November 1997, along with the known literature on health hazards of smokeless tobacco use.

The committee noted that Section 23 of Prevention of Food Adulteration Act 1954, already empowers the Central Government to prohibit the sale of any substance which may be injurious to health when used as food or as an ingredient of food. Taking into consideration all aspects of health hazards of tobacco chewing, the committee recommended a total ban on use of tobacco in pan masala/ gutka or as an ingredient in food item or as such. Based on the recommendations of the CCFS, the Directorate General of Health Services proposed a notification for consideration of the Ministry of Health & Family Welfare, aimed at banning the manufacture, production, sale or trade of smokeless tobacco.

A sudden ban on smokeless tobacco could be considered as a drastic recommendation for a country where it is practiced by about half of tobacco users. As a first step, the Ministry of Health & Family Welfare held discussions with some experts in the field, to reassess the technical confidence in the findings and also the likely implications of such a ban. The change in tobacco use habits because of non-availability of smokeless tobacco was also discussed. It was felt that majority of people may not consider cigarette smoking as an alternative due to the comparative differences in cost of smokeless tobacco products and cigarettes. However, shift to bidi smoking was a distinct possibility. In view of the multisectoral implications of this step, it was decided to discuss the issue in an inter-ministerial meeting. ICMR prepared a background paper for the meeting. It was felt that if a ban on smokeless products was agreed by all sectors concerned with tobacco, the Ministry of Health & Family Welfare would need to establish a large number of tobacco cessation clinics, so that people tend to quit their habit of smokeless tobacco use rather than shift to smoking. It was agreed by all sectors of the government that both tobacco smoking and tobacco use in smokeless form are injurious to human health. It was felt that suitable campaigns need to be initiated to reduce the prevalence of tobacco smoking in the first phase. It was also indicated that the tobacco habit being as addictive as cocaine or heroin, tobacco users would require help in cessation of their habit. Further deliberations on the issue were received from various ministries providing operational issues connected with sudden ban on smokeless tobacco. In view of this, sudden ban of all tobacco habits was not agreed. Some states banned sale of gutkha as a short term measure, in view of the recommendation of CCFS. As the legislative provisions enable such a state action only for a short period, the Supreme Court of India asked the states to repeal the legislation after a few years.

Work at RMRC, Dibrugarh

North East India - Understanding the role of Tobacco (2005-10)

North eastern states of the country have reported high incidence of cancer of many anatomical sites and most of them are associated with use of tobacco. The RMRC, Dibrugarh carried out this study in the six population based cancer registry areas of the NE region with a case control epidemiologic design in collaboration with 6 (six) PBCRs of northeast; NCRP, Bangalore; NIOP, Delhi, ICPO, Delhi and NIOH, Ahmedabad. A total of 2027 newly histologically confirmed 4 (four) tobacco related cancer (TRC) sites i.e., oral cavity, oesophagus, lung and stomach, have been recruited in the study alongwith equal number of controls with age, sex and place of residence matched from 6 (six) population based cancer registries viz. Dibrugarh, Guwahati, Silchar, Imphal, Aizawl and, Gangtok. Of these total cases, 735 were oral cavity (36.3%), 529 were oesophageal (26.1%), 333 were stomach (16.4%) and 430 were lung (21.2%) cancer cases. The summaries of findings are as follows:

(i) Oral Cavity Cancer:

Most cases occur in the age group of 31 to 80 years with male preponderance. Illiterate people were more among oral cancer patients. Smokeless tobacco chewers were more at risk to develop oral cancer.

Particularly chewers of combined chewing item of betelnut and tobacco were at more risk. Early age chewers with a long duration of chewing habits were more prone to develop oral cancer. Retention of chewing quid for long duration and keeping during sleep in mouth was found to be more risky behavior for oral cancer. Smokers who also chew tobacco were at higher risk.

(ii) Oesophageal Cancer:

Cases with male preponderance occur in the age group of 31 to 80 years. Illiterate people are more among oesophageal cancer patients hailing from rural areas. Chewers of betel nut with tobacco had clear dose response effect with the development of oesophageal cancer. Consumption of locally prepared food item like “Kalakhar” (particularly in Assam) had shown significant risk of development of oesophageal cancer.

(iii) Stomach Cancer:

Most cases occur in the age group of 31 to 70 years with male preponderance. Uses of local smoking items like *meizial* or *zozial* etc are at higher risk to develop stomach cancer. The consumption of locally prepared alcohol like *rakzu*, *chang*, *sekmair* etc. were at more risk to develop stomach cancer. The frequent eater of *smoked salted fish* and *meat* are at higher risk for stomach cancer.

(iv) Lung cancer:

The occurrence of lung cancer among bidi smokers was significantly higher than non-smokers. It was also indicated that smokers with drinking habit and drinker with chewing habit were at higher risk to develop lung cancer than persons with no habits. The risk to develop lung cancer among bio-mass fuel users were more than non-biomass fuel users. The use of fire wood and animal dung was found to be exterminating risk factors.

Cancer in North East India-Understanding the Role of Pesticide (2005-10):

This study envisages to investigate the link between pesticide use and the cancers of two anatomical sites in North East and genetic variation including polymorphism/mutations. The study was carried out in the six population based cancer registry area of the north-east with a case control epidemiologic design in collaboration with 6 (six) PBCRs of northeast; NCRP, Bangalore; IOP, Delhi, ICPO, Delhi and NIOH, Ahmedabad. A total of 842 newly diagnosed and histologically confirmed for 2 (two) pesticide related cancer site. Breast and Non-Hodgkin lymphoma were recruited. Of the total cases, 585 were breast cancer and 257 were NHL cancer cases. The summaries of the epidemiological study areas:

Breast Cancer

Most of the female cases were in the age group 31 to 60 years of age and most of them are female farm workers exposed to pesticide showed association with breast cancer.

NHL Cancer

Most of the male cases were in the age group 51 to 60 years of age. Farm workers exposed to pesticides were at higher risk of developing NHL cancer.

Genetic Polymorphism in Drug Metabolizing Enzymes and Oral Cancer & Pre-cancers

Squamous Cell Carcinoma (SCC) of Head and Neck (HNSCC) is an important disease for India, contributing to significant morbidity and mortality. While the causal association is well known, the mechanism(s)

of pathogenesis is still not very clear, more so with regard to the use of smokeless tobacco. Existing studies on CYP1A1 and GSTM1 adopted case-control approach and the results from these studies were equivocal and not able to produce confidence on the role of these genetic variations. Thus, this cohort study on the subject was planned.

The study focused on determining the prevalence of usage of tobacco and chewing products and oral precancerous lesions in Delhi through a population based survey; to determine the frequency of polymorphism of CYP1A1 and GSTM1, in patients with precancerous lesions identified in the above survey and in healthy controls; follow up of patients with precancerous lesions and healthy controls identified in the survey in a prospective cohort study, to assess the relative risk of conversion to cancer or progression of oral lesions; and to determine the influence of at-risk genotypes on progression of precancerous lesions.

Data on tobacco usage and existence of oral lesions were collected from 71,022 individuals using cluster and selective sampling of low and middle income areas of southern Delhi. Any form of tobacco use (ever) was found in 15789 (22.2%) of the population being substantially higher in males (37%) than in females (6.6%). Tobacco smoking was observed more frequently in males (23%) than consumption of the smokeless form of tobacco (18.4%). However in females the frequency of usage of smoking (3.2%) and smokeless (3.5%) forms of tobacco was similar. Bidi was observed to be the most popular form of tobacco smoking accounting for 16.3% in males and 2.8% in females. Among the smokeless forms of tobacco, gutkha (8.7%) and khaini (8.1%) were observed to be more commonly used by males. Among females, pan with tobacco (1.3%) followed by khaini (1.1%) and gutkha (0.8%) were popular form of smokeless tobacco consumed. The proportion of tobacco users was small among persons under the age of 20 years (587/21001, 2.8%). There was a gradual increase in prevalence of tobacco use with age. Of the total population surveyed 2833 (4.0%) were alcohol consumers, of which 70.4% also used tobacco products. The habit of drinking alcohol increased with age predominantly among 30 to 69 years age group.

A total of 3.4% (2389/71022) prevalence of precancerous lesions was observed in the total surveyed population. There were considerable differences between tobacco consumption habits of cases and controls. Most of the cases (92.6%) consumed tobacco in one or another form as compared to 55.9% controls. Genotype analysis showed no significant differences in the allele frequency for GSTM1 null and distribution of CYP1A1 Ile/Val genotype among cases and controls suggesting no significant association with oral precancerous lesions or progression of these lesions.

The longitudinal follow up of patients with oral precancerous lesions and controls was a unique aspect of the study. Of the 1629 patients available for follow up, 227 patients showed progression of the lesions during the study period of 4 years. The progression in lesions was observed most commonly in subjects harboring erythroplakia (20.6%), followed by lichen planus (18.8%). Motivating people to quit tobacco by increasing general awareness about the harmful effects of tobacco by counselling showed significant effect on the progression of the lesions. GSTM1 and CYP1A1 Ile/Val genotype were not significantly associated with risk modification, or alteration alone or in the development and progression of oral lesions in this study cohort.

The study is a successful example of collaboration of principles of epidemiology and laboratory aspects on oral carcinogenesis, which is a priority area for India. Such multi-disciplinary projects can provide important information on the subject.

The study also helped in validating the tobacco use questionnaire developed by ICMR. This modified questionnaire was also used in another task force project, and would be available to the scientific world.

Molecular Cytogenetic Studies in Oral Cancers

Oral cancers are leading cancers in Indian men and are closely linked to the habit of chewing tobacco. Tobacco specific carcinogens are known to interact with DNA causing DNA damage, which may eventually lead to the development of cancers. Genomic damage in target tissues may lead to chromosomal alterations, which have been found to be a driving force in several human malignancies. Oral cancers develop and progress through the accumulation of genetic and epigenetic changes, which ultimately results in gross genomic instability. The chromosomal alterations or genetic imbalances can serve as a marker of genomic instability and damage. Comparative genomic hybridization (CGH) analysis is a direct method for comparing genetic imbalances in DNA from tumor and normal cells, which helps in rapid screening of tumor genomes. At the time of initiation of study (early 2000s) there was limited information about the presence of definite genetic alterations in oral cancers and their relationship with the biological behaviour.

The study aimed at identification of chromosomal markers that may be specific to oral cancers in India at primary tumour site and metastatic lymph node using comparative genomic hybridization and related molecular cytogenetic techniques.

Study identified the genetic alterations in 97 primary oral cancers and investigated their relation with different clinicopathological parameters such as site, size, stage, nodal status and differentiation. All oral cancer samples were first assessed for proportion of malignant cells in the biopsy tissue/surgically resected samples. Only those samples with malignant cell proportion of 60% or more were considered for CGH.

The average gains were higher in moderately and poorly as compared to the well-differentiated oral tumors. A nonrandom pattern of chromosomal alterations was observed with frequent gains ($\geq 25\%$) on 8q, 9q, 11q11-13, 7p, 20q, 3q and 20p and frequent losses ($\geq 25\%$) on 3p, 8p and 18q chromosomes. Gain of 8q was detected as the most frequent alteration and was considered to play an important role in the development of diseases. Chromosomal alterations showed statistically significant associations with both, differentiation and nodal status. Gains of 11q11-13 and 20p chromosomes correlated with moderately and poorly differentiated tumors. Gains of 7p and 19p chromosomes and loss of 8p and 18q chromosomes were associated with the lymph node positive tumors. Multivariate analysis suggested that specific interactions of 7p, 8p, and 19p chromosomes might be required for nodal metastasis. Thus, CGH analysis revealed chromosomal alterations associated with differentiation and nodal status, which are important prognostic evaluators of oral cancer. These alterations may serve as potential biomarkers to complement histopathological analysis.

Analysis of the T Cell Receptor Repertoire ($V\beta$, $V\gamma$ and $V\delta$) and Immune Dysfunction in Patients with Oral Cancer

Immuno suppression appears to be more frequent and more profound in patients with head and neck cancer than in patients with malignancies involving other sites. Decreased or lost expression of signal transducing molecules could be another mechanism underlying functional defects in patient's T cells. Indian studies in early 2000s had shown that these patients exhibit low lymphocyte proliferative responses, decreased intracellular calcium release and altered cytokine (Th1/Th2) profiles when stimulated with phytohaemoagglutinin (PHA), phorbol myristyl acetate (PMA) +ionomycin and anti CD3 MAb. As the stage of the disease progressed further from stage I to stage IV, a marked decrease in the lymphocyte responses was observed.

The project focused on analyzing T cell receptor repertoire (TCR, $V\beta$, $V\gamma$ and $V\delta$ as well as immunophenotypes $\alpha\beta$ & $\gamma\delta$) and clonality in lymphocytes (by heteroduplex analysis) present in

peripheral blood, tumor compartment and in lymph nodes of patients with cancer of buccal mucosa. Other parameters studied in the same materials included, determination of TCR diversity in junctional regions of TCR genes; expression of TCR signaling molecules CD3- ζ chain, lck and ZAP-70 in T cell subsets ($\alpha\beta$ and $\gamma\delta$) using dual color flow cytometry.

The TCR V β , V γ , V δ repertoire was analyzed in peripheral blood (PBL), tumor (TIL) and lymph node (LNL) compartment of oral cancer (OC) patients. A heterogenous and skewed TCR repertoire was observed in PBL, TIL and LIL of oral cancer patients. The CD3 ζ chain defect was narrowed down to specific clonal V β subsets in OC PBL. CD3 ζ chain was restored back in subsets of T cells by stimulating lymphocytes of OC patients by anti TCR V β subset specific MAb and anti CD28 MAb.

Studies further demonstrated that a decreased CD3 ζ chain expression translates into inefficient T cell signaling resulting in a partial or complete loss of T cell function. The expression of CD3 ζ chain mRNA in peripheral blood lymphocytes (PBL) remained unaltered indicating a posttranslational alteration. A decrease in the expression of phosphorylated form of CD3 ζ chain was observed which correlated with the tumor stage. Increased ubiquitination of CD3 ζ chain was also observed in patients with oral cancer.

In order to characterize the factor released by oral tumor cells that cause CD3 ζ chain degradation, the tumor supernatant was analyzed in a sandwich ELISA for the presence of TNF α (tumour necrosis factor alpha). Preliminary results indicated that TNF α might play a role in the degradation of CD3 ζ chain.

In order to establish CD3 ζ chain as prognostic marker, a retrospective analysis of CD3 ζ chain expression at diagnosis was correlated with patients having recurrence of the disease. It was noted that those patients who had a recurrence of the disease showed low expression of CD3 ζ chain at the time of diagnosis. It was also observed that the expression of CD3 ζ chain was restored in patients after surgery followed by radiotherapy as compared to surgery alone.

The study suggested the importance of CD3 ζ chain as a biomarker for prognosis in oral cancer patients. The results indicated that this biomarker should be further studied for specific circumstances like use of specific tobacco products, other polymorphisms, presence and persistence of pre-cancerous lesions, etc.

Oral Cancer in North East India

India has the highest rate of oral cancer in the world. It comprises 30-40% of total malignancies in India and ranks as the leading cancer site in males and the third leading site in females. A retrospective study (from 1993 to 2004) on the prevalence of head and neck cancers in NE India showed significantly high prevalence 54.48%, affecting males more than females in the age group of 40-69 years. Moreover a very high AAR of carcinoma of the tonsil has been reported from at least five districts in Assam State with Kamrup district recording the second highest AAR in India. Approximately 30% of oral cancers are attributed to betel/tobacco chewing alone and an additional 50% to the combined habits of chewing and smoking. Ninety-two percent of oral squamous cell carcinoma (OSCC) in men and 61% of OSCC in women are attributable to tobacco usage. The combined use of peculiar betel/tobacco quid chewing habits, widely prevalent in this region, increases the relative risk of developing oral cancer. The pathogenesis and molecular mechanism implicated in smokeless tobacco consumption is unknown. Currently there are no genetic studies available to explain the alarmingly high prevalence of oral cancers from North east region of India.

The quantitative absorption, distribution, metabolism, and excretion of carcinogenic tobacco constituents

depend on the activity and efficiency of metabolic and enzymatic detoxification pathways. The enzymatic detoxification process is mainly divided into three phases. Phase I enzymes involved in activation of toxic compounds predominantly by oxidation into more reactive intermediates that are neutralized and conjugated by phase II family of enzymes such as glutathione-S-transferase (GST), N-acetyltransferase (NAT). The resultant water-soluble and less-toxic conjugated product can easily be eliminated from the cell by phase III transport mechanisms for the elimination of glutathione conjugates. The detoxification efficiency of GST enzymes is determined by the presence and nature of the isoenzymes coded by GSTT1, GSTM1, and GSTP1 genes. The prevalence of tobacco and betel quid chewing habits as well as the occurrence of tobacco-associated cancers is high in the northeast (NE) region of India. However, the prevalence of polymorphism in GST genes in tobacco-associated cancer patients from this region is not well known.

Frequencies for GST enzyme genes (GSTT1, GSTM1, and GSTP1) polymorphism was analyzed in oral cancer cases and age and sex matched controls from the same ethnic group. The conditional logistic regression analysis revealed that the risk of developing oral cancer significantly increased in tobacco chewers (OR =2.44, 95% CI: 1.47–4.05, $p=0.001$), tobacco smokers (OR=1.72, 95% CI: 1.08–2.73, $p=0.02$), and betel quid chewers (OR=2.20, 95% CI: 1.29–3.76, $p=0.004$). When adjusted for other variables under consideration, no significant association was found for GSTM1 and GSTT1 null genotype independently or in combination with oral cancer risk. Although no significant independent association of oral cancer with null genotypes of GSTT1, GSTM1, and the variant alleles of GSTP1 was found, individuals with both GSTM1 null genotype and variant alleles of GSTP1 were found to have marginal increased risk for developing oral cancer (OR=1.84, 95% CI: 0.91–3.72, $p=0.08$). Study of polymorphisms in three genes [CYP1A1 (Msp1 and Nco1), NAT2 and NQO1] in oral cancer cases and healthy controls from this region showed significantly higher frequency of both heterozygous Pro/Ser (AOR=1.64, 95% CI=1.04-2.58) as well as homozygous Ser/Ser (AOR=1.81, 95% CI=0.98-3.32) NQO1 genotypes in cases than controls suggesting polymorphic variant genotypes of NQO1 are significant risk for oral cancer with homozygous variants genotypes showing a higher risk compared to heterozygous genotypes in univariate analysis. Gene-environment interaction analysis however showed that the variant genotypes of NQO1 did not interact with tobacco consumption habits. Although variant genotypes of NQO1 may play an important role in the genetic susceptibility to oral cancer, its pathway appears to be unrelated to the detoxification mechanism of tobacco constituents. No significant association was found between polymorphisms of CYP1A1 (Msp1 and Nco1) and NAT2 genes with oral cancer risk.

The human TP53 tumor suppressor gene plays a central role in many cellular processes, regulating cell growth, DNA maintenance and apoptosis. It is an important component of DNA repair machinery in response to DNA damage induced by radiation or adduct formation. This might explain the occurrence of the *P53* gene mutation and alteration in about 50% of all cancers, particularly tobacco related cancers. Studies have shown a relationship between tobacco smoke exposures, carcinogen-DNA adduct formation, tumor specific mutation of TP53 gene and cancer risk.

Numerous polymorphism in the wild type P53 have been reported both in coding and non-coding regions. Out of the five polymorphisms described in the coding region, polymorphisms in codon 47 and 72 in exon 4 are functionally well characterized. More common of the two, codon 72 polymorphism is a single base substitution of cytosine for guanine, leading to arginine (A72) being replaced by proline (P72) that has been reported to be associated with the risk of several cancers. Studies on codon 72 polymorphism have revealed striking ethnic differences in frequency of P53 variant allele that varies with latitude, increasing in a linear trend as populations near the equator. Thus ethnicity might be related to allelic distribution of the gene and its determinacy in disease involvement; however some studies do refute the ethnicity-risk confounding relationship.

A case control study was undertaken in oral cancer and normal age matched controls collected from Dr. B. Borooah Cancer Institute, Guwahati, Civil Hospital, Aizawl, and Sir T.N.M. Hospital, Gangtok in three north east states to investigate the role of p53 codon 72 polymorphism and its interaction with tobacco, betel quid and alcohol. Tobacco chewing was the strongest risk factor for oral cancer with a 3 fold increased risk (OR2= 3.05, 95% CI=1.79-5.20; $p<0.001$). Both tobacco smoking (OR2=1.68, 95% CI=1.00-2.81; $p=0.04$) and betel quid chewing (OR2=1.85, 95% CI=1.02-3.33; $p=0.04$) also increased risk for oral cancer. A strong association of risk of oral cancer was seen in patients carrying Arg/Arg and Arg/Pro genotypes (OR2= 8.62 95% CI=2.19-33.93; $p=0.002$ and OR2=5.22 95% CI=1.38-19.75; $p=0.01$ respectively) irrespective of smoking habit. Although tobacco and betel quid chewing emerged as a risk factor yet no interaction of p53 genotypes with the habit showed statistical significance.

Curcumin Modulates Cellular AP-1, NF- κ B and HPV 16 E6 Protein in Oral Cancer.

This study performed on HPV 16-positive oral cancer cell line 93VU147T, demonstrated the beneficial effects of curcumin. Curcumin was not only a potent inhibitor of the activity of transcription factors AP-1 and NF- κ B but it also suppressed transcription of HPV16 E6 oncogene.

This study suggested a therapeutic potential of curcumin in high-risk HPV-infected oral cancers.

Differential Expression and Activation of NF- κ B family of Proteins during Oral Carcinogenesis: Role of High Risk HPV Infection

Human papilloma virus (HPV) has been shown to be associated with oral carcinogenesis in addition to other known risk factors. Reciprocally, cellular transcription factors such as NF- κ B and AP-1 are known to modulate expression of viral and other genes. However, there was a lack of data on NF- κ B expression in relation to HPV infection in oral cancer.

This study was aimed at evaluating the DNA binding activity and expression pattern of NF- κ B family of proteins in different stages of oral cancer and its correlation with HPV infection. A total of 110 fresh tissue biopsies were included comprising of 10 controls, 34 precancer and 66 untreated oral cancer lesions. Diagnosis of HPV was done by both consensus and type-specific PCR. Electrophoretic mobility shift assays, western blots and immunohistochemical analysis were performed to assess the binding activity and expression pattern of NF- κ B family of proteins (p50, p65, p52, c-Rel, RelB and Bcl-3) in oral tissue biopsies. Of the oral cancer biopsies, 27% (18/66) showed the presence of exclusive high-risk HPV type 16. A high constitutive activation of NF- κ B with concomitant upregulated expression of all the NF- κ B members in oral cancer tissues was observed. Expression of NF- κ B components gradually increased as the severity of lesion increased from precancer to invasive cancer. NF- κ B p50 was found to be the major DNA binding component, which is indicative of homodimerization of p50 subunits. Interestingly, in HPV16 infected oral cancers although p50 showed high binding activity, p65 also showed a partial involvement as evidenced in supershift assay. Both by western blotting and immunohistochemistry, a differential overexpression and nuclear localization of p50, p65 and partially of Bcl-3 were observed in HPV16 positive oral cancer patients that also showed an over-expression of p21.

This study demonstrated a constitutive activation and differential expression of NF- κ B proteins, which change as a function of severity of oral lesions during development of oral cancer. It was hypothesized that the involvement of p65 in HPV infected oral cancer may be linked to improved differentiation and better prognosis of the disease when treated.

Comparative Study Between HCII test and PCR Based Assay for Detection of HPV DNA in Oral Submucous Fibrosis and Oral Squamous Cell Carcinoma

Oral malignancy is a major global health problem. Besides the main risk factors of tobacco, smoking and alcohol, infection by human papillomavirus (HPV) and genetic alterations are likely to play an important role in these lesions. The purpose of this study was to compare the efficacy of HC-II assay and PCR for the detection of specific HPV type (HPV 16 E6) in OSMF and OSCC cases as well as find out the prevalence of the high risk HPV (HR-HPV) in these lesions.

The overall prevalence of HR-HPV 16 E6 DNA positivity was nearly 26% by PCR and 27.4% by the HC-II assay in case of potentially malignant disorder of the oral lesions such as OSMF. However, in case of malignant oral lesions such as OSCC, 32.4% HPV 16 E6 positive by PCR and 31.4% by the HC-II assay. In case of OSMF, the two test gave concordant result for 42 positive samples and 154 negative samples, with an overall level of agreement of 85.4% (Cohen's kappa = 66.83%, 95% CI 0.553-0.783). The sensitivity and specificity of the test were 73.7% and 92.05% ($p < 0.00$). In case of OSCC, the two test gave concordant result for 61 positive samples and 152 negative samples, with an overall level of agreement of 88.3% (Cohen's kappa = 79.29, 95% CI 0.769-0.939) and the sensitivity and specificity of the test were 87.14% and 92.76% ($p < 0.00$).

This study concluded that slight difference was found between the positivity rate of HR-HPV infection detected by the HC-II and PCR assay in OSMF and OSCC cases and the HC II assay seemed to have better sensitivity in case of OSCC.

Functional Polymorphism of MMP-1 Promoter in Potentially Malignant and Frankly Malignant Head and Neck Lesions

Matrix metalloproteinases (MMPs) are a family of zinc-dependent proteases that degrade the entire component of the extracellular matrix. This study explored the association of the MMP1 gene promoter (-1607 1G/2G) polymorphisms in oral submucous fibrosis (OSMF) and head and neck squamous cell carcinoma (HNSCC) in an Indian population. The MMP1 single-nucleotide polymorphism (SNP) was genotyped by polymerase chain reaction-restriction fragment length polymorphism analysis in 412 patients with OSMF, 422 with HNSCC and 426 controls. Results showed that the frequency of 1G/2G or 2G/2G promoter genotypes having the 2G allele was associated with higher enzymatic activity and significantly increased in OSMF ($p < 0.001$) and HNSCC cases ($p < 0.00$). In this study, results concluded that SNPs in the MMP1 promoter region may be associated with susceptibility to OSMF as well as HNSCC in an Indian population and addiction habits such as areca nut chewing and alcohol abuse may enhance the expression of the 2G allele of MMP1 genes in OSMF and HNSCC cases.

Synergistic Effect of Stromelysin-1 (MMP-3) Promoter Polymorphism in Oral Submucous Fibrosis and Head and Neck Lesions

Matrix metalloproteinases (MMPs) are enzymes that degrade all the components of extra cellular matrix and collagen. Various types of MMPs are known to be expressed and activated in patients with oral submucous fibrosis (OSMF) as well as head and neck squamous cell carcinoma (HNSCC). The purpose of this study was to assess the association of the single nucleotide polymorphism (SNP) adenosine insertion/deletion polymorphism (-1171 5A->6A) in the MMP-3 promoter region in these lesions.

MMP-3 SNP was genotyped by polymerase chain reaction-restriction fragment polymorphism (PCR-RFLP) analysis in a case control study consisting of 362 participants; 101 cases of OSMF, 135 of HNSCC and 126 controls, compared for age, sex and habits. ROC distribution was plotted to assess the

contributions of genetic variation in MMP-3 genotypes with relation to age.

Analysis of MMP 3 (-1171 5A->6A) polymorphism revealed the frequency of 5A allele in OSMF, HNSCC and controls to be 0.15, 0.13 and 0.07 respectively. A significant difference was found in 5A genotype frequency between OSMF (5A genotype frequency = 0.15, $p = 0.01$, OR = 2.26, 95% CI = 1.22-4.20) and in controls (5A genotype frequency 0.07) as well as HNSCC (5A genotype frequency 0.13, $p = 0.03$, 95% CI = 1.06-3.51) and controls (5A genotype frequency = 0.07). In this study, 5A genotype had greater than two fold risk for developing OSMF (OR = 2.26) and nearly the same in case of HNSCC (OR = 1.94) as compared to controls. In patients with OSMF as well as HNSCC, the ROC analysis between the MMP-3 genotype and age, 6A/6A allele was found to be significant in patients both over and under 45 years of age; while the 5A/5A carrier alleles showed an association only in patients less than 45 years of age.

This study concluded that the expression of MMP-3 genotype associated with the 5A alleles may have an important role in the susceptibility of the patients to develop OSMF and HNSCC.

Study of Genetic and Molecular Epidemiology of Oral Cancer in 2 States (Assam & Meghalaya) of North Eastern Region of India (2009 - 2012)

The study included a total of 343 study subjects (case: 171; control 172). Biopsy was the predominant (82%) mode of diagnosis of the cases. Major cancer sites were gum (39%) followed by buccal mucosa of cheek (31%). Major clinical presentation of the cases were ulceration in the mouth (39%) followed by swelling (19%). The cases from illiterate and lower income group from rural areas were found to be more prone to develop oral cancer. Betel nut chewer had synergistic effect on oral cancer. Among the genotypes studied; GSTM1 null genotype and MSP1 mutant genotype of CYP1A1 gene were independent risk factors of oral cancer. Risk of oral cancer in subject with GSTM1 null genotype was found to be enhanced by habit of tobacco chewing and smoking. Risk of oral cancer in subjects with MSP1 mutant genotype was also found to be increased by GSTT1 null genotype.

Risk Factors of Lung Cancer in Manipur and Mizoram, India: a Case-Control Study (2008 – 2012)

The incidence of lung cancer is very high in the state of Manipur and Mizoram in comparison to the other parts of the country. Particularly, incidence among women superseded the counterpart in the state of Mizoram. A case-control study has been initiated to find out the epidemiological risk factors in the development of primary lung cancer in Manipur & Mizoram, India and estimation of lung cancer risk imposed by inheritance of polymorphic alleles of GSTM1 and CYP1A1 genes by comparing allele frequencies among lung cancer patients and appropriate control population. This study revealing significantly higher risk for environmental exposure like users of Meiziol, a locally made cigarette (OR=2.25; $p=0.006$) as compared to general cigarette users (OR=1.68; $p=0.171$) among women in Mizoram. Current users of *Tuibur*, a liquid tobacco (OR=1.59; $p=0.039$) was also significantly associated with increased risk for lung cancer among women in Mizoram. Traditional dietary practices in Mizoram such as consumption of smoked fish (OR=2.57; $p=0.006$), smoked meat (OR=3.03; $p=0.001$) and soda (OR=7.96; $p<0.001$) were more prone to develop lung cancer among women in Mizoram. Exposure of cooking oil fumes (OR=2.56; $p<0.003$), wood as heating source for cooking (OR=1.50; $p=0.044$), kitchen inside living room (OR=1.79; $p=0.001$), improper ventilated house (OR=2.10; $p=0.003$) were also significantly prone to develop lung cancer among women in Mizoram. GSTM1 null genotype and CYP1A1 homozygous mutant were significantly associated with the increase risk of lung cancer in both Manipur (GSTM1 OR=2.59; $p<0.001$ & CYP1A1 OR=4.12; $p<0.001$) and Mizoram (GSTM1 OR=2.95;

$p < 0.001$ & CYP1A1 OR=2.23; $p=0.003$). Gln/Gln alleles of both XRCC1 and XPD genes appear to amplify the effects of household exposure, smoking and betel quid chewing on lung cancer risk in the study population of both the states. Gln/Gln genotype was higher among cases than control groups in both XRCC1 (8.5% vs 6.3%) and XPD genes (12.9% vs 10.5%). Significant protective effect for lung cancer was observed for intake of bamboo shoots (OR=0.15; $p < 0.001$) and egg (OR=0.05; $p < 0.001$) in both Manipur and Mizoram. In addition to these among the traditional dietary habits of Manipur consumption of citrus fruits (OR=0.07; $p < 0.006$) and garlic (OR=0.01; $p < 0.001$) also plays an inverse association for lung cancer risk specifically in Manipur.

Study of Genetic and Molecular Epidemiology of Stomach Cancer in Mizoram of North Eastern Region of India (2009 - 2012)

Tobacco smoking, especially meiziol (local cigarette) is the important risk factor for stomach cancer in Mizoram. GSTM1 and GSTT1 genes modify the effect of tobacco habits. This study revealed that p53 codon 72 polymorphism and dietary and tobacco habit interactions influenced stomach cancer development in Mizoram, India. Although no direct association between CYP2E1 RsaI polymorphism and stomach cancer was observed, relations between CYP2E1 RsaI polymorphisms and different tobacco and dietary risk habits in terms of developing stomach cancer exist in this high risk population of north-eastern part of India. However further elaborate study recruiting larger population is required to shed more light. This study was a first step in understanding the epidemiology of stomach cancer in Mizoram, India.

Study of Genetic and Molecular Epidemiology of Oral Cancer in 2 States (Assam & Meghalaya) of North Eastern Region of India (2009 - 2012)

The study included a total of 343 study subjects (Case: 171, Control 172). Biopsy was the predominant (82%) mode of diagnosis of the cases. Major cancer sites were gum (39%) followed by buccal mucosa of cheek (31%). Major clinical presentation of the cases were ulceration in the mouth (39%) followed by swelling (19%). The cases from illiterate and lower income group from rural areas were found to be more prone to develop oral cancer. Betel nut chewer had synergistic effect on oral cancer. Among the genotypes studied GSTM1 null genotype and MSP1 mutant genotype of CYP1A1 gene are independent risk factors of oral cancer. Risk of oral cancer in subject with GSTM1 null genotype was enhanced by habit of tobacco chewing and smoking. Risk of oral cancer in subjects with MSP1 mutant genotype was increased by GSTT1 null genotype.

Study of Genetic and Molecular Epidemiology of Breast Cancer in 2 States (Assam & Meghalaya) of North Eastern Region of India (2009 - 2012)

After establishing network with the different hospitals, recruitment of staffs, finalization of epidemiological questionnaire (breast cancer) and procurement of different laboratory chemicals, the epidemiological information and biological samples were collected as per protocol. Genetic polymorphism of tumor suppressor gene p53 and XRCC1_Arg399Gln was found to be associated with increased risk of Breast Cancer in this study. Family history was also found to be associated with Breast Cancer. Ten cases had family history of Breast Cancer and out of 10 cases, only 1 had mutations in BRCA1.

Comparative Study of Genetic, Clinical and Epidemiological Factors of Breast Cancer in Rural and Urban Area of India (Mizoram & Tripura) (2009 - 2013).

A total of 978 blood samples were collected which included blood sample from 310 breast cancer (BC) cases and 668 controls. Biopsy samples from the breast cancer patients were also collected. Betel nut

chewing with or without tobacco was found to be an important risk factor of BC in Tripura but not in Mizoram. Smoking habit was found associated with BC in both states. Genetic mutations in BRCA1 gene was found in 12% cases of Mizoram but only 2.44% of cases from Tripura. Carriers of mutations in XRCC1 and XRCC3 and TP53 genes significantly increased the risk of breast cancer subjects from Mizoram and Tripura. Next Generation Sequencing data along with TaqMan allelic discrimination facilitated in identification of a novel mutation in PKHD1 gene which predisposed carriers of the variant allele to increased risk of breast cancer (OR = 3.1; CI = 1.4 – 7.2; $p = 0.007$). As a proof of concept the present study based on Next Generation Sequencing technology revealed that NGS has a significant utility for healthcare diagnostic especially by quickly screening mutomes which will be specific for different ethnic groups.

Genome-wide Analysis of Genetic Alterations in Patient's Esophageal Cancer from North-East India Using Single Nucleotide Polymorphism Arrays (2011 - 2014)

Epidemiological Part:

The epidemiological part of the project was carried out at RMRC, Dibrugarh. On completion of project, the information of 140 esophageal cancer cases and 140 controls was collected. 65.8% of the cases were males while 34.2% were females. Majority of male cases belonged to the age group of 60 - 70 yrs. Male cases were mostly cultivators i.e. 44.3%; while among the female cases, majority of them were housewives i.e. 65.8%. Topography inclination indicated more towards C15.4 (middle-third of esophagus) among both the sexes. Squamous cell carcinoma was mostly predominant (90.7%). The pathological grade of most of the cases was moderately differentiated (G2) showing 81% of the total cases irrespective of sex. Chewing of tobacco was significantly associated with increased risk of esophageal cancer among the study subject (OR=2.17; $p=0.040$). Significant risk association was also observed for those who consumed betel nut leaf with lime *katha* and *zarda* (OR=7.30; $p<0.001$). Smoking of bidi (OR=2.18; $p=0.018$) and those who smoke both bidi and cigarette (OR=3.21; $p=0.020$) were also significantly prone to develop esophageal cancer. Significant risk was also observed for those who consumed alcohol (OR=2.23; $p=0.001$). While among the type of alcohol consumption significant risk was observed for those who consumed whisky (OR=3.41; $p=0.002$) and rice beer (OR=2.75; $p=0.011$). From the dietary habits, significant risk association for esophageal cancer was observed for the intake of *kalakhar* (extract made from banana leaf used as food additives) (OR=2.77; $p=0.029$). In this study, significantly protective effect was observed for the intake of curd among the study subject (OR=0.40; $p=0.018$).

Epigenetic Studies in Esophageal Cancer in High Risk Region of Northeast India (2011 - 2014)

Epidemiological Part :

The epidemiological part of the project was carried out at RMRC, Dibrugarh. A total of 75 esophageal cancer samples and 75 controls were enrolled. Among the cases; 74.6% were males and out of which majority of them belong to the age group of 51-60 years. On the other hand female percentage was 25.3% and majority of the cases belonged to the age group of 51-60 yrs. Most of the cases had the family income in the range of Rs.1000-5000 and among controls majority had their income in the range of Rs.5000-10000. Male cases were mostly cultivators (42.85%) followed by business (19.64%), service (16.07%), unskilled (12.5%) and skilled (8.92%). Among the female cases majority were housewives (78.9%) followed by cultivators (10.52%) and service (10.52%). Squamous cell carcinoma type which is mostly predominant (94.7%) followed by adenocarcinoma (5.3%). Most of the tumors were found to be moderately differentiated (66.7%) compared to well differentiated (20%) and poorly differentiated (13.3%). Tobacco smoking and tobacco and betel nut chewing was higher among the cases. 70.7% cases were

smokers and 93.3% cases were chewers. Alcohol drinking habit was also more among the differentiated cancer cases (45.3%).

Pattern of Survival and Quality of Life of Esophageal and Stomach Cancer Patient in North Eastern Region of India (2014–2017)

The epidemiological information was collected from 100 cases of esophageal cancer which include 18 cases from PBCR Dibrugarh, 42 cases from PBCR Guwahati, 24 cases from PBCR Silchar, 16 cases from PBCR Aizawl and 55 cases of stomach cancer which included 5 cases from PBCR Dibrugarh, 37 cases from PBCR Guwahati, 7 cases from PBCR Silchar and 6 cases from PBCR Aizawl. Out of the total 100 case of esophageal cancer; male were preponderance with an estimated rate of 68% and maximum of the cases belonged to the age group of 50-59 yrs, whereas in case of stomach cancer female were preponderance with the estimation rate of 53% out of total 55 cases belonging to the age group of 50-59 yrs. As far as caste/ethnic group is concerned; the non tribe was found to be the main victim of both esophageal and stomach cancer. Geographically the esophageal cancer was more prominent towards urban area with an estimation of 48% whereas stomach cancer was more prominent in rural area with 52.7% out of total cases. Maximum of the esophageal cases are found to be service holder with education upto middle level whereas in stomach cancer house wife is the most victim and they are found to be generally illiterate. The nuclear families are the main sufferer of both esophageal and stomach cancer. It was observed that the maximum of the esophageal cases were from the middle class group and the maximum of the stomach patient were from the lower category. In case of anatomical stage; the cancer has spread towards stage IIIA and in both esophageal and stomach cancer. In esophageal cancer; radiotherapy was used for treatment by most and in stomach cancer chemotherapy was used by many for treatment. 60% of the esophagus cancer and 50.9% of the stomach cancer patient had been given treatment prior to the Reporting Institution (RI). 90% of the esophagus cancer patient and 53% of the stomach cancer patient were continuing the prescribed treatment. Maximum of the esophagus and stomach cancer patient chose to take treatment in their own state than go out of state for treatment. It was found out that 78% of the esophageal patient was alive and 22% of them were dead, whereas in stomach cancer 65% of the patient was alive and 34.5% of them were dead till date. In context to clinical information; most of the esophageal cases are squamous cell carcinoma with pathological grade G1: Well differentiated and the topography inclination indicated more towards C15.5: Lower-third of esophagus. In case of stomach cancer, adenocarcinoma was the most common type of cancer with pathological grade G1: Well differentiated and the topography inclination indicated more towards C16.3: Gastric antrum. The overall average survival probability of esophageal cancer patient is 10.81 month and stomach cancer patient is 8.96 months are found. It was observed that EORTC QLQ C30 questionnaire gave reliable results on quality of life of both esophageal and stomach cancer patients. Results showed that almost all the multi-item scales of the questionnaire registered high value of (> 0.70) signifying that the scales were internal consistent.

Risk Factors of Hepatocellular Carcinoma (HCC) in Sikkim and Arunachal Pradesh, India (2014–2017)

A total of 31 HCC cancer samples and 31 controls were collected till August 2015 from both Arunachal Pradesh and Sikkim. Male cases were reportedly higher than the female cases in both Arunachal Pradesh (61.9% vs. 38.1%) as well as in Sikkim (60% vs. 40%). Majority of the cases in both the states (for both male and female) were in the age group of 51 – 70 years. Majority of the cases were illiterate and cultivators in both the states. In Arunachal Pradesh; 76.2 of the cases were cultivators, 14.3% were in service or business and the remaining 9.5% were housewives while in Sikkim 50% of the cases were cultivators, 40% of the cases were either in service or business, and housewives and unskilled persons occupied 20%

of cases each. Regarding family income it was seen that in both the states 50% of the cases had family income in the range of Rs.500-1750 and among controls majority had their income more than Rs.4000. From the data collected till date it was found that the most common type of Hepatocellular Carcinoma is “Hepatocellular Carcinoma, NOS(C22.0)” in both the states. Most of the tumors were found to be moderately differentiated (85.7%) followed by well differentiated (14.3%) in Arunachal Pradesh while in Sikkim most of the tumors were poorly differentiated (50%) followed by moderately differentiated (30%). Further majority of the cases were diagnosed with HCC through FNAC while the remaining cases have done AFP in both the states. Cases were mostly suffering from abdominal pain (32%), inability to eat (30%) and weight loss (27%). Hepatitis B positive was found to be the common history of illness in both the states. Chewing of tobacco (Khaini) was significantly associated with increased risk of Hepatocellular Carcinoma among the study subject). Risk was also observed in case of smoking but it is not significant in either of the cases. Alcohol consumption was also significantly associated with increased risk of HCC among the study subject especially those who consume government approved local drinks and rice beer. Diet mostly included rice, pulses and green leafy vegetables along with non vegetarian foods which comprised of eggs, meat, and fish (dried and smoked). 89.5% of the people consumed non-vegetarian as well as leafy green vegetables. Consumption of fermented and non fermented bamboo shoot was widely preferred by the people of Arunachal Pradesh. Fruits were occasionally consumed which included citrus fruits, apples, jackfruit, etc.

Monitoring Unit of North East Population Based Cancer Registries (NEPBCR) at RMRCNE, Dibrugarh, Assam

Cancer incidence rate is generally expressed as age adjusted or age standardized rate (AAR) (according to world standard population) per 100,000 persons. Aizawl district of Mizoram state showed the highest AAR in both males and females. In males, Aizawl district is followed by East Khasi Hills district of Meghalaya state and in females by Kamrup urban district not only in North East but in India. Among males, cancer of the oesophagus was the leading site in the registries in the states of Assam and Meghalaya. Stomach cancer was the leading site in Sikkim and Mizoram while cancer of the nasopharynx was the leading site in Nagaland. Among females, cancer of the gallbladder and oesophagus followed by cancer of breast as the leading site in Dibrugarh and Kamrup urban respectively. Lung was the leading site in Manipur and Mizoram (particularly among females). East Khasi Hills district of Meghalaya had the highest relative proportion of tobacco associated cancers for males and females at 69.3% and 43.0% respectively when compared to all sites of cancers. The data of the newer PBCRs that are being reported for the first time in report are Nagaland, Meghalaya including Khasi Hills district and Tripura. Among these new PBCRs, males in Nagaland show the highest AAR of 21/100,000 for nasopharyngeal cancer. East Khasi Hills district and Meghalaya state as a whole showed the highest AAR in cancers of the oesophagus (in both males and females) and cancers of the hypopharynx and larynx in males. The states of the North East had already reported high incidence rates of cancers of the upper aero digestive tract that would include anatomical sites such as oral cavity, pharynx, hypopharynx, nasopharynx, larynx, oesophagus and even stomach. The international comparisons of the incidence rates of these sites of cancer revealed that the PBCRs at Meghalaya (especially East Khasi Hills district) and Mizoram (Aizawl district) and Kamrup Urban in Assam had the highest AARs of cancers of these sites, particularly oesophagus, hypopharynx and larynx.

Sites of Cancer Associated with Use of Tobacco

This chapter indicated the impact of the use of tobacco on the occurrence of cancer through overall proportions and specific anatomical sites of cancer.

Table 6.5 gives the number and relative proportion of Tobacco Related Cancers (IARC - 1987) in different Population Based Cancer Registries (PBCRs) established by Indian Council of Medical Research through National Cancer Registry Program. East Khasi Hills district of Meghalaya had the highest relative proportion with 69.5% and 45.0% for males and females respectively. Among males, the lowest proportion of tobacco related cancers (TRCs) was in Naharlagun excluding Papumpare district (18.3%) whereas in females the lowest proportion for TRCs was observed in Pasighat PBCR (6.9%).

Fig. 6.14 depicts the proportion of tobacco related cancers relative to all sites of cancer for 27 PBCRs.

Fig. 6.15 illustrates the proportion of specific sites of cancer among all tobacco related cancers (TRCs) for 27 PBCRs.

In males, lung cancer was the leading site in 11 registries among them, its contribution ranging from 21.5% in Sikkim PBCR to 48.2% in Manipur state. Oesophageal cancer was the leading site in 8 registries, ranging from 22.1% in Cachar PBCR to 47.7% in Meghalaya PBCR. Mouth cancer was the leading site in 7 registries, its contribution ranging from 23.1% in Barshi Expanded PBCR to 36.1% in Ahmedabad Urban PBCR.

In females, lung cancer was the leading site among tobacco related cancers in 11 registries. Cancer of the oesophagus is the leading site in 8 registries and that of the mouth was the leading site in 7 registries. Lung cancer's contribution ranged from 24.1% in Chennai PBCR to 72.2% in Manipur PBCR. Oesophageal cancer ranged from 25.7% in Barshi Rural PBCR to 55.0% in Meghalaya PBCR and mouth cancer ranged from 24.1% in Ahmedabad Urban PBCR to 31.4% in Wardha PBCR.

Table 6.5 : Number (#) and Relative Proportion (%) of TRCs

Registry	Male		Female	
	#	%	#	%
Bangalore (2012)	1301	34.0	649	14.3
Barshi Rural (2012-2014)	166	36.6	70	14.7
Barshi Expanded (2012)	350	38.8	116	10.3
Bhopal (2012-2013)	937	54.5	289	16.6
Chennai (2012-2013)	2217	40.7	859	13.8
Delhi (2012)	4185	41.2	1141	11.9
Mumbai (2012)	2598	39.4	1027	15.2
Cachar District (2012-2014)	1233	46.2	432	20.6
Dibrugarh District (2012-2014)	773	51.6	307	22.8
Kamrup Urban District (2012-2014)	1525	49.7	576	24.1
Manipur State (MR) (2012-2014)	743	35.7	490	19.3
Imphal West District (2012-2014)	238	37.2	152	18.5
MR - Excl. Imphal West (2012-2014)	505	35.0	338	19.7
Mizoram State (MZ) (2012-2014)	1125	43.8	496	23.7
Aizawl District (2012-2014)	612	48.0	282	26.5
MZ - Excl. Aizawl (2012-2014)	513	39.7	214	20.9
Sikkim State (2012-2014)	237	33.5	129	19.0
Ahmedabad Urban (2012-2013)	3085	56.3	817	19.8
Aurangabad (2012-2014)	601	53.5	187	16.7
Kolkata (2012)	1285	46.3	434	16.7
Kollam District (2012-2014)	2393	43.2	707	12.9
Nagpur (2012-2013)	1105	49.4	441	18.2
Pune (2012-2013)	1381	40.4	602	16.3
Thi'puram District (2012-2014)	2835	37.1	869	10.9
Meghalaya (2012-2014)	1717	65.2	684	42.3
East Khasi Hills District (2012-2014)	1129	69.5	445	45.0
Tripura State (2012-2014)	1953	53.8	563	20.8
Nagaland (2012-2014)	323	39.6	69	12.6
Wardha District (2012-2014)	539	41.3	258	18.1
Naharlagun (NH) (2012-2014)	179	24.4	87	12.4
Papumpare District (2012-2014)	99	33.1	50	15.0
NH - Excl. Papumpare (2012-2014)	80	18.3	37	10.0
Pasighat (2012-2014)	51	29.1	11	6.9
Patiala District (2012-2014)	1091	38.2	411	13.0

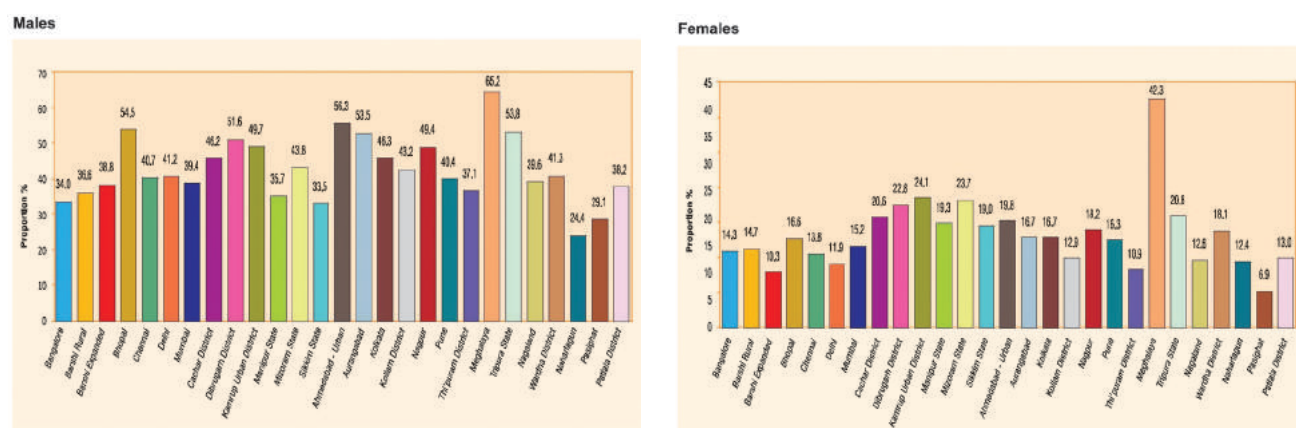


Fig. 6.14: Proportion (%) of Tobacco Related Cancers (TRCs) Relative to All Sites of Cancers.

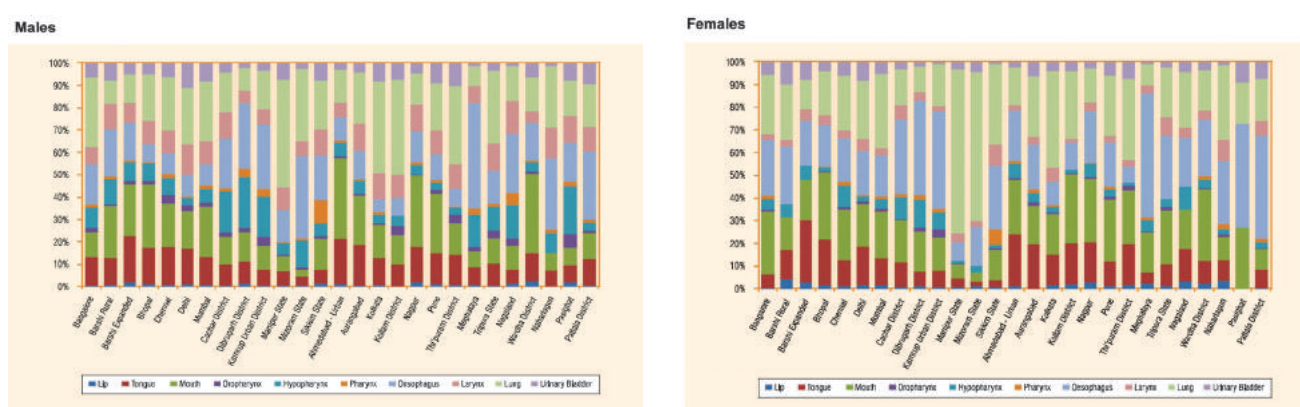


Fig. 6.15: Proportion (%) of Specific Tobacco Related Sites relative to all Tobacco Related Cancers (TRCs).

National Institute of Pathology (NIP), Delhi:

In North East states in India, a very high incidence of several cancers associated with tobacco consumption and pesticide exposure has been reported and NIP has under taken several studies on 'Cancers in North East Region of India' to investigate the contribution of genetic factors and gene- environmental interactions.

Gene expression profile and copy number variation was studied by genome-wide approaches in esophageal cancer patients from Assam and Kamrup districts showing high incidence (AAR of 33/100,000) and familial clustering (29%) with aim to identify the molecular functional pathways and link between tobacco use and genetic variation. Up-regulation of B-cell receptor signaling pathway genes (PRKCB1) and down regulation of Natural killer cell mediated cytotoxicity genes (NFAT5, KLRC1) in familial ESCC suggested that immune response contributed significantly to familial esophageal cancer. Down regulation of genes associated with xenobiotic metabolism viz. GST & EPHX1 and mutations in BRCA2 gene were found to be associated with risk for familial ESCC and could help in screening. Up-regulated GPCR activity gene NPY and MAPK activity gene FGF12 and down-regulated ribosome proteins S4, L32 and cytokeratin genes KRT4 were found important prognostic factors for both familial and non-familial tobacco and betel quid associated esophageal cancer. DHFR gene was found to be important drug target for this cancer. A high through put tissue micro array containing 120 esophageal squamous cell carcinoma from the patient pool was constructed to validate the gene expression profile.

Studies Conducted in North East Region are Described under the Cancer Research in North East India

Gastric Cancer in North East India

Frequency of gastric cancer (GC) is very low in India compared to that in Japan and China. However in Mizoram, one of the NE states in India, a very high age adjusted incidence of stomach cancer was recorded. Epidemiologic studies indicated that the risk of gastric cancer was significantly elevated among current smokers and in *meiziol* (a local cigarette) smokers (OR>2). The risk increased with increasing cumulative dose of tobacco smoked. Use of locally prepared *Tuibur* (tobacco smoke-infused water) in Mizoram, seemed to increase the risk of stomach cancer (OR, 2.1). There was no report so far on the genetic profile of the patients with gastric cancer from this high-incidence region of Mizoram in NE India.

Although CYP and GST enzymes are involved in the activation and detoxification of N-nitrosamines which are some of the most potent carcinogens present in smokeless tobacco, snuff and tobacco smoke and related compound, studies on the relationship between genetic polymorphisms of CYP1A1, GSTT1, and GSTM1 and the risk of gastric carcinoma (GC) have not been carried out in this high risk region of India so far and the previous reported studies from other part of the world are conflicting.

To evaluate the association between polymorphic detoxifying genes and GC risk, a case-control study including histological confirmed gastric cancer patients and 282 healthy controls was conducted in a gastric cancer high-risk area in North east region of India to screen the GST, CYP1A1(2A,2C) genes and p53 codon 72 polymorphisms.

The frequency of GSTT1 and GSTM1 null genotypes did not show significant differences between cases and controls. Variant genotypes (Ile/Val and Val/Val) of GSTP1 were found more frequent in cases of gastric cancer as compared to controls but the difference was not statistically significant (OR=1.25, 95% CI 0.77-2.03, P=0.37). No significant difference in frequency distribution of CYP1A1 2A genotypes (ww, wv, vv) and CYP1A1 2C genotypes (II, IV, VV) was found in cases and controls, suggesting no associated risk for gastric cancers with GST and CYP1A1 genes. Conditional logistic regression analysis of frequency distribution of p53 codon 72 Arg/Arg, Arg/Pro and Pro/Pro in the cases and controls showed higher risk associated with Arg/Pro genotype whereas the Pro/Pro appeared to be a protective genotype. These observations were represented both by OR1 and OR2 but lacked statistical power. No significant interactions of risk habits was found with P53 genotypes, however, interaction of betel quid chewing with Arg/Pro genotype reached a near significance level with a 2 fold risk for gastric cancer (OR2=2.40; 95% CI=0.91-6.26; p=0.07)

Risk estimates for *H. pylori* status, tobacco, betel quid and alcohol consumption habits in 112 cases and 66 healthy age matched controls showed significant association of tobacco chewing with risk of development of GC, while there was no significant association with *H. pylori* as reported earlier. However, tobacco chewers who were also *H. pylori* positive showed a seven fold risk (OR2=7.12, 95% CI=5.02-36.31, p 0.0001) whereas betel quid chewers who were also *H. pylori* positive showed a four fold risk (OR2=3.52, 95% CI=1.16-10.86, p=0.026) for developing gastric cancer.

To understand the molecular mechanism underlying the development and progression of GC in this high-risk region in NE India, the gene expression profiles of tumor tissue from 5 patients having history of tobacco/ betel quid chewing, alcohol consumption and sero positivity for IgG antibody against *H. pylori* were studied and compared with normal appearing pooled gastric mucosal tissues. A total of 170 genes were found to be differentially expressed. Of the 108 significantly up-regulated genes, genes involved in apoptosis (RYK, SH2D2A, CSNK1E, LRP12, HSPE1, FGB, PXN, CHP), proinflammatory response

(IL9), angiogenesis (SH2D2A, ANG), cell adhesion (CD151, PXN, WASF1), cell proliferation (CDC2L1, JAG2, IL9), regulators of I-kappa B kinase / NF-kappa B cascade (CXXC5), Wnt signaling (CHP, CSNK1E, RYK), Notch signaling (JAG2), Hedgehog signaling (CSNK1E, BMP7), metabolic pathway (LDHB), thyroid cancer (RET) and epithelial cell signaling in *H. pylori* infection (ATP6V1G2) were found to be biologically relevant in tumorigenesis. Of the 62 significantly down-regulated genes, genes involved in base-excision repair (RAD51L3), inhibition of cellular proliferation (INSM1), immune response (RFX1, CXCL10, C3AR1, MALT1), viral response (ATP6V1G2, HBXIP, ACE2), epithelial cell signaling in *H. pylori* infection (JAM3) were found to be relevant in tumorigenesis. Analysis of the molecular functional pathways by Genowiz™ identified four different potential relevant molecular functional pathways that were deregulated in these patients including mainly hedgehog signalling (CSNK1E, BMP7), wnt signalling (CHP, CSNK1E, RYK), notch signalling (JAG2) and epithelial cell signaling in *H. pylori* infection pathways (JAM3, ATP6V1G2). It was reported that a balance between wnt-fgh-notch and bmp-hedgehog signaling network was important to regulate the homeostasis among stem and progenitor cells and disruption of the signaling network results in cancer.

Validation of LDHB, PXN, RYK, RET, ANG, BMP7, RAD51L3 and CXCL10 genes by real time PCR in 37 cases of gastric cancer including 26 *H. pylori* positive and 11 *H. pylori* negative cases showed significant down-regulation in expression of RAD51L3 ($p=0.0013$), CXCL10 ($p=0.0038$) and BMP7 ($p=0.0002$) genes whereas significantly up-regulation in expression of PXN ($p=0.0077$), ANG ($p=0.0015$), LDHB ($p=0.019$), RYK ($p=0.044$) and RET ($p<0.0001$) genes in *H. pylori* positive cases compared to adjacent normal tissue. The *H. pylori* negative cases showed a significant up-regulation of expression of RET ($p=0.024$) and a significant down-regulation of expression of BMP7 ($p=0.018$) as compared to its adjacent normal tissue. The gene expression of PXN ($p=0.039$) was alone found to be significantly up-regulated in *H. pylori* positive Gastric Cancer cases on comparison to *H. pylori* negative Gastric Cancer cases. A significant direct correlation of expression of PXN with RET (0.523, $P=0.05$) was observed in *H. pylori* positive patients while *H. pylori* negative cases did not show any correlation in the expression of any of the genes studied.

The diverse patterns of expression of tumorigenesis related genes in *H. pylori* positive and negative patients suggested the probability for the existence of different and distinct molecular pathways in the pathogenesis of Gastric Cancer. It was possible that factors other than *H. pylori* infection including high tobacco consumption and food habits such as high salt intake interact with and contribute significantly to the initiation of inflammation in Gastric Cancer. This is a novel protumorigenic mechanism of inflammation in Gastric Cancer in which tumor-related pathways such as Wnt (CHP, CSNK1E, RYK), Notch (JAG2), Hedgehog (CSNK1E, BMP7) and epithelial signaling pathway in *H. pylori* (ATP6V1G2, JAM3) may play an important role in the development of Gastric Cancer in *H. pylori* infected and non-infected individuals via distinct molecular pathways.

Global Knowledge Hub on Smokeless Tobacco (KHSLT)

As per recommendation of Conference of Parties in its sixth conference (COP6) Global Knowledge Hub on Smokeless Tobacco (KHSLT) was established in National Institute of Cancer Prevention and Research by WHO Framework Convention on Tobacco Control (WHO FCTC) Secretariat and Ministry of Health and Family Welfare, Government of India in 2016.

The first objective of KHSLT was to generate and share expertise, information and knowledge on smokeless tobacco through information collection, analysis and exchange of information on smokeless tobacco and dissemination of these information through web portal, individual and group consultation, meetings and international conferences.

Second objective of the KHSLT is to assist Parties of the WHO Framework Convention on Tobacco Control (FCTC) in the area of SLT control by identifying laboratories and investigators working on SLT in the South-East Asia region and beyond; Strengthen communication and collaboration between laboratories through developing platforms for sharing data, information and expertise; nucleate and support collaborative projects on molecular and related social and economic aspects of SLT; develop standard operating procedures for analysis of SLT products; and generate and assess the existing data on specific SLT products and their constituents.

The second project was aimed at searching for the most recent literature and preparing a background paper to address smokeless tobacco products and the policies to control their use; the paper proposed draft recommendations on how to strengthen the control of smokeless tobacco products in the South-East Asia region; engage with Parties from the South-East Asia region, discuss the draft recommendations from the paper and develop strategies for their implementation in each Party of the region through an expert meeting organized for the Parties in the South-East Asia region.

The KHSLT is released through Global Web Portal. A policy document has been prepared. Several scientific and review papers have been submitted to different journals worldwide.

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The ICMR's National Institute of Cancer Prevention and Research (erstwhile Institute of Cytology and Preventive Oncology) has carried out research related to prostate and urinary bladder. Prostate cancer (CaP) is the fifth most common cancer worldwide and is the sixth leading cause of death from cancer in men. Older age, African ancestry and a positive family history of prostate cancer have long been recognized as important risk factors, only the early stage of unraveling the complex genetic and environmental influences on this disease. Over the past 20 years, the body of evidence suggests that genetics plays a key role has grown immensely, ranging from familial aggregation and twin studies, to family-based linkage studies, to detection of likely functional genes via mutation screening. However, the evidence also points toward a much more complex genetic basis of CaP than initially anticipated.

Androgens are implicated in the development of CaP and benign prostatic hyperplasia (BPH). The genes involved in androgen pathway and metabolism have been reported to contribute considerably to CaP risk. The increased genetic instability may play significant role in molecular carcinogenesis of CaP and DNA repair genes may serve as a biomarker for prognosis, predictor of therapy response, or indeed as target for novel therapy.

The study of gene expression and polymorphisms in androgen signalling (AR, PSA) and metabolizing (CYP19) genes in prostate tumours was carried out.

By regulating cell proliferation, differentiation and apoptosis the androgen receptor (AR) plays a pivotal role in CaP progression, as well as in normal prostate development. AR acts as a transcription factor regulating the prostate-specific antigen (PSA) gene by interactions with AREs. Aromatase encoded by CYP19 is a key enzyme that converts testosterone into estrogen in males is suggested to play an important role in the development of Benign Prostatic Hyperplasia (BPH) and CaP. The present study thus investigated the role of polymorphisms in AR, PSA and CYP19 in CaP risk. The study also aimed at investigating gene expression of AR and AR regulated PSA in CaP in Indian population. Functional consequence of risk genotypes of AR and PSA on gene expression was also evaluated along with clinicopathological correlation. The AR CAG and CYP19 TTTA repeat were genotyped by GeneScan analysis and validated by direct sequencing (Fig. 7.1 & 7.2). The PSA G-158A polymorphism was detected by PCR-RFLP assay. Gene expression of AR and PSA was evaluated using quantitative Real Time PCR (qRT-PCR) (Fig. 7.3 & 7.4).

Results showed a significant association of short AR CAG repeats (≤ 24) with risk of CaP (OR, 2.98, $p < 0.001$) and BPH (OR, 1.96, $p = 0.01$). PSA G-158A SNP was found to be significantly associated with risk of CaP (AA:OR, 2.68, $p = 0.016$ and GA:OR, 2.07, $p = 0.018$) p -trend 0.031 and BPH (AA:OR, 3.46, $p < 0.001$ and GA:OR, 2.47, $p = 0.03$) p -trend 0.009 respectively. Synergistic effect of short AR CAG repeat length (≤ 24) with PSA-158 GA genotype in increasing the risk of CaP was observed. PSA-158 GA genotype independently increased the risk of developing BPH [OR, 16.37 (95% CI 2.82-19.83, $p < 0.001$)] irrespective of AR CAG repeat length. Significant upregulation of AR and PSA gene expression in CaP comparison to BPH was detected. While, short AR CAG (≤ 24) repeats were associated with higher

AR expression in CaP ($p=0.002$), PSA SNP did not correlate with its expression. A positive correlation of PSA gene expression with AR was observed which provided further evidence supporting the concept that PSA was an AR regulated gene. Significant risk for BPH with A2 allele of CYP19 TTTA repeats was also observed. However, CYP19 TTTA repeat genotypes was not found to be associated with disease phenotype ($p>0.05$). Significant increase risk for CaP with the combined effect of short AR CAG (≤ 24) repeats with A2A2 and A2A3 (most common genotypes) genotypes of CYP19 TTTA was observed.

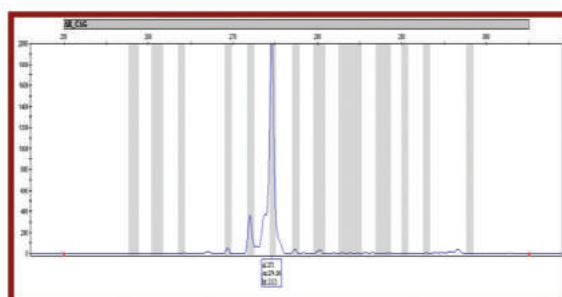


Fig. 7.1: Genotypes Plot showing the Gene Scan Analysis for CAG repeats in AR gene in CaP.

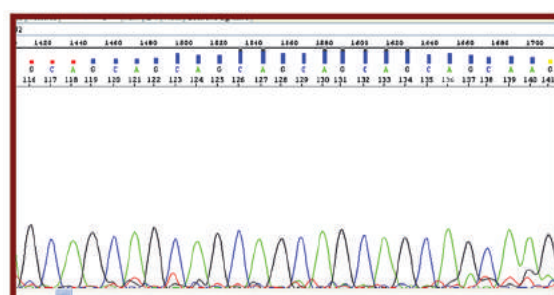


Fig. 7.2: Electrogram of CAG repeats in AR gene: CAG repeat length was found to be 23 in CaP case.

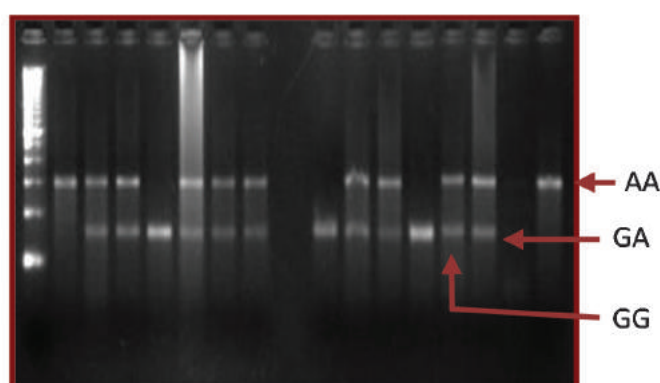


Fig. 7.3: PCR-RFLP analysis of *PSAG-158* Apolymorphism showing different genotypes.

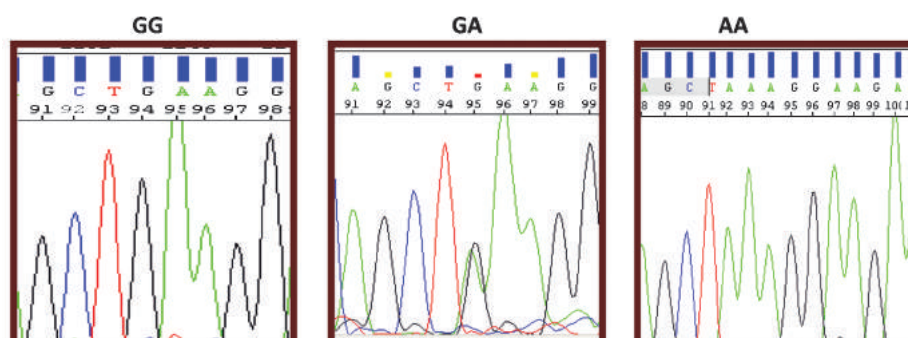


Fig. 7.4: Electrogram showing the sequence analysis of different genotypes of -158 G/A PSA polymorphism in CaP samples.

Study of TP53 Gene Polymorphisms/Mutations and Correlation with Microsatellite Instability in Androgen Receptor Gene

In majority of the cancers, TP53 dysfunction is caused through a direct mutation within the DNA-binding domain of the gene. Four mutation “hot spots” have been identified in exons 5 to 8, which coincide with the four most highly conserved regions of TP53 gene. In addition, exon 4 of TP53 harbours a common G/C nucleotide variation encoding the amino acids arginine (CGC) or proline (CCC) at codon 72 (Arg/Pro). This most frequent TP53 polymorphism at codon 72 lies in a proline-rich region, which is important for its ability to induce apoptosis. The codon 72 TP53 polymorphism was analysed in cases of CaP, Benign Prostatic Hyperplasia (BPH) and normal healthy controls using PCR-RFLP assay. Identification of mutations in TP53 gene, was performed on DNA extracted from formalin fixed paraffin embedded (FFPE) tissue of CaP and BPH.

A somatic C/A variation at the intronic boundary of exon 7 in TP53 gene was observed in 2% of CaP and 4% of BPH biopsy samples (Fig. 7.5). The study also showed a significant association of codon 72 TP53 Pro/Pro genotype with the risk of CaP (OR, 2.59, $p = 0.02$) and BPH (OR, 6.27, $p < 0.001$). Analysis of combined effects of AR CAG repeats and TP53 codon 72 genotypes showed that patients with short AR CAG (24) and Pro/Pro genotype of TP53 codon 72 polymorphism demonstrated a significant increased risk for CaP [OR, 5.33 (95% CI 1.47-19.30; $p = 0.01$)].

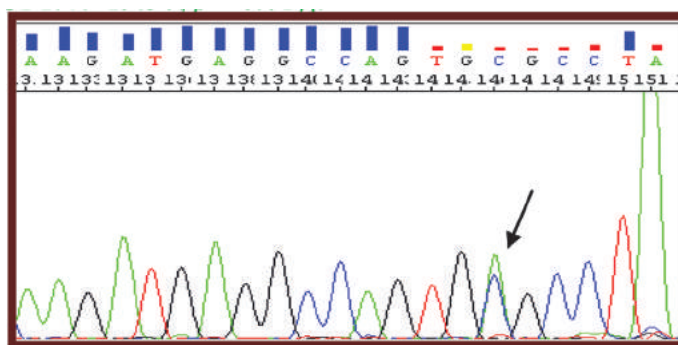


Fig. 7.5: C/A variation at the intronic boundary of exon 7 of TP53 gene.

The evidence suggested that TP53 was involved in androgen signaling. Additionally, studies have suggested that mutations in the TP53 gene were associated with human CaP progression. Moreover, mutations in TP53 may be a poor prognostic factor in CaP. Also literature suggested that the majority of metastatic CaP-derived cell lines harbor mutations of AR and/or TP53. These observations together suggested an important functional relationship between the TP53 and AR in the progression of human CaP. The risk assessment for the AR CAG repeats and TP53 polymorphism with CaP risk was performed by multinomial logistic regression analysis. Interestingly, patients having longer AR CAG repeats (>24) with TP53 codon 72 Pro/Pro genotype also demonstrated significant risk for developing BPH [OR, 4.55 (95% CI 1.26-16.44, $p=0.02$)] suggesting that the TP53 codon 72 Pro/Pro genotype was also an independent risk factor for BPH irrespective of status of AR CAG repeats.

Study of Gene Expression Profile of Mismatch Repair Genes and Correlation with the Prostate Carcinoma

The DNA mismatch repair (MMR) plays a role in promoting genetic stability by repairing DNA replication errors, inhibiting recombination between non-identical DNA sequences and participating in responses to DNA damage. While the role of MMR in prostate carcinogenesis remained unclear, MMR deficiency in CaP could prove to be clinically significant. Thus, the present study investigated the gene expression profile of six major MMR genes viz. hMLH1, hMSH2, hPMS1, hPMS2, hMSH3 and hMSH6 and polymorphism in hMLH1 and hMSH2 in CaP in Indian population. Further, correlation

with clinicopathological parameters was evaluated to establish their role as potential prognostic marker. The expression of the MutL homologues hMLH1, hPMS1 and hPMS2 and of the MutS homologue hMSH2, hMSH3 and hMSH6 were examined by quantitative Real-Time PCR (qRT-PCR) methodology and immunohistochemistry. Multiple endogenous control strategy using GAPDH, 18S RNA and TBP as endogenous gene was utilized to enhance the accuracy of expression analysis. A significant down regulation of hMLH1, hMSH2 and hPMS2 gene expression was observed in CaP compared to BPH (Fig. 7.6).

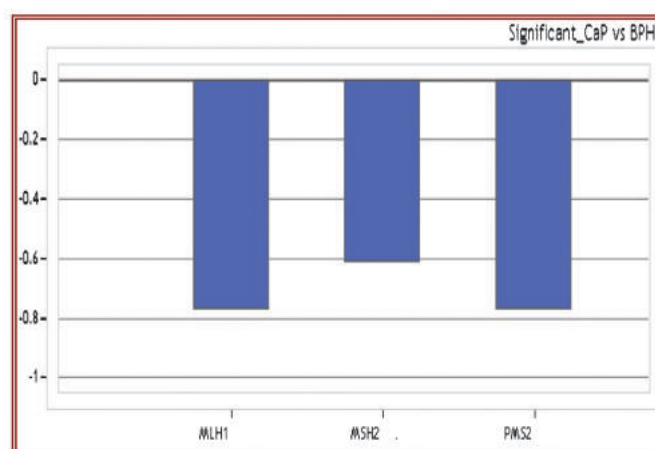


Fig. 7.6: Significant down regulation of hMLH1, hMSH2 and hPMS2 gene expression in CaP compared to BPH.

Similarly, immunohistochemical analysis in CaP showed loss of one or the other MMR protein, as profile of hMSH2+/hMLH1+/hPMS2+/hPMS1+ was not observed in a single case. Further, analysis of immunohistochemical patterns revealed that 33 (41.3%) cases of CaP had hMSH2+/hMLH1+/hPMS2-/hPMS1- profile, which was found as the most frequent immunohistochemical pattern. The greater loss of hPMS2 protein in poorly differentiated tumors was demonstrated which was in concordance with significant inverse correlation of hPMS2 gene expression with gleason score indicating its significance as marker for CaP progression (Fig. 7.7).

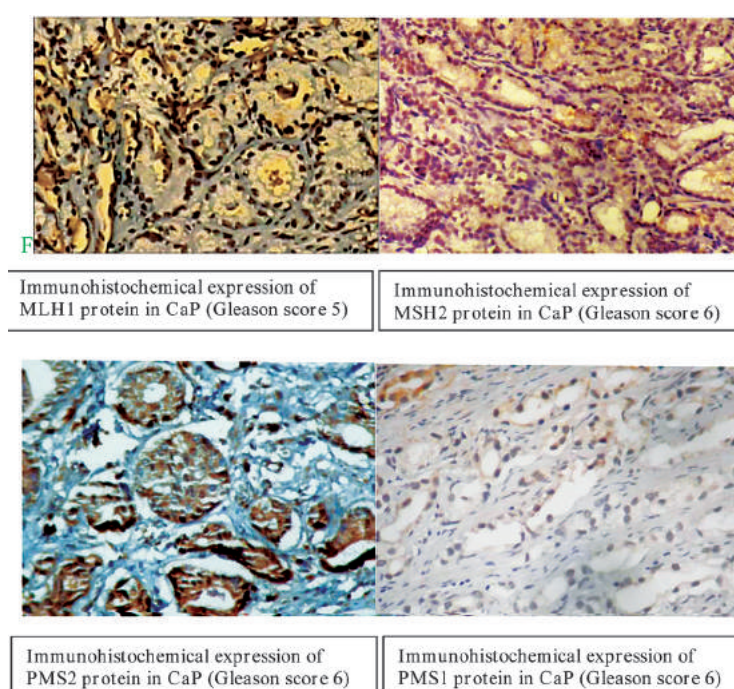


Fig. 7.7: Immunohistochemical expression of MLH and PMS gene.

An important association of hMLH1 93G>A polymorphism with the risk of CaP was also identified (Fig. 7.8).

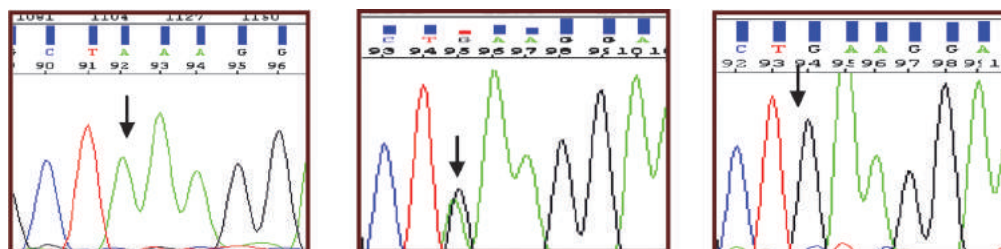


Fig. 7.8: Electrogram showing the sequence variants -93 position of the core promoter region of *hMLH1* gene.

In conclusion, the results suggested significant association of androgen signaling gene polymorphisms with risk of CaP and BPH and provided evidence for putative functional role of AR CAG repeat in regulating its expression. TP53 codon 72 polymorphism may play significant role in the pathogenesis and susceptibility to CaP and BPH. The present study also provided the evidence that TP53 and AR interact with each other to alter the risk of developing CaP. The results also suggested that an altered MMR had important biological and clinical significance in CaP in Indian population.

Molecular studies conducted by ICMR's National Institute of Pathology, Delhi showed that prostatic neoplasia showed interesting leads on the role of tumor markers such as p53 tumor suppressor gene in neoplastic and pre neoplastic diseases, cyclin genes in localized prostate cancer and mismatch repair genes and androgen receptor genes in advanced hormone resistant prostate neoplasia. Down regulation of MMR genes in Indian patients with carcinoma prostate (CaP) was reported for first time. Loss of hPMS2 expression served as significant prognostic marker for CaP. A novel MLH1 93G>A polymorphism in CaP patients may serve as potential diagnostic marker

URINARY BLADDER CANCER:

Urothelial (Transitional cell) cancer (UC/TCC) encompasses the spectrum of non-muscle-invasive (NMIUC), muscle-invasive (MIUC), and metastatic disease with an age-adjusted incidence rate of 21.1 per 100,000 population per year. The extent of disease determines clinical behavior, treatment and prognosis. NMIUC or superficial TCCs account for 80-90% of urothelial cancers. They can be diagnosed early and removed easily by transurethral resection of bladder tumour (TURBT), but management is complicated by recurrences in 70–75% of cases within 6-12 months. Adjuvant chemotherapy and immunotherapy are used to reduce recurrences, but do not reduce it to a significantly low level. Currently, various cytotoxic drugs e.g. doxorubicin, thiotepa, mitomycin-C, and cisplatin, and immunomodulating agents such as BCG and interferon- α 2b have been used in an attempt to reduce recurrence, with varying success. These reports indicated that, for superficial TCC, there is a need for individualized treatment, according to tumour-cell sensitivity.

In vitro Cytotoxicity Assays of Bladder Cancer Cells for Personalized Treatment

In vitro cytotoxicity assays were performed on autologous tumour cell cultures from patients with superficial TCC to select drugs with maximum cytotoxicity which were then delivered intravesically. Drug dosage was also decided based on the most effective dosage i.e. causing maximum cytotoxicity. The patients were followed up for recurrences. The chemotherapeutic drugs used in the study included Mitomycin C, Doxorubicin and Cisplatin and the immunomodulating agents used were BCG and Interferon 2b. The utility of treatment based on *in vitro* assessment was compared with the administration of BCG (120 mg) alone. The patients treated with chemotherapeutic/immunomodulator agents based on *in vitro* cytotoxicity assays had fewer recurrences (30%) compared to the control group (38%) and Kaplan–Meier

analysis showed a longer recurrence-free survival in the *in vitro* group (75%) than in the control group (49%) at 48 months of follow-up (Fig. 7.9). Further reduction of recurrence rate (15%) was achieved by repeating the *in vitro* cytotoxicity assays and revising their treatment accordingly. The repeat assay helped to identify the susceptibility of the recurrent tumour cell clones, and when the modification of dose or alternative/additional cytotoxic agent was added to the therapeutic protocol, it further reduced the recurrence rate from 30% to 15%. The failure to respond in the patients might be due to initial under staging of occult muscle-invasive disease, large tumour burden (>2.5 cm) and inability of the host to mount an adequate immune response.

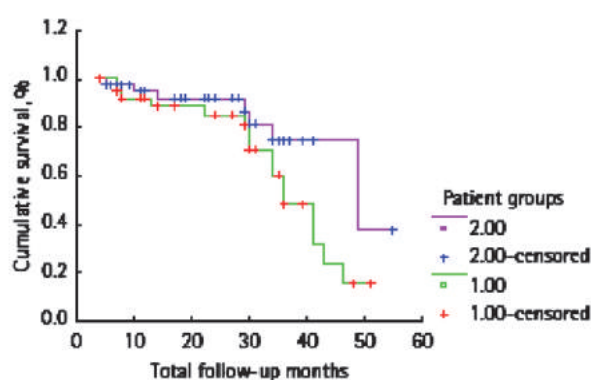


Fig. 7.9: Kaplan-Meier survival curve showing the comparison of cumulative survival in the two groups of patients.

To identify biomarker for predicting recurrent clones of bladder cancer cells, tumour suppressor genes (p53 and p21waf1) and angiogenesis factors (VEGF and CD105) were studied by immunohistochemistry. Recurrence of tumour [overall:39(43.3%) and low grade(LG):26(54.2%)] was found to be significant with p53 and VEGF expression and the profiles p53/VEGF, p53/CD105, VEGF/CD105, p53/p21/CD105, p53/VEGF/CD105 and all four together were significantly associated with recurrence in both groups. A mutually modulating effect of tumor suppressor and angiogenesis markers may be active, especially in low grade pTa non muscle invasive bladder carcinoma, with implication on change in therapeutic strategies. Concomitant expression of various markers in low grade pTa tumors could possibly indicate the need for stringent follow-up and efficient therapeutic management, in this subset of tumors which were clinically accepted to be less aggressive. Focus on the multiple pathways and their interactions may help in identification of a combined profile and contribute significantly in enriching knowledge regarding the biological behavior of non muscle invasive bladder cancer, in general and low grade/pTa cases, in particular.

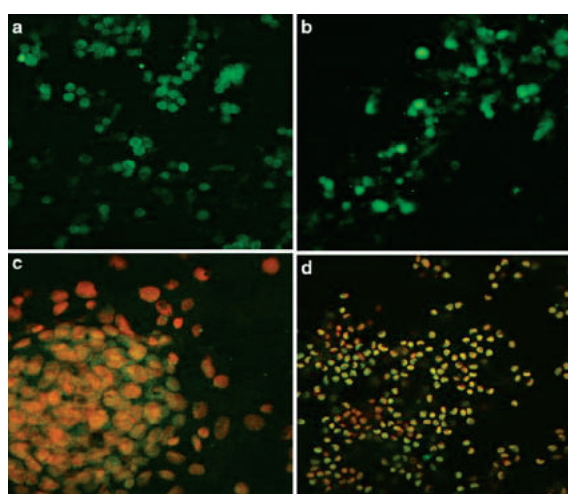


Fig. 7.10: The expression of the tumor markers is seen by Immunofluorescence staining. a) p53 nuclear expression in cell spots of bladder cancer b) p21waf1 nuclear expression c) cytoplasmic expression of VEGF with propidium iodide nuclear counterstain, d) CD105 expression in cytoplasm of bladder cancer cells.

Host Immune Factors and their Role in Recurrence and Progression

Earlier studies have suggested that patients with malignancies are immunodeficient, which manifests as impaired cell-mediated and humoral immunity, as well as by impaired nonspecific host defence mechanisms. Dysregulation of the immune response caused by the imbalance of Th1/Th2 cytokines may be responsible for survival of a tumour cell sub-clone despite intensive therapy. The reduction of recurrences in NMIUC by adjuvant immunotherapy suggests an association of molecular alterations of host immune factors in bladder cancer but the role of immunological dysfunction and tumour cell escape from immuno-surveillance has not been well documented.

To study the host immune response, the Th1 and Th2 balance was evaluated in Peripheral Blood Mononuclear Cells of urothelial cancer patients by flow cytometry. A significantly lower cell surface marker (CD4+, CD3+, CD8+ and NK) expression was found in patients compared to healthy controls and also in recurrent cases compared to non-recurrent cases.

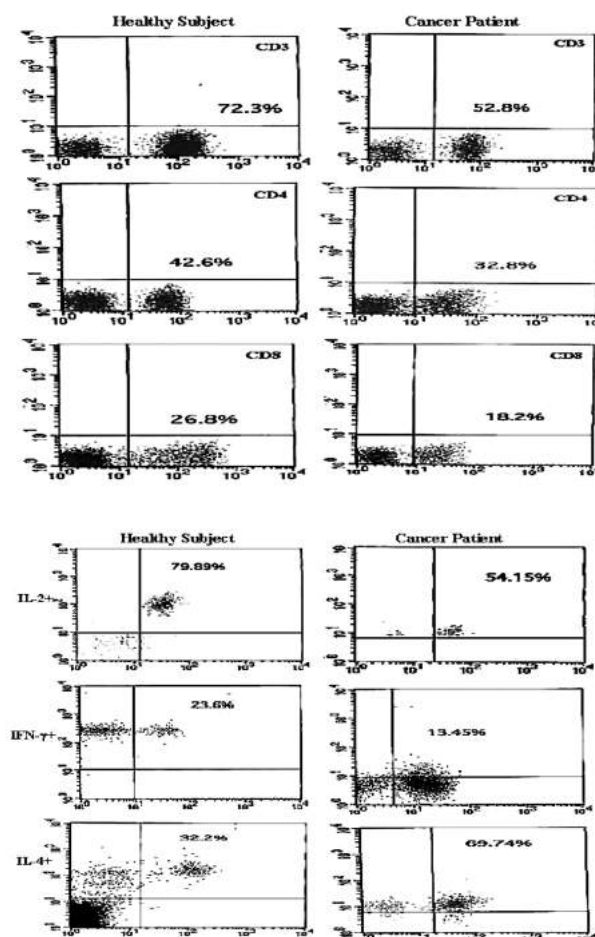


Fig. 7.11: Dot plots in a bladder cancer patient and a healthy control subject showing the a) phenotypic characterisation and the b) frequency of CD4+ cytokine positive T-lymphocytes by flow cytometry. The frequencies of CD4+ IFN γ +, IL2+ and TNF + T cells were reduced whereas CD4+ IL4+, IL6+ and IL10+ were higher in patients with TCC than in healthy controls.

In addition, mean levels of IFN- γ , IL2 and TNF were reduced and IL4, IL6 and IL10 were increased in patients compared to healthy controls thus showing a Th2 dominant response in Urothelial cancer patients. The response of patients to combination immunotherapy with intravesical Bacillus Calmette-Guerin (BCG) plus interferon- 2b (IFN- 2b) appeared to be immune dependent and activation of Th1 immune response required for clinical efficacy.

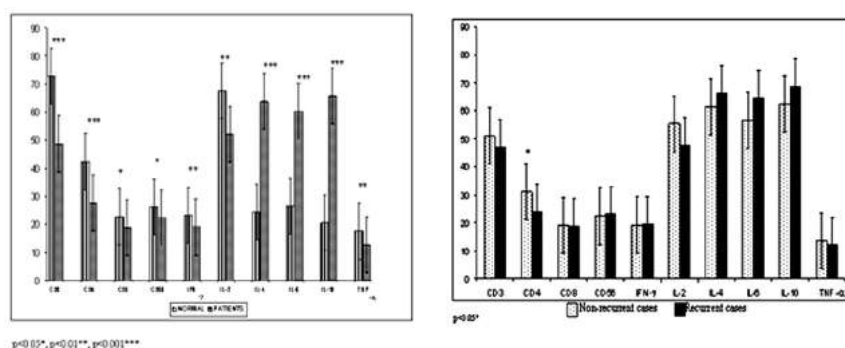


Fig. 7.12: Phenotypic characterization and the frequency of CD4+ T-cells producing Th1 and Th2 cytokines in a) healthy controls and bladder cancer patients; b) in recurrent and non-recurrent bladder cancer patients.

Evaluation of circulating serum cytokine profiles (Th1/Th2 cytokines IFN- γ , IL-2, TNF- α , IL-4, IL-6 and IL-10) using cytokine bead array (CBA) was performed in 41 bladder cancer patients prior to transurethral resection of tumor (TURBT) (pretherapy), and following intravesical combination immunotherapy (post-therapy) and their association with recurrence. Mean levels of IL-2 and TNF- α were significantly reduced while IL-4, IL-6 and IL-10 were significantly enhanced in pre-therapy samples as compared to controls. Mean levels of IFN- γ , IL-2 and TNF- α were significantly increased whereas, IL-4 and IL-10 were significantly reduced in patients after instillation of combination immunotherapy. The systemic immune response mounted by the host revealed a Th2 polarisation with tendency to reversal following therapy.

The study of local host immune factors responsible for tumour progression was evaluated on tumour tissue by Real-time PCR using SYBR green and compared with normal adjacent urinary bladder mucosa. The association of transcription factors of NF κ B pathway was also evaluated in the urothelial cancer patients and fold change of gene expression showed a Th2 dominant status in urothelial cancers overall, Th2 polarisation in muscle-invasive bladder cancers and Th1 polarisation in non-muscle invasive bladder cancers. Network visualization revealed the essential core genes affected to be Stat1, Fos, Jun and Egr1 in both invasive and non-invasive cancers. Gene signature inferred from centrality values of networks included Socs1, Tbk1, Gata3, and Rip1 in non-muscle invasive and Akt1, Gja1, IFN γ and NF κ B1 in muscle-invasive bladder cancers.

Further studies are in progress to evaluate the local and systemic immune response in an attempt to understand the role of host immune factors in modulation of therapy, progression and recurrence.

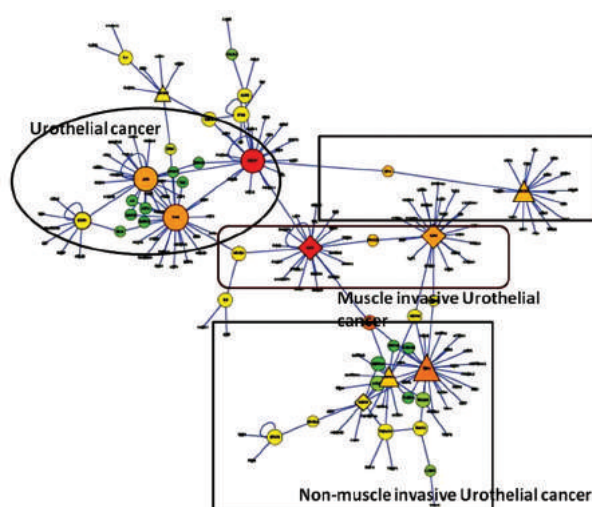


Fig. 7.13: Network visualization of cytokines and transcription factors altered in Urothelial cancer overall and in non-muscle invasive as well as muscle invasive cancer alone

At ICMR's National Institute of Pathology, Delhi, isolation and characterization of urinary bladder cancer cells and short term culture has been established. The model was tested for assessment of cytotoxicity induced by combination of cytotoxicity drugs with immunomodulators in the management of superficial bladder cancer. The role of Th1 and Th2 cytokines in the immunopathogenesis of superficial and invasive bladder cancers have been analyzed. A Th2 dominant host immune profile was shown to have association with recurrence of tumour in bladder cancer patients. Combination panel of immunohistochemical markers (p21waf1/VEGF/CD105) was more effective in predicting recurrence of bladder cancer than a single marker.

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Awareness on the role of diet and cancer has increased in the last few decades. The Indian Council of Medical Research's National Institute of Nutrition (NIN) has taken up some studies in the late 1980s and 90s. The role of diet in the causation of cancer has always been difficult to study and quantify, partly because the diet encompasses a wide variety of foods, dietary traditions, habits and a complex mixture of nutrients and non-nutrients.

Diet is composed of macronutrients like protein, carbohydrate, fats and micronutrients namely vitamins, minerals and innumerable phyto-chemicals. Diet can be a source of agents that promote carcinogenesis as well as inhibitors of the process. Carcinogens bind to the cell macro molecules namely, DNA, RNA and protein and result in mutagenic events leading to cell transformation and neoplastic changes. Some phytochemicals prevent these changes from occurring either by directly binding to the carcinogens and/or their metabolites or by metabolizing and eliminating toxic xenobiotics. These are known as anti-mutagens/ anti carcinogens.

NIN's research in cancer started with studies *in vitro*, in animals and subsequently progressed to applied human studies. In the early 80's, the research was directed to Aflatoxins B1 (AFB1) and Drug Metabolizing Enzymes in undernourished rats. The results indicated that diet restriction *per se* may tilt the balance of reactions towards formation of toxic metabolites. These may be mutagenic/ carcinogenic metabolites. Certain drugs which were in use in humans like rifampicin were studied and have been concluded to be at risk of developing mutagenic metabolites.

Research on the anti-mutagenic effects of spices carried out in animal models demonstrated that turmeric has anti-mutagenic effects mediated through induction of detoxifying enzymes system particularly GSHT.

Mustard, a spice has effect on conjugation systems and has significant impact on mutagen load produced by the ubiquitous pollutants like Benzo(a)pyrene (B (a) P). At the same time, a sensitive assay for detecting DNA adducts was standardized and applied for studying tissue samples. This ³²P post labeling was tested in cigarette smoking volunteers for DNA adducts, and was successfully demonstrated to be higher than in non-smokers (P<0.05).

An experimental study was undertaken in animals induced with iron deficiency anaemia where there is decrease in activating liver enzymes and decrease in conjugating enzymes. This possibly led to impaired detoxification process resulting in persistence of carcinogen/metabolites in body.

Another *in vitro* study carried out indicated that the spices and sauces contain sodium nitrate much above the specified limit of 200 ppm in such foods. The nitrites are known to convert nitrates into nitrosamines. Nitrosamines were detected in various substances using a Gas Chromatograph attached to Thermal Energy Analyser (GC-TEA). Carcinogenic nitrosamines, many a time, are present in malt blended alcoholic beverages. For this reason, many chronic alcoholics are at risk for developing cancers.

A study revealed that the alcoholic beverage especially beer in our country contained high amounts of nitrosamines (3.6 ppm) as compared to products in most Western countries (<0.6). In other countries, meat and meat products have been shown to contain nitrosamines.

SPICES AS ANTIMUTAGENS/ANTICARCINOGENS

Turmeric

Turmeric is used extensively in Indian culinary practice for its color, taste and flavor. It is also added to food as a preservative. Curcumin, the active principle in turmeric, has strong antioxidant and anti-inflammatory potential. Its potent anti-mutagenic effects were demonstrated against well known carcinogens in vivo rat model. In rats, turmeric intake was observed to elevate levels of tissue xenobiotic enzymes that are protective in nature against carcinogen metabolism. There was decreased binding of liver DNA to the carcinogen benzo (a) pyrene as quantitated by ³²P post label assay. In oral cancer model, turmeric/curcumin painting of Syrian Golden Hamster cheek pouch resulted in retardation of tumors induced by exposure of cheek pouch to carcinogen dimethyl benzanthracene (DMBA). In mice, fore-stomach tumors were induced by benzo (a) pyrene and efficacy of turmeric and curcumin as anti-initiator and anti-promoter was tested. While turmeric inhibited both the phases, curcumin could only inhibit post-initiation process.

DNA repair is one of the important mechanisms of protecting the system from the toxicity of environmental toxicants. Protective effect of curcumin against induction of single strand breaks in DNA was observed in yeast cells treated with UV radiation, 8 methoxypsoralen and BP. Anti-mutagenicity of turmeric was demonstrated in human smokers who were known to excrete mutagens. The excretion of urinary mutagens was reduced at the end of 15 days of turmeric ingestion (1.5 g/day orally for 30 days) and further by 30 days time.

Alliums

Among the vegetables, those belonging to the allium family have received increased attention in recent times. Onion and garlic are commonly consumed in Indian diets. They contain sulphur compounds like diallyldisulphide, allium and ajoene. Prior feeding of onion at 1 and 5 % levels and garlic at 0.1, 0.5, 1 % levels through diet showed decreased excretion of urinary mutagens by rats exposed to mutagens. Stimulation of the activity of liver cytosolic glutathione-S-transferase, increase in activity of the antioxidant enzyme quinine reductase in liver and lung microsomes was observed in garlic fed rats. In onion fed rats also, stimulation of detoxification enzymes was reported. These findings suggested that toxicity due to exposure of endogenous or exogenous carcinogens could be averted by consuming alliums regularly.

Treatment with Allium Sativum, Lactobacillus rhamnosus GG and NSAIDs prevented DMH induced histopathological alterations and tumour formation in the colonic tissue; and inhibited Wnt and β -catenin expression. Microarray experiment demonstrated that PI3-kinase/Akt and Wnt/ β -catenin pathway mediate key signals for intestinal epithelial cell proliferation and inhibition of apoptosis in an experimental model of colon cancer. Inhibition of Wnt and β -catenin pathways and induction of GSK-3 β , which may eventually activate apoptosis and block key signals for intestinal epithelial cell proliferation, appeared to have a potential role molecular therapy.

Ginger

Ginger is a condiment used in various forms throughout the world. In traditional medicine ginger in the form of dried powder, raw form and juice have been prescribed to ameliorate conditions like rheumatic disorders, inflammation and gastrointestinal discomforts.

Cytogenetic damage to tissue is considered as one of the biomarkers of genotoxicity. Human blood lymphocytes of male smokers, male non-smokers and female non-smokers were exposed to trans-stillbene oxide in the presence and absence of ginger. The cytogenetic damage in terms of micronuclei was significantly reduced due to the presence of ginger. Ginger was also observed to counteract induced DNA damage in human peripheral blood lymphocytes. The excretion of urinary mutagens by benzo (a) pyrene exposed rats that were fed ginger was reduced significantly. Ginger feeding also stimulated the levels of detoxification enzymes and antioxidant enzymes proving that it can be an effective anti-mutagenic and anti-cancer agent.

Effect of cooking on anti-mutagens in spices

The studies on anti-mutagenic potential of spices were conducted using raw forms or powders. Most of the spices are usually subjected to boiling or frying while cooking food and may therefore undergo thermal denaturation. Turmeric, curcumin, allium, few vegetables, ginger were tested for their anti-mutagenic potential after subjecting them to heat treatment to simulate cooking conditions. The spices and their active principles did not show alteration in their anti-mutagenic properties suggesting that they can liberally be incorporated during cooking and consumed.

Human Studies

Case Control Studies: One of the ten most important cancers in the world is cancer of oral cavity and pharynx. In India, oral and pharyngeal cancers are the most common forms of cancers among males and the third most common among females. In South East Asia, including India, the habit of chewing betel leaves, tobacco with other ingredients is one of the main risk factors in the causation of oral cancer. Use of tobacco in other forms like smoking is also implicated in oral cancers.

There is a marked variation in the incidence of cancers of the oesophagus and stomach around the world. A case control study was conducted to investigate the role of dietary factors in the incidence of oropharyngeal and oesophageal cancers. Detailed personal and dietary history was obtained and dietary nutritional correlates of cancers were evaluated. The studies revealed that intake of green vegetables and fruits were significantly low in cases as compared to controls. The food frequency questionnaire showed higher risk association with less frequent intake of vegetables and higher consumption of fried foods. The odds ratios calculated revealed high risk estimates with low intake of beta carotene, thiamin, riboflavin, folic acid, iron, magnesium and copper. The odds estimates were also higher for low levels of hemoglobin, albumin, folate, zinc and vitamins A and E and selenium in plasma. Similar observations were seen in oesophageal cancers too.

The low intake of vegetables including roots and tubers and the low levels of some micronutrients in case control studies prompted further research trials to elicit the cause and effect relationship.

Intervention studies: Studies were conducted to understand the effects of nutrient cocktail comprising of vitamin A, riboflavin, selenium and zinc on oral precancerous lesion caused by reverse *chutta* (rolled tobacco leaf) smoking in rural areas of coastal Andhra Pradesh. The results suggested a positive role in the remission of oral lesions. The micronuclei were shown to decrease by 25% after the intervention (for 1 year) in the lesion group, while the non-lesions showed no such changes. There was a positive correlation on reverse *chutta* smokers with oral palatal lesions. The micronuclei which were shown to be a cytological marker of carcinogenesis decreased significantly after supplementation.

The DNA adducts also decreased significantly in the lesion group, while the non-lesion group did not show any change even after 1 year of intervention. These findings added strength to the fact that micronutrient supplementation has beneficial effects on pre-cancerous oral conditions.

A similar study in the above population with B-carotene as nutrient intervention agent supplemented for a year did not show any beneficial effect on oral precancerous lesions. But turmeric when supplemented for a year was beneficial and showed a remission of oral precancerous lesions. The micronuclei and DNA adducts which did not decrease on beta carotene supplementation, decreased on supplementation with turmeric.

Other Studies: Female breast cancer patients at different stages of the disease were studied for any changes in lipid profile, and it was shown that different grades of cancer were not related to cholesterol and triglycerides levels. This indicated that these parameters may not be related to breast cancers.

A study undertaken for the role of nitrosamines, *in vitro* under simulated gastric conditions, was carried out in a population residing in high risk area. Some foods consumed by them were found to get nitrosated to a greater extent than corresponding foods consumed by population residing in low risk areas. This may make the former population more at risk of developing cancers.

Mucosal damage is the common side-effect in cancer chemo/radiotherapy. A study was undertaken at the Institute on the apoptosis of rat intestinal cells and the mechanism involved when the animals were deprived of 50% vitamins in their diet. Apoptosis was induced by an anticancer drug, cisplatin. It showed oxidative stress, apoptotic index and caspase – 3 activity to increase and antiapoptotic factors were not expressed. With vitamin supplementation, oxidative stress and apoptotic rates were decreased, while Bcl – 2 expressions (anti-apoptotic) was very high.

Type of dietary fat in the etio-pathogenesis of carcinogenicity with respect to breast cancer was looked at, in experimental animals. It revealed that n6 fatty acid rich diet has deleterious effects in breast neoplasm while diets rich in only n3 fatty acid had a better outlook.

In a more recent study, when the effect of dietary fatty acids (saturated vs unsaturated vs trans fats) was assessed in development of Dimethylbenzanthracene (DMBA) induced experimental mammary tumours in female fisher rats, it was found that the saturated fatty acid (SFA) level was not affected by the type of dietary fat administered and total mono-unsaturated fatty acid (MUFA) level was highest in the group fed with trans fats. Long chain n-6 PUFA levels were not significantly different among the groups. Number of tumours was found to be least in trans fat group and highest in n-6 PUFA group. Adenocarcinoma was observed as predominant tumor type and metastatic tumours, the least common type and metastasis was noted only in saturated fat group. The study indicated that the fatty acid composition to some extent reflected the plasma phospholipid fatty acid composition. Trans fats caused development of mammary tumours. N-3 PUFA was associated with low and n-6 PUFA was associated with high risk of cancer development.

Aldose Reductase (AR), the rate-limiting polyol pathway enzyme that converts glucose into sorbitol and involved in secondary complications of diabetes, is usually upregulated in many cancer cells and is thought to be involved in resistance to chemotherapeutic drugs. A study looked into the specific activity of AR in RBCs; and its specific activity and expression in tumour and non-tumour areas. When whole blood samples of cancer patients and controls were collected and their histopathological evaluation was done for typing and grading the tumours, it was noted that the specific activity of AR in RBCs of cancer patients was significantly increased compared to those in controls. This showed that AR could possibly serve as a biomarker for identifying the potential cancer development.

Another study to elucidate the importance and implication of apoptosis (programmed cell death) in prognosis of medullary carcinoma (MC) of breast indicated that a high frequency apoptosis may be related to a favourable prognosis in medullary carcinoma even though it demonstrated high mitotic count with a

rapid cell turnover. Since apoptotic cells can be counted with a good reproducibility in H and E stained tumour sections, the apoptotic index could possibly be used as an additional prognostic indicator in invasive breast cancer.

In a study that assessed the role of dietary phytate-mineral interactions in suppressing colon cancer, it was observed that the body weights were comparable among the groups although feeding phytate along with Azoxymethane [but not azoxymethane (AOM) alone] decreased food intake in rats. Incidence of aberrant crypt foci (ACF), an identifiable, early marker for the onset of colon cancer was seen in all AOM treated but not in other groups. Protective effects of phytates against colon cancer are indicated in the preliminary findings

A study that assessed the efficacy of polyphenol-rich dietary ingredients as proteasome inhibitors (The proteasomal degradation pathway is essential for many cellular processes, including the cell cycle, the regulation of gene expression, and responses to oxidative stress) and their role as anticancer agents. It was observed that *Murraya koenigii* leaf (curry leaf) extract caused decrease in the proteasome activity and induced cell death in a panel of human cancer cell lines.

Conclusion:

Data on incidence of cancer indicate that the rates are growing over the years. Urbanization, western life style and food habits, increasing usage of chemical fertilizers and pesticides, the habit of tobacco smoking or chewing have all been implicated in the etiology. The morbidity and mortality is causing a huge burden on the health front. Moreover the cost of treating cancer is enormous and our country cannot afford it.

On the other hand results from experiments in animals and epidemiological studies on humans provide unequivocal evidence that chemopreventers, in the levels found in human diet are safe. Diet rich in chemo-preventers will have beneficial health effects, cost of treatment of cancer are higher than cost of providing diets rich in phytochemicals and modifying eating habits. While there is a need to identify these substances, tendency to take supplements must be avoided as against a back drop of poor diet intake, no nutrient can produce the desired outcome.

A more appropriate approach is one, in which the importance of foods in the diet is stressed. Nutrition education will promote public awareness of the dietary guidelines for promotion of healthy life.

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The Indian Council of Medical Research (ICMR), the apex body in India for formulation, co-ordination and promotion of biomedical research is one of the oldest medical research bodies in the world. The Council promotes biomedical research in the country through intramural as well as extramural research. Extramural research is promoted by ICMR through setting up (i) centers for advanced research in different research areas around existing expertise and infrastructure in selected departments of medical colleges, university and other non-ICMR research institutes (ii) task force studies which emphasize a time bounded, goal oriented approach with clearly defined targets, specific time frames standardized and uniform methodologies and often a multicenteric structure (iii) open-ended research on basis of application for grants-in-aid received from scientists in non-ICMR research institutions located in different parts of the country.

The Human Resource Department in biomedical research is encouraged by ICMR through various schemes such as (a) Research Fellowships and Research Associateship (i.e. Junior, Senior Research Fellowships and Research Associates); (b) Short term visiting fellowships (which allow scientists to learn advanced research techniques from other well established research institutes in India); (c) Short term Research Studentship for research studentship (for under graduate medical students to encourage them to familiarize themselves with research methodologies and techniques); (d) various training programmes and workshops conducted by ICMR.

Regarding Man Power Development under the aegis of Indian Council of Medical Research, special efforts were taken to train staff.

The ICMR Bulletin is a monthly in-house periodical of the Council under the Division of Publications & Information. It is published with the main aim of disseminating to various target groups, the results of medical research and information on major research activities being carried out under the aegis of ICMR. It is not only becoming more popular with various categories of biomedical scientists and medical students, but it has also registered increasing popularity among the media personnel.

Apart from publishing regular features on topics of current biomedical interest, the ICMR Bulletin also features review articles, bring out special issues commemorating important events and national and international days of relevance such as No Tobacco Day, World AIDS Day, World Health Day, Breast Feeding Week, etc. ICMR reproduce articles/ publish extracts of the Bulletins of previous year are also available.

The extramural research conducted through various research institutes across India, funded by ICMR through various projects in oncology and published in form of abstracts from 1971- 2011 has been compiled to mark the achievements of Council in promoting cancer research in country.

BREAST CANCER RESEARCH

Breast Cancer forms the second most common cancers among women. As the number of women diagnosed with breast cancer each year is increasing worldwide, research into breast cancer's origin, pathology and treatment option becomes inevitable. There are numerous studies which report role of sex steroid hormones in etiology of breast carcinomas. Physiologically, human female breast is under primary control of different hormones, the role of estrogen being pivotal. Worldwide major studies about different types of breast carcinomas are being studied and mammothly funded to bring down the increasing numbers of breast cancer patients. Breast cancer patients are classified into four pathophysiological subgroups based upon expression of Estrogen Receptor (ER + or PR+), overexpression of members of Epidermal Growth Factor receptor family (HER2 +ve) or none falling into these- known as triple negative breast cancer (TNBC). A subgroup with poor prognosis is triple negative breast cancer. In this, tumor cells lack three important receptors which are found in breast cancer. Research in this direction is already being going on. Targeted therapies for above three may still be present but for the fourth and remaining cancers, none exists.

Numerous studies on estrogen receptor and progesterone receptor have been undertaken to determine their role as prognostic marker in disease outcome.

According to a study done in early decades of cancer research in country, ER analysis was performed to determine the receptor positivity in particular ethnic group of society (South Indian population). It was observed that majority of tumors studied were positive for ER with increased positivity in postmenopausal women. The study demonstrated requirement of hormones (estradiol and progesterone) to maintain tumor growth.

In another study done in late 90s; estrogen and progesterone status and its relationship to endocrine profile in breast cancer patients was studied. Risk factors along with hormonal imbalance play a major role in development of breast cancer. Circulating levels of Luteinizing Hormone (LH), Follicle Stimulating Hormone (FSH), Prolactin, Estradiol, Progesterone and Testosterone were studied. According to a study ER and PR receptor measurement helped in selection of proper treatment modality in primary and metastatic breast cancer patients where receptor rich tumor responded to endocrine therapy whereas receptor negative tumor responded well to chemotherapy.

Another study undertaken at the end of 20th century, working on prognostic indicators in human breast cancer, studied the expression of factors like Epidermal Growth Factor Receptor (EGFR), cathepsin D and p53 oncoprotein. It was concluded that immunohistochemistry was useful technique for detection and study of overexpression of p53, cathepsin D and EGFR.

In another study, presence of prolactin (PRL) receptor in crude membrane tissue and its role in mammary tumorigenesis was studied. In this study, FNAC showed consistent immunostaining in carcinoma cells as compared to benign tissues.

The expression of antimetastatic marker-nm-23 and PRL in human breast cancer along with other prognostic markers was investigated. It was observed in a separate study that hyperprolactinaemia in patients with advanced stage cancer with recurrence / metastatic disease and decreased in patients who responded to the therapy. The levels of PRL remained elevated throughout the course of disease in patients who did not respond to adjuvant therapy. The study concluded showing that PRL was superior marker for monitoring the disease whereas nm-23 was proved to be a marker of tumor aggressiveness.

A study on the role of c-erb2 oncoprotein as prognostic indicator in Breast Cancer was carried out on patients with histologically identified carcinomas of Breast Cancer. In the study the role of new prognostic

factors viz AgNOR (Silver Binding Nucleolar Organizing Regions), expression of proto-oncogene c-erb B2, vimentin expression along with Estrogen Receptor and Progesterone Receptor was evaluated. The findings indicated that c-erb2 marker can be used to predict disease progression, vimentin helped in tumor detection in ER and PR –ve tumors. Vimentin and c-erb2 oncoprotein helped in detection of tumors which did not respond to routine management. However, no relation was shown between AgNOR counts, c-erb2 oncoprotein and histologic grade. Thus, AgNOR alone could not be used as marker for tumor proliferation.

Study of Serum Fucose level (SFL) in benign & malignant lesions of breast cancer may also be of value in identifying patients “at risk” of developing recurrences, as it was concluded in the study where levels of SFL increased with progression of malignancy from stage I to IV, with mean SFL higher in patients with metastasis.

In a study monoclonal antibodies to estrogen receptor and its application in study of breast cancer was undertaken. The study was done on MCF-7 cell lines. Estrogen receptor was isolated and monoclonal antibodies were generated. ERE5, ERE10, ERE 7 were found to recognize cytosolic ER. These monoclonal antibodies not only serve as diagnostic tools but also serve to understand the molecular mechanisms by which estrogen stimulate cell proliferation.

CERVIX CANCER

In India, cervical cancer in female is the second most common cancer of reproductive tract. Surveys showed that India has nearly 1/3 of world's cases of cancer. It is associated with persistent infection with oncogenic human papilloma virus (High risk HPV 16 and 18 type). Herpes Simplex Virus was found to be a co-factor. The concept of screening healthy and asymptomatic women for early *in situ* detection of cervical cancer has proved to be useful in decreasing the mortality from this type of malignancy.

Most of the cervical cancers (90%) are squamous cell carcinomas. Remaining 10% are adenocarcinomas (originates in mucuous producing cells near cervix). This is one of the few types of cancer where there are clear stages called precancerous stage of cervical intraepithelial neoplasia (CIN) which can be diagnosed via Pap smear test. These may form lesion called dysplasia or squamous intraepithelial lesion (SIL). Dysplasia being the common condition often disappear without treatment. However, precancerous cells can become cancerous.

In the year of 1980s, a study was conducted for the identification of atypical reserve cells in smears of patients with premalignant and malignant changes in the squamous and glandular epithelium of uterine cervix, pure ARC hyperplasia was found to be important precursor of malignant lesions particularly of the epimoid type.

As per treatment is concerned, for years, staging of cervical cancer was based on clinical examination and treated via radiotherapeutic doses. In a study, thermoluminescence dosimetry (TLD) carried out by LiF (Lithium Fluoride) appeared to be ideal choice for intracavitary treatment for cancer cervix by radiation doses to rectum and bladder.

With the advancement of modern technologies in research and well accepted follow up of the patients after the treatment leads to advance clinical applications to study the malignancy of the disease.

During 1990's, studying association of cell kinetics with response of tumor to radiotherapy resulted to know the proliferative activities of argyrophilic nucleolar organizer regions (AgNORs) and this was considered to be a positive indicator of prognosis. Other Tumor Growth Factor (TGF) associated markers were also studied. Disease free patients showed increased AgNORs and TGF, mild to moderate TGF- β ,

moderate bcl-2 while patients with residual or recurrent disease showed intense bcl-2 –ve for TGF's. EGF and EGF-R showed significant pre-treatment co relations with final disease outcome. As per the advance researches the causes and symptoms of cervical cancer were evaluated. Many immunological studies paved a way to know the risk factors associated with cervical cancers.

An immunological study using rHPV fusion proteins to determine humoral and cellular immune response against HPV-16 in patients with benign and malignant lesions of uterine cervix was done. Peptide PI and PII were found to be major immunodominant B cell epitomes of HPV 16 E7, PII being more than PI. HPV 16 E7 T cell epitome reactivity was also found to be depressed in HPV 16 +ve cervical cancer patients by lymphoproliferative assays.

Another immunotherapeutic study was undertaken to understand the role of T cells response to chronic infection and natural Tregs i.e. T regulatory cells were encountered in progression of premalignant lesions to malignant lesions in squamous cell carcinoma (SCC) in addition to TGF – β secreting Tregs. CD 25-FOX P3+ population of cells were also seen in tumor microenvironment. Effector T cells belonged to both subsets and were found in invasive disease.

OVARIAN CANCER

The research studies on ovarian cancer led to identification of p16 playing an important role in human ovarian cancer progression and blood DNA methylation of p16 promoter region is a weak predictor of tumor tissue methylation status in human ovarian cancers.

In a study conducted in late 1990s; significant role of tumor markers in ovarian neoplasms was undertaken. It was found out that AFP was found in endodermal sinus tumors and embryonal carcinoma whereas hCG in embryonal carcinoma. Estradiol was found in granulosa cell tumor and serves as ideal marker for sex and stromal tumors. Keratin was found to play an important role in epithelial neoplasms and germ cell tumors showing extra embryonic differentiation.

Further, in a study undertaken during the late 90s; role of p-53 and bcl-2 genes and its cellular manifestations in ovarian cancer was studied. The objectives of the study were to determine extent of apoptosis in ovarian tumors. The study led to conclusion that bcl -2 gene was expressed in benign cystadenomas and endometroid carcinomas whereas it was not expressed in serous tumors, clear cell carcinomas and undifferentiated carcinomas. The presence of PCNA (Proliferative Cell Nuclear Antigen) and p53 protein was co-related. Role of p53 in regulation of cell proliferation in ovarian tissue was determined. However, no co-relation was shown between bcl-2 and PCNA. Therefore, bcl-2 was not found to be a major regulator of cell proliferation.

HEPATOCELLULAR CARCINOMA

Hepatocellular carcinoma is cancer of the liver and is the most common type of malignancy of cancer which occurs in population worldwide. It is responsible for significant morbidity and mortality across the world. It is eighth most common cancer in women and fifth most in men worldwide. The main aetiology being Hepatitis B and C infections, smoking and alcohol consumption also accounted for disease progression. Although surgical resection or liver transplantation remained standard procedure for treatment, advances in molecular pathogenesis of liver could pave the way for improved diagnosis and treatment regimes for management of disease.

In the projects conducted during 1970s, an attempt was made to determine the protective effect of two known enzyme inducers namely DDT and sodium barbitone against a carcinogen DENA (Diethylnitrosamine). The effect of first inducer DDT could not be worked out as the animals treated with DDT died in 2 and 10

weeks of injection but second inducer was tolerated well for more than 7 weeks. Phenobarbitone treatment seemed to protect the liver against chronic damage by the carcinogen. It was observed that induction of hepatic microsomal enzymes brought about by phenylbarbitone administration detoxified DENA in liver and protected the organ against injuring action of this agent. Liver cell detoxified substance by use of microsomal enzymes. Based upon this mechanism, beneficial effects of phenylbarbitone with regard to DENA toxicity were concluded. In a study done in early 1990's, alterations in liver fibrosis were studied in adult male albino rat. The study of DMN induced liver revealed enhanced anabolism and catabolism of hepatic collagen. Study on mineral and trace elements during progression of hepatic fibrosis revealed significant decrease in Ca, Mg, K, Na, Se, Zn in serum. Results indicated exacerbation of hepatic fibrosis with ascites played a major role in alteration of essential elements which further aggravates disease. During research conducted in mid 1990s, frequency of hepatocellular alterations of preneoplastic nature in liver of population groups with different incidences of carcinoma in wister strain albino rat and rhesus monkey to hepatocarcinogenic agents was undertaken. The study concluded that chemical carcinogenesis induced multicentric tumors. This might be related to neoplastic transformation of hepatocytes.

The projects aimed to determine chemical and biological effects of oxygen radicals produced by exposure to benzene and antioxidant potential in liver and serum during oxidative stress was studied. Such Fe (II) was possibly made available due to reduction of Fe (III) by polyphenols and further oxidation. It also concluded that glutathionyl hydroquinone (GHQ) as a potent pro-oxidant, capable of enhancing iron dependent formation of superoxide radical and also due to its faster auto-oxidation to glutathionyl benzosemiquinone may facilitate formation of adducts with DNA which was one of the prerequisite of benzene toxicity and carcinogenesis.

In the late 1990s, role of 'S' transactivator in hepatocarcinogenesis; *in vitro* expression and purification of transactivating proteins was done. The study was aimed to elucidate the effect of 'S' transactivator on various promoter enhancer systems; raise monoclonal/ polyclonal antibodies against this transactivator protein prepared either by rDNA method and localize the protein in HCC cells. Possible role of HBV in hepatocellular carcinoma was suggested. Insertional mutagenesis, transactivation of truncated X / preS2/ S regions and activation of growth regulated genes or oncogenes have been suggested as possible mechanism for carcinogenesis.

The research studies conducted in 2000s highlighted the involvement of virological, serological tumor markers and serum pro and anti-angiogenic factors in spectrum of chronic liver diseases and HCC; role of sensitivity and specificity of alpha-fetoprotein (AFP) and Des gamma carboxyprothrombin (DLP) as diagnostic marker of HCC. Angiogenic and antiangiogenic factors like VEGF (vascular endothelial growth factor), basic fibroblast growth factor (bFGF), angiogenin and endostatin patients with HCC. The studies suggested that increased serum levels of VEGF, bFGF, angiogenin and endostatin were observed in patients with HCC. High VEGF levels linked possible invasion into portal vein.

ORAL CANCER

Oral Cancer is one of the most fatal health problems faced by the man kind today. It is reported that India accounts 86% of total oral cancer cases across the world, ranking number one in terms of incidence among men and third among women (men are twice as susceptible than women). A number of anatomic sites are included in the category of Oral Cancers including mouth, oral cavity (lips, lining of lips, cheeks, teeth, tongue, floor under the tongue, hard palate & gums), oropharynx (back of tongue, soft palate, tonsils). The statistics of cancer patients showed 40% buccal mucosa, 30% alveolus cancer, 15% lip cancer and 13% tongue cancer patients.

The attributes for oral cancers are many risk factors like addictive habits of tobacco & tobacco related products, alcohol, cigarette smoking & various genetic and hormonal factors; tobacco chewing at the top in case of India, because Indians chew tobacco more than smoking (14% smoking tobacco users and 26% chewing tobacco users). Geographical and sexual differences may also be the factors that should not be ignored.

In early 1980s, various biochemical studies were designed to determine cell-mediated immunity and oncogenic factors associated with the oral cancer patients.

In a cell line study undergoing chromium release assay technique, the T-suppressor cells were found to be increased with concomitant decrease in T-helper cells in the peripheral circulation. Leucocytes were found to be adhered on the glass surface, lymphocytes showed no responses except when tested with autologous lymphocytes which was specific to antigen. Three-fold increase in glucosaminoglycan composition was observed mainly in dermatin sulfate & chondroitin sulfate fraction. In another study, during similar time fluorescent studies showed intracytoplasmic Herpes Simplex Virus (HSV) antigen in oral cancer patients. So, HSV was found to be an oncogenic virus associated with oral cancer.

In late 1980s; clinical examinations and clinical trial studies were undertaken to determine the prognosis and treatment of the disease.

A study reported that though the patients were able to mount the immune response but with certain impairments due to increased number of suppressor cells in circulation and presence of immune complexes and other blocking factors in serum. Some tumors were also found to escape mechanisms along with sharp decline in cell numbers following radiotherapy which took more than six months to attain normalcy.

Simultaneously, various therapeutic studies were also designed for the treatment of squamous cell carcinoma of oral cancers.

In a study, it was found that levamisole specifically stimulate T-cell population but had found no significant stimulatory effect on proportion of B1TG & TM lymphocytes. It was also reported that with the increase in Circulating Immune Complexes (CICs) level; the patient becomes prone to recurrence. Levamisole was found to decrease the level of CICs.

In context of other parts of oral cavity specifically in buccal mucosa cases, in late 1980s, a similar study was designed to evaluate BLM+HBO+radiation & radiation + HBO+ placebo in comparison to BLM + radiation in treatment of T3, T4 oral squamous cell carcinoma. Short term follow up of the patients show radiation+BLM gave best results as compared to the radiation with addition of HBO and group of radiation + HBO+placebo. Another biochemical study was done to study the tumor host interaction of buccal cancer. Impaired immune response was observed by delayed hypersensitivity to Purified Protein Derivative (PPD) and Phytohemagglutinin (PHA). Even primary stages of cancer showed depression in absolute lymphocytes count, absolute T-cell count & E-rosettes. These clinical parameters showed inverse co-relations with tumor stage, size, and lymph node involvement. IgA was also found to be elevated during advance stages. Total serum protein and albumin was also found to be decreased along with tumor iso antigen.

Various chemotherapeutic studies were also designed during the years 2010-11. Chemopreventive effects by dietary administration was done on hamster buccal pouch carcinogenesis by bovine lactoferrin (bLF) and black tea polyphenols. DMBA (7,12-dimethylbenzaanthracene) induced genotoxicity in HBP (Syrian Hamster Buccal Pouch) carcinogenesis was found to be inhibited by dietary administration of bLF (0.2%) and P-B (0.05) by modulating carcinogen metabolizing enzymes CYP1B1 and CYP1A1. It was also

observed that combination act synergistically to inhibit cell proliferation by downregulation of PCNA, GST- η , NF- κ and mutant p53. It also induced apoptosis by downregulation of Bcl-2 and upregulation of Bax, Fas and caspase-3 expression.

A clinicopathologic study in 1990s determined prognostic significance of tumor markers in tongue carcinoma. It was found that immunoreactive PRL mRNA (prolactin mRNA) was produced from tongue tumors and proved to be a better marker than bioactive prolactin; differentiating patients with aggressive tumors. Cathepsin D was also found to be overexpressed in aggressive tumors, so it can serve as potential biomarker.

LEUKEMIA

Leukemias account for 2% of all cancers. It is regarded as disease of common form of childhood cancer. However, leukemias affect far more adults than children. Leukemia begins in the immature cells of the bone marrow in central cavities of bones. The WBCs that don't mature normally hinders the production of other blood cells & eventually accumulates. The main types of leukemia are myelogenous, lymphocytic and each has an acute (rapidly processing) and chronic (slowly progressing form).

Acute Lymphoblastic Leukemia (ALL) is the most common type of leukemia in young children (also affects adults, especially those of age 65 and above) and treated with chemotherapy and radiation. Chronic Lymphoblastic Leukaemia (CLL) is the second most common cancer that affects adults over age of 55 (sometimes in younger adults, never in children). It is incurable, but there are some effective treatments. Acute Myeloid Leukemia (AML) occurs more commonly in adults than in children and if left untreated, progress quickly. Chronic Myeloid Leukemia (CML) occurs mainly in adults. It occurs due to genetic abnormality called *philadelphia chromosome*.

The various risk factors identified play a role in the development of leukemia: exposure to ionizing & medical radiations, history of Down Syndrome and other genetic factors, cigarette smoking, exposure to certain chemicals such as benzene, viral infections (HTLV-1 i.e. Human T-cell Leukemia Virus-1), Myelodysplastic syndromes.

In early 1980s; the role of prostaglandins in the pathogenesis of leukemia was studied and it was observed that Prostaglandins (PGs) would modify normal and leukemic Leukocyte alkaline phosphate (LAP) activity as well as cell membrane bound enzyme activity by modulating fluidity and gene action of cell membrane. Activity of LAP in CML was also studied in correlation with low cyclic AMP level in leukocytes. Lower LAP concentration in CML cells was due to lower cAMP content which may be due to altered prostaglandin system.

In late 2000s; a chromosomal and cytogenetic analysis study of bone marrow samples was done to evaluate the prognostic effects of karyotypic changes and telomerase activity in pre-leukemic and leukemic conditions. Broad spectrum of karyotypic abnormalities and heterogeneity in telomerase activity in different FAB subtypes of MDS and ANLC patients was observed. It was reported that telomerase expression can be used as biomarker. So pre treatment status of telomerase expression could be considered as novel genetic determinants of prognosis, apart from karyotypic abnormalities.

A study conducted in 2011 showed up-regulation of GST-Pi and MDR in AML and also poor prognosis was reported in T-ALL patients. DHFR expression found to be up-regulated and co-related with resistance to MTX in AML. The p53 was reported to be up-regulated in both AML and T-ALL but not co-related with drug sensitivity. Patients undergone *in vitro* drug assays and had achieved complete remission (CR) had low white blood count and CD34+/- CD10+ phenotype.

BRAIN TUMOUR

During 1996, role of neural modulation of immune response in patients of intracranial tumor, immune parameters were studied for surgery treated patients with benign tumors and surgery + RT + CT treated patient with malignant tumors. The study confirmed the altered innate, cell-mediated and humoral immunity in brain tumor patients before and after treatment. The immune changes observed were related to cell mediated immunity and were immunosuppressive. Though the pituitary region tumors were benign, patients with these tumors showed either marked increase or decrease in levels of immune parameters studied than observed in those with tumor in other areas. The immunomodulatory effects of malignant tumors were more marked than those of benign tumors.

Cavernous Angioma, a blood vessel abnormality occurs anywhere in Central Nervous System. In a study titled "Diagnostic parameters of vascular neoplasms by immunohistochemistry" endothelial cell markers FVIIIIRAg (Factor VIII Related Antigen) and UEA-I Lectin (Ulex Europens Agglutinin -I Lectin) were studied in tumor tissue in the year 1991. The study was done using 103 tumor tissues mainly of vascular origin. The study showed that these two markers are of immense value in study of vascular neoplasms.

In 1990s, patients with pituitary adenomas were studied using light and electron microscopy and serum hormone estimation to elucidate hormonal status and biological behavior of different types of pituitary adenomas. According to study, the hormone responsible for hyperfunction was always raised in serum and localized in tissue. Growth Hormone (GH) was observed in all tumors from patients with acromegaly (somatograph adenomas), prolactin in all tumors from patients with Cushings Syndrome. 30% tumors were found to be pleurihormonal, GH and prolactin being the commonest combination. However, all hormones localized by histochemistry in pleurihormonal adenomas were not detected in serum or reflected clinically.

In a study, during 1987, utility of various tumor markers viz. β HCG, LDH, H/M ratio, glucosaminoglycans (GAG), β -endorphin was studied among patients who underwent surgical removal followed by radiotherapy to establish simple tumor marker for early detection and management of brain tumors. The study concluded that despite discordance between disease activity and marker levels, there was a value in markers level that correlated of tumor. More frequent determinates of tumor markers would be useful especially during time of greatest tumor change (due to initiation of treatment) and to provide an indication of response to therapy.

HEAD AND NECK CANCERS

In a study of 1984, combination treatment in moderately advanced head and neck cancer was undertaken to compare conventional treatment against combination of radiotherapy, chemotherapy and surgery in moderately advanced HNCC. The study included a control group and that undergoing combination therapy with radiotherapy and chemotherapy. It was observed that tumor regression was faster in group with combination therapy as compared to control group.

OESOPHAGUS

In another study of 2011, a case controlled study involving neo-adjuvant chemotherapy downstage the disease and found to be much beneficial for the patients with bulky growth to reduce the growth. Survival advantage was seen only for the patients who underwent preoperative chemotherapy not surgery alone.

In a study of 2006, carcinoma associated antigens (Tn, s-Tn and T); efficacy in detecting and monitoring patients with squamous cell carcinoma of oesophagus (OSEC) was undertaken with the aim of analyzing the early glycobiological changes in the OSEC using specific lectin. Peanut Agglutinin (PNA) binding

Transcription Factor antigen (Galbt3GalNAC) was used as marker for study. Increased circulating levels of TF-Ag were observed in sera of OSEC patients compared to healthy controls. Higher levels of simple sugar and mucin were found in OSEC patients. The study clearly revealed a trend in the altered expression of TF Ag analyzed by PNA with tissues as well as sera of OSEC patients.

NON-HODGKIN'S LYMPHOMA

The project on "Overexpression of p53 and cell proliferation indices in NHL" assessed morphological classification of Non-Hodgkin's Lymphoma. Proliferating cell nuclear antigen (PCNA) and Argyrophilic nucleolar organizer regions (AgNOR) showed higher labeling indices in high grade lymphomas with high p53 positivity in tumors. Finding of the study indicated that p53 expression in malignant lymphomas was related to the grade of lymphomas and that p53 positivity may also be a marker for cell proliferation.

In another study during late 90s, tumor cell kinetics in gestational trophoblastic disease as a predictive index of tumor aggressiveness was studied. The study correlated markers of cell proliferation with regression pattern of tumors, level of β -hcg and histological nature of the tissue on clinical aspects. PCNA and EGFR expressed strongly by molar placenta of all gestational ages unlike normal placenta where expression was restricted to 1st and 2nd trimesters. Ki67 was also found to be strongly expressed by molar placenta. Also, PCNA in combination with Ki67 would serve better marker in the assessment of prognosis of trophoblastic tumors.

In another study during initial years of 90s, detection and assessment of molecular markers associated with multidrug resistance (mdr) in high grade non Hodgkins Lymphoma was undertaken. Lymph node biopsies and fine needle aspirates from patients with high grade NHL were analyzed to study the expression of MDR gene and MDR gene product, P-glycoprotein in high grade NHL and correlation of findings with course of disease. Of 109 patients, 73 remained disease free and 36 showed residual and recurrence of disease. 26 out of 36 (72%) with recurrent disease expressed P-glycoprotein and recurrence of disease. Co-relation analysis showed significant relationship between prognosis and detection of P-glycoprotein by immunohistochemistry, western blot and mdr gene expression by southern blot. In order to rule out false positive results, P-glycoprotein analysis accompanied by western blot and southern blot analysis was found to be a prognostic indicator for high grade NHL.

HODGKIN'S DISEASE

In late 1990s, an *in situ* hybridization study of Epstein Barr virus in Hodgkin's disease was studied. The objective of the study was to demonstrate presence of EBV -RNA in malignant cells and to localize the EBV encoded latent membrane protein -1 (LMP-1) in malignant cells. The data from the study strongly suggested that when present, EBV was not merely a silent passenger in Hodgkin's disease but contributed directly to the pathogenesis. Some neoplasms were suspected to be EBV related. The demonstration of LMP-1 along with detection of EBER-1 sequences was found to have important implications since it was one of the latent gene products which produce B-lymphocyte transformations.

COLON CANCER

In late 1990s; the project entitled "Fragile sites and mutagen sensitivity in colon cancer families" was studied to determine presence of constitutional chromosomal markers and chromosomal fragile sites in familial and non-familial colon cancer patients and their relatives to see whether any member showed sensitivity to mutagen or resistance with regard to chromosomal breakage. It was observed that genetic susceptibility risk in familial colon cancer was transmitted from generation to generation. Good correlation between bleomycin induced chromosomal hypersensitivity and cancer susceptibility was seen. The results

indicated that bleomycin induced chromosome sensitivity analysis can be used as biomarker for higher risk members in cancer families.

In late 1990s, c-myc localization in pre-malignant and malignant lesions of human colon was studied with the aim to seek co-relation of expression of c-myc oncogene with degree of differentiation of pre-malignant and malignant lesions of colon using immunohistochemistry methods (using monoclonal antibodies against c-myc protein) and morphometry.

GALL BLADDER CANCER

A hospital based study undertaken on 200 newly diagnosed patients with carcinoma of gall bladder was undertaken during 2011 with special reference to nutritional risk factors. According to the study, the major risk factors encountered were preference for fried foods, consumption of mustard oil and family history of gallstone disease. In females, early age of marriage, number of live birth (more than 3) and post menopausal status were found to be major risk factors for gall bladder cancer. Tea and coffee were found to show protective effect.

PANCREATIC CANCER

A multivariate logistic regression analysis study was undertaken to study risk factors for cancer of pancreas in 2010. The risk factors according to study were found to be chronic pancreatitis, diabetes for more than 4 years, smoking and family history. Dietary risk factors found were non-vegetarian diet, higher intake of carbohydrate and proteins. β -carotene and chromium containing foods were protective. The study can be considered to target high risk individuals in order to reduce pancreatic cancer related mortality.

URINARY BLADDER

In study conducted in late 1980s, cytogenetic study in patients with solid tumor was undertaken. The study was done with solid tumors in esophagus, breast ovary and retinoblastoma and normal controls. Analysis of C- Banding positive chromatin of chromosomes 1,9, 16 revealed that size of C-Band region was significantly larger for all three chromosomes in patients with carcinoma breast, esophagus and retinoblastoma whereas only for chromosomes 1 & 9 in carcinoma ovary patients. 50% of cancer patients carried complete or partial inversion of C-Band region in chromosomes 1, 9 & 16 as compared to only 15% controls. The study concluded that in cancer patients, C-Band heterochromatin was observed with significantly higher degrees and in particular C- Bands of chromosome 1 were most noteworthy.

In study conducted in early 1980s; presence of circulating immune complexes was studied in 65 patients of transitional cell carcinoma of urinary bladder. The circulating immune complexes (CICs) were detected by 5 techniques viz. Rheumatoid Latex Agglutination Inhibition (LAI), Polyethylene Glycol Precipitation (PEG), Anti-Complementary Activity (ACA), Rheumatoid Factor, Cryptoglobulins. It was observed that CICs presence correlated with clinical staging and histological grading (Broder's classification) of tumor. CICs were found in all types of malignancies of genitourinary tract but difference between them was not statistically significant. Increased amounts of rheumatoid factor associated with progression of tumor were observed. Cryptoglobulins did not show any special trend.

ASCITES FIBROSARCOMA

Studies on effect of tumor growth and radiation on immune surveillance of mice in relation to modern therapeutic markers was undertaken. The study was aimed with objectives to delineate the immune status favorable for growth of particular tumor line in an animal, to identify immune parameters that are important for resisting or rejecting the growth of tumor and to study effect of radiation on these parameters. Immune status in syngenic and allogenic mice with respect to Ascites Fibrosarcoma Tumor were clearly

demonstrated. Study revealed greater resistance of allogenic mice to AFS tumor. Comparative studies on immunological status in tumor bearing syngenic and allogenic mice revealed a depressed cell mediated immunity reaction with particular reference to structural and functional changes as given by rosette formation and cytological potential. According to study, limitation for using higher dose of radiation for clinical and other purposes has been established to avoid risk of immune suppression leading to increased susceptibility to neoplastic development. Membrane transformation changes of splenic lymphocytes in tumor bearing animals could be used as important marker for assessment of neoplastic growth and be of diagnostic value in clinical field.

DMBA EXPOSURE

A study to experimentally confirm the abrogation of carcinogenic response in chick embryo model and role of thymus and bursa of fabricus as factors involved in carcinogenesis was undertaken in 1991. The study was carried out to confirm whether abrogation of carcinogenic response was possible by pre-treatment of chick embryo with 9, 10, dimethyl 1, 2-benzanthracene (DMBA) and also to delineate role of thymus and bursa of fabricus in process of carcinogenesis. Study reported that tumour incidence was recorded to be very low when thymectomised and bursectomized chicks were injected with DMBA dissolved in PEG as none of 51 thymectomised chicks developed tumor after single intramuscular dose of 4 mg DMBA in 0.1 ml of PEG.

According to a study done in the same year, effect of Zinc supplementation on DMBA induced tumor in chicks indicated a possible protection against DMBA induced tumors by zinc supplementation. Incidence of tumor in animals supplemented with 50mg ZnSO₄/kg body weight and injected with DMBA in 0.3 ml DMS, was 46.4% as compared to 80.1% in control group. Tumor always occurred at site of injection indicating DMBA itself as carcinogenic and not its metabolites.

BIOCHEMICAL MARKER STUDIES

In 1990s, studies on immunohistochemical localization of keratin proteins in tumor diagnosis were undertaken. Paraffin sections of various tumor tissues stained by three stage immunoperoxidase technique using commercially available DAKO rabbit polyclonal anti human keratin antiserum to detect keratin helped to distinguish tumors of similar morphology and also assessed the usefulness of technique in routine reporting of surgical specimens. A majority of squamous cell carcinomas were positive for keratin, while lymphomas were constantly negative, adenocarcinomas showed wide range of staining from totally negative to faint positive but majority displayed negative for keratin.

Metabolic profiles in sera of cancer patients were studied in the year 1990. Levels of cholesterol and lipid associated sialic acid (LSA) in sera of patients of uterine cervix and breast cancers were determined using agarose gel electrophoresis and precipitation methods. Elevated LSA were found to be reduced in chemotherapized patients so as to serve as marker. It was observed that cholesterol in LDL increased and HDL decreased in uterine cervix cancer. No change in very low density lipoprotein (VLDL) was observed. Cholesterol in low density lipoprotein (LDL) and high density lipoprotein (HDL) was found to be increased in breast cancer. The study concluded showing good positive correlation between electrophoresis and precipitation methods for lipoprotein analysis. Electrophoresis method is however recommended for rapid routine analysis as conventional precipitation procedure is time consuming.

In studies conducted in late 1980s; immunohistochemical studies using 12 histochemical markers as an important aid in diagnosis of undifferentiated soft tissue sarcoma was undertaken using Vimentin, Desmin, Actin, Myoglobin, Keratin, epithelial membrane Antigen (EMA), S100 protein, Alpha-1 Antitrypsin (A1AT), Alpha-1 Antichymotrypsin (A1ACT), FVIIIIRAg, Collagen Type IV and UEA-1 using peroxidase

and antiperoxidase and avidin-biotin complex technology with Di-aminebenzidine tetrahydrochloride (DAB) as substrate. The results of study concluded with immunohistochemistry being considered as a reliable tool in diagnosis of undifferentiated soft tissue sarcoma.

Studies in mid 1980s were undertaken for evaluation of aspiration cytology in diagnosis of lymphadenopathies. The study assessed the fine needle aspiration cytology (FNAC) for diagnosis of metastatic lymphadenopathies. The accuracy of FNAC in diagnosis of metastatic lymphadenopathies was 96.57%. FNAC was found to be safe, simple and quick diagnostic method.

In an attempt to discover diagnostic tools for early detection of cancer, biochemical estimations in serum (fucose, proteins, phosphohexoisomerase-PHI, alkaline acid phosphatase, glutamate oxylate transaminase-GOT, glutamate pyruvate transaminase-GPT, lactate dehydrogenase – LDH, CL and PLs) were carried out. Serum fucose levels showed a significant increase in cancer patients. The increase was directly proportional to clinical stage of disease. The other parameters (serum PHI, phosphatase, GOT, GPT, LDH, CL and PLs) were also significantly elevated in cancer patients. Serum protein levels were decreased in cancer patients. Based upon the study, serum fucose estimation appeared to be helpful in diagnosis of early stages of malignancy and may be used as follow up tool for determining effectiveness of therapy.

TREATMENT OF MDR CANCERS

In search for novel resistance modifying agent (RMA) deactivating resistance causing protein (p-gp, MRP) and Glutathione (GSH), a number of novel Schiff's bases and their metal chelates were synthesized and characterized in a project entitled "Study of mechanism of action of novel resistance modifying agents overcoming multi drug resistance in cancer" in the year 2010. Copper (II) N-(2-hydroxyacetophenone) glycinate (CuNG) was observed to be most potent in overcoming Multi Drug Resistance. CuNG increased ROS generation and reduced MRP1 in drug resistant EAC/DOX cells. It was observed that level of copper in serum was significantly higher in MDR cancer bearing animals and also in cancer patients unresponsive to drugs. CuNG treatment resolved MDR cancers through induction of apoptogenic cytokines (IFN- γ /TNF- γ) from splenic mononuclear cells or patients peripheral blood mononuclear Cells (PBMC). The results of study showed usefulness of CuNG in immunotherapy of multi drug resistance through programming of tumor associated macrophages (TAM). Modulation of behavior of TAM by reactive oxygen species (ROS) generating metal chelate like CuNG may pave way of developing new drugs for treatment of cancer irrespective of drug resistant phenotypes

In a cell line study undertaken in the year 2006, non-toxic resistance modifying agents i.e. RMAs were identified and developed. RMAs were found to be of maximum potentiality in *in vivo* conditions of various cell lines (EAC/DOX, S180/DOX, 3-LL, HL-60/AR mrp-expressing). RMAs such as Copper N-(2-hydroxy acetophenone) glycinate increases life span of drug resistant cancer (swiss mice) to nearly 323%. Cobalt N-(2-hydroxy acetophenone) glycinate (CoNG) overcomes MDR in *in vivo*. Other RMAs like Oxalyl bis (N-phenyl) hydroxamic acid (OBPHA) was found to be effective in P-gp mediated multidrug resistance (MDR). Metal chelates were also found to be as potential as RMAs. These RMA showed glutathione (GSH) depleting properties and affecting the cellular expression of GST, MRP1, MRO2, gst-pi, gst-mu in various cell lines and thus showing the reversal of MDR.

In mid 1980s, correlation of *in vitro* and *in vivo* sensitivity patterns of breast cancer tissue against various anticancer drugs was studied. According to study, an *in-vitro* assay system was developed to facilitate the establishment of a patient specific chemotherapeutic regimen for management of breast cancer. *In-vivo* and *in vitro* drug response showed a positive correlation of 45%. The drug assay system used a technically feasible approach to determine *in vitro* chemo sensitivity of tissue.

DRUG THERAPY

In late 1990s; role of some non-steroidal anti-inflammatory drugs in carcinogenesis was studied. The study carried out in male weanling inbred swiss mice to evaluate role of non-steroidal anti-inflammatory drug (NSAIDs viz. piroxicam, Diclofenac and Naproxen in carcinogenesis. Animals were fed NSAIDs through diet. For tumor studies highest doses of the NSAIDs were used with carcinogen Nitrosodiethylamine (NDEA). Diclofenac (375ppm) inhibited NDEA induced lung tumorigenesis whereas Naproxen (1500ppm) stimulated promotional phase of carcinogenesis. Ornithine decarboxylase (ODC) which is rate limiting enzyme in mammalian polyamine biosynthesis is required for tumor promotion and cell proliferation in various organs was inhibited by Diclofenac, Piroxicam but was stimulated by Naproxen indicating this enzymes as good markers for tumorigenesis and agents with anti-carcinogenic properties. Prostaglandin E2 release, involved in stimulation and tumour promotion was decreased to a variable degree in plasma, liver and lung by all three NSAIDs listed. Increased activities of glutathione-s-transferase, glutathione reductase and reduced glutathione by NSAIDs, may detoxify xenobiotics more efficiently. The enhanced tumorigenesis and ODC levels produced by Naproxen required further evaluation for its safety.

The study undertaken during early 80s, studied the role of adjuvant combination of chemotherapy in connective tissue malignant tumors with special reference to assessment of cell-mediated immunity (CMI). In the study total of 53 patients of connective tissue tumors (27 osseous sarcomas, 26 soft tissue sarcoma) were studied for cell mediated immune status before and after adjuvant chemotherapy with different combinations of antitumor drugs prior to or after surgery. 82% of patients were allergic to PPD and DNCB before treatment showing depressed CMI in diseased state. 83.3% showed positive response to DNCB after therapy. According to study, chemotherapy was showing promising results in terms of remission, duration of remission and disease free period. The maximum disease free period in patients with both types of tumors was found to be 30+ months.

DIETARY INTAKE

In 2010, a study on potential role of Galectin-3 and dietary Galectin inhibitors in tumor spread and preventive response was undertaken. Systematic analysis of galectin-3 levels in various normal and metastatic cell lines indicated co-relation between Galectin-3 level expressions vs. stage of cancer. Series of dietary sources examined for potential Galectin 3 blockade that can halt or prevent metastasis. Screening studies indicated dietary sources including pectin polysaccharides from swallow root (SR) and carrot (CR) expressed potent Galectin 3 inhibitory properties. The data suggested critical involvement of Galectin-3 in metastatic spread. Galectin inhibitors exerted their effect at molecular levels as evidenced by modulation of tumour-suppressor gene product (p53), transcription factor (NFkB) necessary for proliferation, caspases (responsible for induction of apoptosis), matrix metalloproteinases (responsible for cell invasion) and bcl-2 (an apoptotic protein) in both *in vitro* and *in vivo* models. Overall, data from the study emphasized the promising role of dietary galectin inhibitors against metastasis.

NATURAL PRODUCT THERAPY

Cancer therapeutics is an upcoming research area in treatment of cancers and disease progression. As cancer is a lifestyle related disease and its number is increasing in urban population and rural population by many folds in coming time; finding novel strategies along with conventional ongoing treatment regimes for various cancers may form a base for upcoming research in cancer therapeutics in decades to come. There have been many new research studies undertaken by using nature derived products. Conventional therapies produce side effects and sometimes the disease may resurrect. Chemotherapeutics may produce toxic side effects, which is the main cause patients stop medication and disease progresses and ultimately

leads to the death of the patient. Phytochemicals derived from plants or plant products or any other natural source may avert such toxic side effects and help in curing the disease with better ease. With the advent of technology and advancement in natural product therapy, cancer can be effectively cured without any harmful side effects and pain. Major research in this direction is focus of the hour. In future, these remedies based upon the use of natural product / derived products may pave the way for effective cure for cancer.

In a study entitled “Search for anti-tumour drugs from Indian plants-Phytochemical study” role of *Trichosanthes Roxb* (roots, fruit, and seeds), *Kigella pinnata* (fruits), *Potamogeton indicum* (whole plant), *Vermonia roxburghii* and *Elephantopus scaber* (whole plant) in Sarcoma-180, Ehrlich's sarcoma in mice was studied. In the study, only *Roxb* was found to be active possessing antitumor activity against Sarcoma 180 and ascites marginal activity in leukaemia. The petrol-ether extract of root yielded yellow sweet smelling oil and showed antibacterial activity against *Bacillus subtilis*.

In a similar kind of study, response of Sarcoma-180 to *Plumbago rosea* (chitraka), radiation and hyperthermia was studied. The study was undertaken with aim of studying tumour growth inhibiting and radiosensitizing effects of alcoholic root extract of *Plumbago rosea* along with hyperthermia (43°C) and gamma radiation (10Gy). It was observed that *Rosea* could be promising drug for human cancer therapy as it showed inhibition of exponentially growing sarcoma-180. It has remarkable radiosensitizing effect.

A study on snake venom as anticancer agent was undertaken. The study was done on animal model and different human cell lines like liver cells, splenic, lymphoblasts, leukemia cells. Venom treated FAC cells (liver) inoculation showed cell growth reduction. Antioxidant activity was increased showing anticarcinogenic activity. Increased macrophage count and reduction in tumor growth was observed. It was concluded that NKV (*Naja Kaoutnia*) is most potential venom.

Role of neem leaf preparation in function of chemokine receptor and its ligand in terms of chemotaxis and tumour cell toxicity was undertaken, where it was seen that neem leaf glycoprotein (NLGP) upregulated CCR5 expression from evidenced studies on monocytes / macrophages of peripheral blood from head and neck cancer patients. The NLGP upregulated HLA-ABC, CD80, and CD 86 on CCR5 + macrophage via better antigen presentation and co-stimulation of effector T cells. The present study suggested new approach in cancer immunotherapy by modulating dysregulated CCR5 signals from monocytes/macrophages.

Role of Eugenol from clove oil was studied in order to evaluate its chemo-preventive potential on N-methyl N-nitrososoguanidine induced gastric carcinogenesis in wistar rats. According to the study phytochemicals such as Eugenol (4-allyl-methoxy phenol), a natural phenolic constituent of clove oil, cinnamon, basil and nutmeg that were capable of targeting NF- κ B signaling pathway by manipulating equilibrium between pro and anti-apoptotic proteins as well as delicate balance between stimulators and inhibitors of invasion and angiogenesis were attractive candidates for tumor prevention.

